

Biomechanics of the knee joint

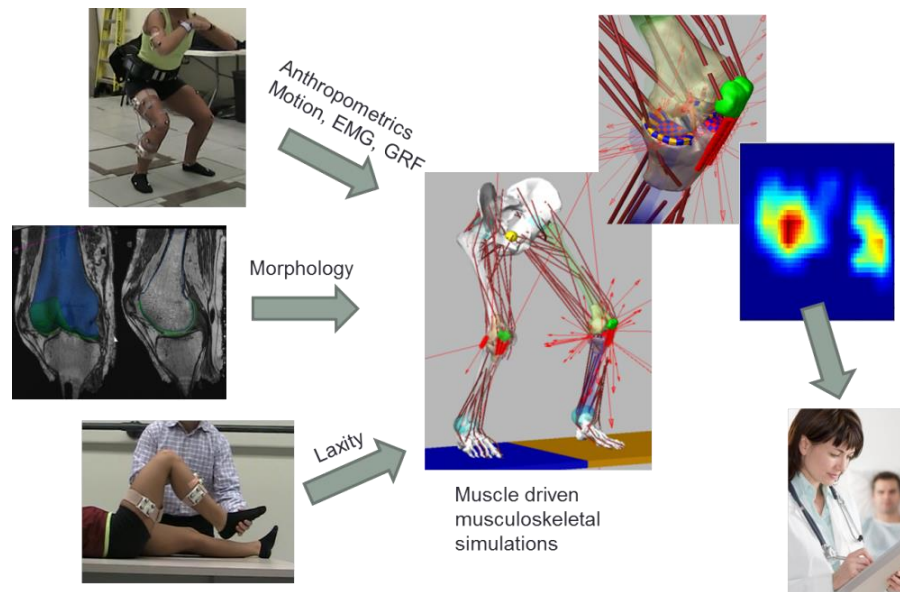
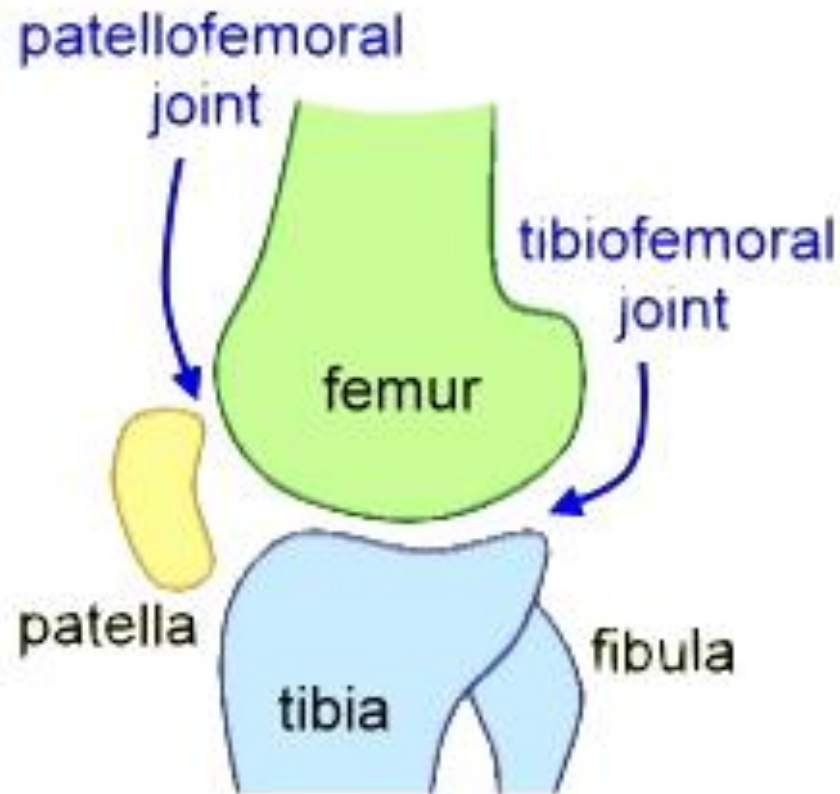


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Knee joint complex



Right Knee Complex
Medial View

Main biomechanical roles of the knee joint complex

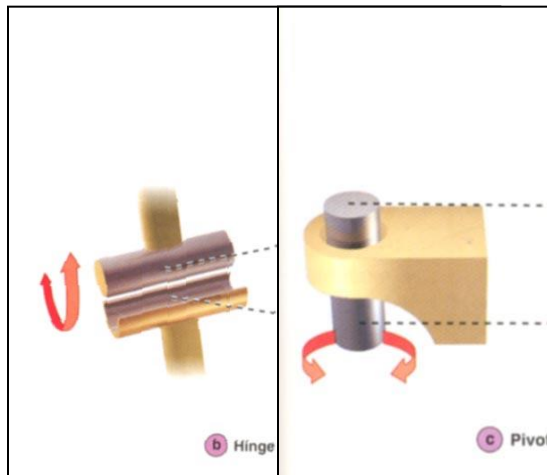
(by Masouros S.D. et al. Orthop Trauma, 2010, 24: 84-91)

- **To allow locomotion with (a) minimum energy requirements from the muscles and (b) stability, accommodating for different terrains.**
- **To transmit, absorb and redistribute forces caused during the activities of daily life**

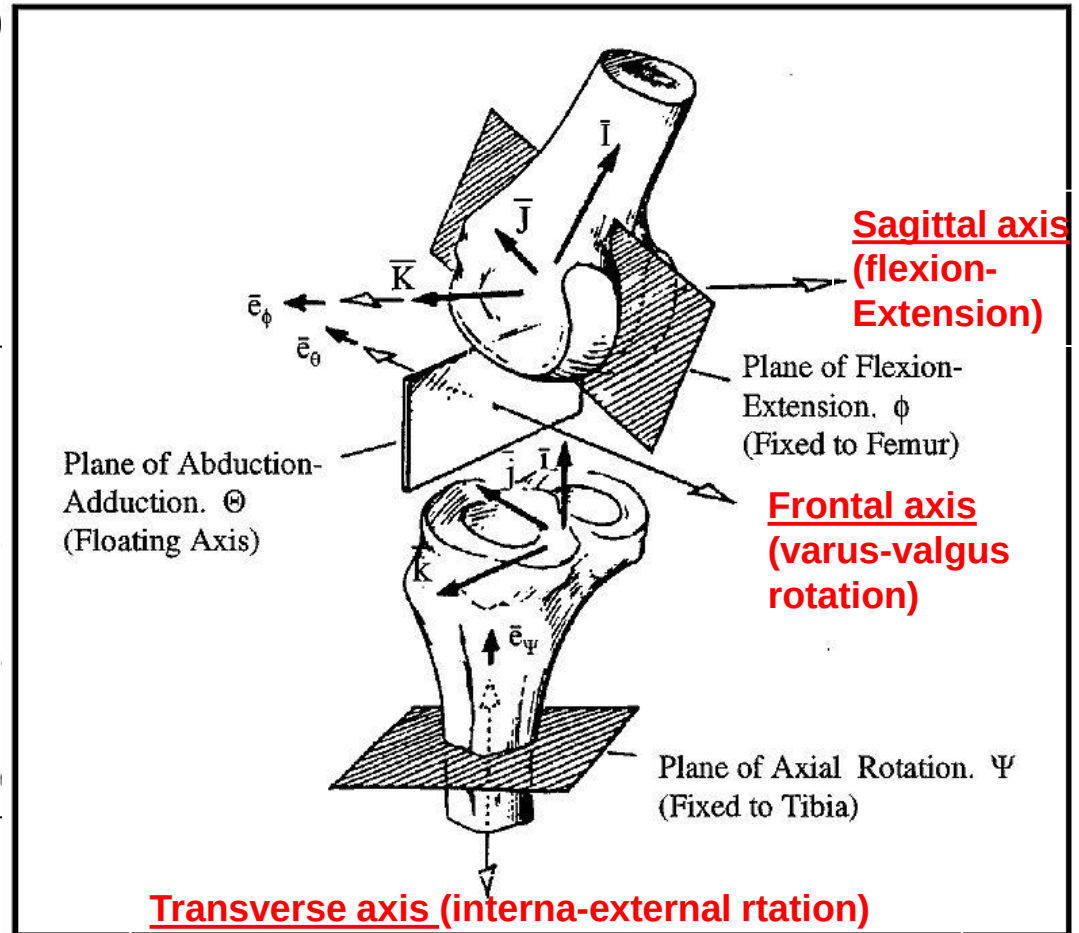
Arthrokinematics of the knee

Knee joint

- Ginglymus (**hinge**)
- Arthodial (**pivot**, gliding)
- 6 degrees of freedom
 - 3 rotations
 - 3 translations



Plokk-ratasliiges



Rotation and translation in knee joint

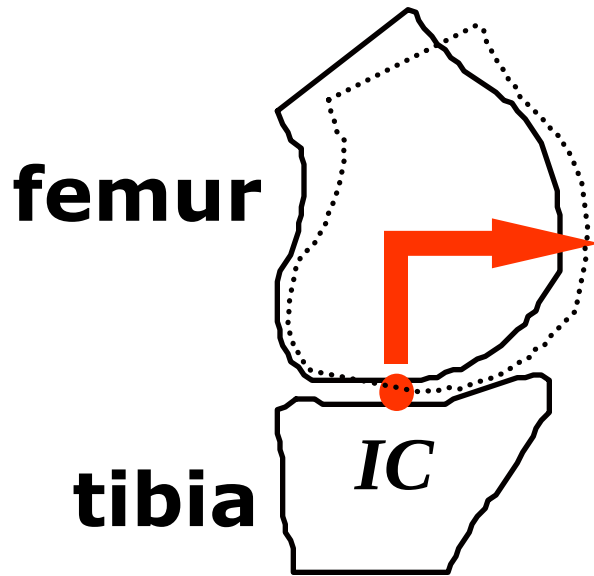
- **Rotation:**

- flexion-extension: up to 160 deg of flexion
(up to -5 deg flexion – hyperextension)
- varus-valgus: 6-8 deg in extension
- internal-external rotation: 25-30 deg in flexion

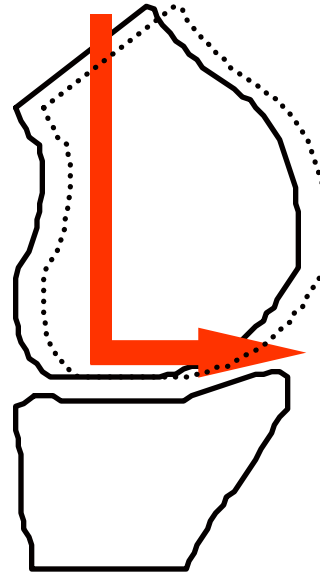
- **Translation:**

- anterior-posterior: 5–10 mm
- compression: 2–5 mm
- medio-lateral: 1-2 mm

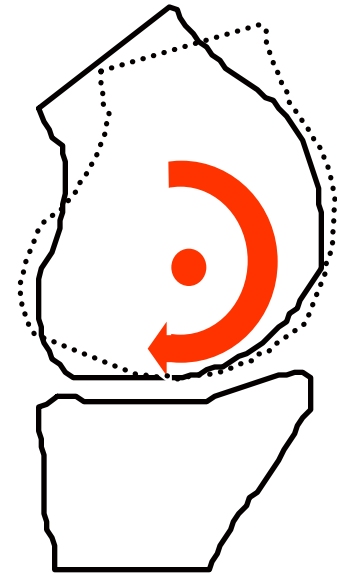
Centre of rotation for femur motion during flexion-extension (sagittal view)



Rolling

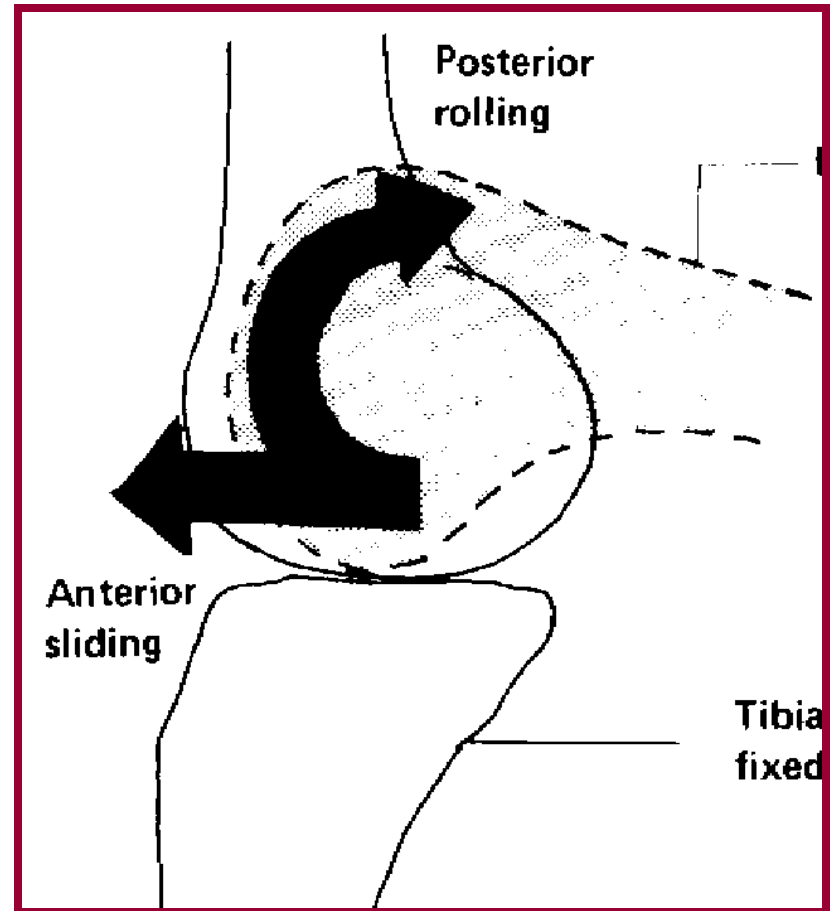
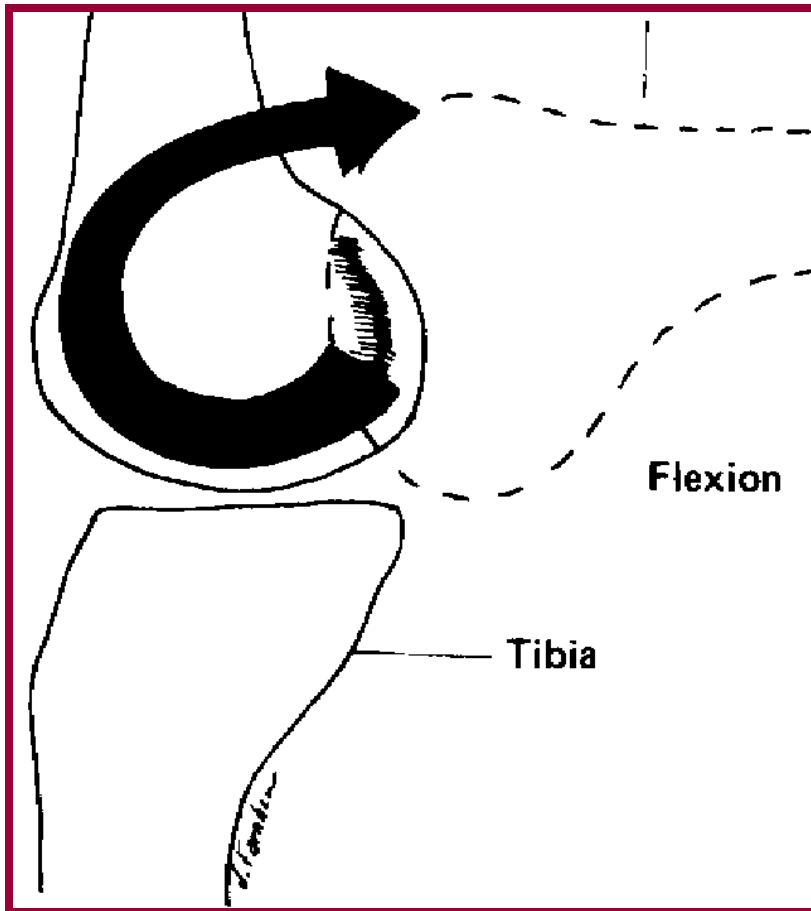


Sliding

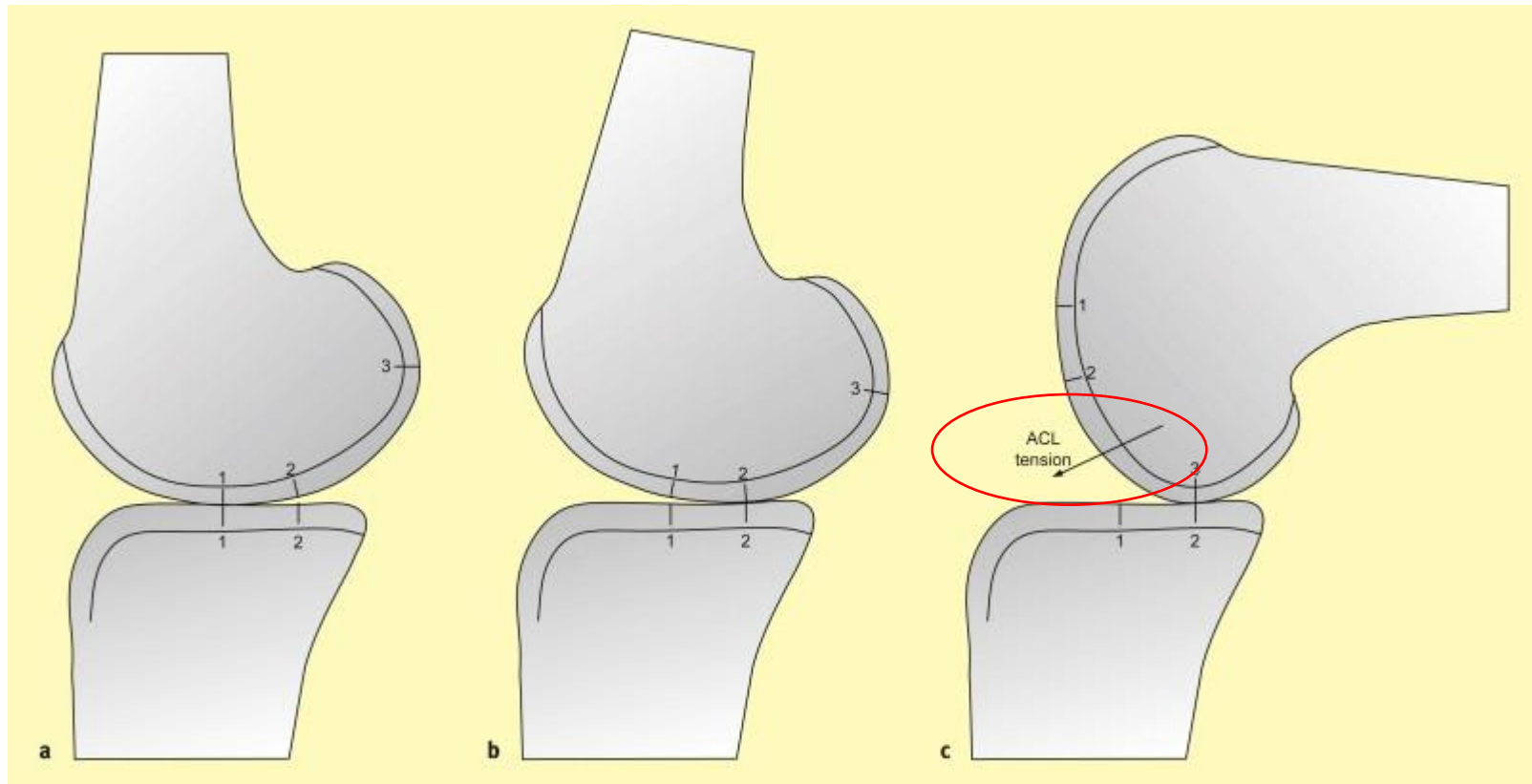


Pure rotation

Femoral condyles in flexion

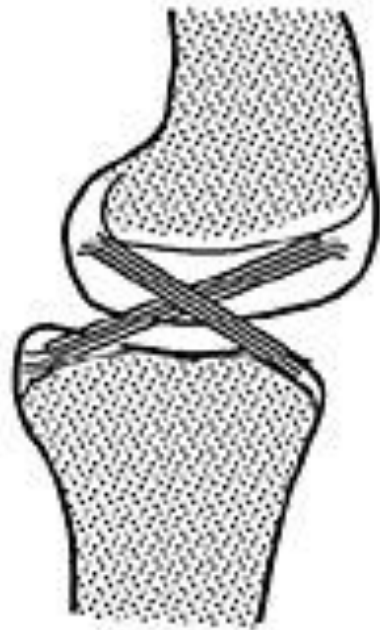


1st 25° - mainly roll >25° roll and ant glide

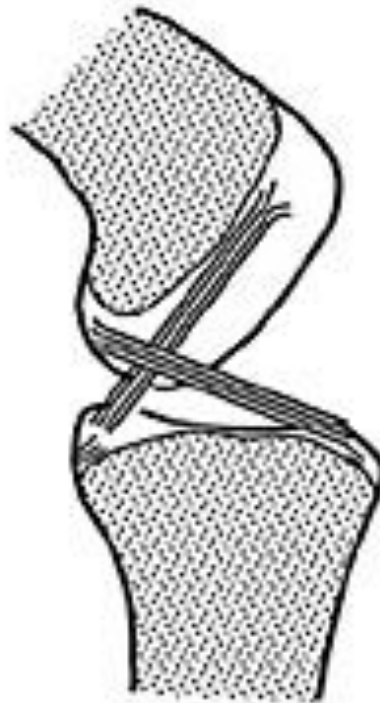


Knee joint kinematics in the sagittal plane during gait.

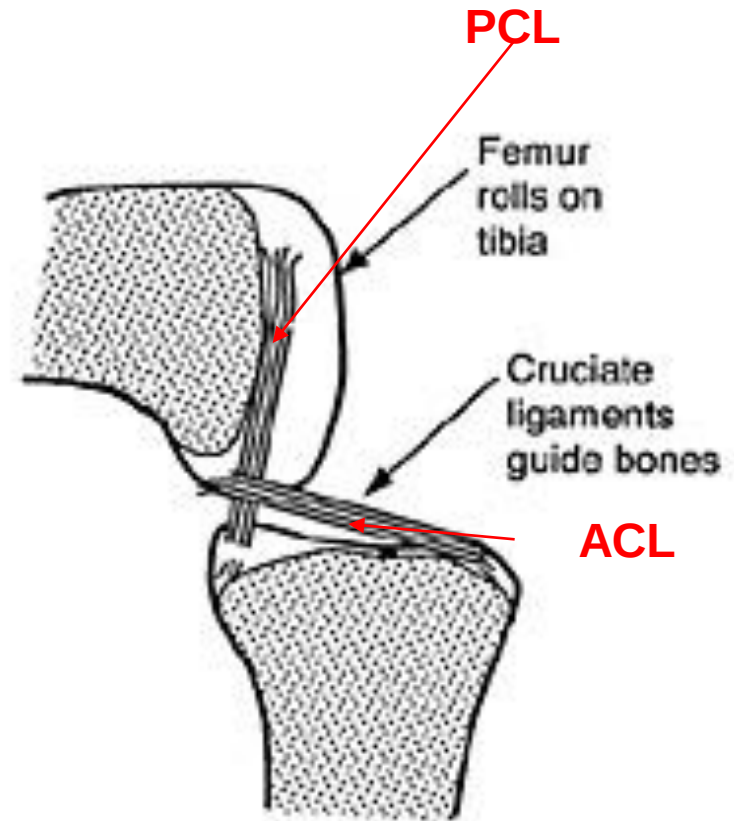
a Extension: contact is located centrally. b Early flexion: posterior rolling; contact continuously moves posteriorly. c Deep flexion: femoral sliding; contact is located posteriorly; the unlocking of the ACL prevents further femoral roll back. (Masouros S.D. Et al. Orthop Trauma, 2010, 24: 84-91)



(a) Straight leg



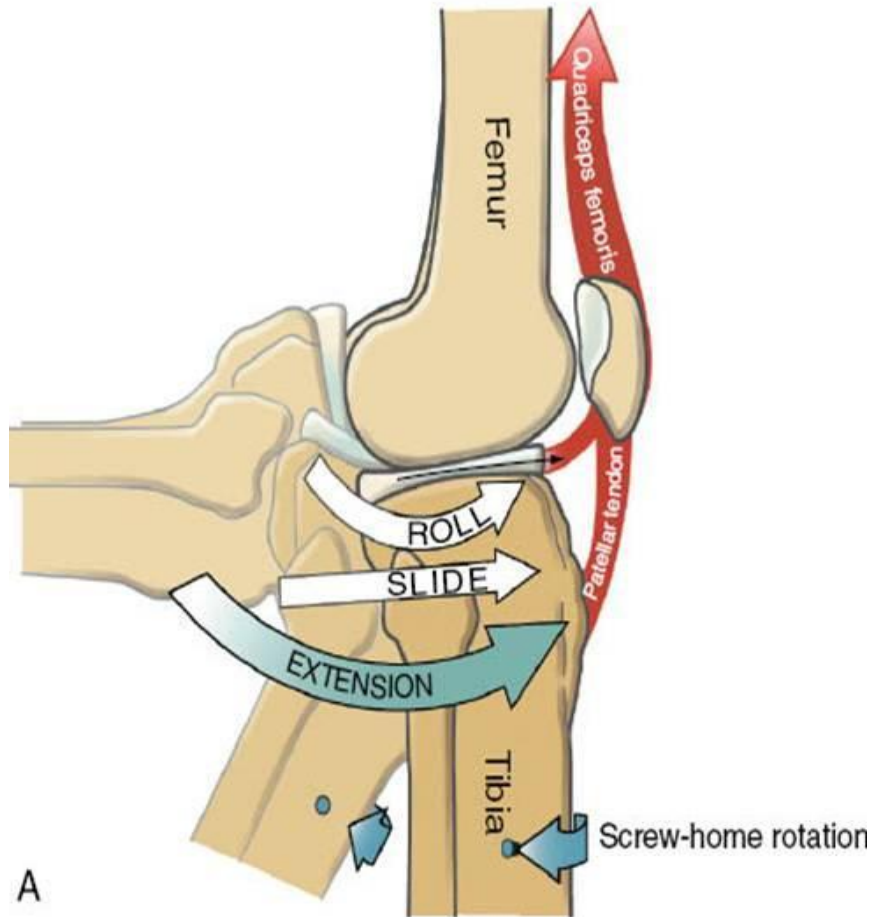
(b) 45 degrees bent



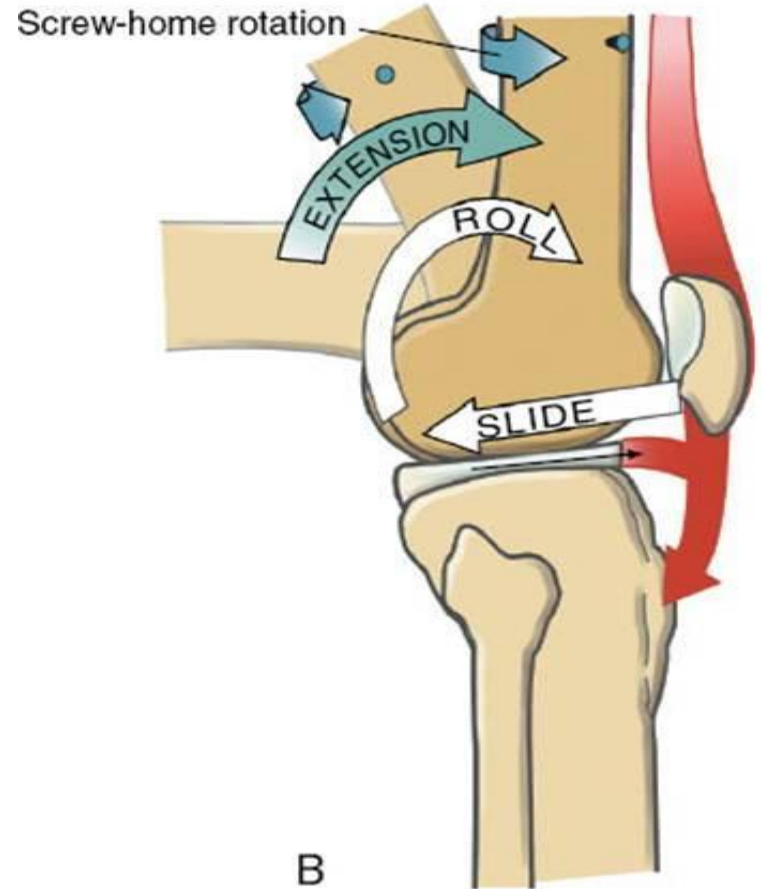
(c) 90 degrees bent

The irreducible mechanism of the knee (bones cut away to show cruciate ligaments).

Tibial-on-femoral extension



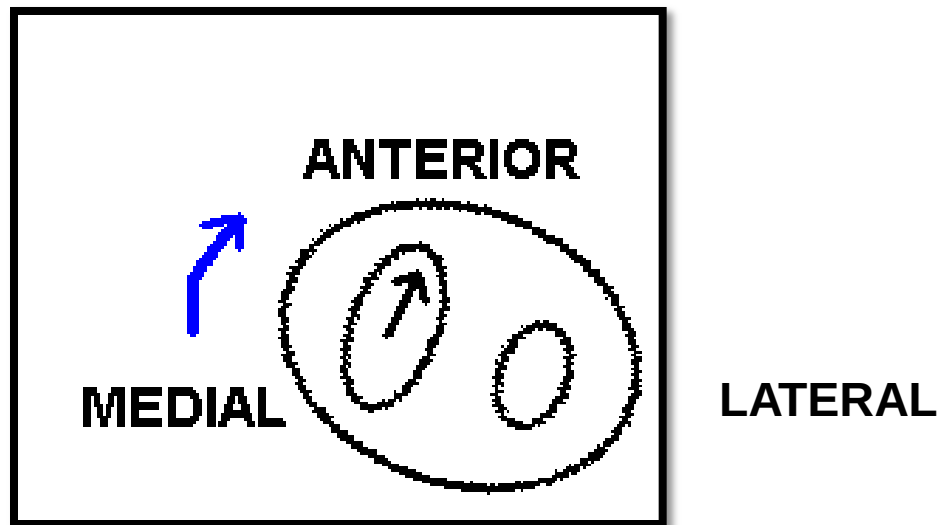
Femoral-on-tibial extension



Screw-home during knee extension

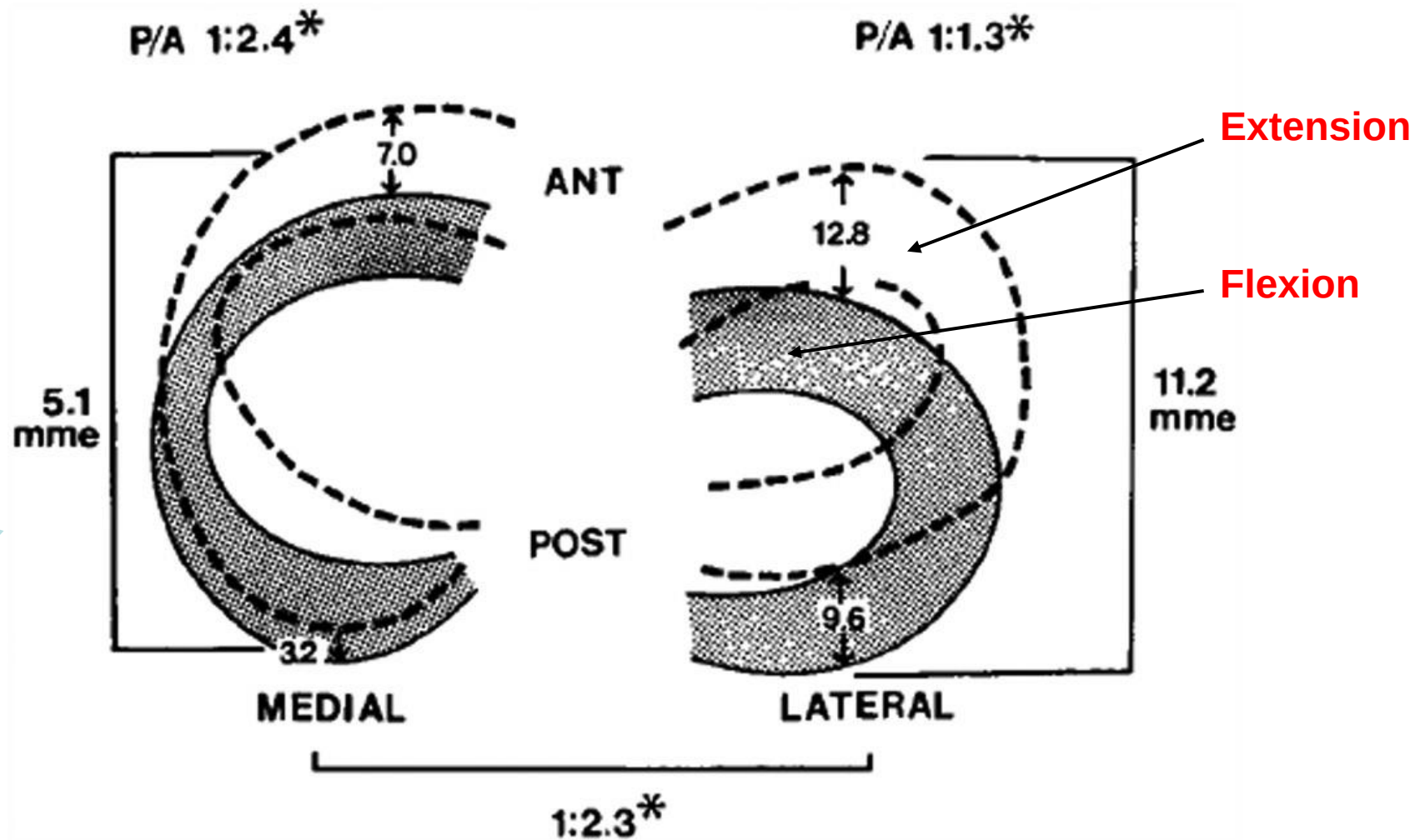
During the last 20 degrees of knee extension

- anterior tibial glide persists on the tibia's medial condyle because its articular surface is longer in that dimension than the lateral condyle's.
- Prolonged anterior glide on the medial side produces external tibial rotation, the "screw-home" mechanism.



Functional range of motion (ROM) at the knee

<u>Activities</u>	<u>Knee flexion</u>
◆ Normal gait/level surfaces	60°
◆ Stair climbing	80°
◆ Sitting/rising from most chairs	90°
◆ Sitting/rising from toilet seat	115°
◆ Advanced function	> 115°

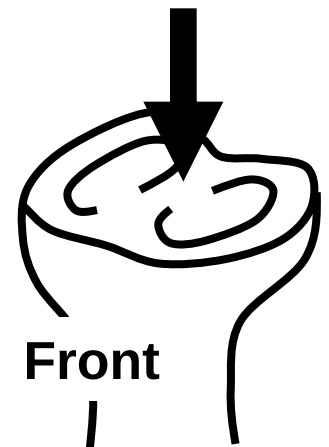


Diagrams showing the mean movement (mm) in each meniscus during flexion (shaded) and extension (hashed). ANT, anterior; POST, posterior; mme, mean meniscal excursion; P/A, ratio of posterior to anterior meniscal translation during flexion. (Thompson W.O. Am J Sports Med, 1991, 19: 200-216)

Knee joint loading

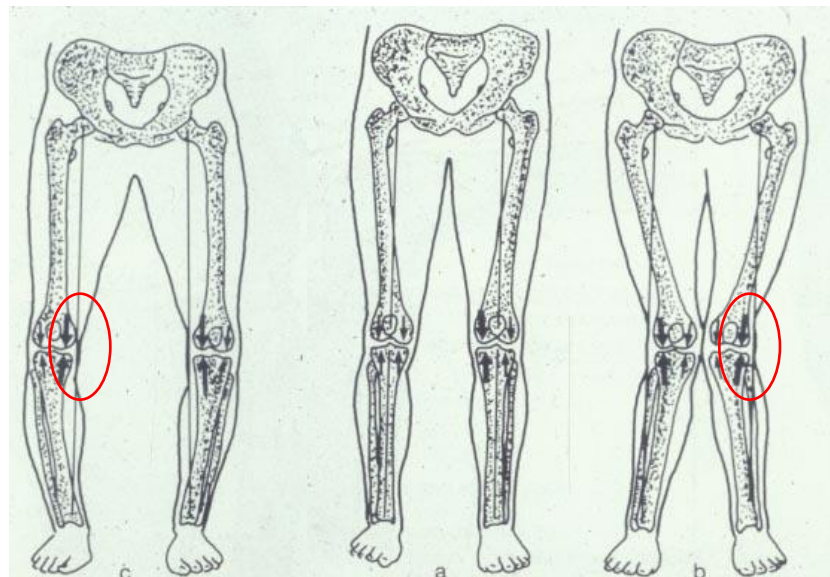
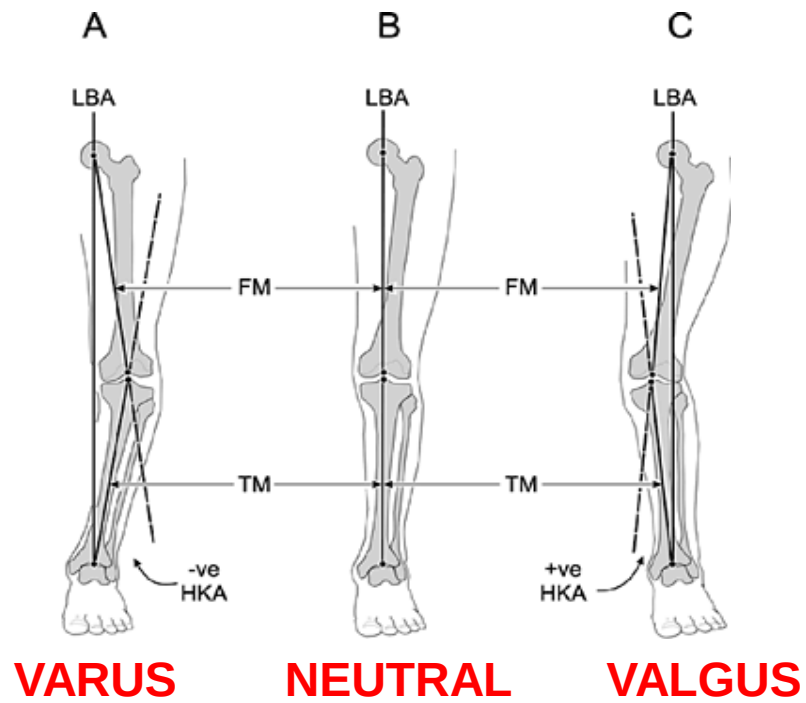
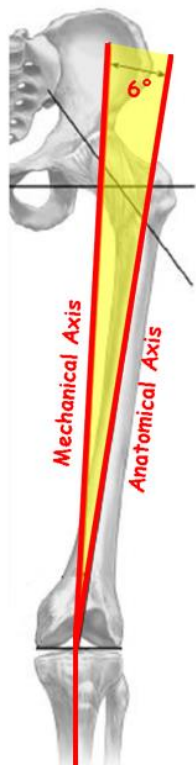
Tibiofemoral joint activities: Flexion & compressive load

• Cycling	60-100 deg	1.2BW
• Walking	15	3.0
• Stairs	60	3.8
• Stairs	45	4.3
• Squat-rise	140	5.0
• Squat-down	140	5.6



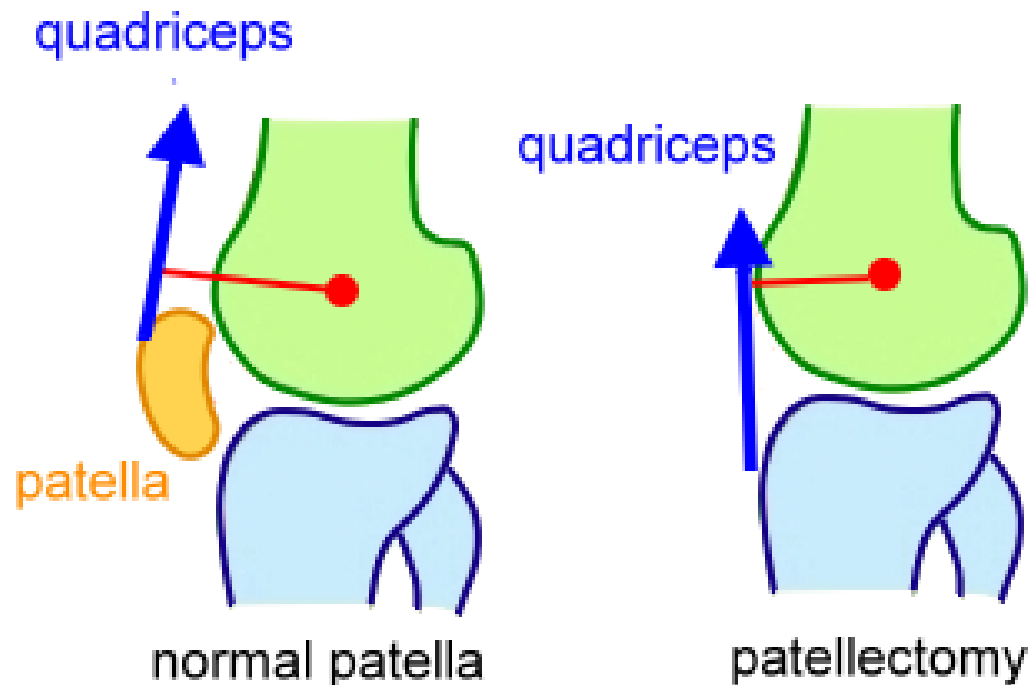
[Mow & Hayes, 1991]

BW = body weight



Function of the patella

- Largest sesamoid bone in the body
- Location: embedded in quadriceps tendon
- **Function:** to increase mechanical leverage of the quadriceps



Function of the Patella

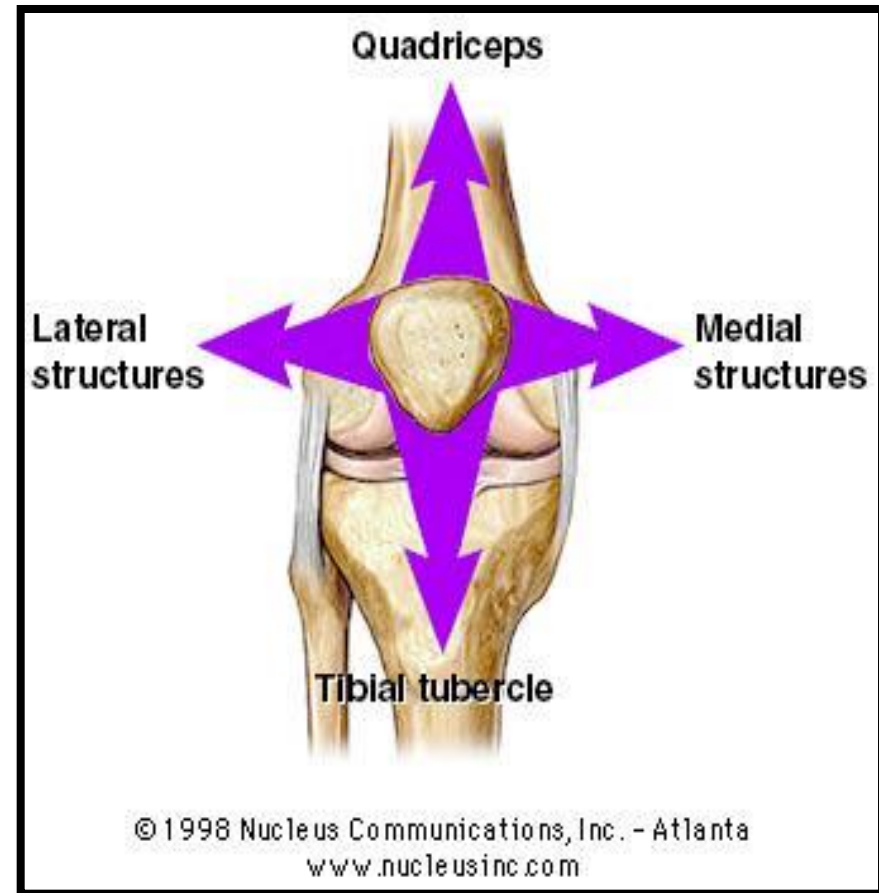
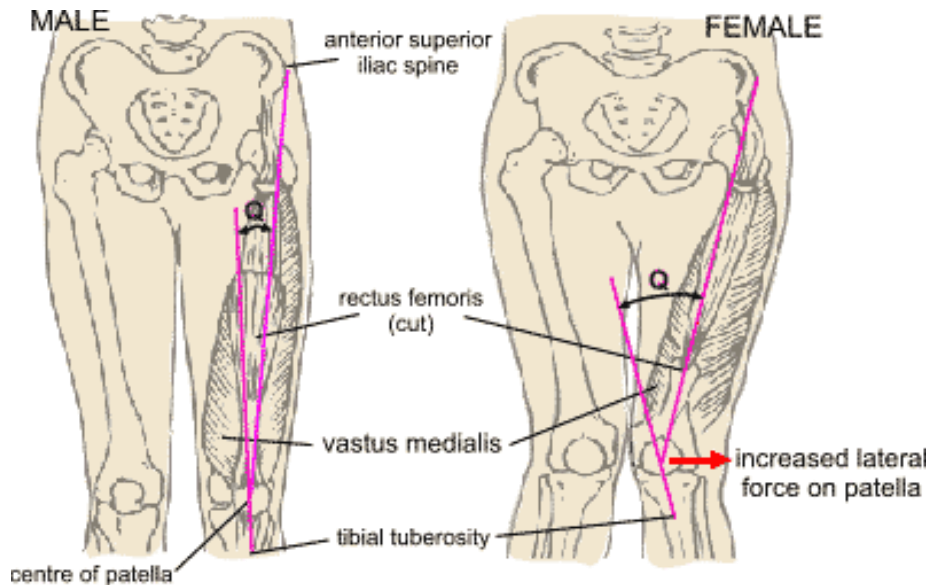
Forces acting on the patella:

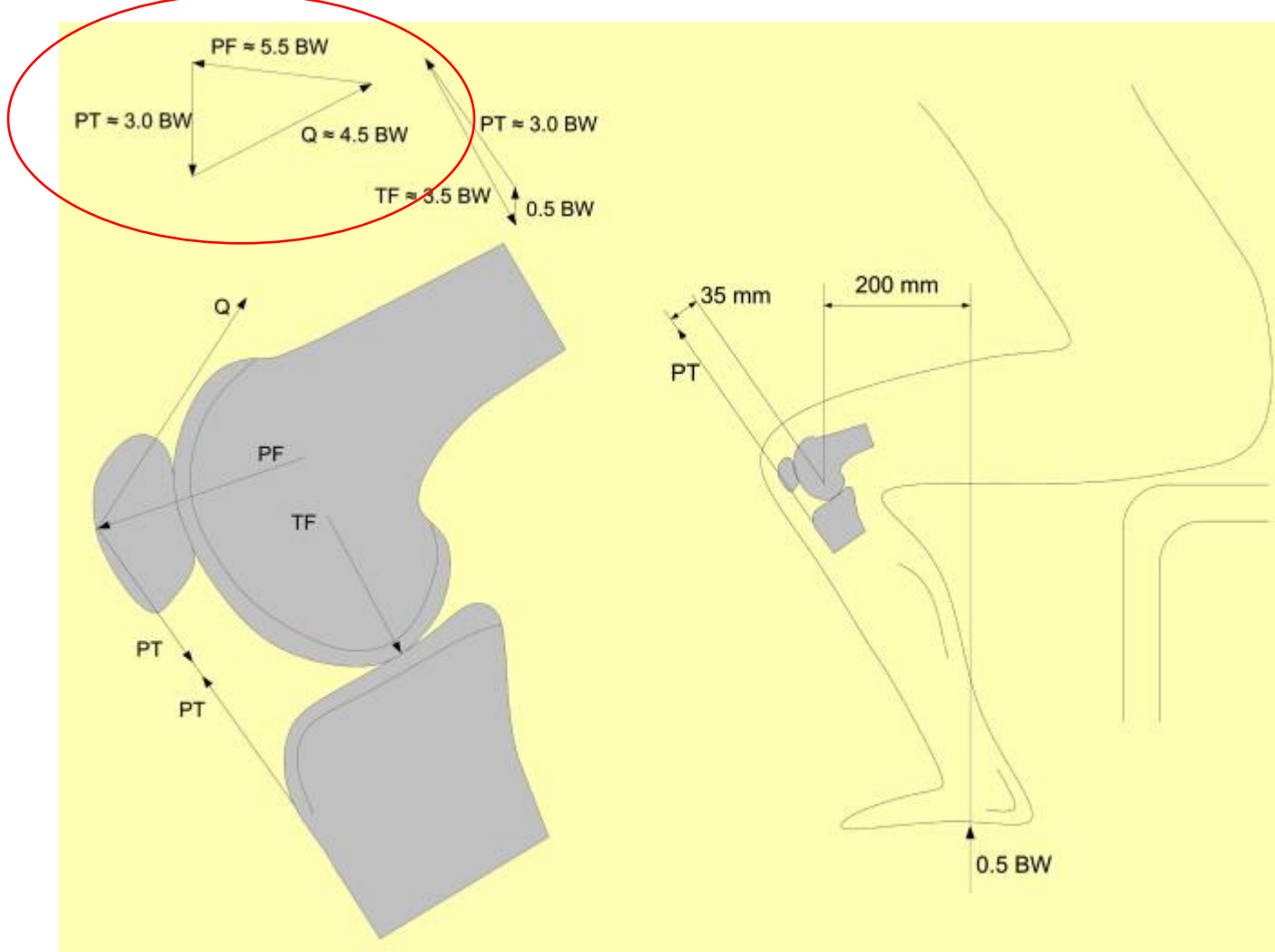
Laterally- lateral retinaculum, vastus lateralis m, iliotibial tract.

Medially- medial retinaculum and vastus medialis m.

Superior- Quadriceps via quadriceps tendon.

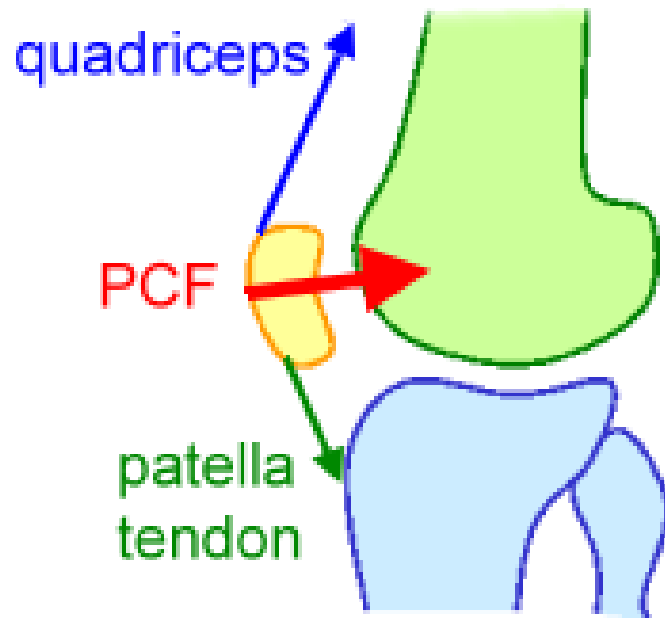
Inferior- Patellar ligament



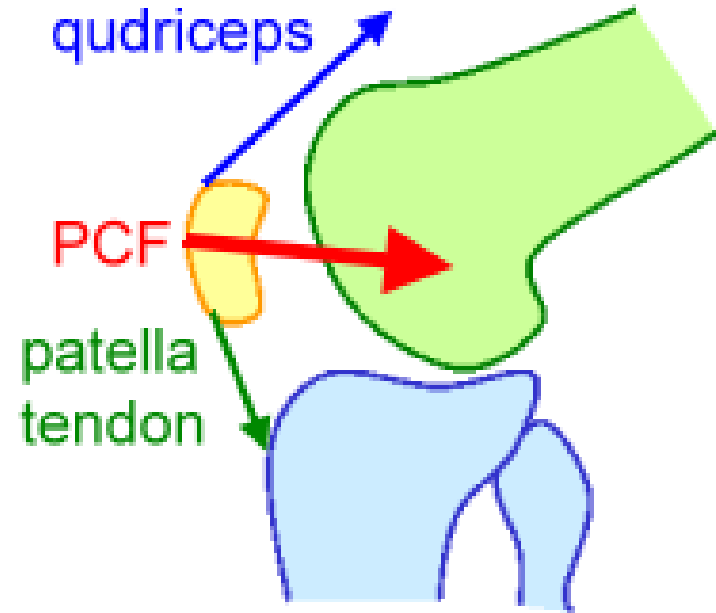


Rising up from a chair. Assuming that 0.5 BW is transferred through each leg applied at the foot, then its line of action is approximately 200 mm posterior to the joint centre; also, the PT line of action is approximately 35 mm off the joint centre. Moment equilibrium requires $PT \times 35 \text{ mm} = 0.5 \text{ BW} \times 200 \text{ mm}$, which results in $PT \approx 3.0 \text{ BW}$. At 90° flexion it has been shown that $PT \approx 70\% Q$, therefore $Q \approx 4.5 \text{ BW}$. Force equilibrium requires the triangle of forces acting at each joint to be closed; this results in joint forces of $PF \approx 5.5 \text{ BW}$ and $TF \approx 3.5 \text{ BW}$ (BW: body weight, PT: patellar tendon, Q: quadriceps muscle, PF: patellofemoral, TF: tibiofemoral). **(Masouros S.D. Et al. Orthop Trauma, 2010, 24: 84-91)**

Patellofemoral joint reaction forces



Knee Extension



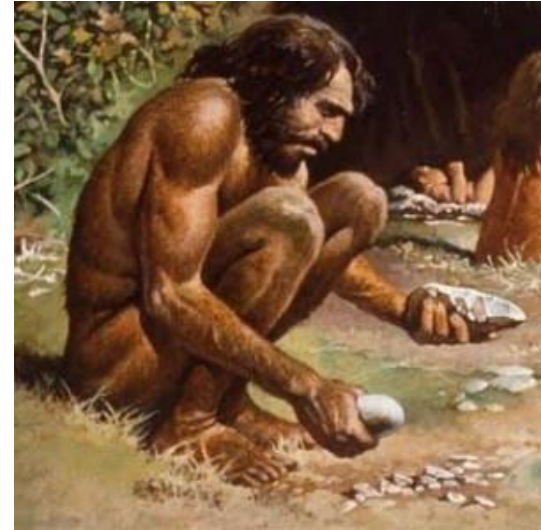
Knee flexion

Patellar Compression Force

Patellar compression force

Activity	Force	% Body weight
Walking	850 N	$1/2 \times \text{BW}$
Bike	850 N	$1/2 \times \text{BW}$
Stair ascend	1500 N	$3.3 \times \text{BW}$
Stair descend	4000 N	$5 \times \text{BW}$
Jogging	5000 N	$7 \times \text{BW}$
Squatting	5000 N	$7 \times \text{BW}$

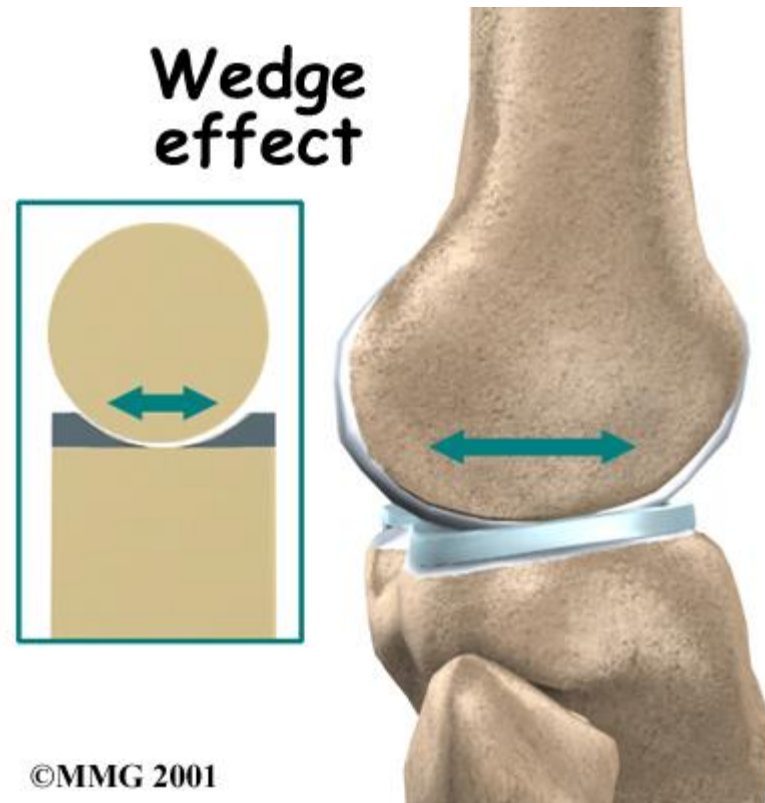
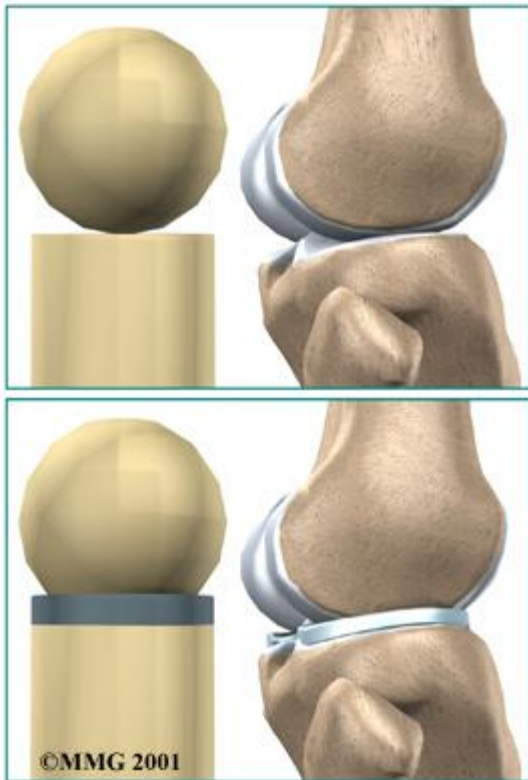
- Squat exercise stressful to the knee complex, produces a patellofemoral joint reaction force 7.6 times body weight.
- It one-half of body weight during normal walking, increasing up to over three times body weight during stair climbing.



Menisci

- **Two functions:**
 - **load bearing**
 - **stability**
 - **also, joint lubrication**
 - **prevent capsule, synovial impingement**
 - **shock absorbers**

Menisci



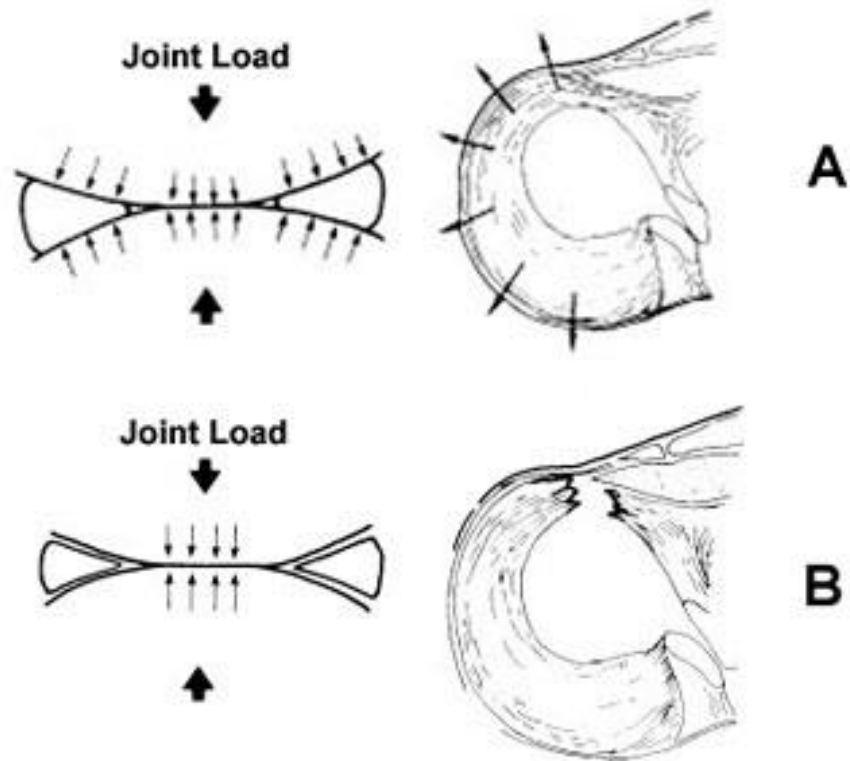
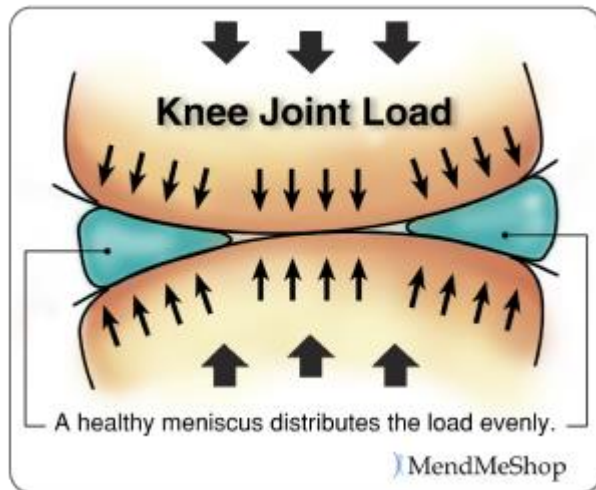
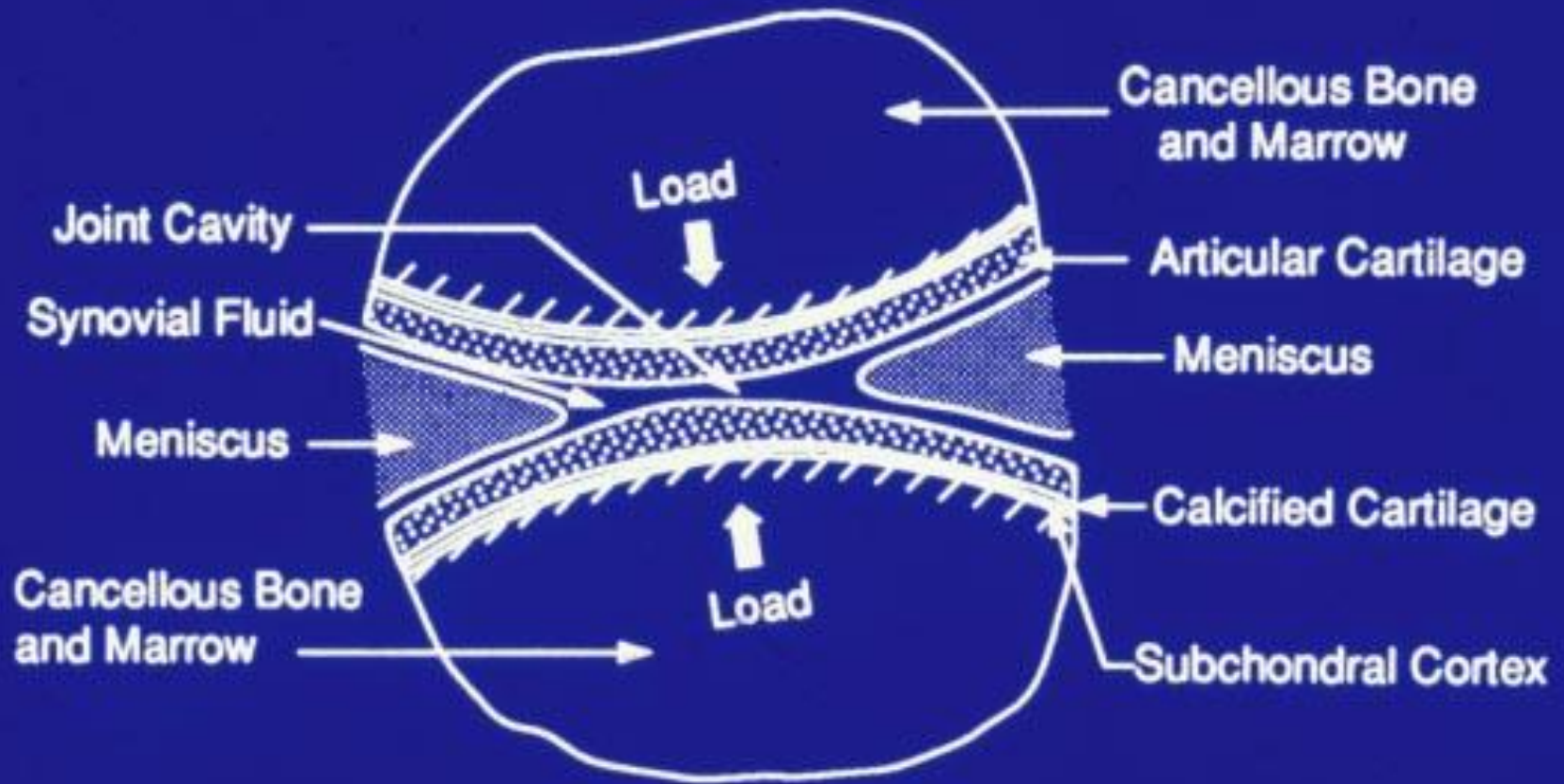
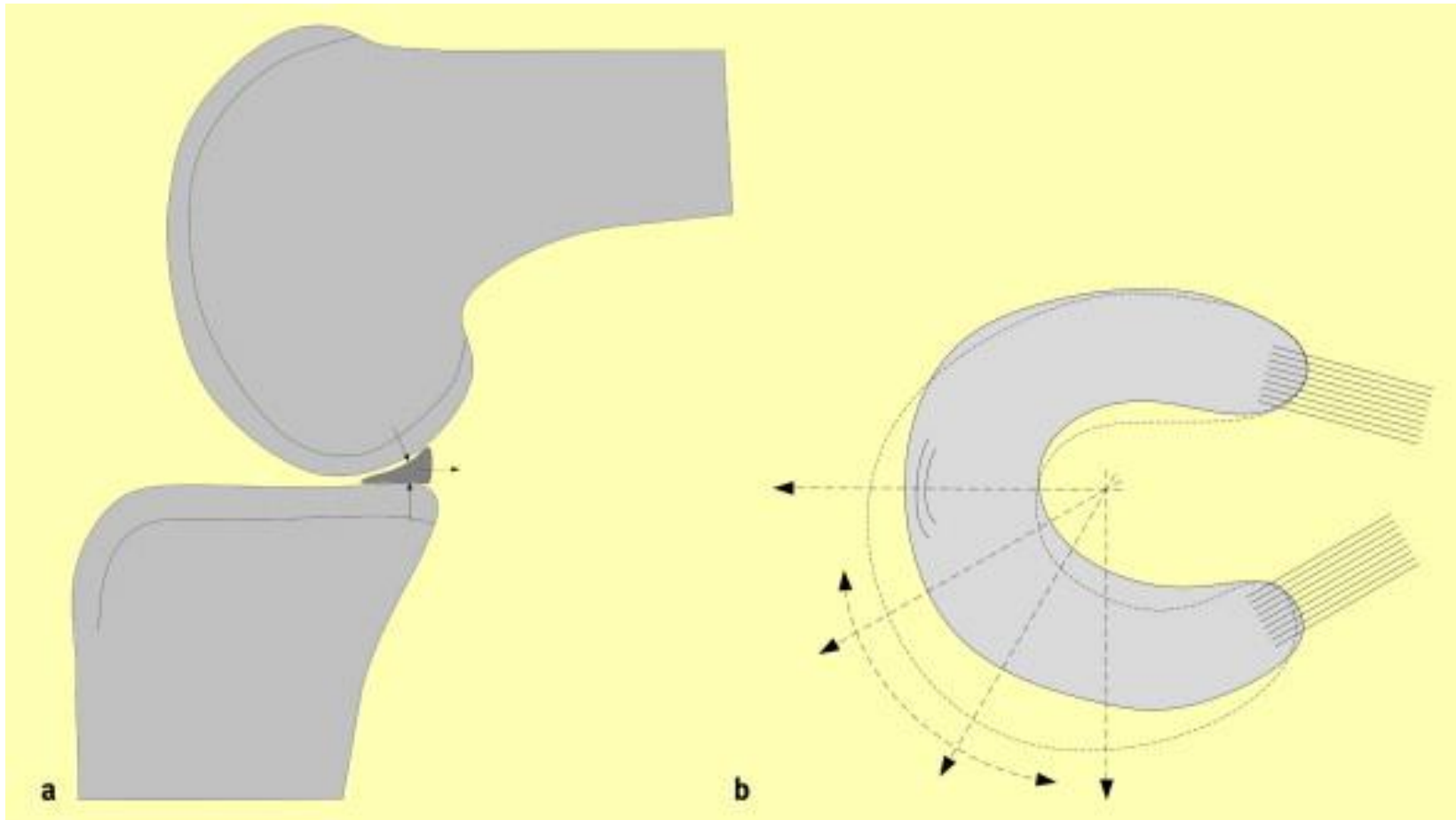


Diagram demonstrates the importance of intact meniscal entheses for the load distribution function of the meniscus. (A) With intact entheses the load (thick arrows) is transmitted via the menisci and articular cartilage through a large contact area (left hand picture, small arrows). Part of the load is transformed to hoop stresses (right hand picture, long arrows). (B) When the insertional ligaments are transected, the menisci will extrude from the knee joint during loading, and the load is mainly transmitted via articular cartilage through a reduced contact area (small arrows). (Grood ES: Adv Orthop Surg 7:193,1984.)

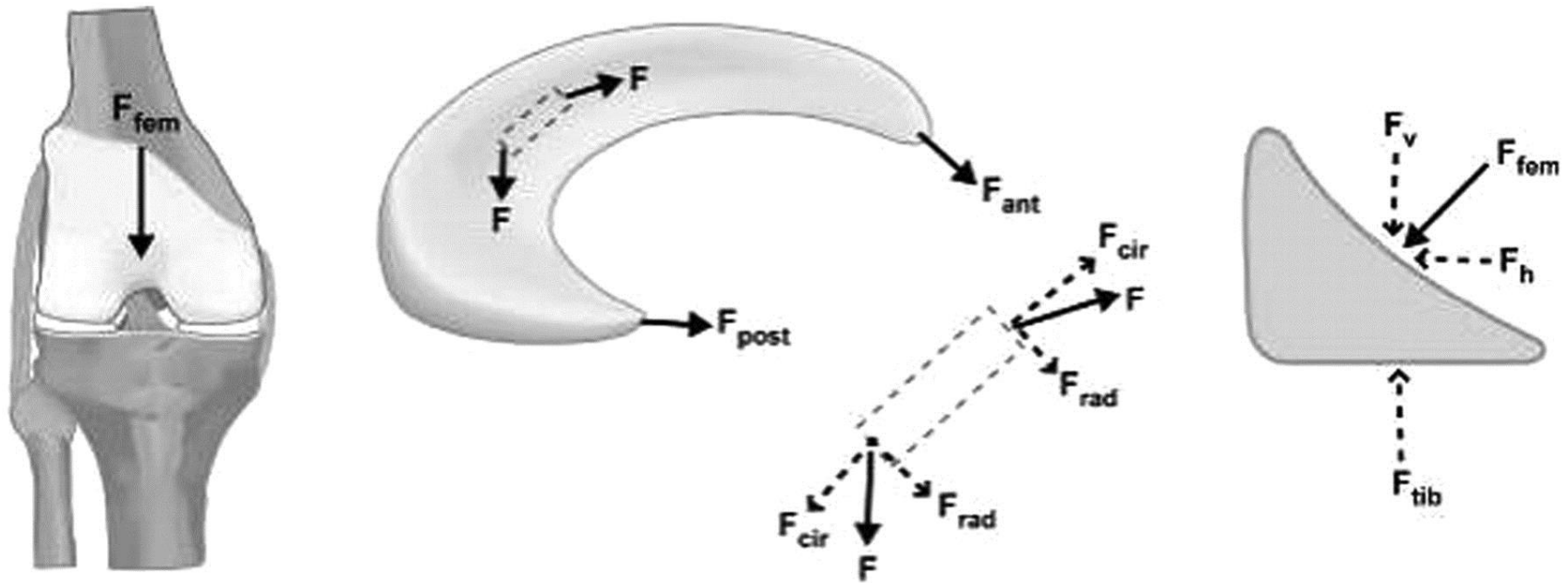
Load-bearing function of the menisci



(b) Load Bearing Region



The load-bearing mode of the menisci. **a** Sagittal section: as the femur compresses through the meniscus onto the tibia, it pushes the meniscus out of the joint cavity; the meniscus deforms to conform with the femoral condyle and allows for the contact to be distributed over a larger area. **b** Top view: the meniscus increases its circumference and moves radially outwards and posteriorly with knee flexion, following the rolling and sliding of the femoral condyle with flexion. (Masouros S.D. Et al. Orthop Trauma, 2010, 24: 84-91)



Free body diagram of forces acting on the meniscus during loading. As the femur presses down on the meniscus during normal loading, the meniscus deforms radially but is anchored by its anterior and posterior horns (F_{ant} and F_{post}). During loading, tensile, compressive, and shear forces are generated. A tensile hoop stress (F_{cir}) results from the radial deformation, while vertical and horizontal forces (F_v and F_h) result from the femur pressing on the curved superior surface of the tissue. A radial reaction force (F_{rad}) balances the femoral horizontal force (F_h).

(Athanasίου K.A., Sanchez-Adams J. Engineering of the knee meniscus. 2009)

Knee biomechanics during gait

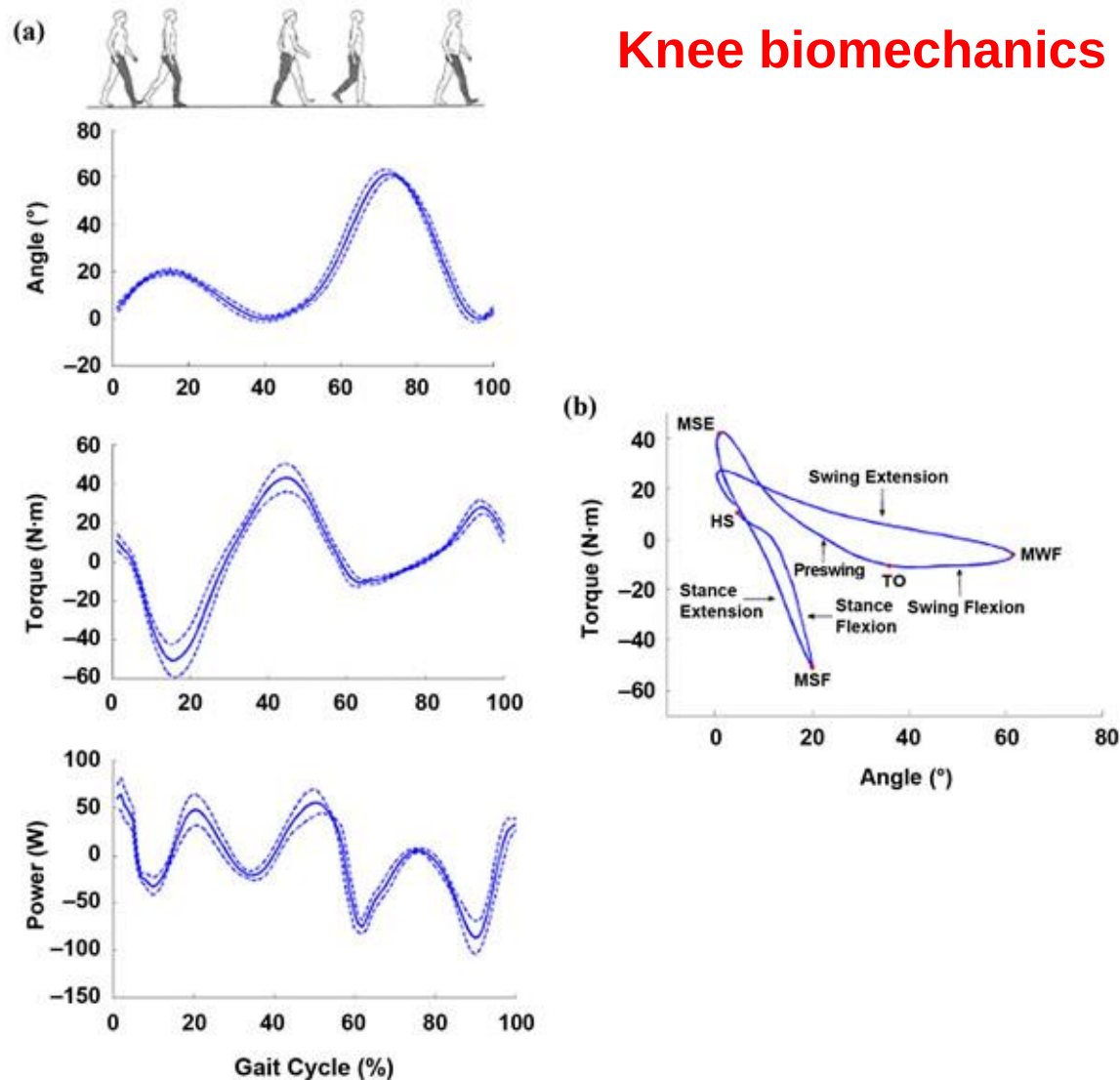
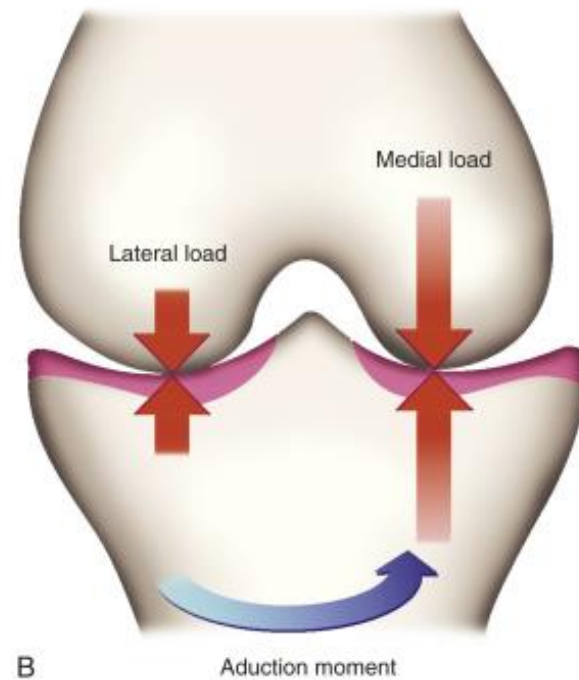
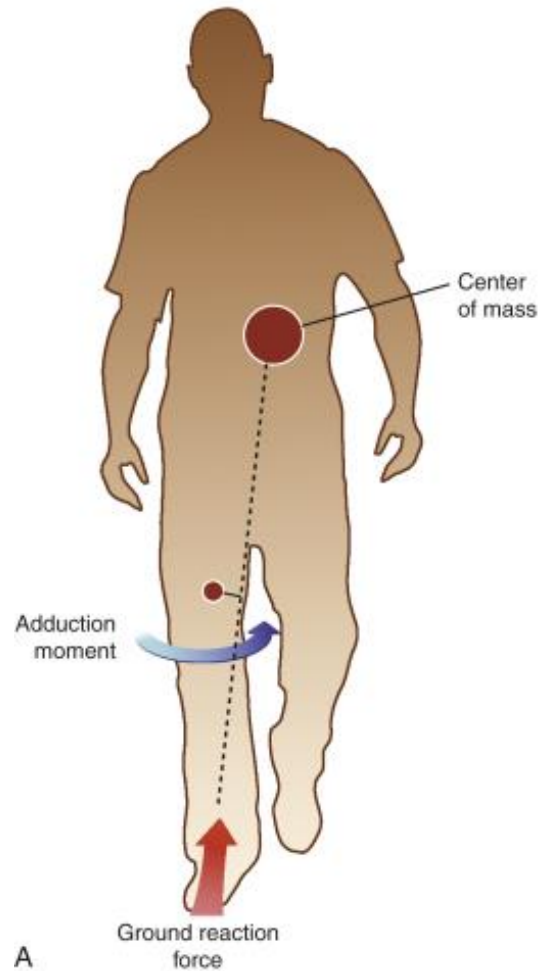
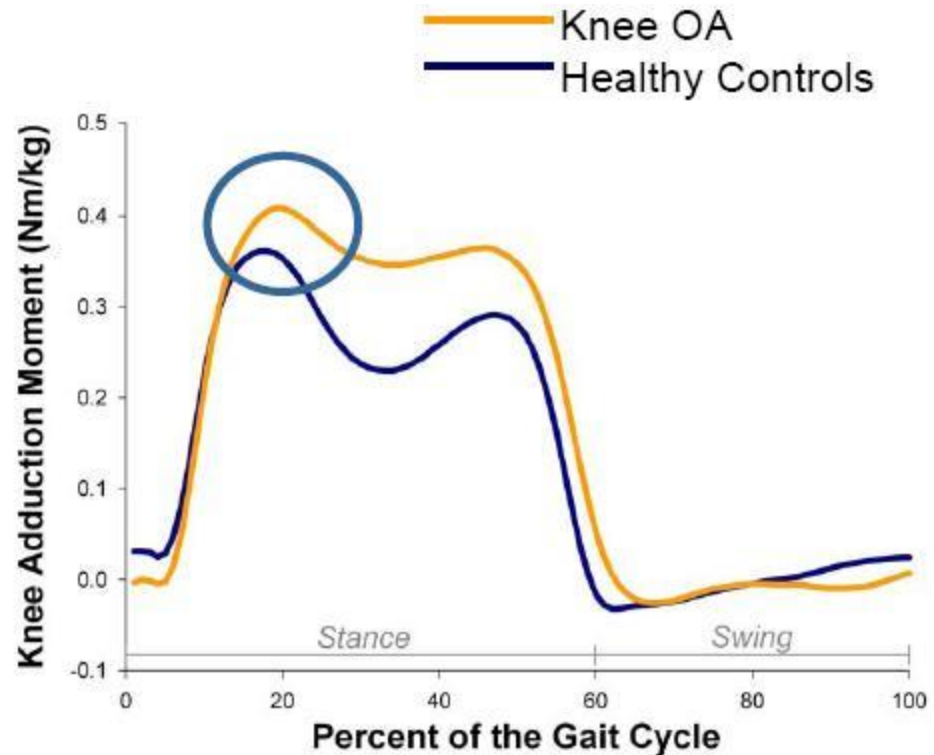
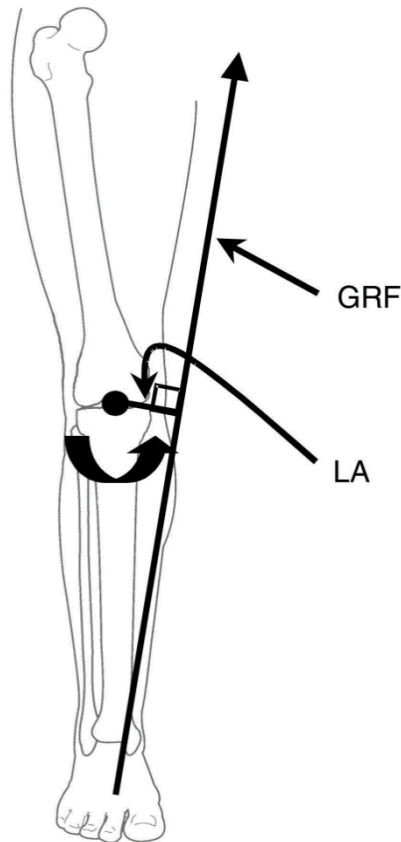


Figure 1.

Representative knee biomechanics in level-ground walking. (a) Knee angle, torque, and power curves of male study participant (mass = 81.9 kg) are plotted against percent gait cycle during level-ground walking at self-selected speed (1.31 m/s). Plotted are mean (solid line) ± 1 standard deviation (dashed lines) for $n = 10$ gait trials. (b) Knee torque is plotted versus knee angular position showing five phases of gait. Key gait events separating five phases are heel strike (HS), maximum stance flexion (MSF), maximum stance extension (MSE), toe-off (TO), and maximum swing flexion (MWF). Detailed description of how these gait data were collected and analyzed is presented in the "Methods—Intact-Limb Walking: Data Collection and Analysis" section (p. 365).

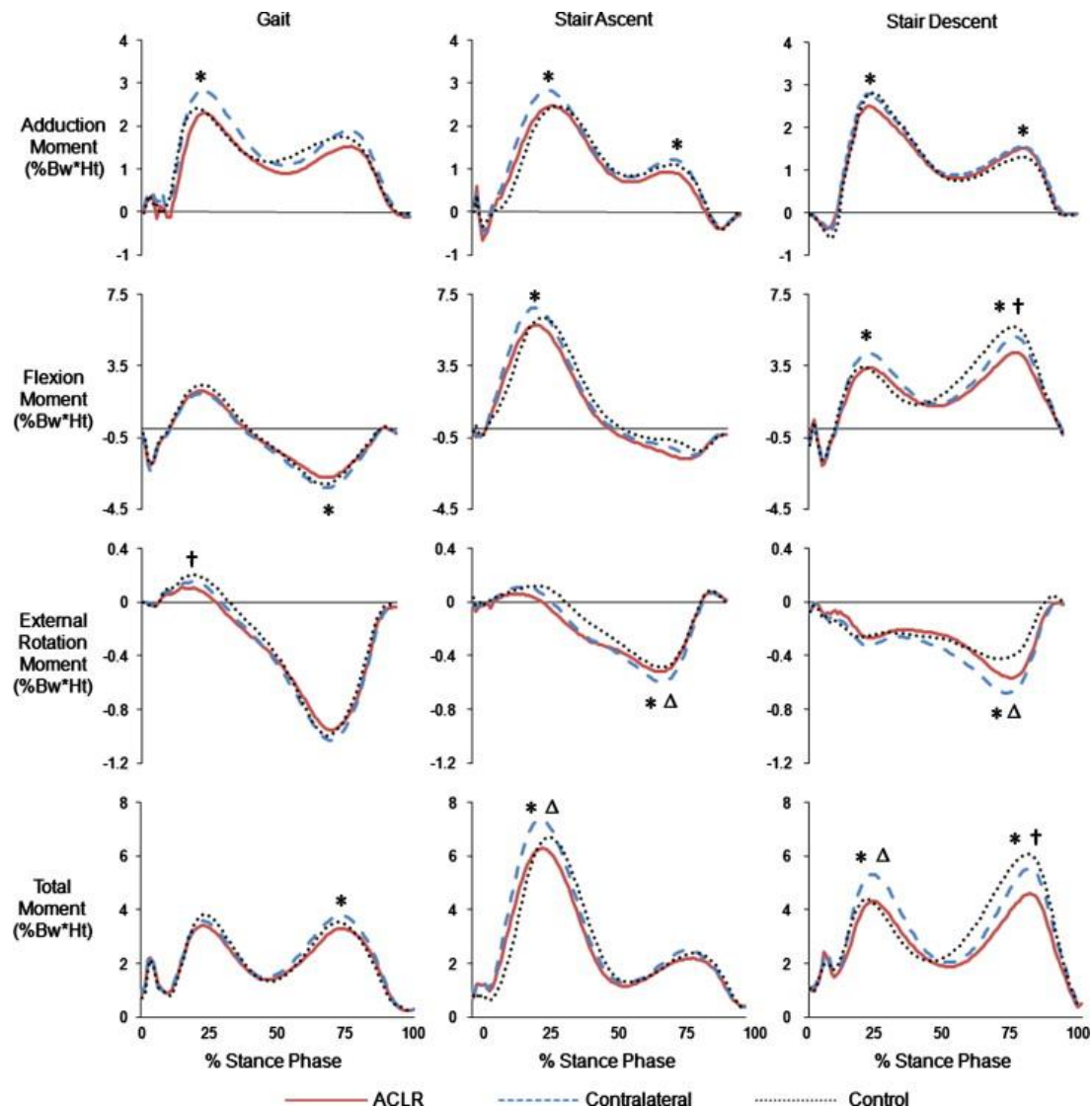
Knee adduction moment and compressive loads during gait





The knee adduction moment is a result of the magnitude of the ground reaction force (GRF) times the distance (i.e. moment arm) from the center of rotation ($GRF \cdot LA$). The graph of the knee adduction moment during a gait cycle in a patient with knee OA is characterized by an increase in the peak and impulse (the area under the curve) of the moment (Presented by Mali M., 2007).

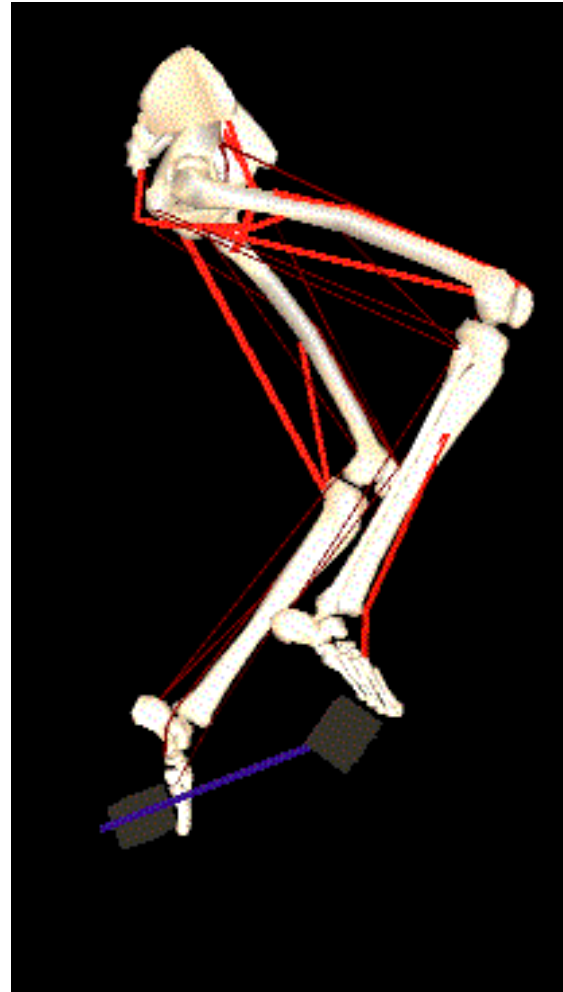
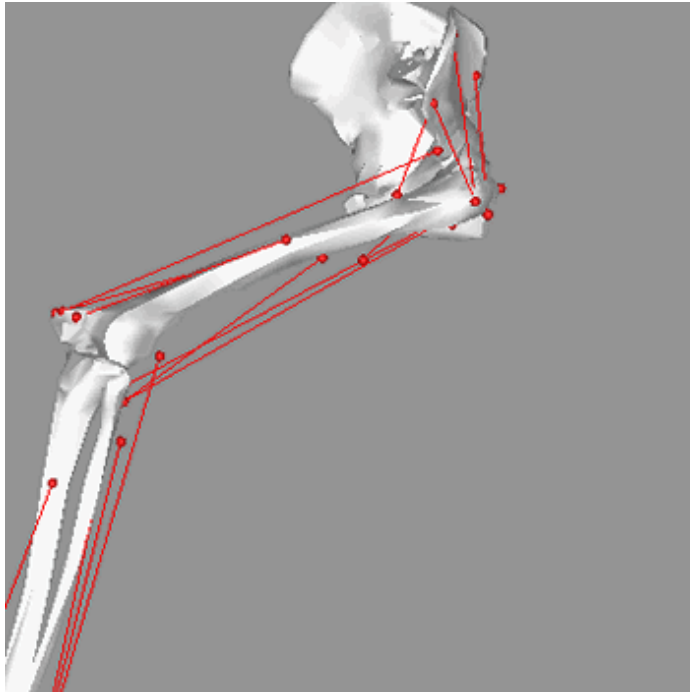


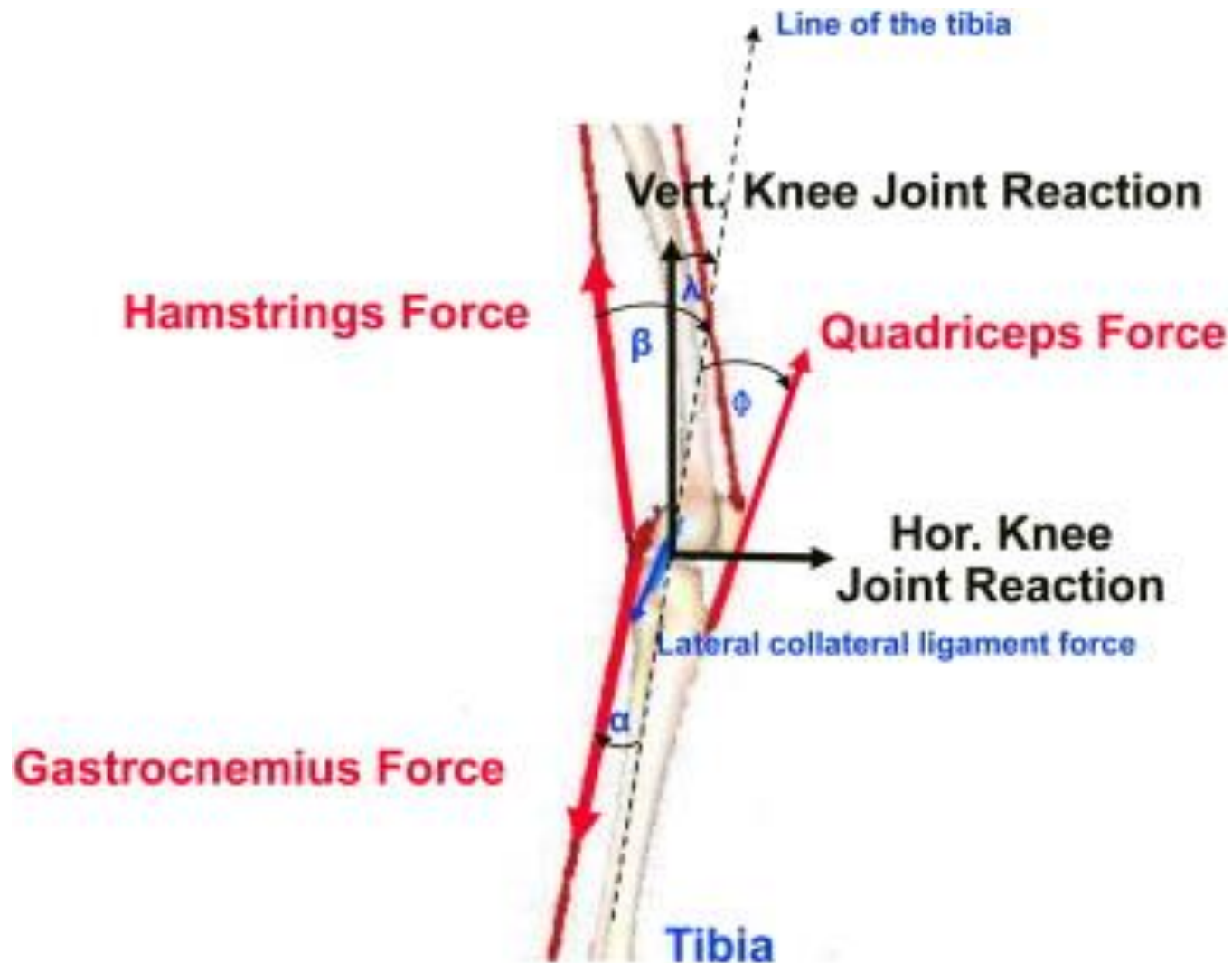


All three moments and the total moment at the knee during gait, stair ascent, and stair descent for ACLR, contralateral, and control knees. (*) indicates significant difference between ACLR and contralateral, (Δ) indicates significant difference between contralateral and control, and (†) indicates significant difference between ACLR and control knees.

Zabala M.E. et al. J Biomech 2013, 46: 515-520

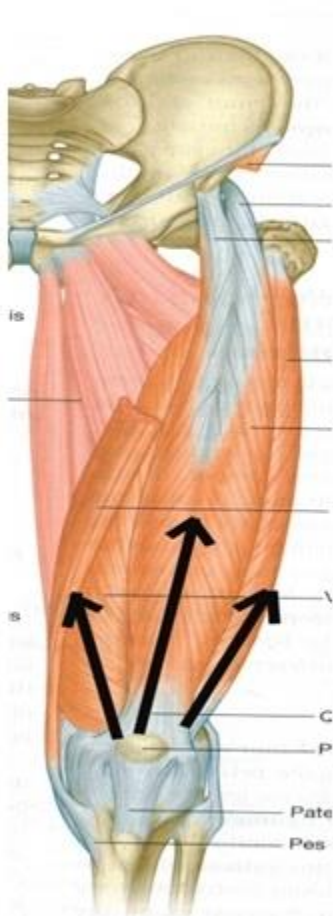
Soft tissue mechanics



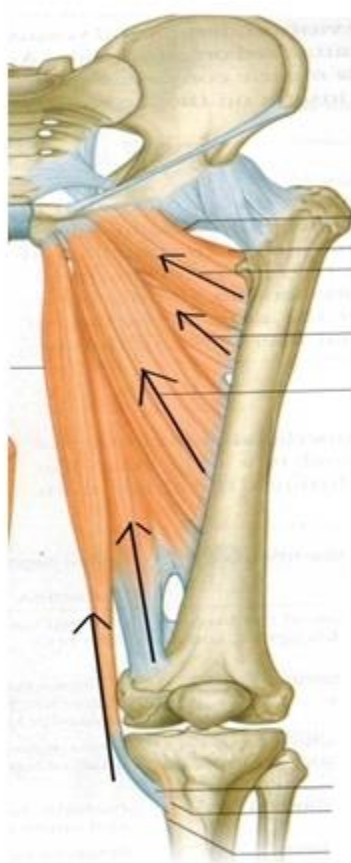


Schematic representation of the biomechanical musculoskeletal knee model used to calculate knee joint loads and muscle forces

Messier S.P. et al. Osteoarthritis Cartilage 2011, 19: 272-280



Quads direction
of pull



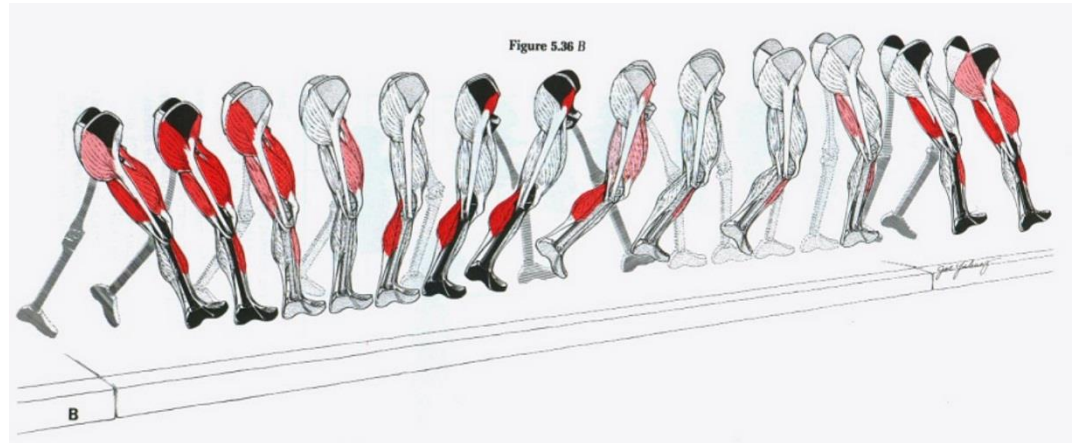
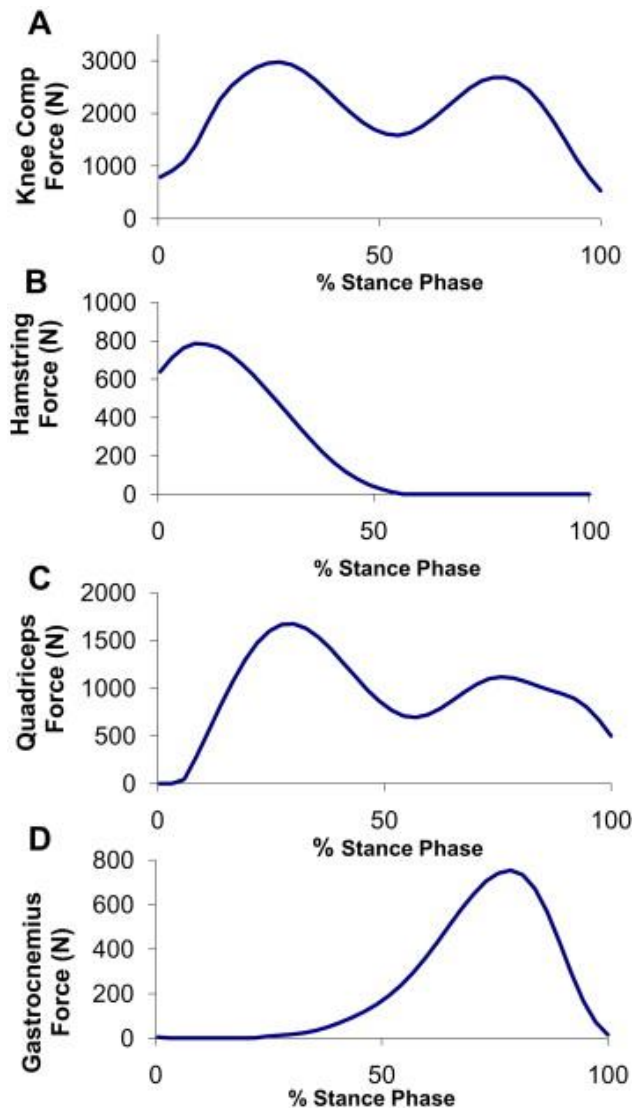
Adductors line of
pull



Hamstrings line
of pull

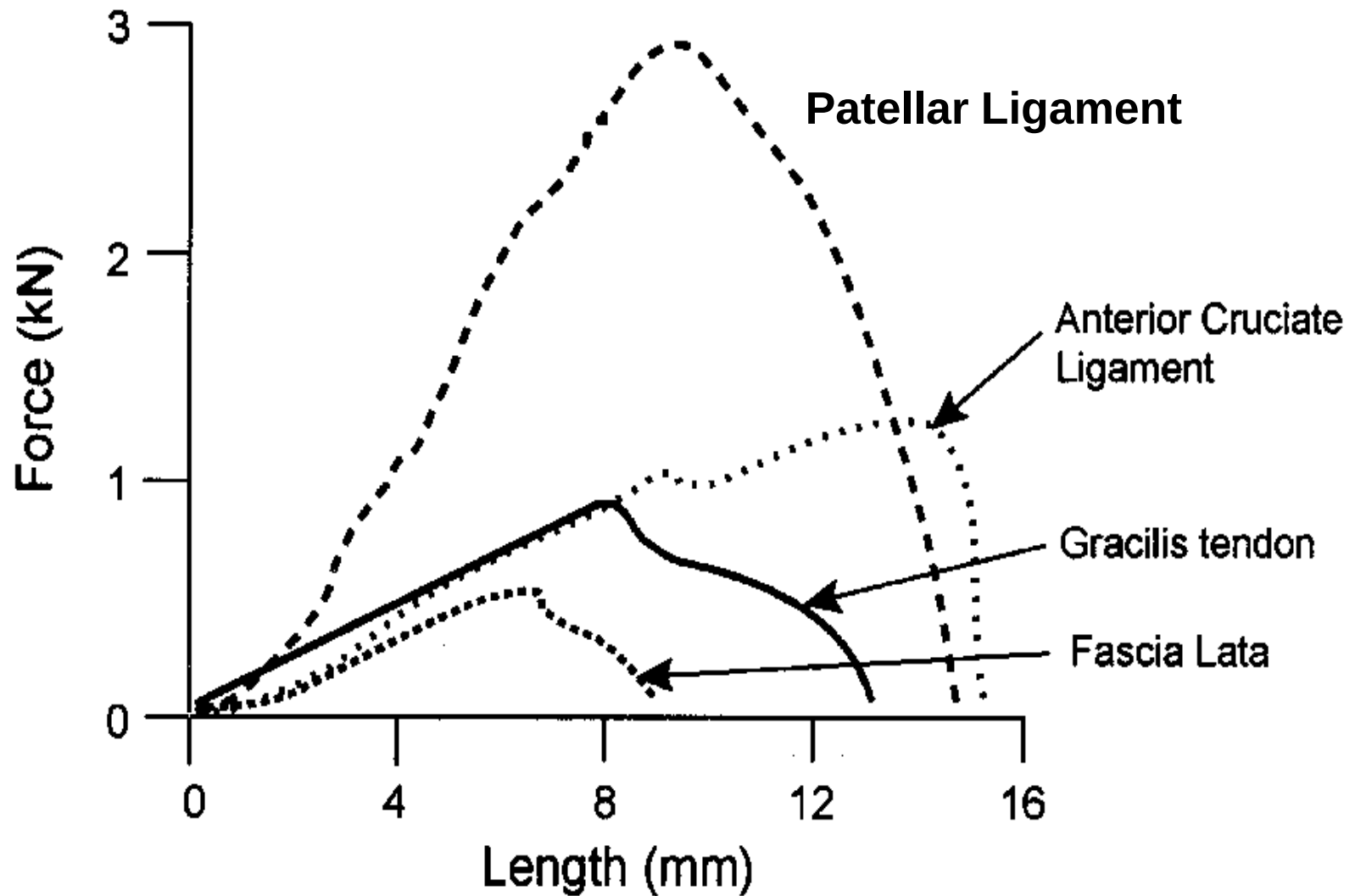


All muscles



(A) Knee compressive, and (B) Hamstring, (C) Quadriceps, and (D) Gastrocnemius muscle forces of a complete stance phase of a typical participant. Note: 1 body weight = 923.1 N (94.1 kg).

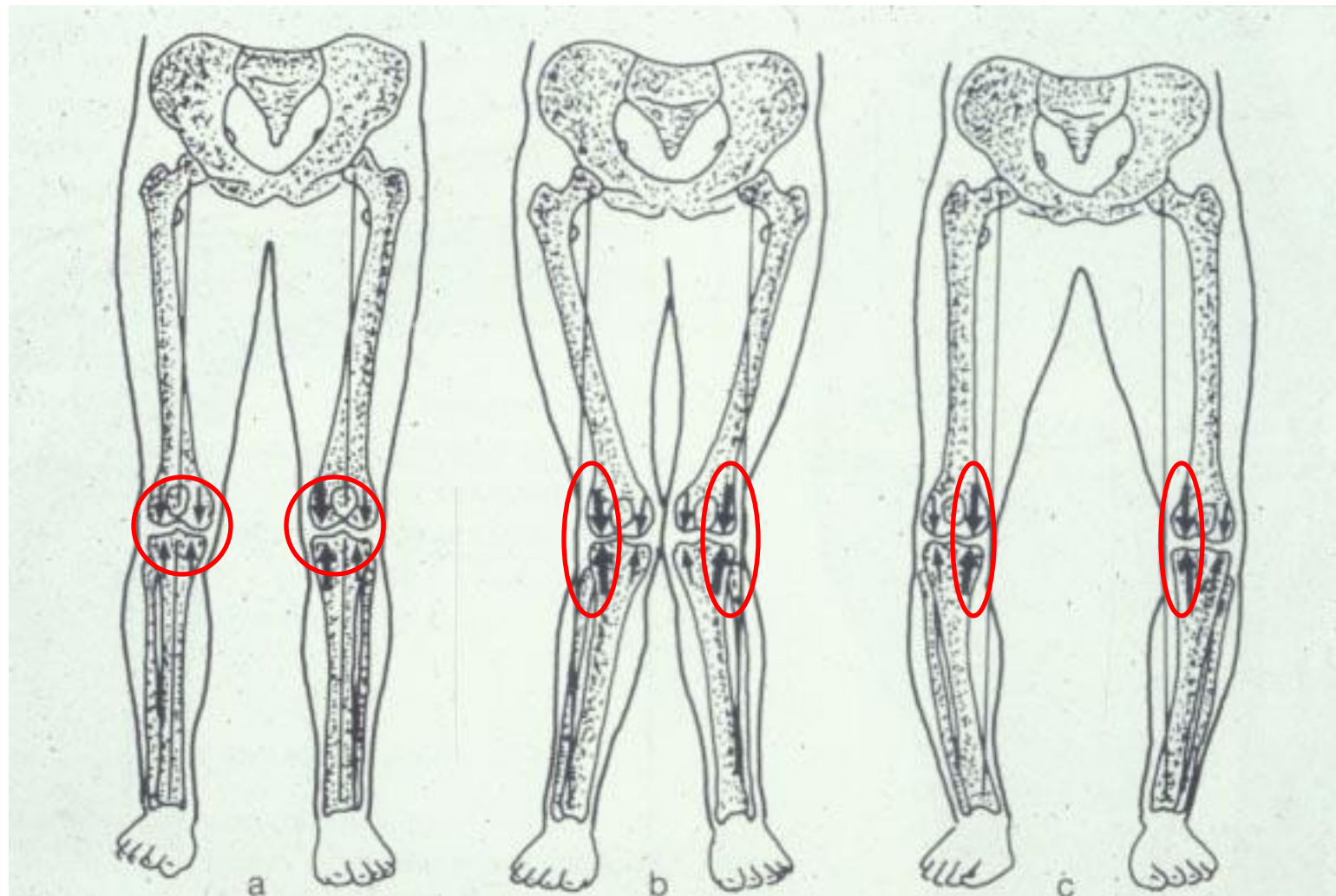
Force-length diagramm of ligaments and tendons



Questions



KOORMUS PÕLVELIIGESELE SÕLTUB ALAJÄSEMETE ANATOOMILISTEST ISEÄRASUSTEST



Normaalne

X-jalad

O-jalad

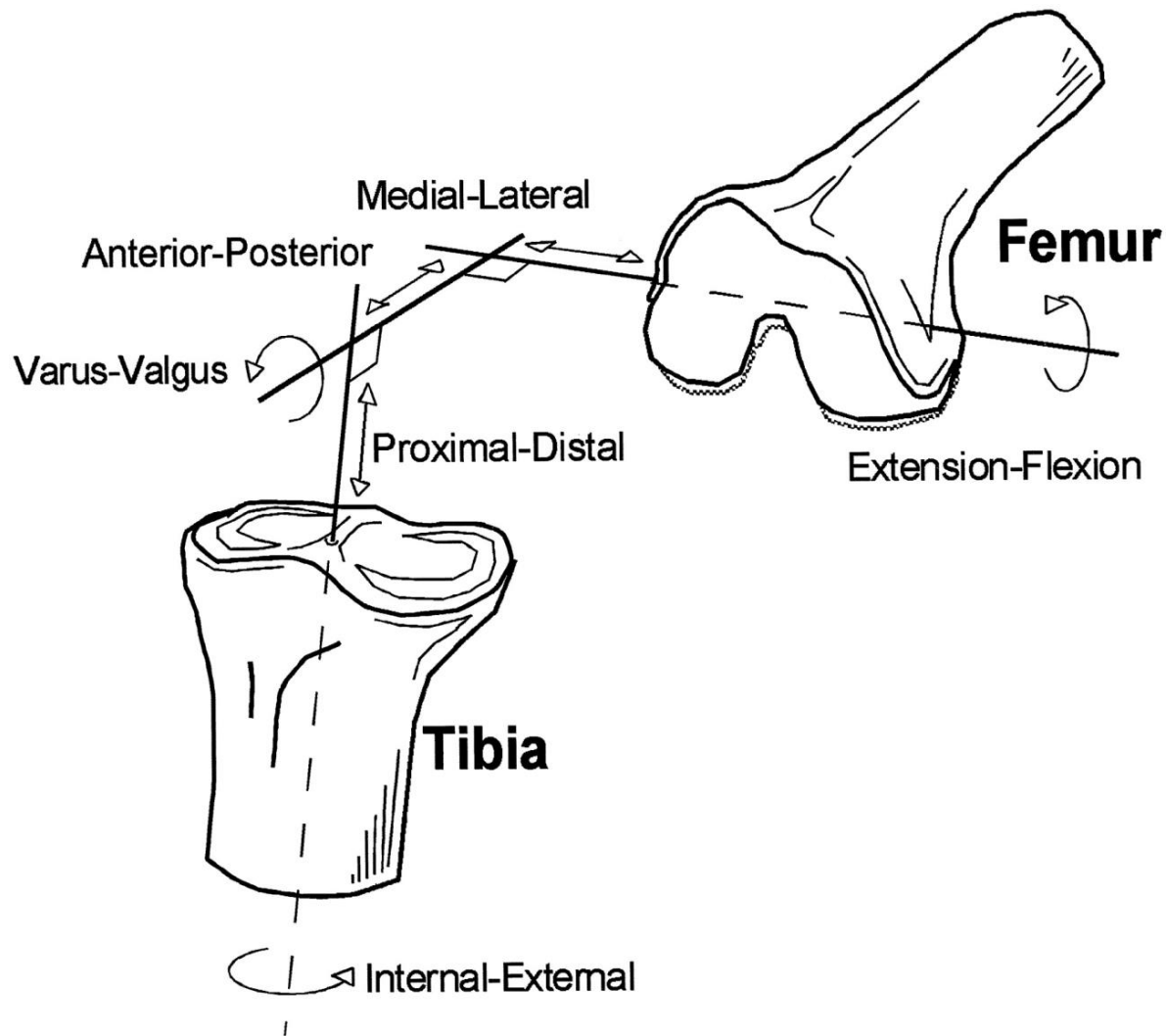
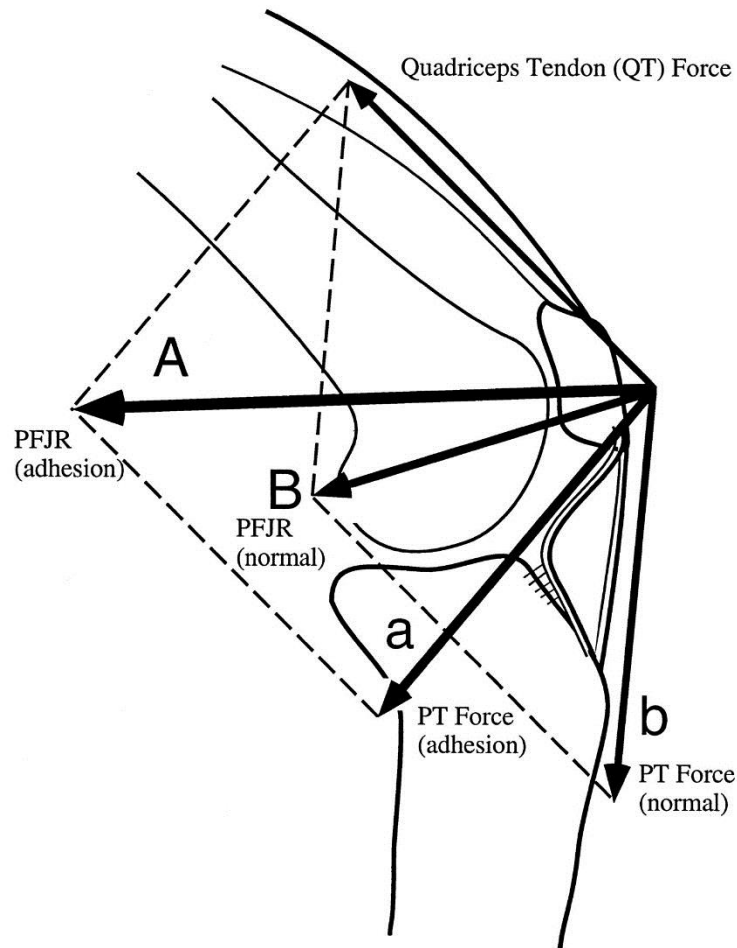
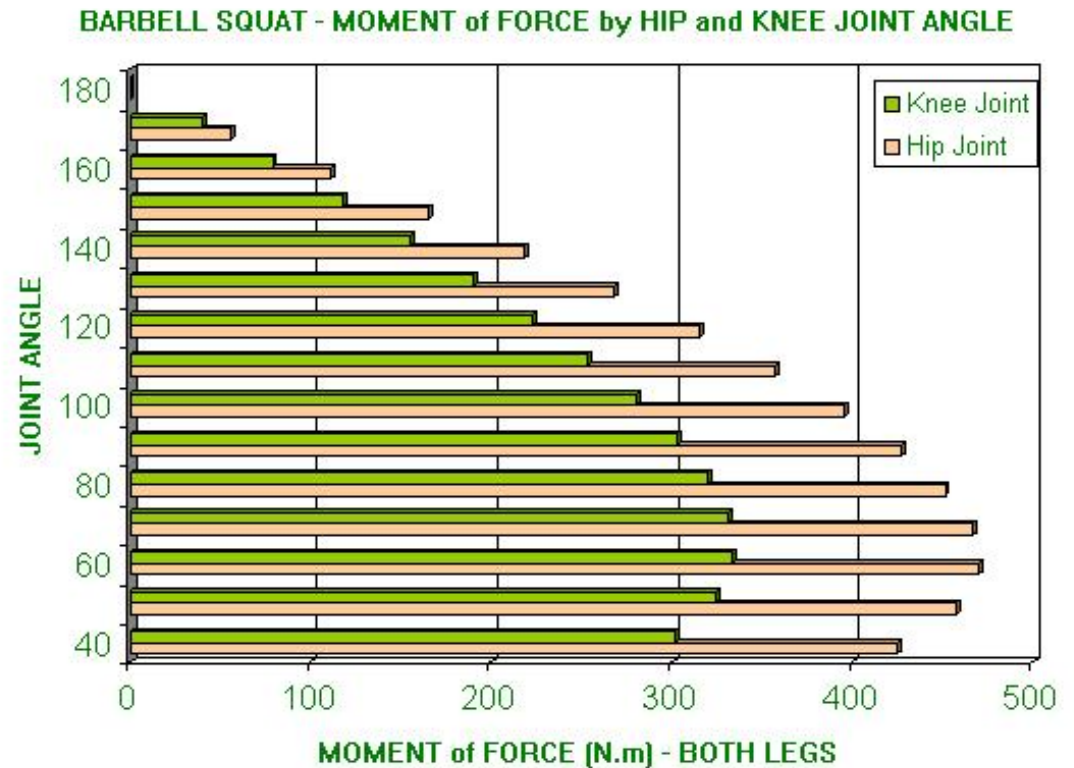
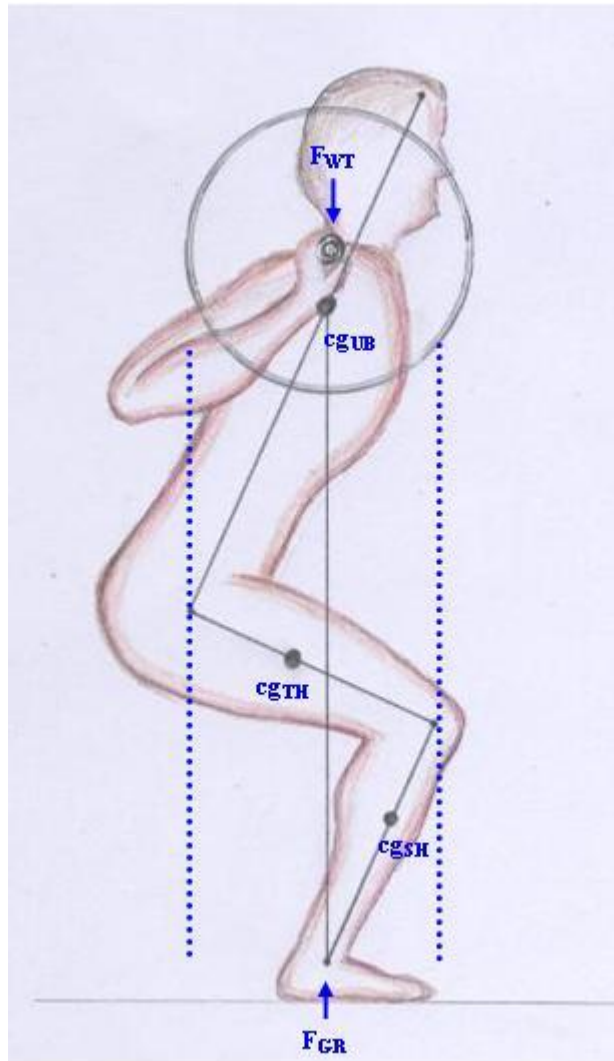


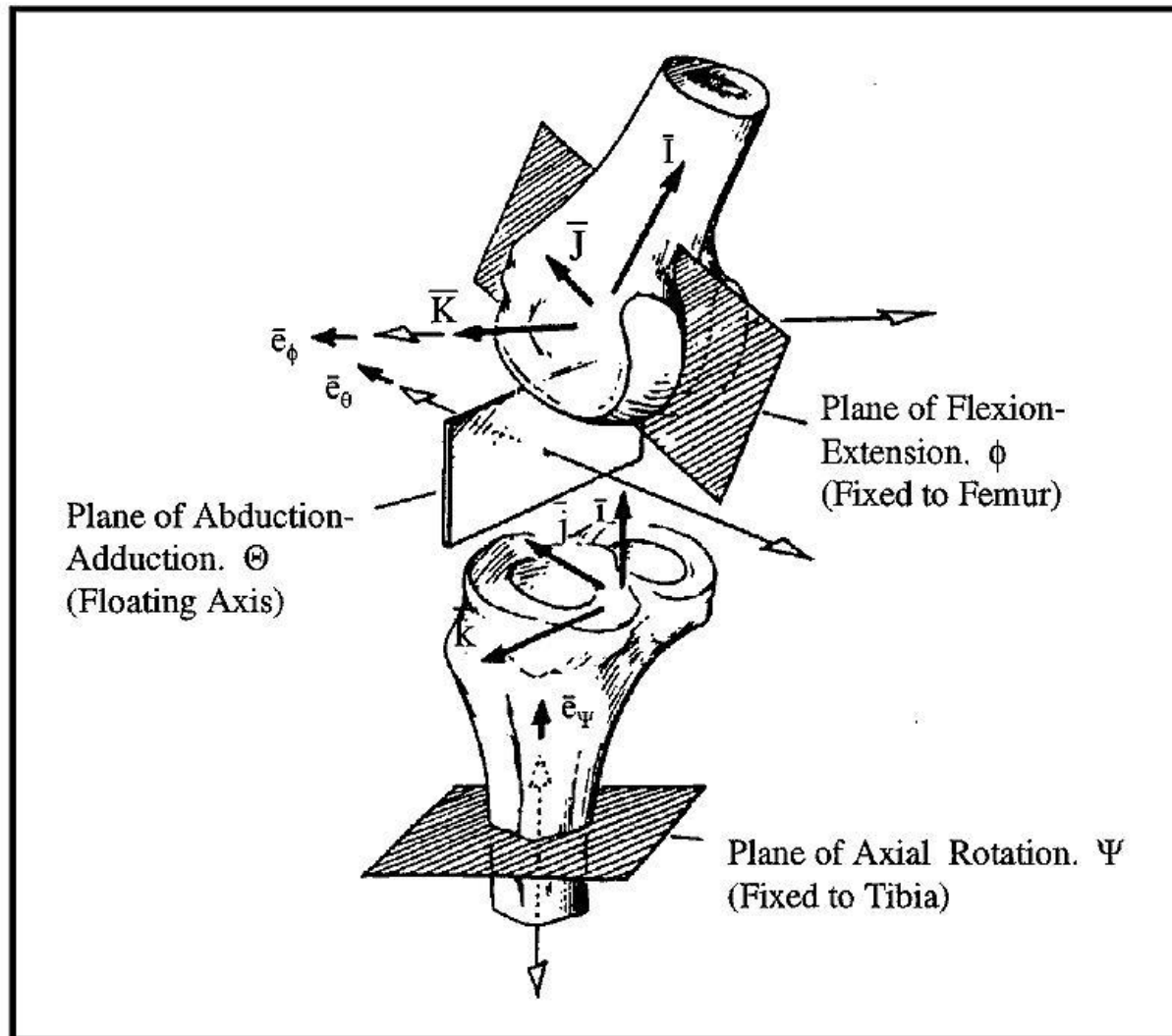
Figure 1 Schematic diagram illustrating the six degrees of motion of the human knee joint. (Reproduced with permission from Woo SL-Y, Livesay GA, Smith: Kinematics, in Fu FH, Harner CD, Vince KG (eds): *Knee Surgery*. Baltimore, Williams & Wilkins, 1994, pp 155–173.)



Patellofemoral joint reaction forces (Ahmad C.S. et al. Am. J. Sports Med. 1998, 26: 715-724)

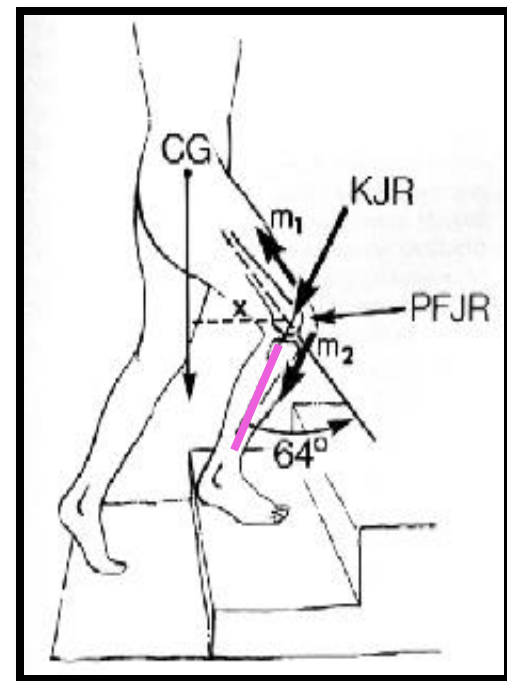
A biomechanical model for estimating moments of force at hip and knee joints in the barbell squat





Ascending stairs

- The actual degree of knee flexion required to ascend stairs is determined not only by the height of the step, but also by the height of the patient.
- For the standard 7" step approximately 65° of flexion will be required.
- In climbing stair , lever arm can be reduced by leaning forward. Also, in stair climbing the tibia is maintained relatively vertical, which diminishes the anterior subluxation potential of the femur on the tibia.



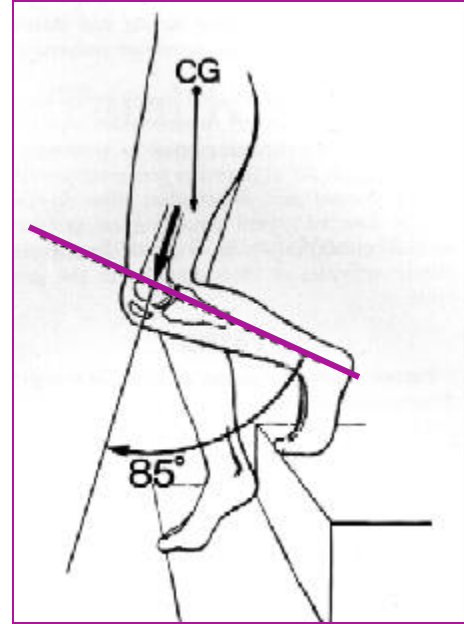
Knee joint

- Knee provides mobility and support during dynamic and static activities
- Support during weight bearing
- Mobility during non-weight bearing
- Involved with almost any functional activity of the lower extremity

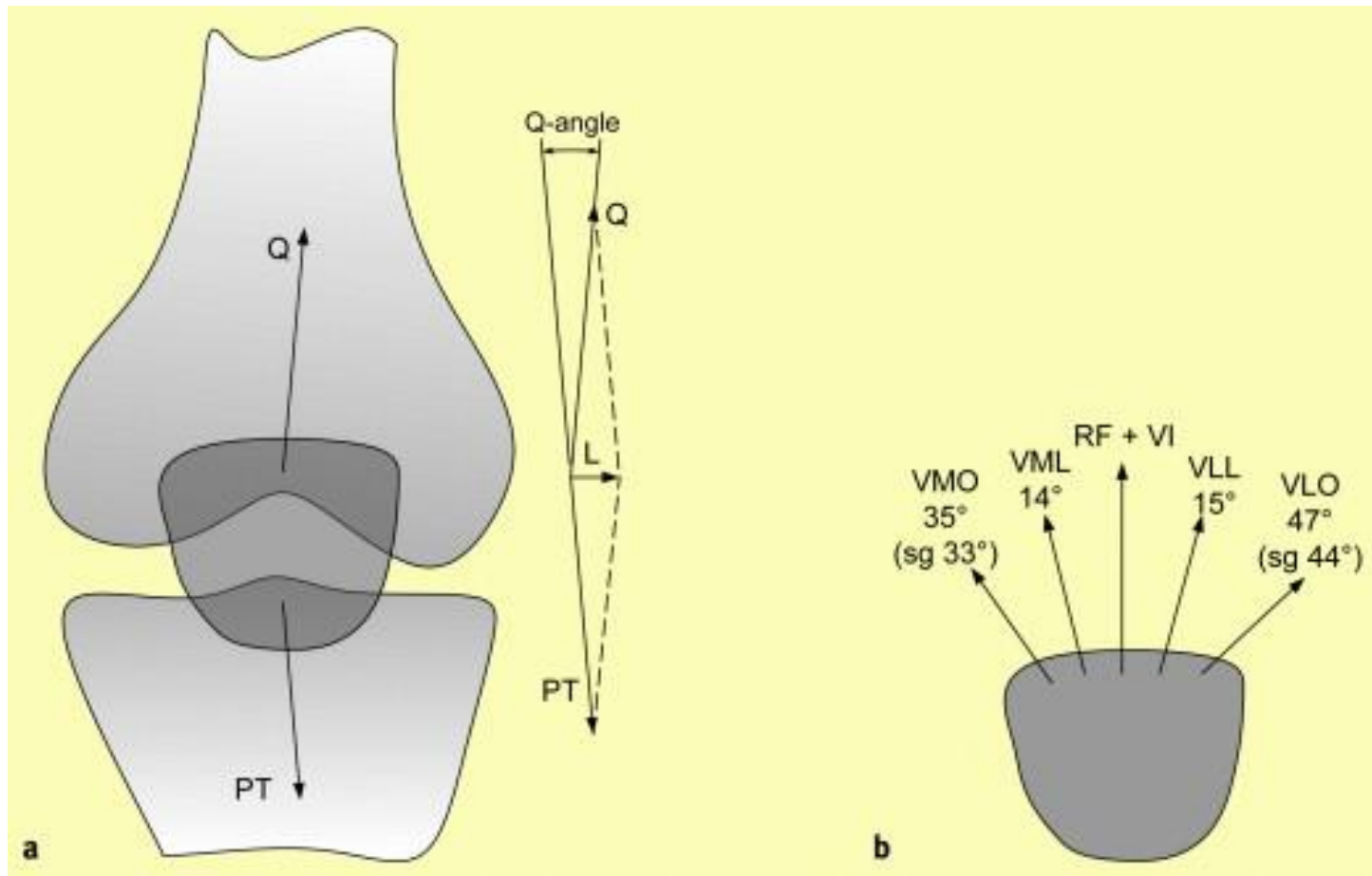


Descending stairs

- In standard step 85° of flexion is required.
- The tibia is steeply inclined toward the horizontal, bringing the tibial plateaus into an oblique orientation.
- The force of body weight will now tend to sublux the femur anteriorly.



This anterior subluxation potential will be resisted by the patellofemoral joint reaction force, and the tension which develops in the posterior cruciate ligament.



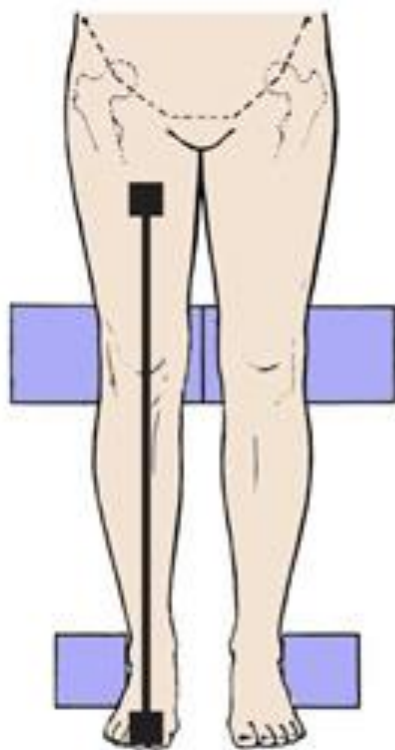
The knee joint in the frontal plane. **a** The Q-angle is defined as the angle between the line of action of the patellar tendon, PT and the line of action of the resultant quadriceps muscles, Q. The Q-angle effect results in a lateral force on the patella, L. **b** The quadriceps muscle force results from summing direction and magnitude of the quadriceps components acting on the patella. The lines of action of each component lie on the frontal plane, except for the obliquus muscles, which describe an angle with the sagittal plane as well⁸ (RF: rectus femoris, VI: vastus intermedius, VML: vastus medialis longus, VMO: vastus medialis obliquus, VLL: vastus lateralis longus, VLO: vastus lateralis obliquus). **(Masouros S.D. Et al. Orthop Trauma, 2010, 24: 84-91)**

Patellofemoral Joint Activities: Flexion & Load

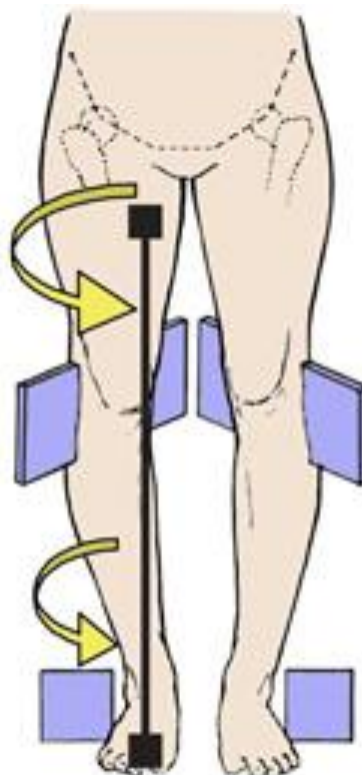
• walking	10 deg	0.5BW
• leg raise	0	0.5
• stairs	60	3.3
• isometric	90	6.5
• squat	120	7.6
• jumping	---	20
• tendon rupture	90	25

[Mow & Hayes, 1991]

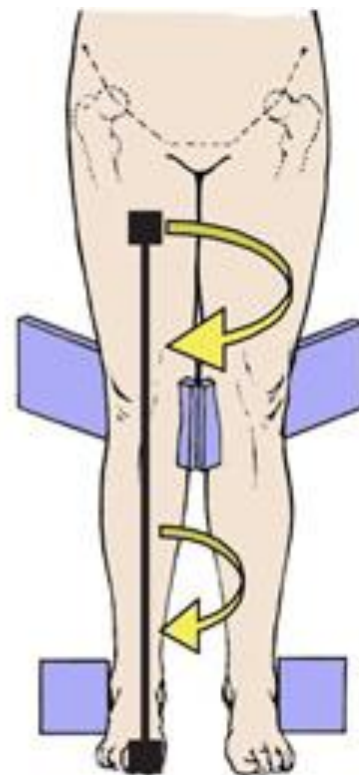
BW = body weight



Neutral subtalar position with neutral knee alignment



Pronation with internal rotation of the knee



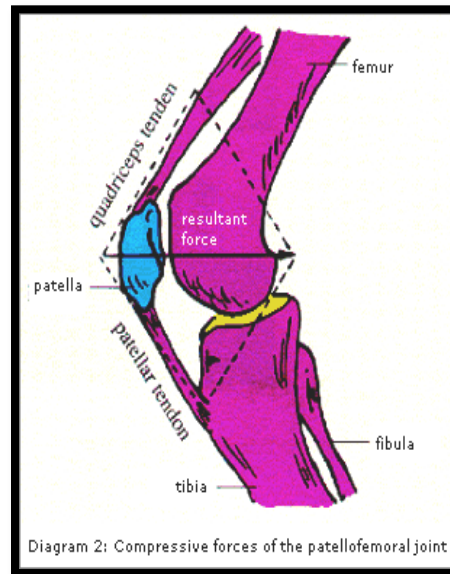
Supination with external rotation of the knee

❖ Compressive force is additional force at patellofemoral joint.

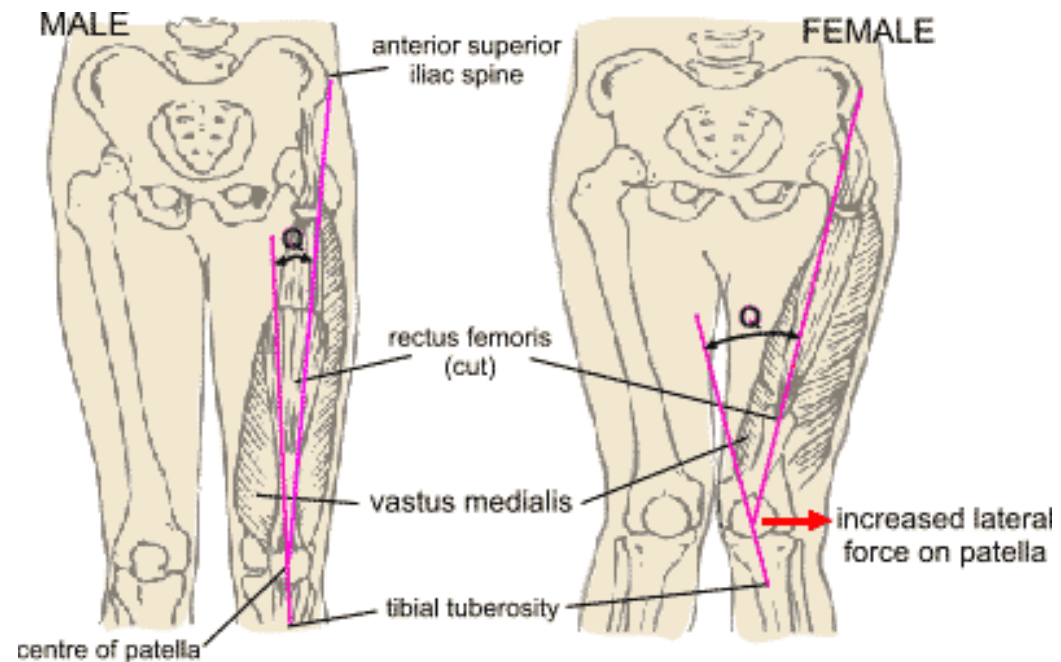
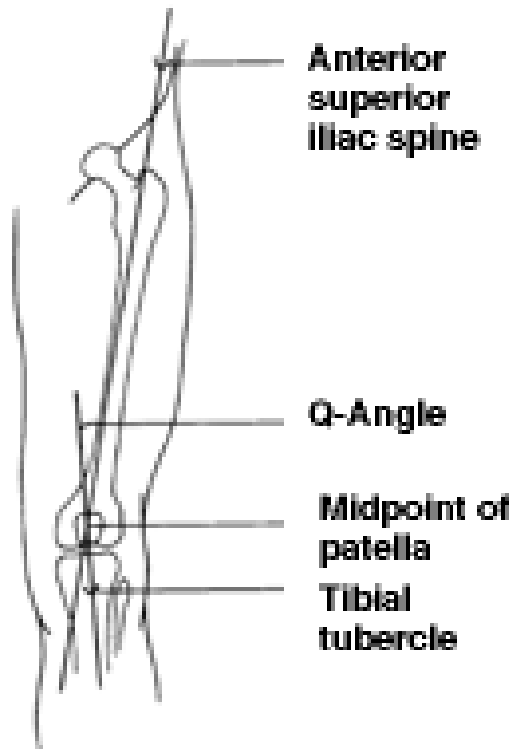
PF Compressive Force Function

- Stabilizes patella in trochlea groove.
- Patella assures “some” compression in full extension.

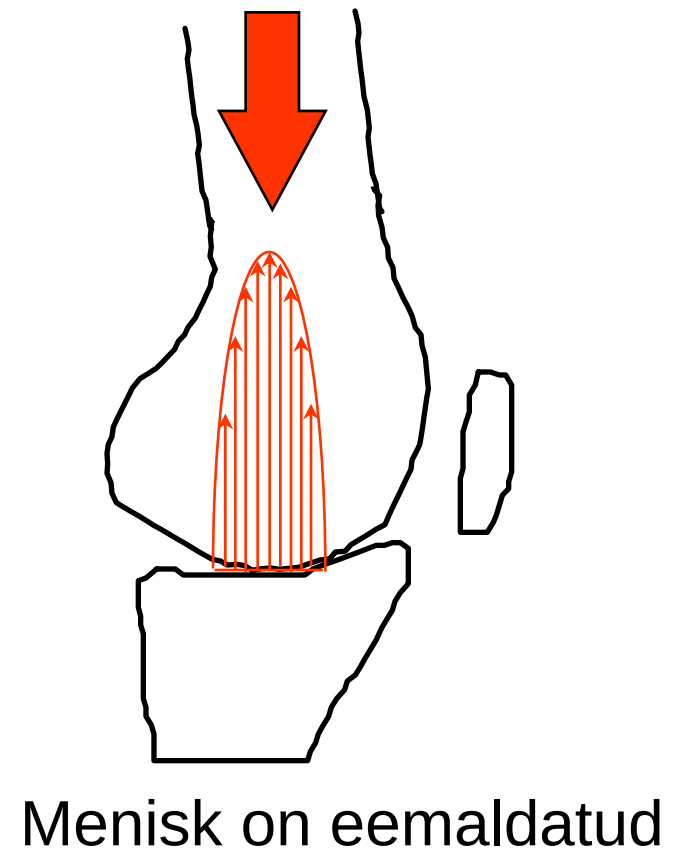
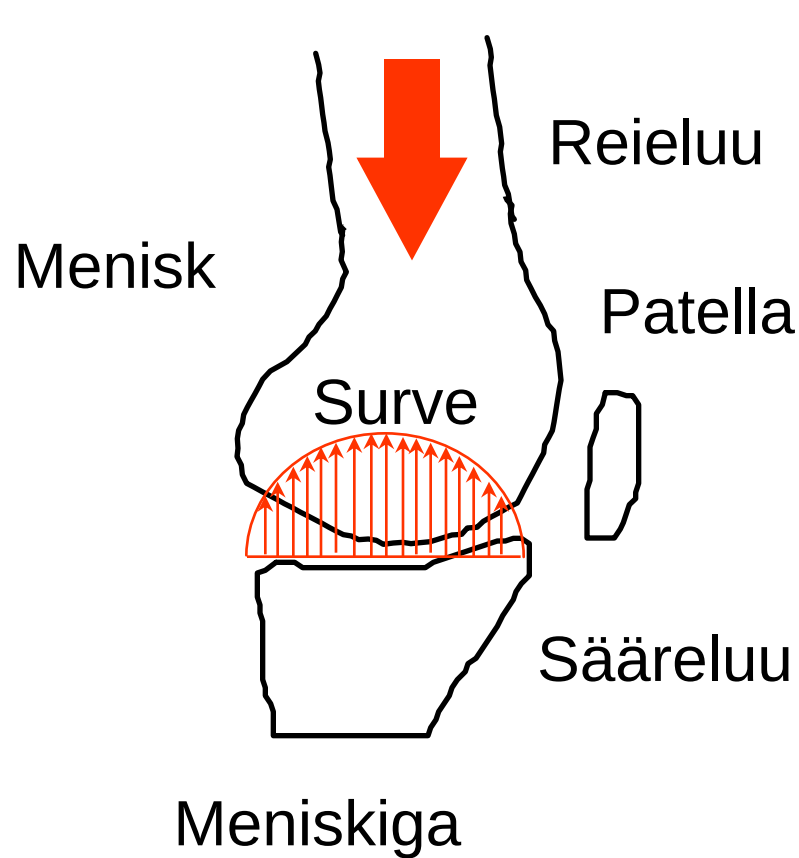
Patellofemoral compression with knee flexion during weight bearing, because of as flexion increases, a large amount of quadriceps tension is required to prevent the knee from buckling against gravity.

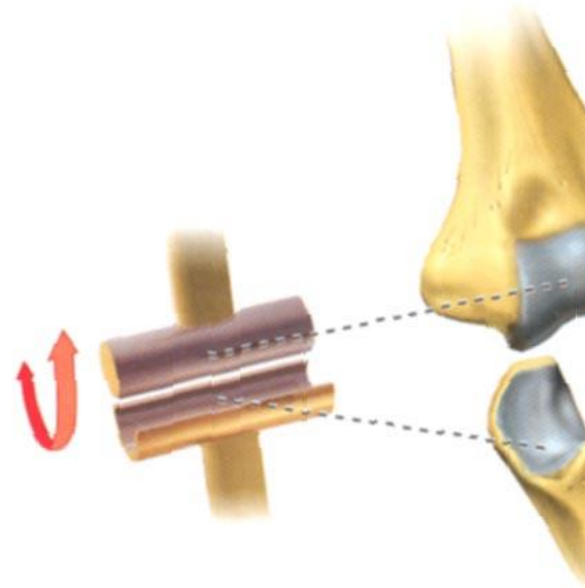


Q-angle



MENISKITE ROLL SURVEJÕUDUDE JAOTAMISEL SÄÄRELUULE



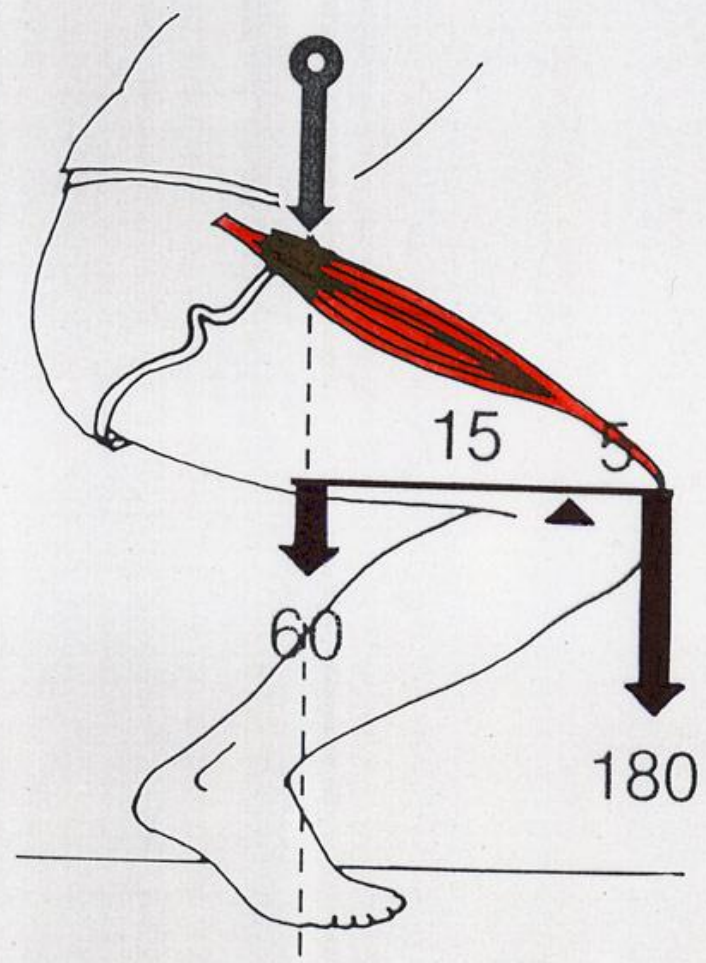
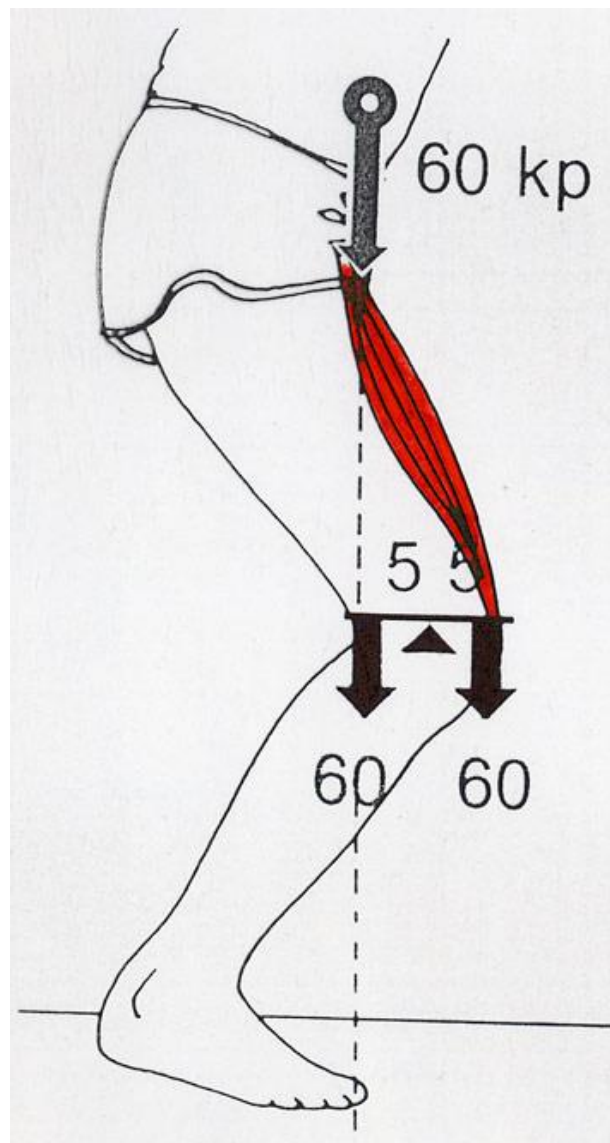


b Hinge joint

Plokkliiges

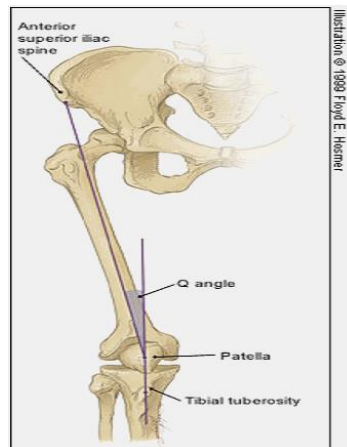


Ratasliiges



Eesmine ülemine niudeoga

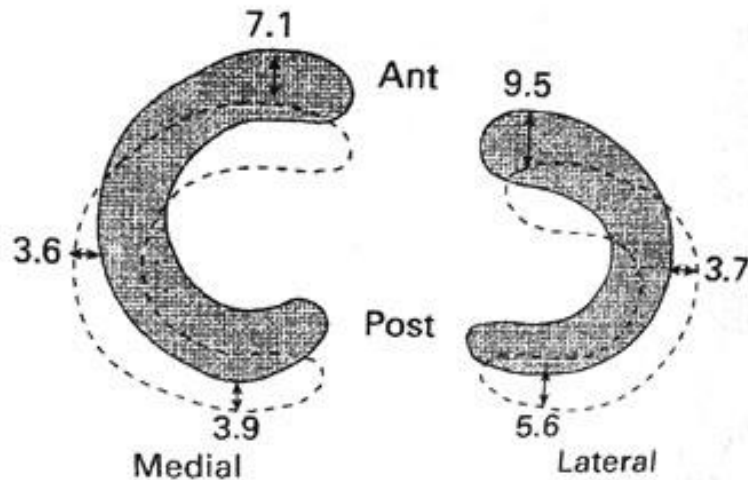
Q-nurk



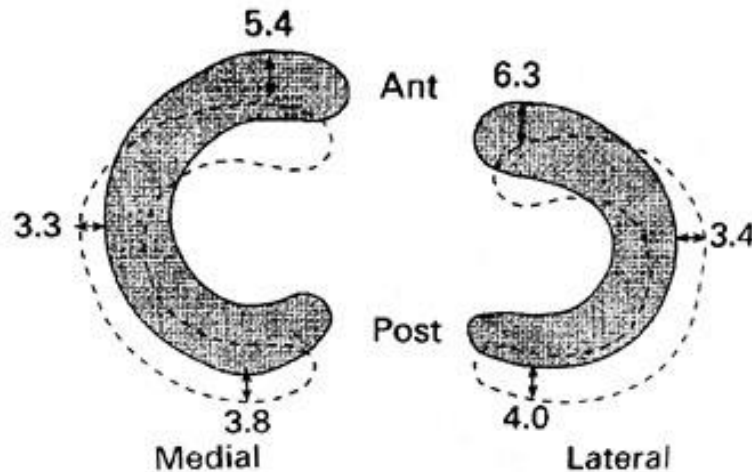
Q-nurk

Põlvekeder

Sääreluuköprus



The mean movement (mm) in each meniscus during flexion on a weight-bearing knee. (From Vedi et al. J Bone and Joint Surg Br. 1999;81-B:37-41)



The mean movement (mm) in each meniscus during flexion on a sitting non weight-bearing knee. (From Vedi et al. J Bone and Joint Surg Br. 1999;81-B:37-41)

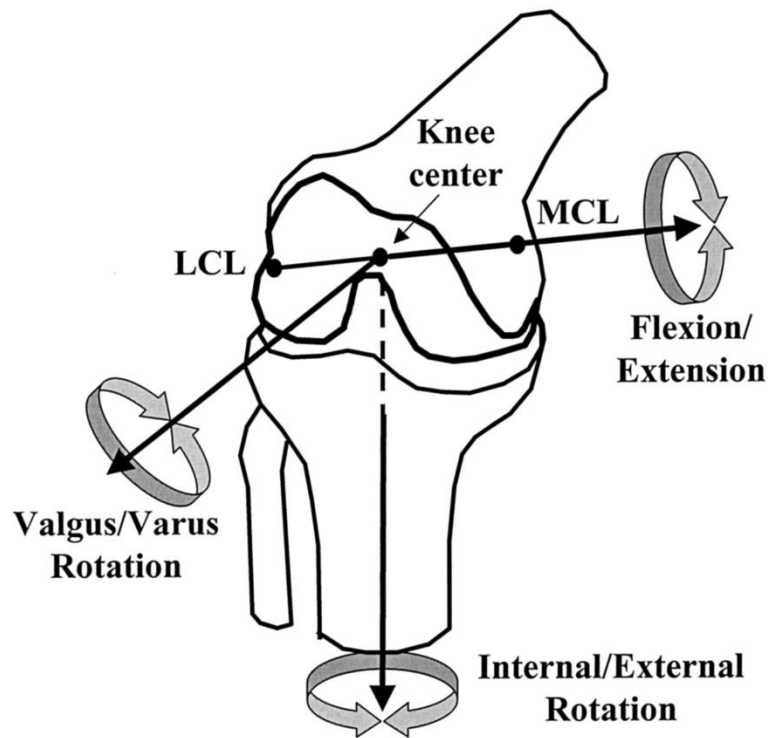


Figure 2. Definition of axes used in the study. The *transepicondylar line*, as defined in this study, is the line formed by the insertions of the lateral (LCL) and medial collateral ligaments (MCL), respectively.

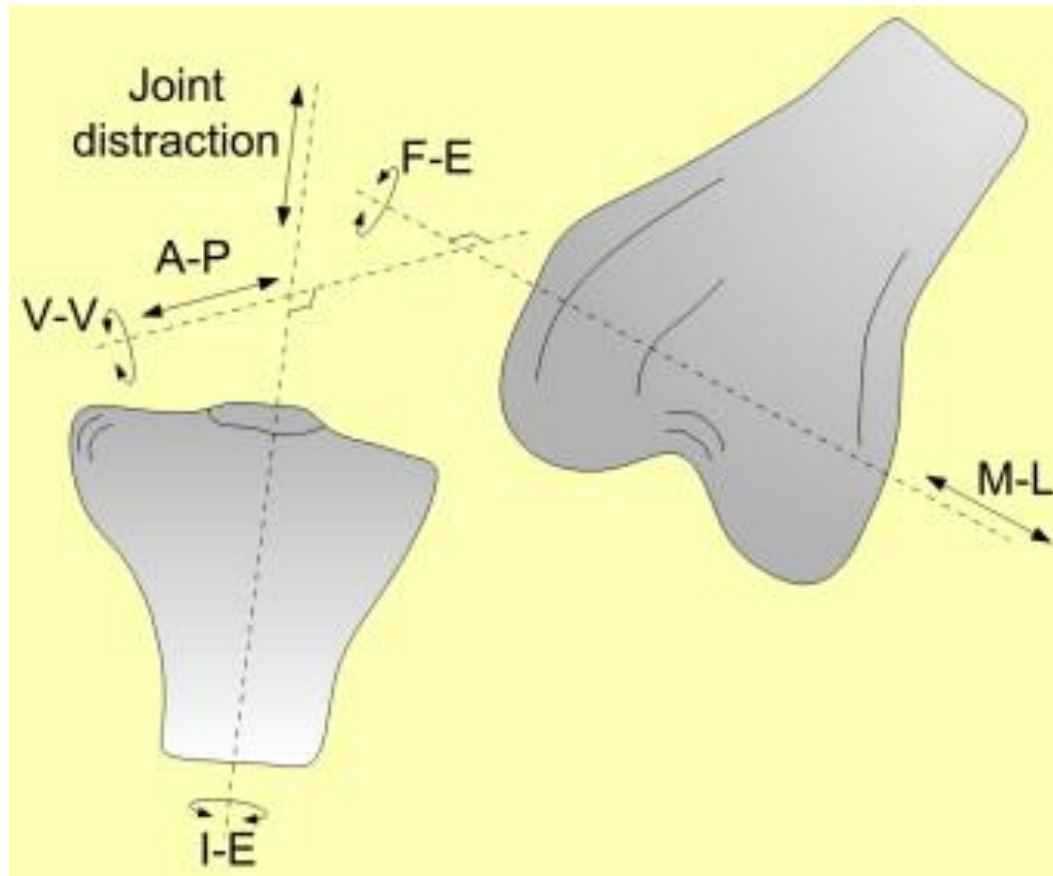
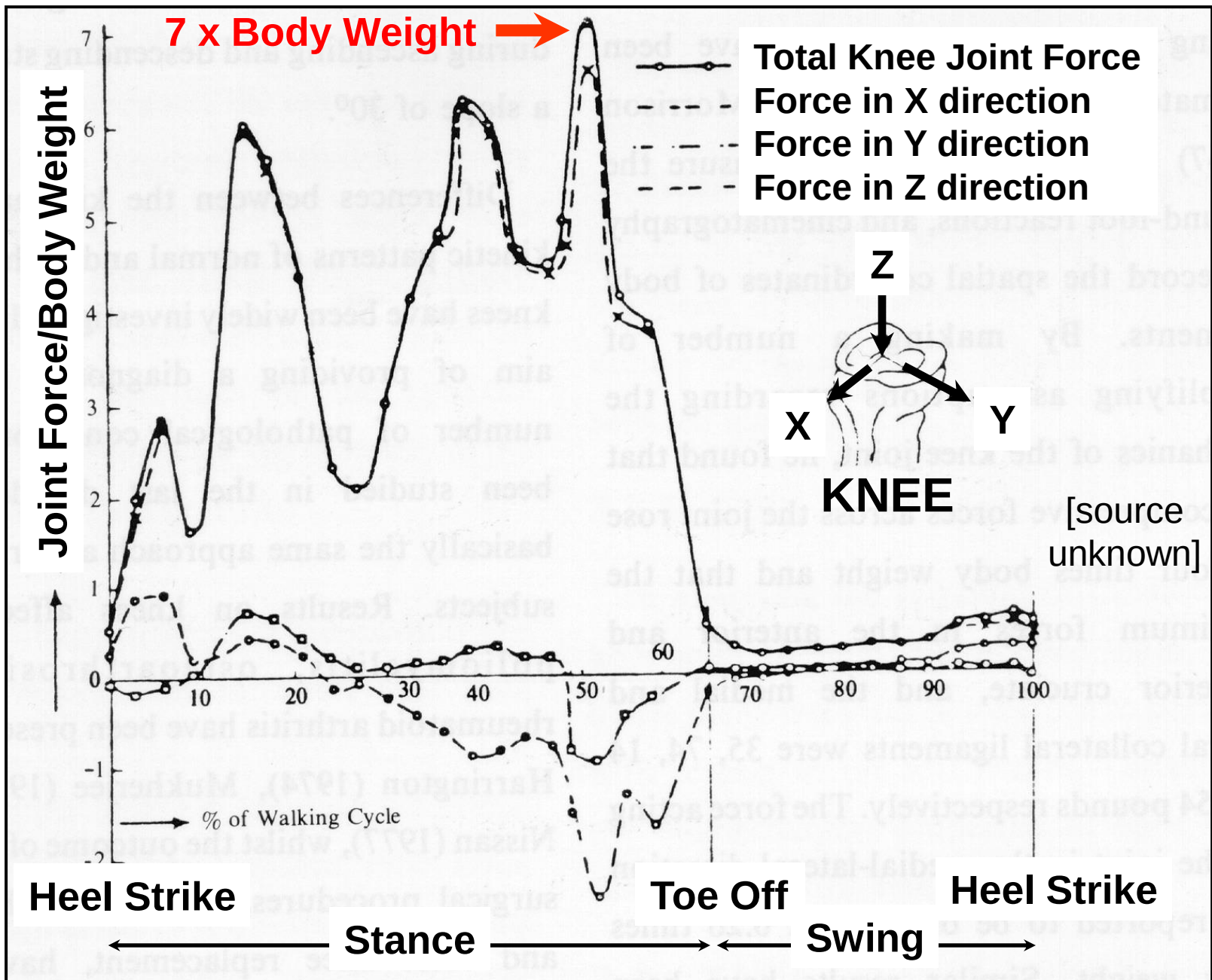


Figure 1. The six degrees of freedom of TF joint motion expressed in a clinical joint coordinate system. Mediolateral translation (M-L) and flexion-extension (F-E) occur along and about an epicondylar femoral axis. Joint distraction and internal-external rotation (I-E) occur along and about a tibial long axis. Anterior-posterior translation (A-P) and varus-valgus (V-V) (or adduction-abduction) rotation occur along and about a floating axis, which is perpendicular to both femoral epicondylar and tibial long axes. (Masouros S.D. Et al. Orthop Trauma, 2010, 24: 84-91)



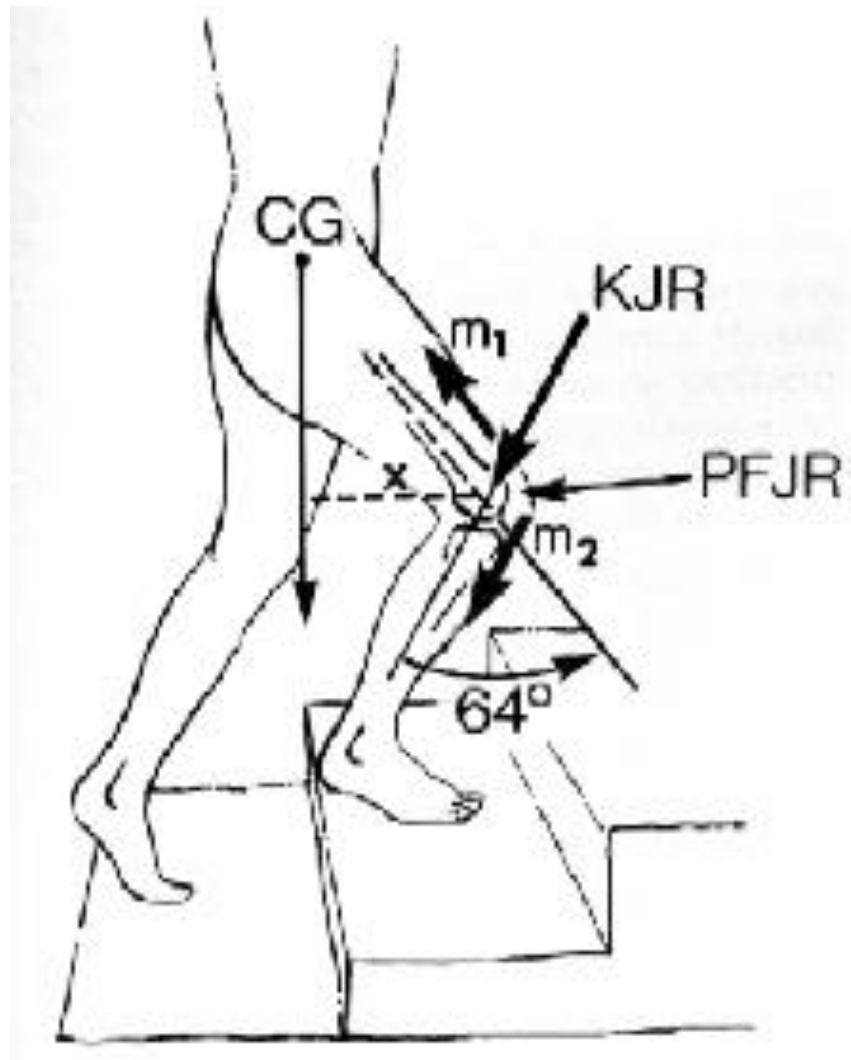
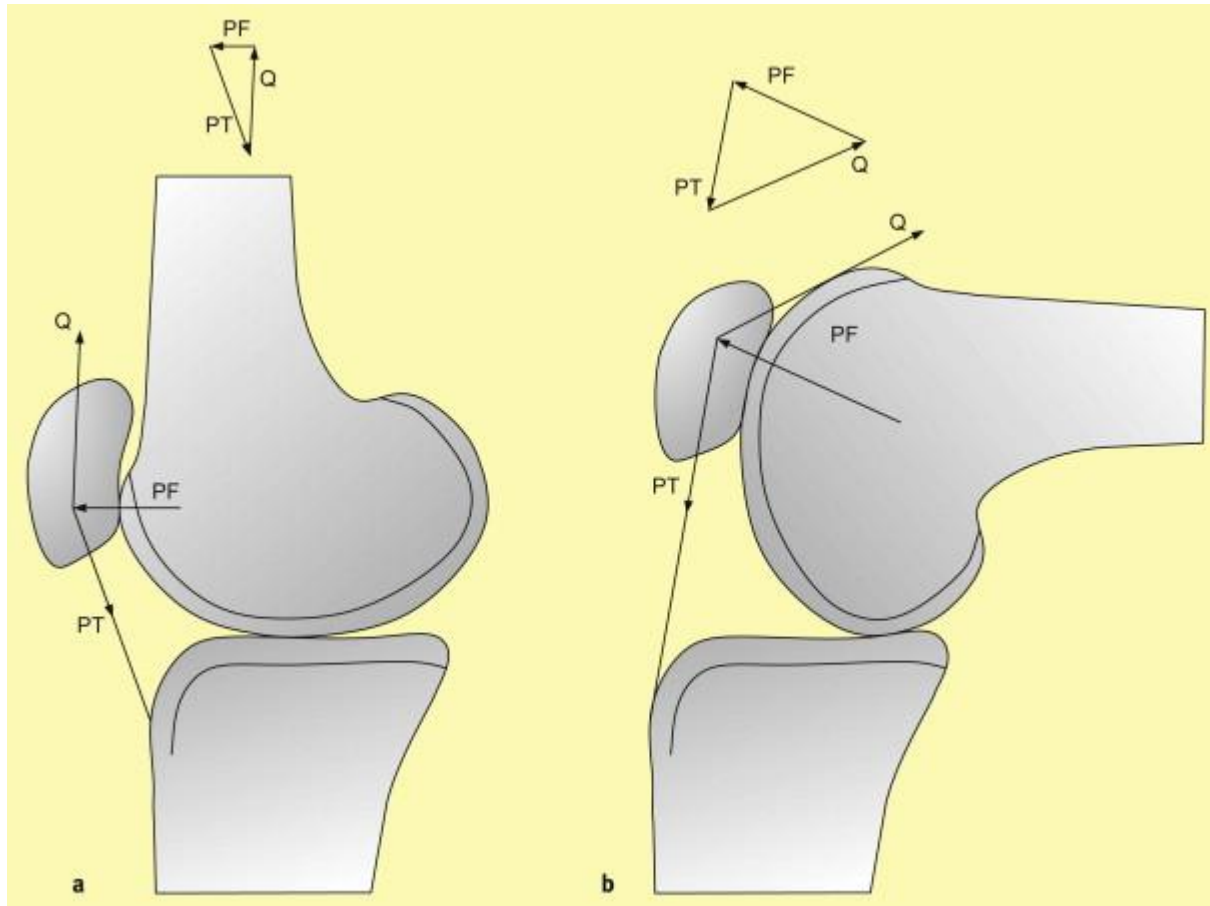
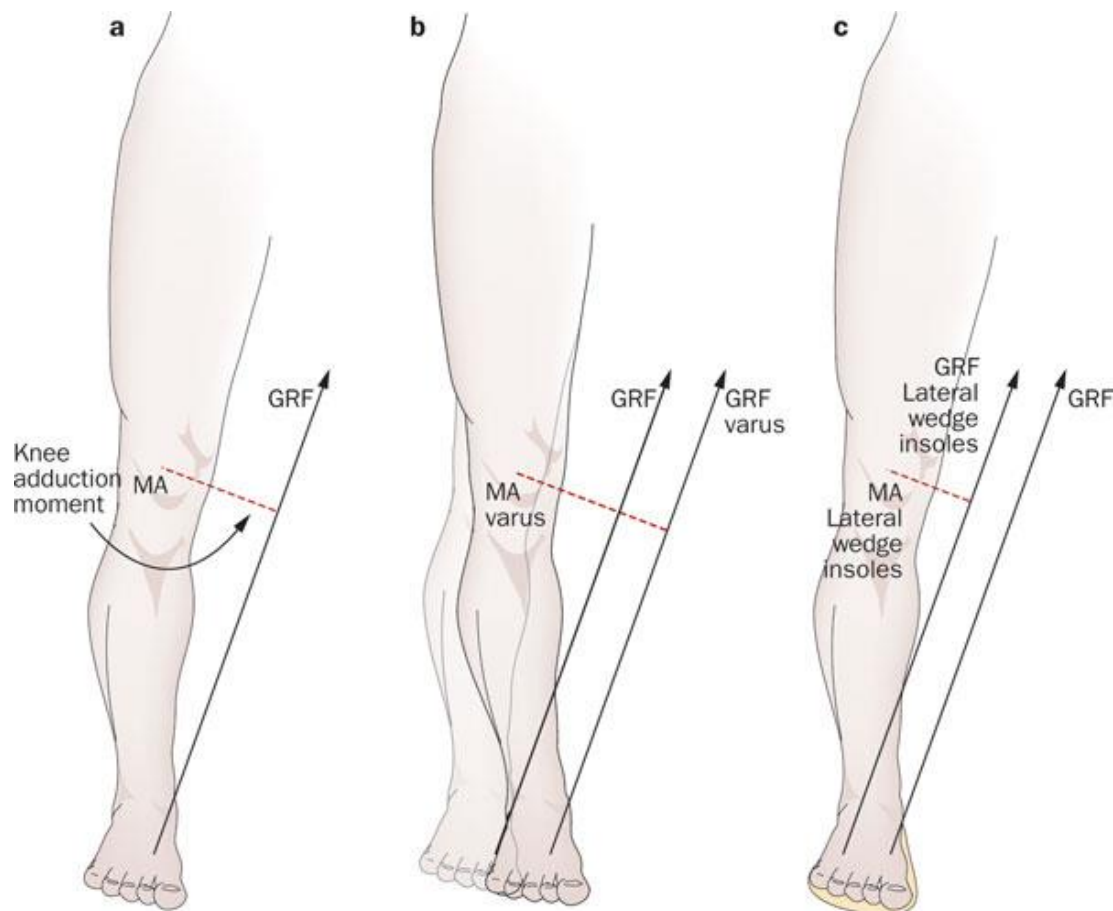


Figure 3.19. Free body diagram of a patient ascending a step, m_1 = quadriceps force; m_2 = patellar tendon tension; KJR = knee joint reaction; $PFJR$ = patellofemoral joint reaction; CG = center of gravity; x = flexor lever arm.

(Reproduced with permission from V.T. Inman et al.: Human Walking, Williams & Wilkins, Baltimore, 1981 (2).)



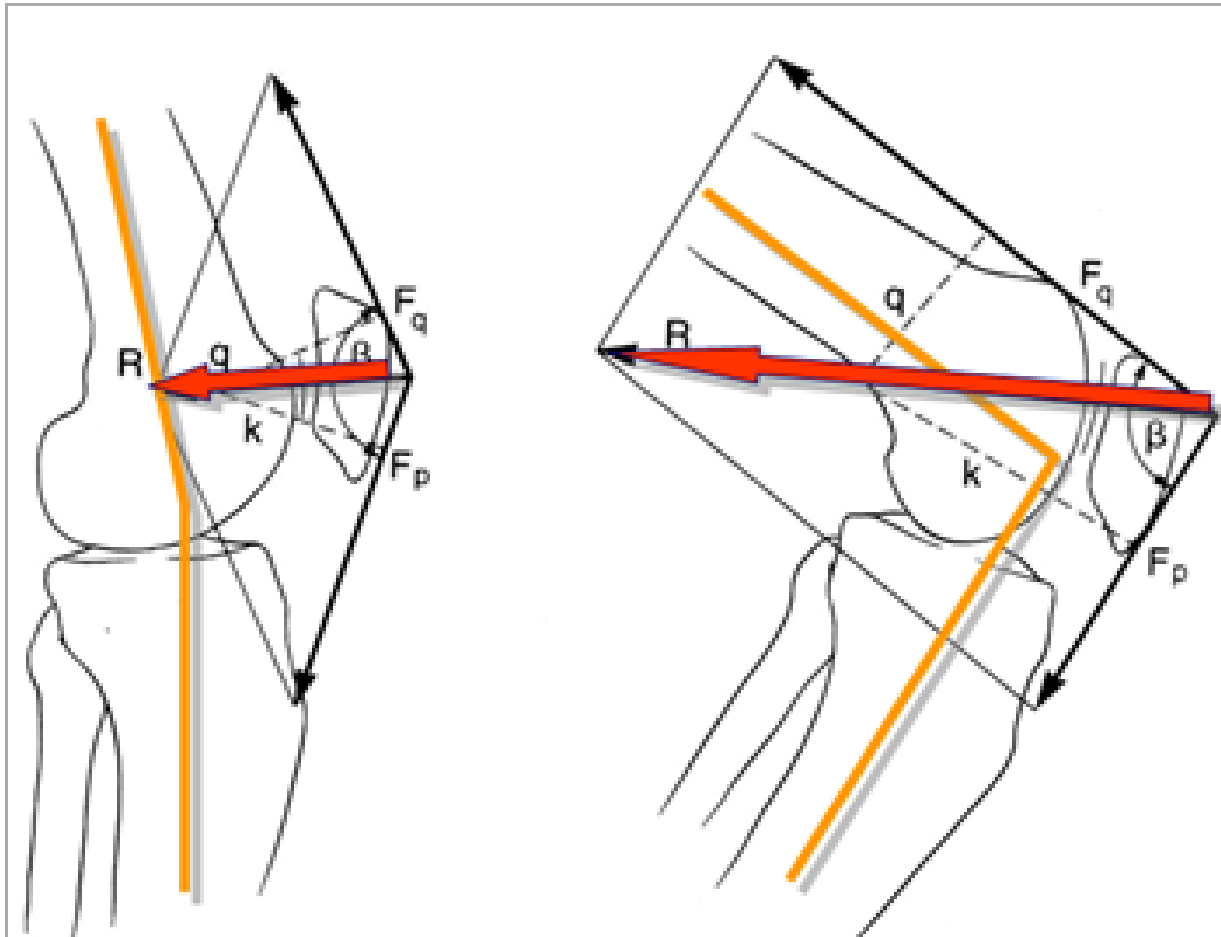
PF joint force at a extension and b 90° flexion, showing geometrically the increase of PF joint force with flexion. (PT: patellar tendon, Q: quadriceps muscles, PF: patellofemoral, TF: tibiofemoral). (Masouros S.D. Et al. Orthop Trauma, 2010, 24: 84-91)

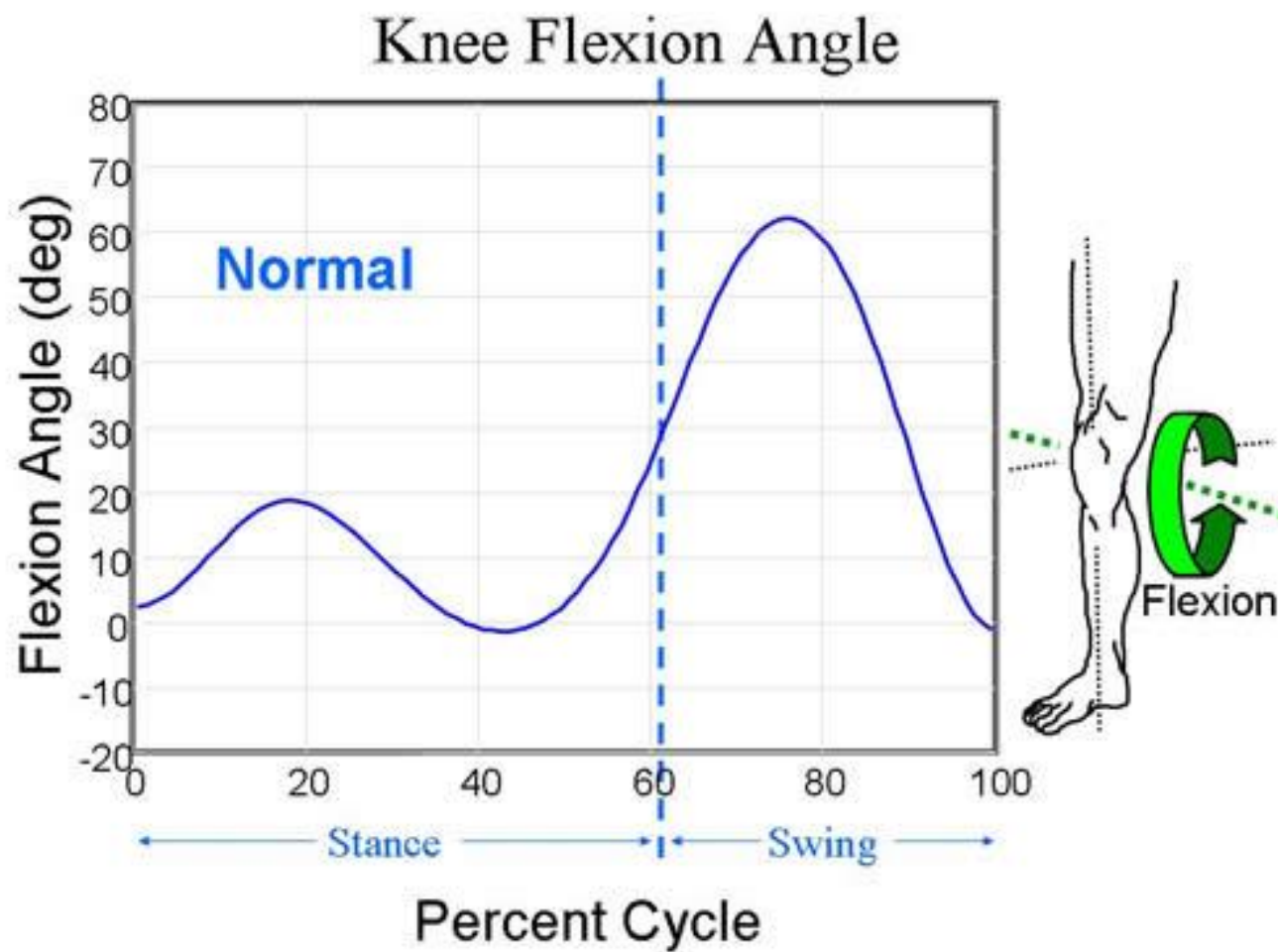


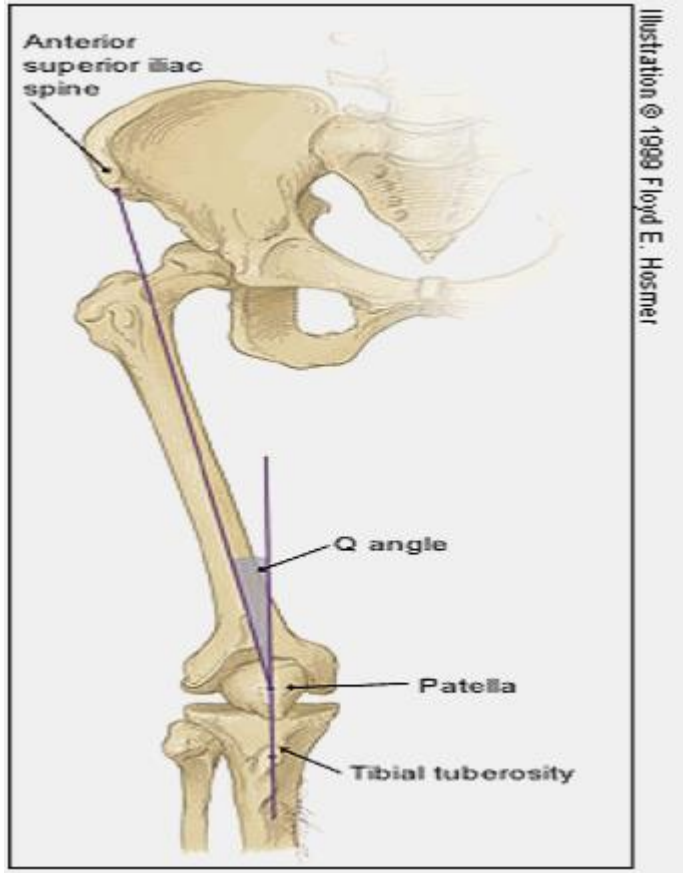
The magnitude and direction of the GRF are shown by the height and direction, respectively, of the straight arrows. The length of the MA of the GRF acting about the knee joint is indicated by dotted red lines. a | The knee adduction moment increases if the length of the MA increases, or the GRF magnitude increases, or both. b | A varus knee deformity (dark-shaded leg) is superimposed over a neutrally aligned knee (light-shaded leg). In the varus-aligned knee, the center of pressure (indicated by the origin of the GRF vectors) has shifted in the medial direction, increasing the MA of the GRF and, therefore, the knee adduction moment. c | Lateral wedge insoles shift the center of pressure, causing the GRF to pass closer to the knee joint center (assuming the GRF angle remains constant). This effect decreases the MA of the GRF about the knee and reduces the knee adduction moment compared with the situation without lateral wedge insoles.

Abbreviations: GRF, ground reaction force; MA, moment arm. (Reeves N.D., Bowling F.L. Net. Rev. Rheum. 2011, 7:

Patellofemoral joint reaction forces







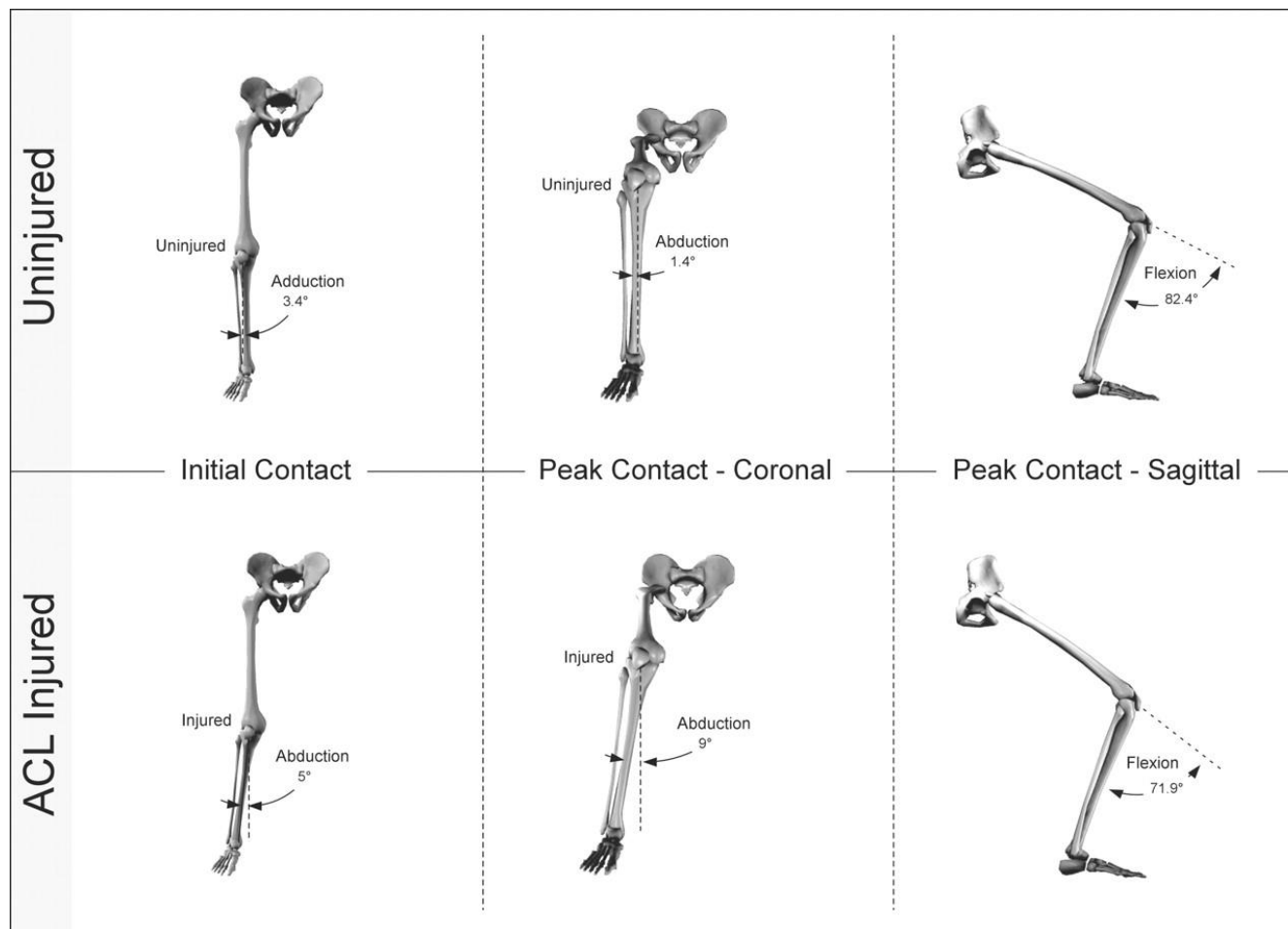
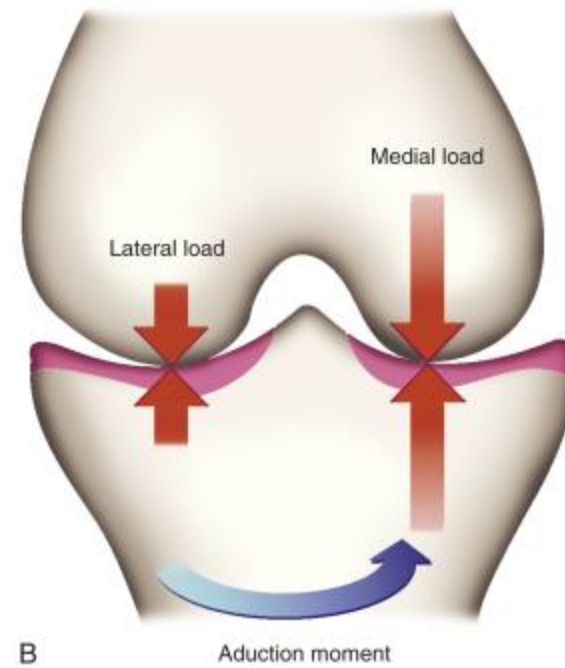
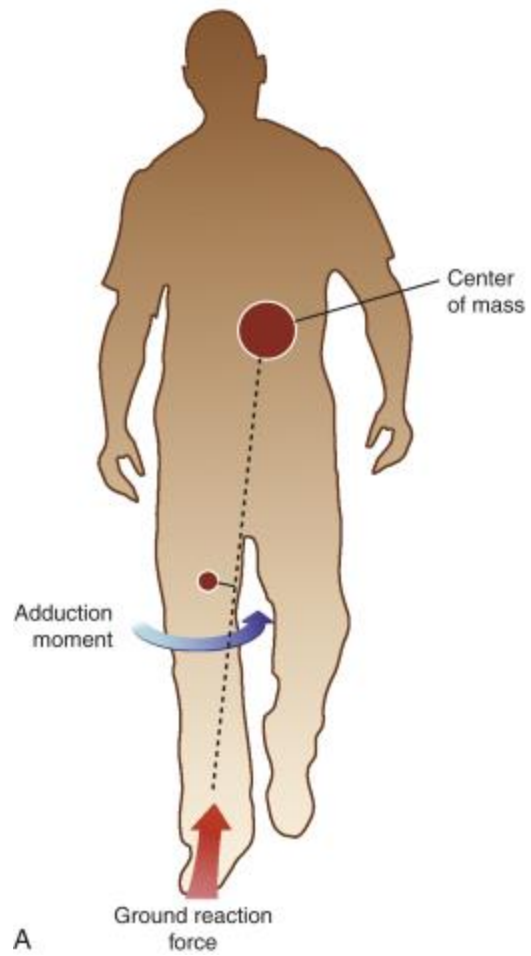


Figure 5. Biomechanical model depicting mean knee joint kinematics during the drop vertical jump at initial contact and maximal displacement in the ACL-injured and uninjured groups (n = 9 knees and n = 390 knees, respectively). Left, coronal plane view of knee abduction angle at initial contact in the ACL-injured and uninjured groups. Center, coronal plane view of maximum knee abduction angle in the ACL-injured and uninjured groups. Right, sagittal plane view of maximum knee flexion angle in the ACL-injured and uninjured groups. (Hewett T.E. Et al. Am J Sports Med 2005, 33: 492-501)



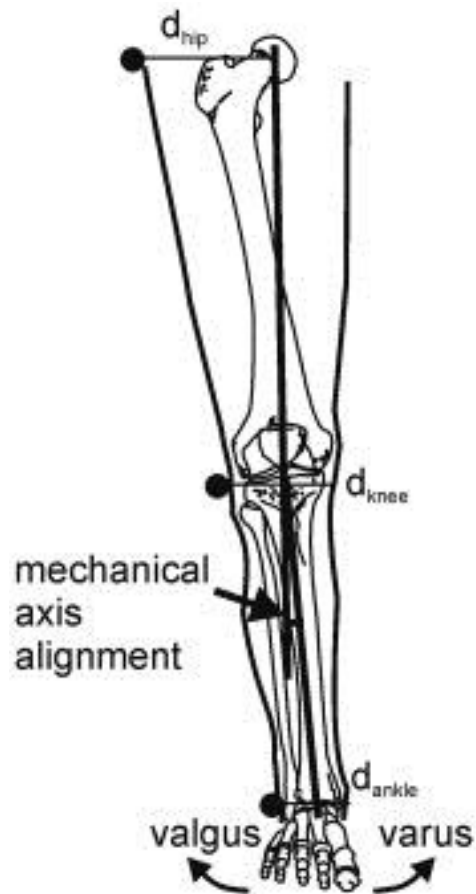


Fig. 1. Mechanical axis alignment was defined as the angle between a line from the center of the femoral head to the center of the femoral intercondylar notch, and a line from the center of the tips of the tibial spines to the ankle talus

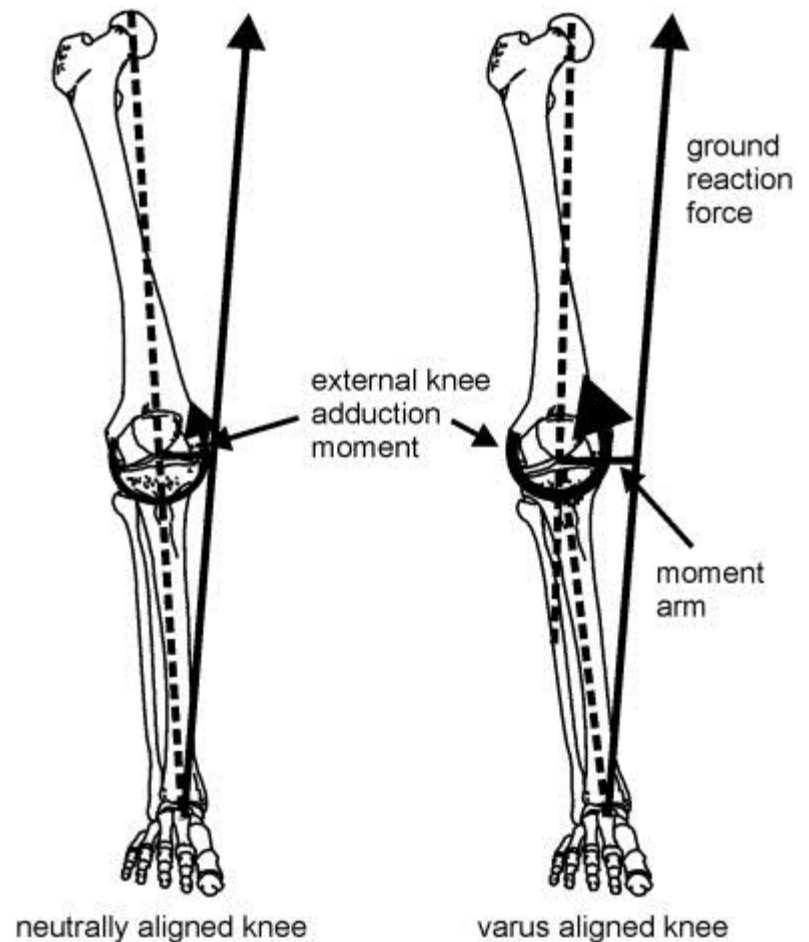


Fig. 2. The external knee adduction moment tends to be greater in varus aligned knees (right) compared to neutrally aligned knees (left) due to a greater moment arm of the ground reaction force about the knee joint center (Mündermann A. et al. The Knee, 2008, 15: 480-485)

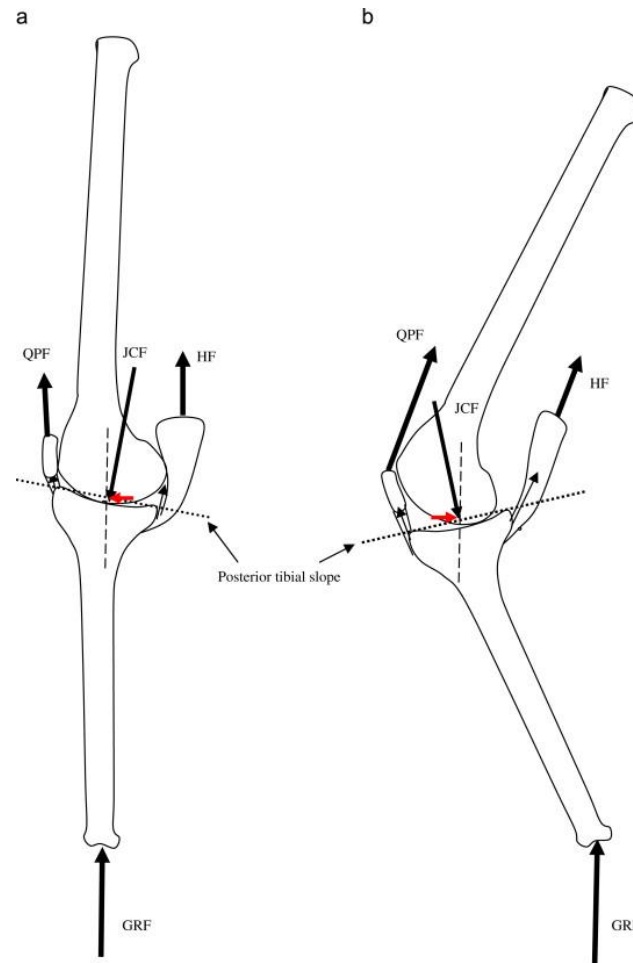


Fig. 1. The role of the slope and orientation of the tibial plateau on the direction of the joint compressive force (JCF). (a) At full extension JCF is leaning anteriorly and (b) at moderate flexion JCF is leaning posteriorly. QPF, quadriceps patellar tendon force; HF, hamstring force; GRF, ground reaction force. (Hashemi J. et al. J Biomech, 2011, 44: 577-585)

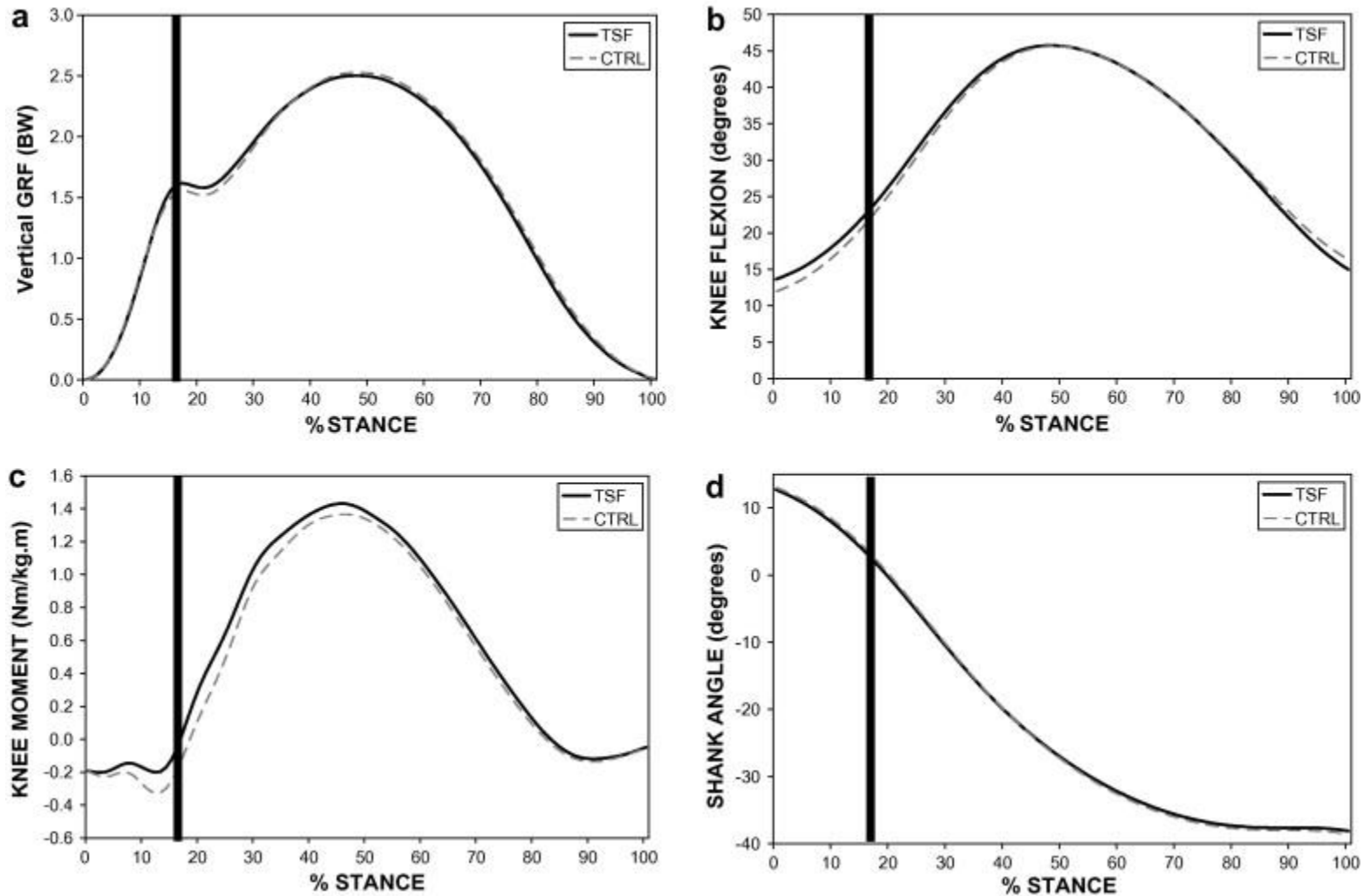
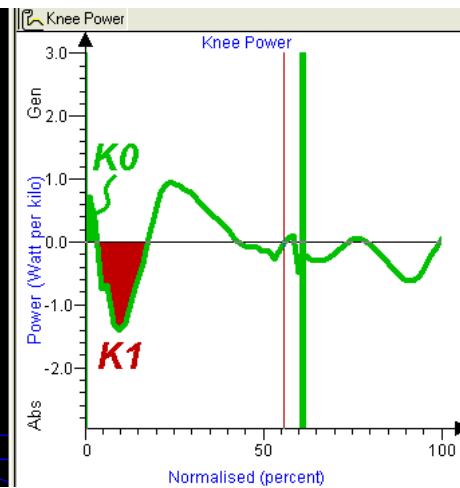
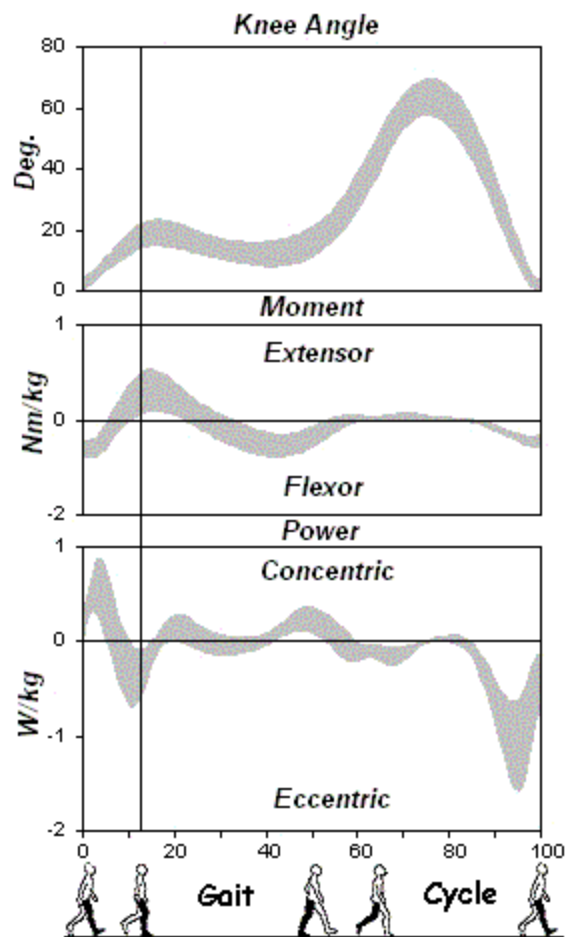
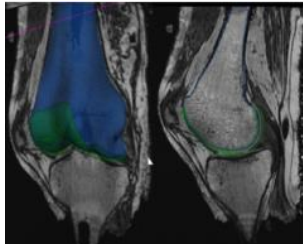


Fig. 2. Ensemble average curves for the TSF and CTRL groups for (a) vertical ground reaction force; (b) knee flexion angle; (c) sagittal plane knee moment and shank angle during the stance phase of running: the initial loading period is from footstrike to the vertical line. Knee flexion angle, internal knee extension moment and distal end of shank anterior to proximal end are positive. (Milner C.E. et al. Clin Biomech, 2007, 22: 697-703)





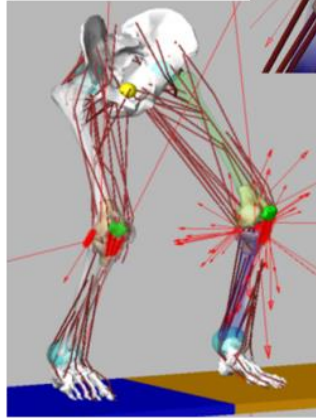
Anthropometrics
Motion, EMG, GRF



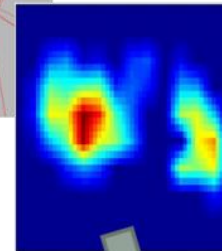
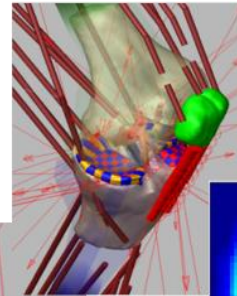
Morphology



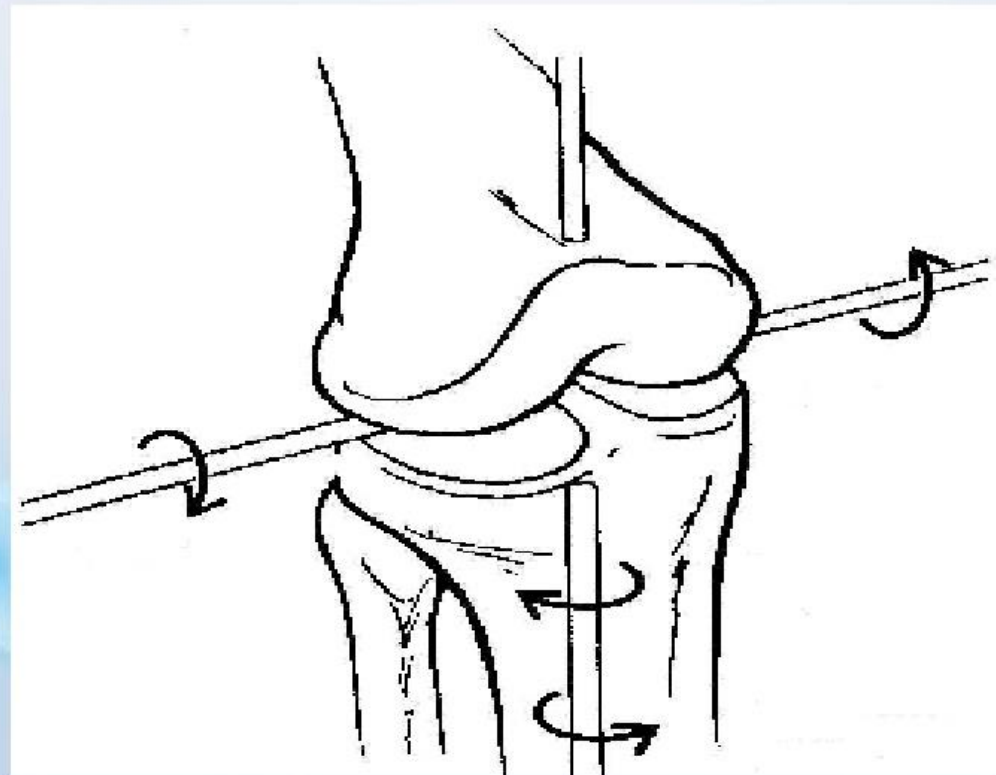
Laxity

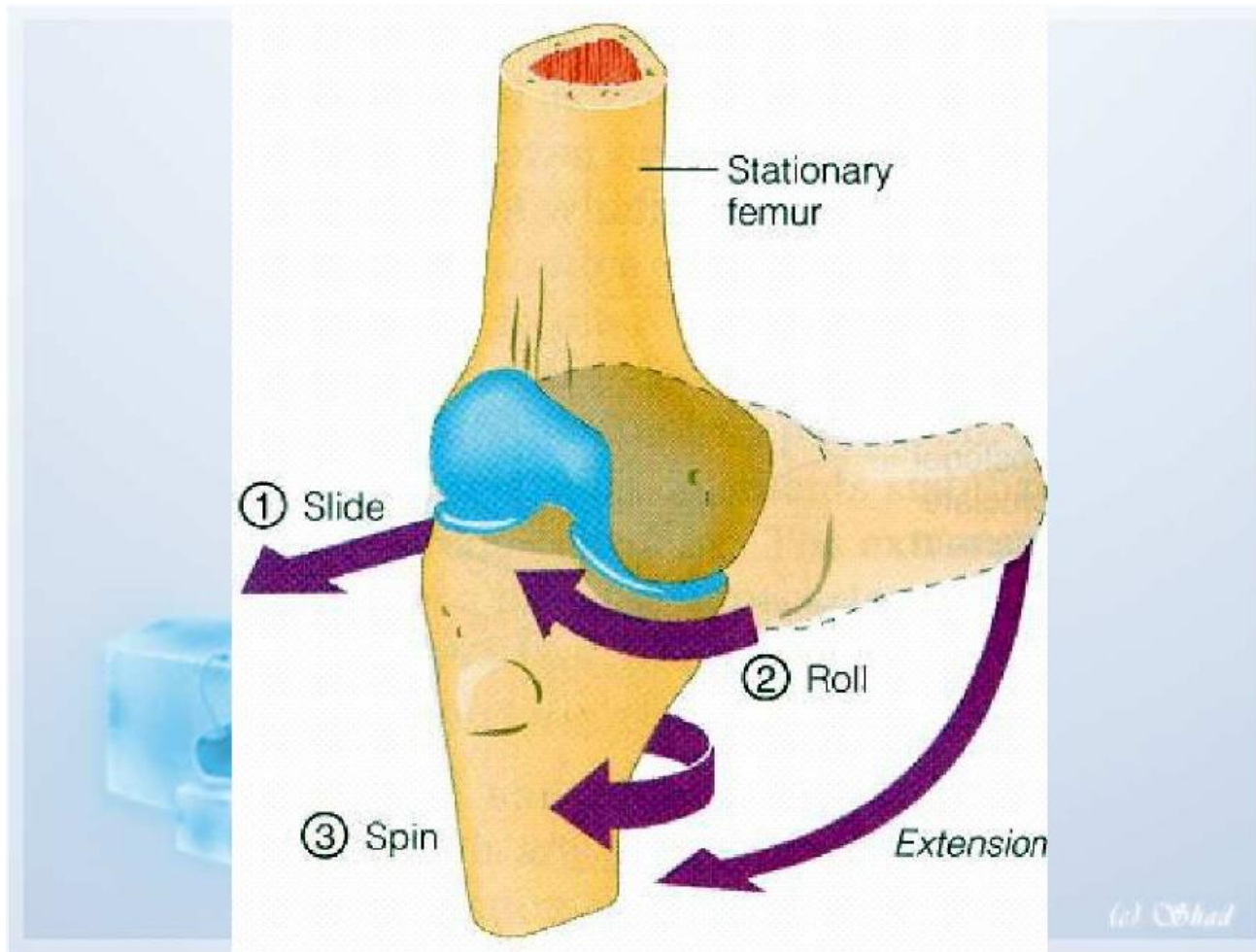


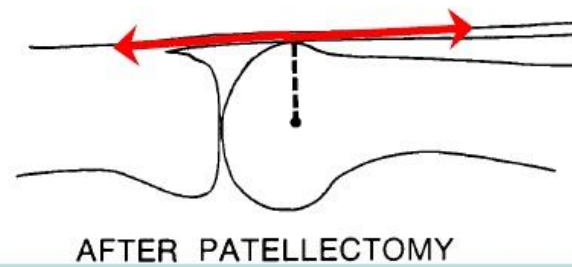
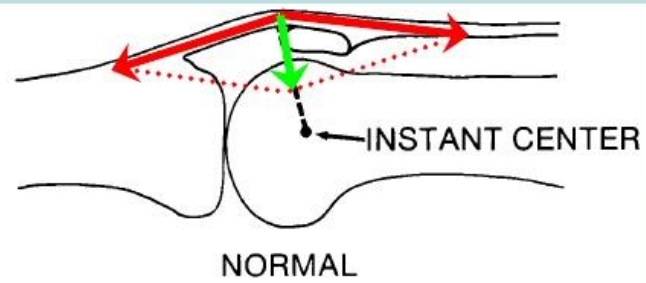
Muscle driven
musculoskeletal
simulations



Knee rotation

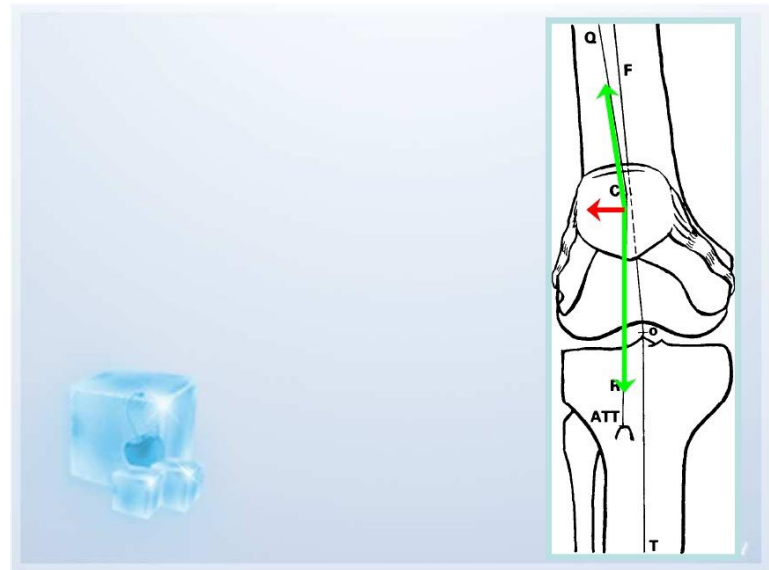
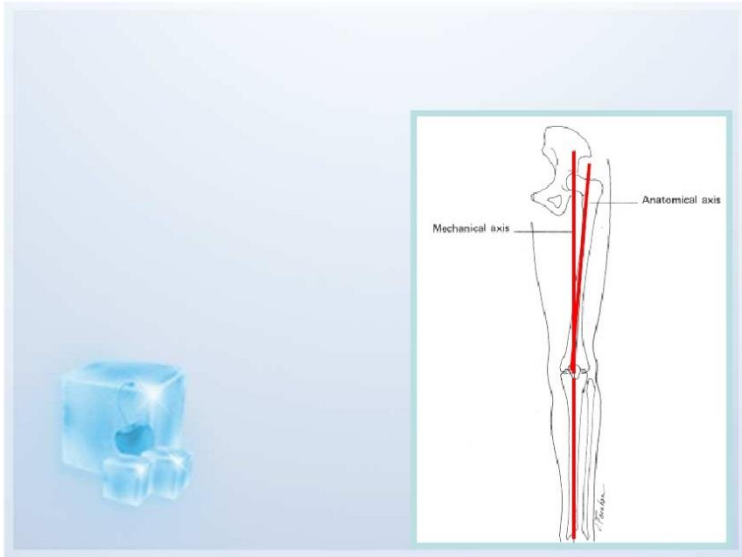




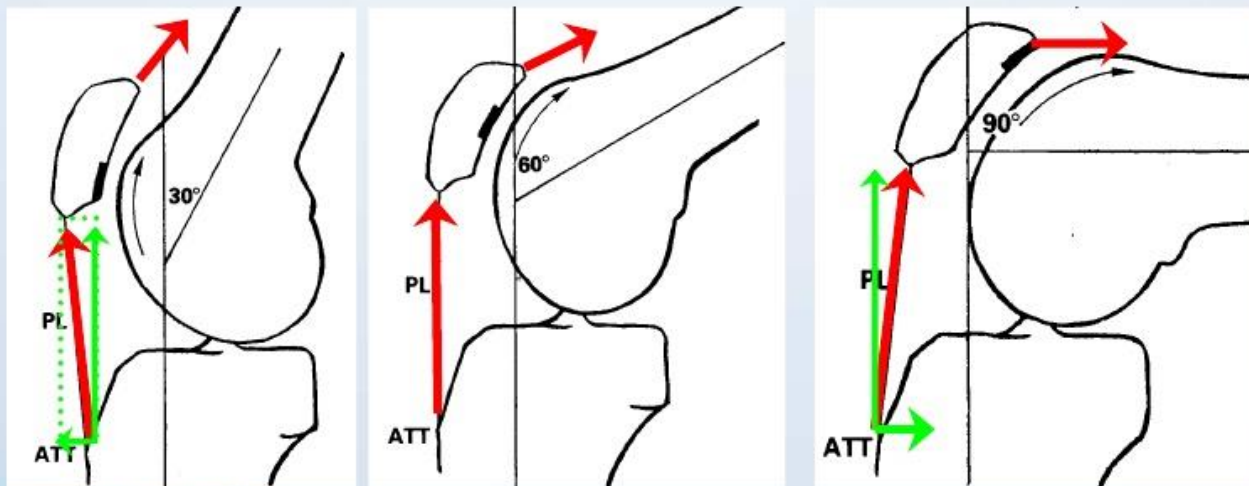


let's Chat

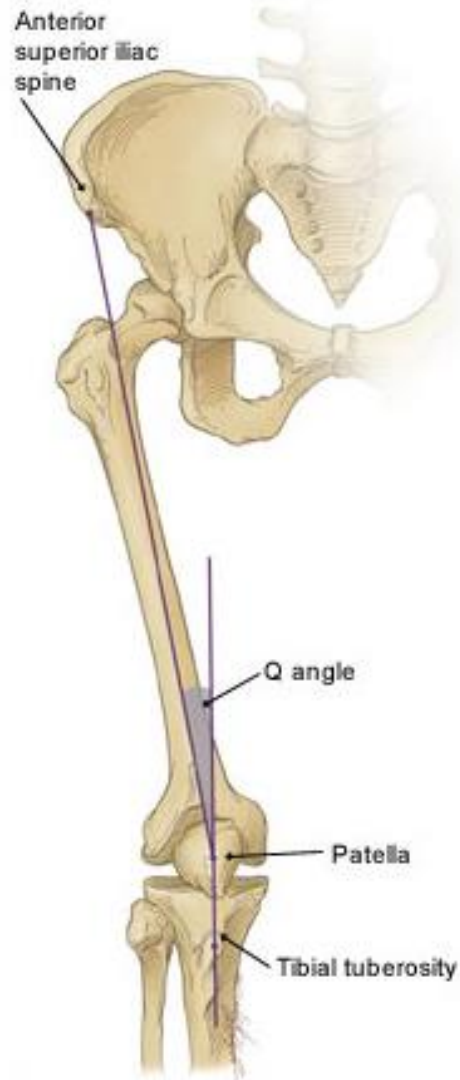
Resultant for
 e has a tenden
 c
 y to laterally translate the patella



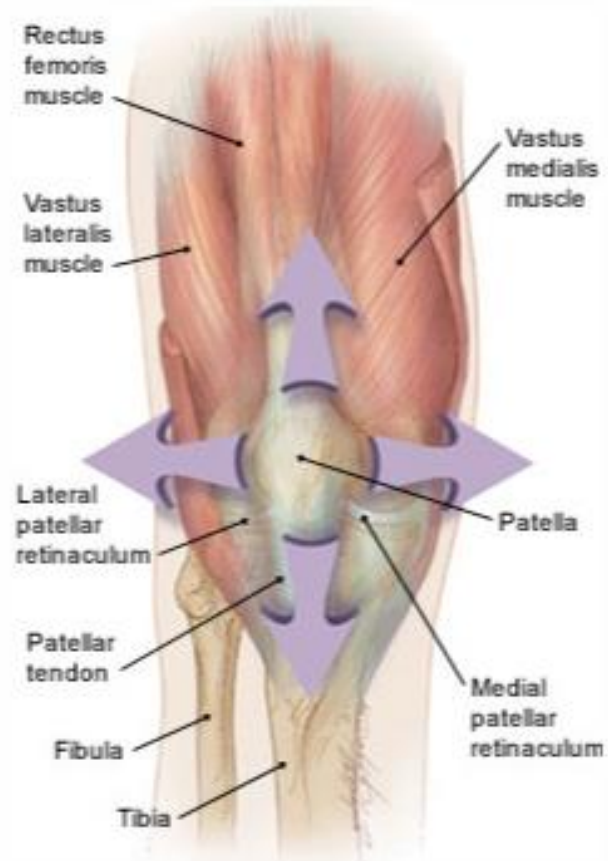
Patellar translation





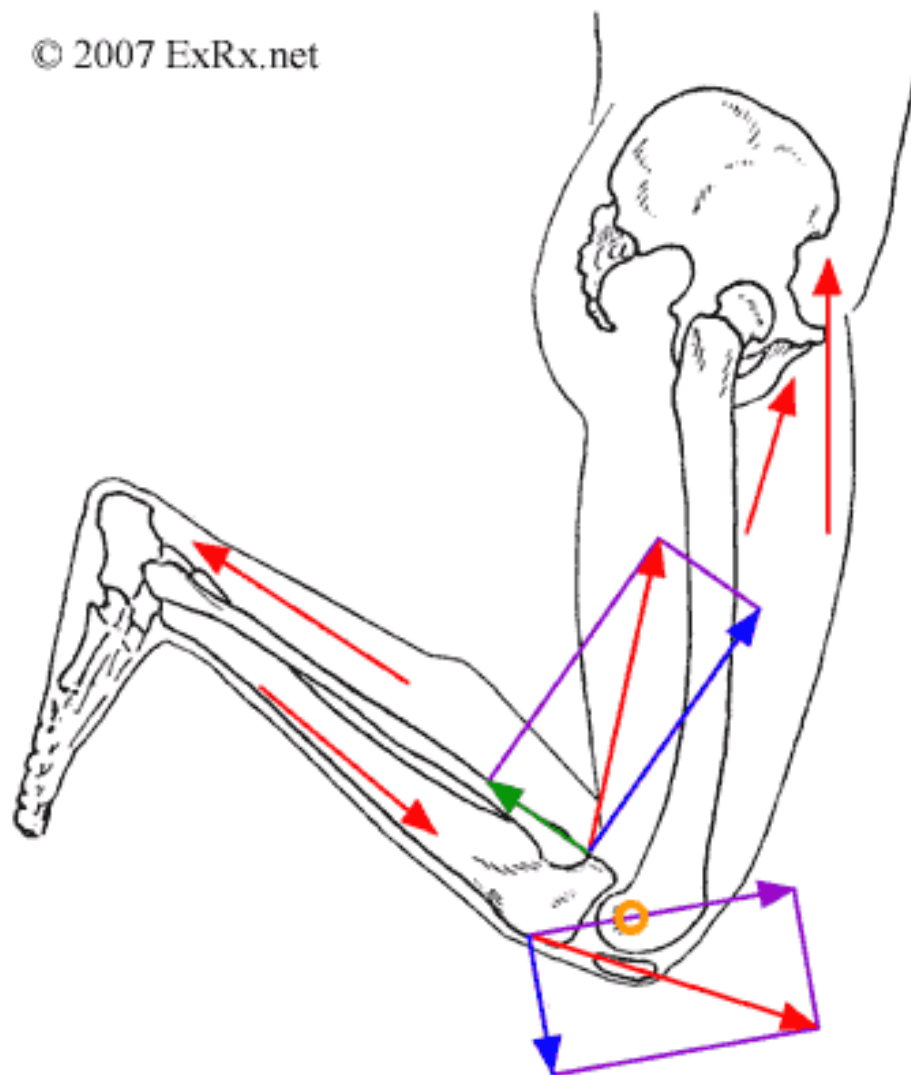


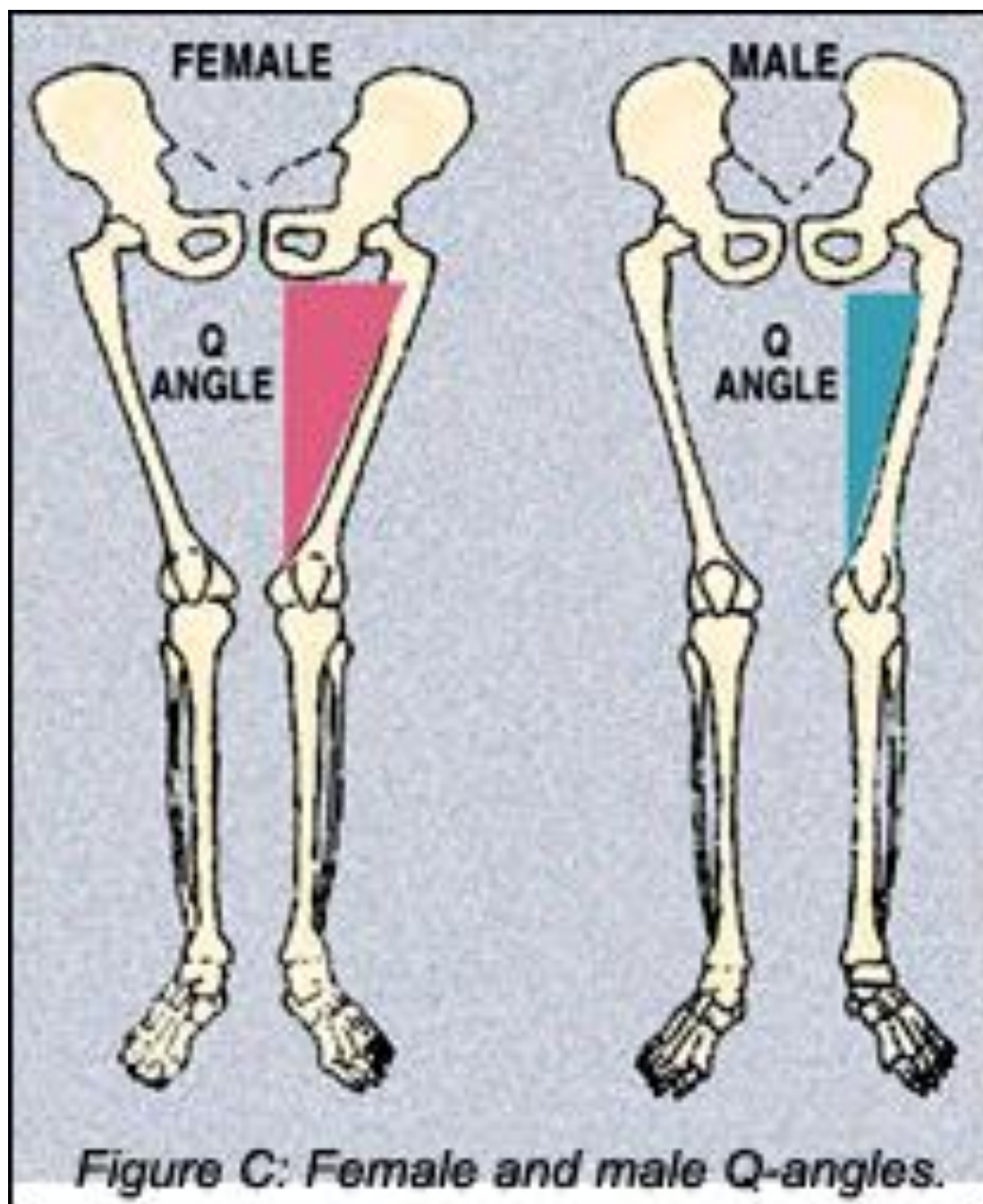
Q Angle measurement



Stabilisers of the Knee

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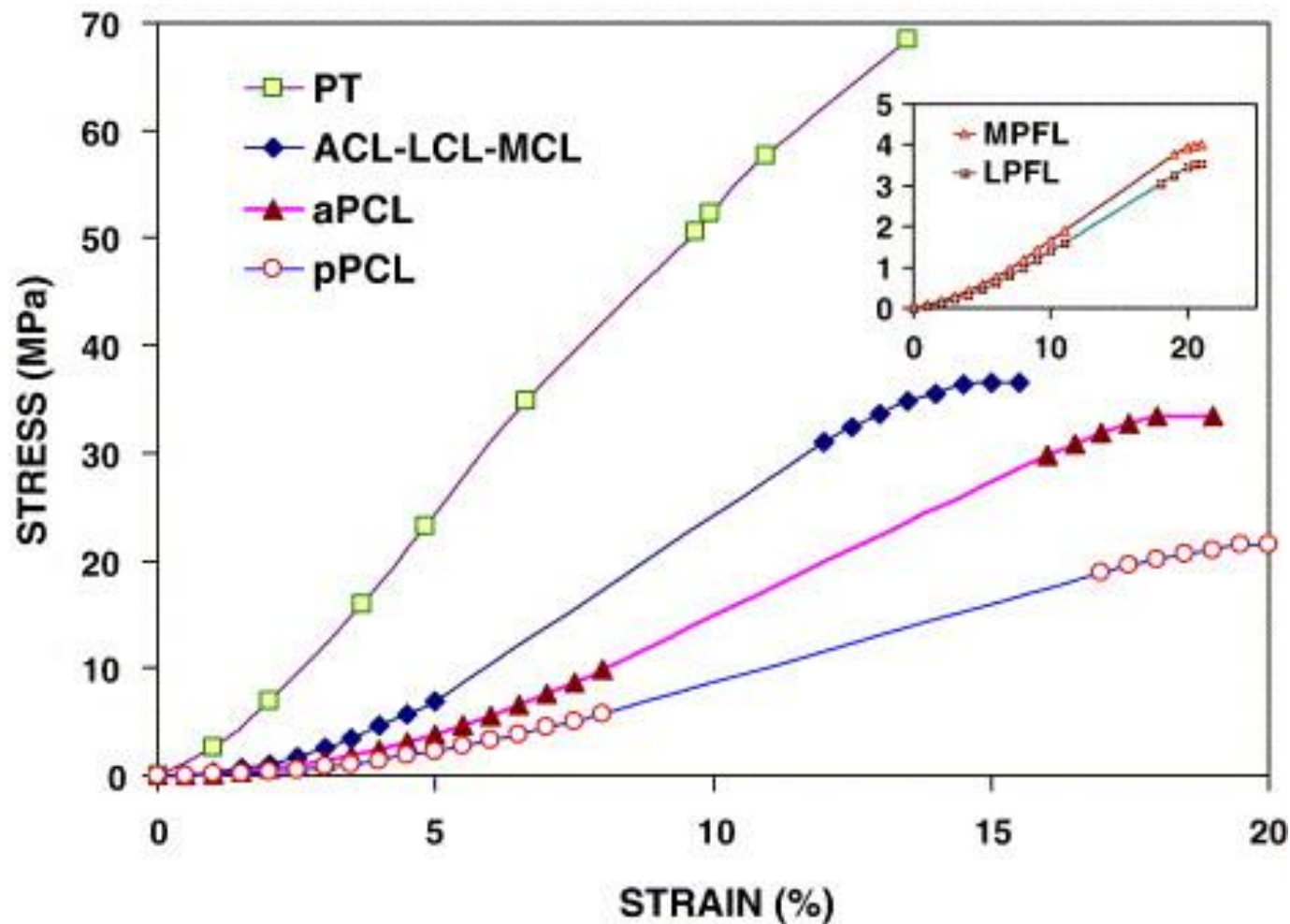


Fig. 2. Stress–strain curves for various ligaments used in the model and the patellar tendon, PT. ACL: anterior cruciate ligament, LCL: lateral collateral ligament, MCL: medial collateral ligament, aPCL/pPCL: anterior/posterior bundles of posterior cruciate ligament, MPFL/LPFL: medial/lateral patellofemoral ligaments.

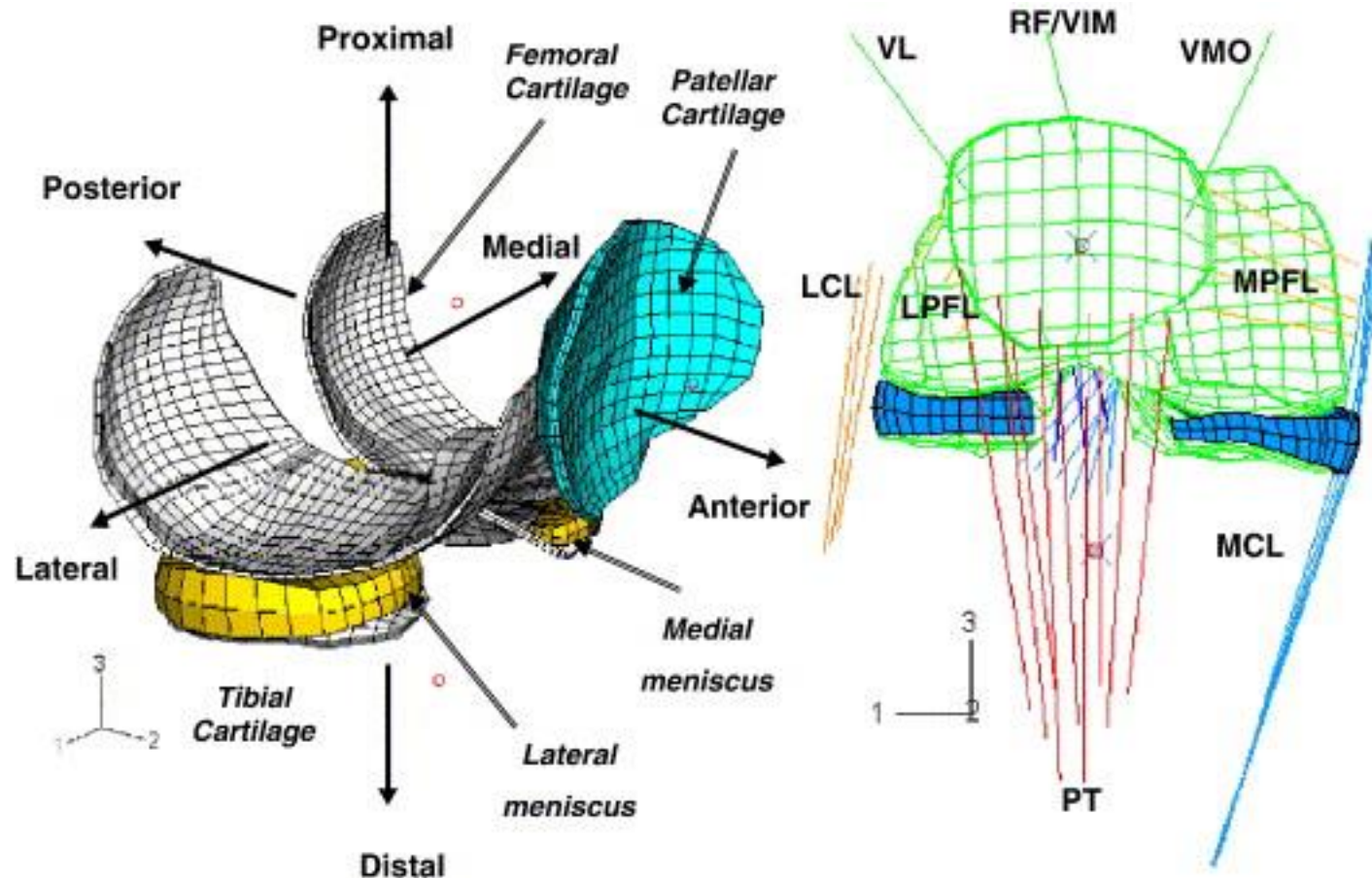
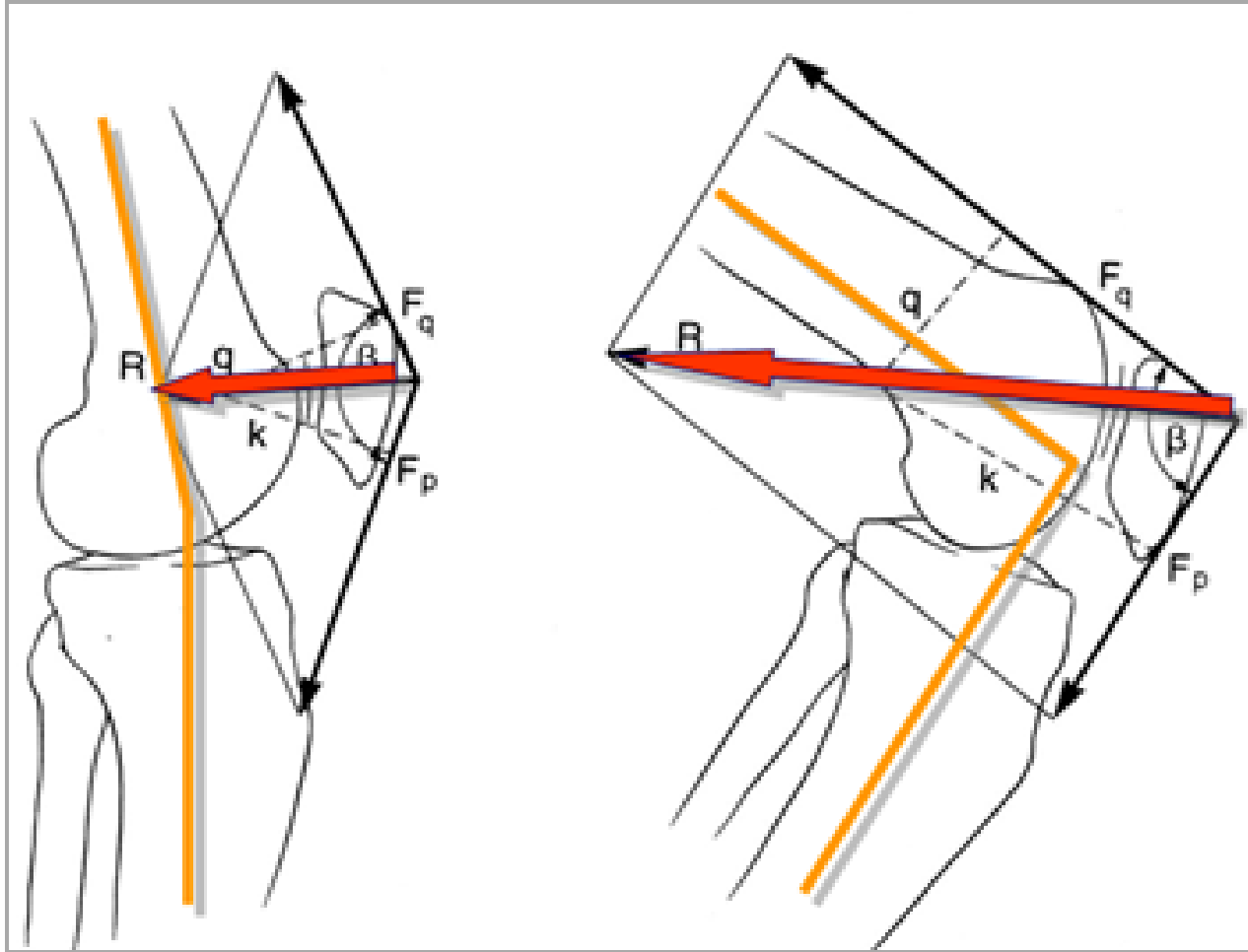
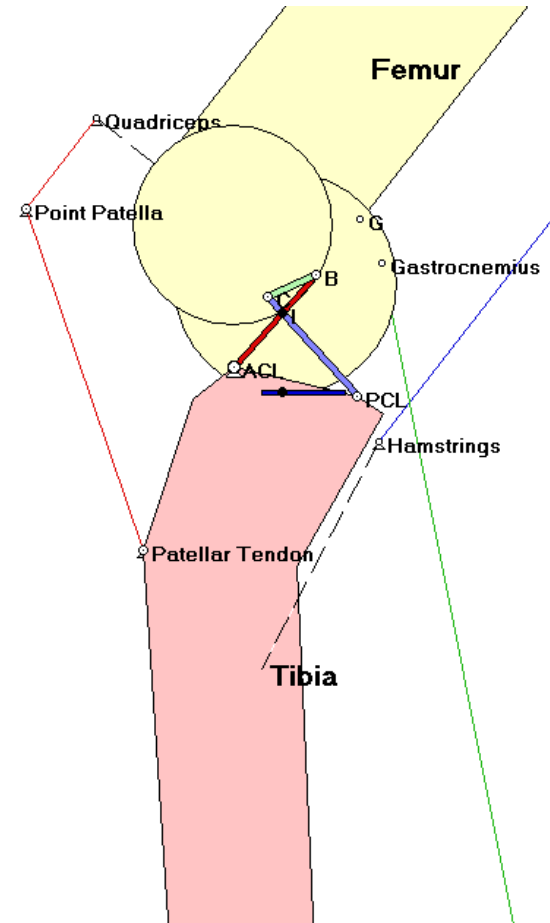
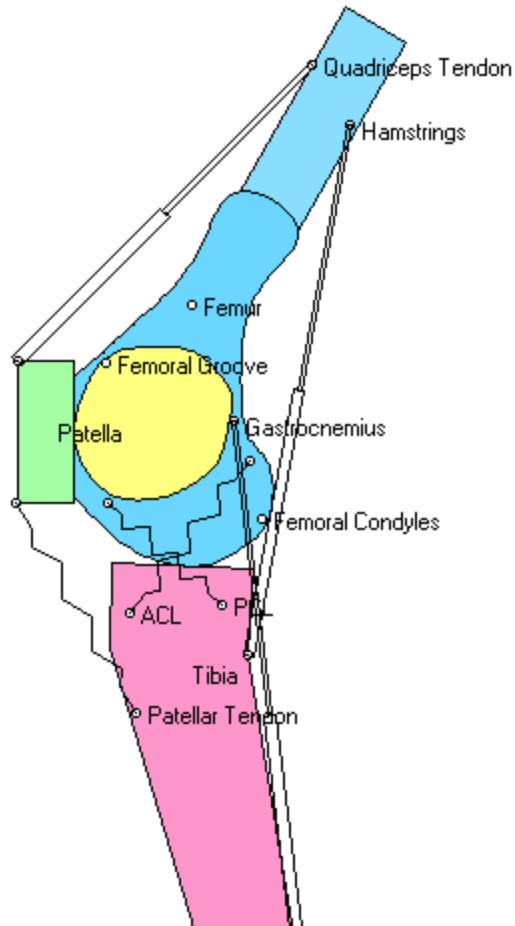


Fig. 1. The knee joint finite element models showing cartilage layers, menisci, ligaments, patellar tendon, and quadriceps muscles. Bony structures are shown only by their primary nodes. Quadriceps components considered are VMO: vastus medialis obliquus, RF: rectus femoris, VIM: vastus intermidus medialis, and VL: vastus lateralis (VL). LPFL: lateral patellofemoral ligament, MPFL: medial patellofemoral ligament.



Questions?



Synovial Joints: Knee –

Other Joints

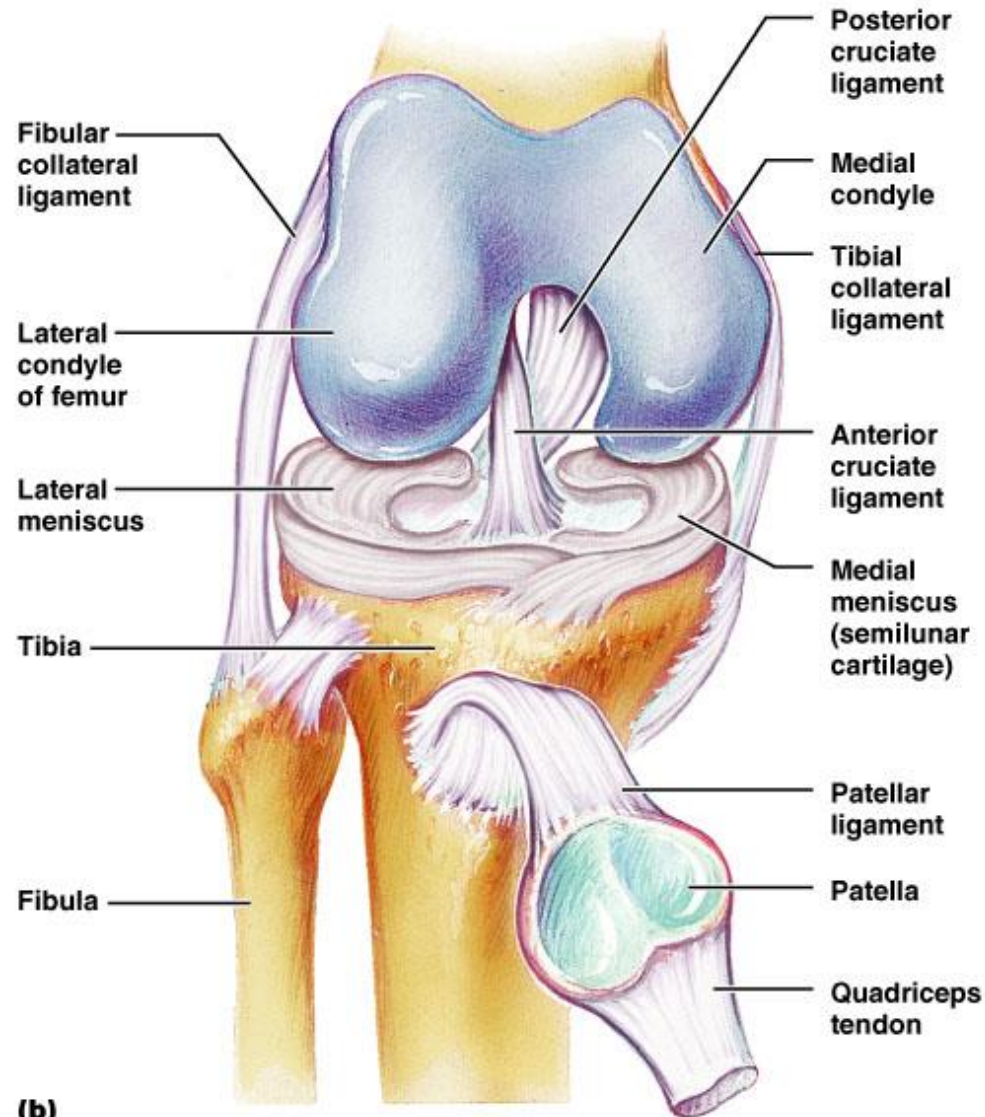


Figure 8.8b

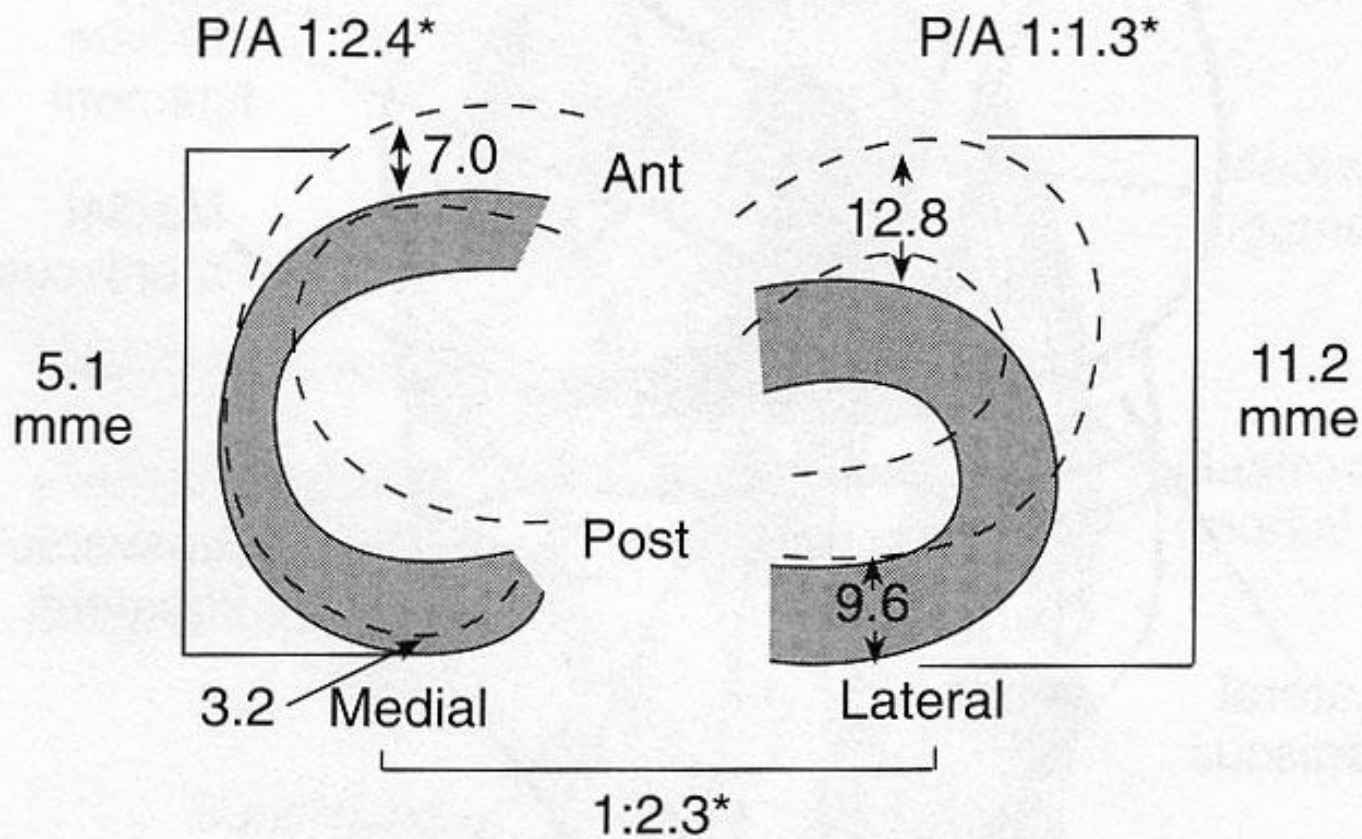
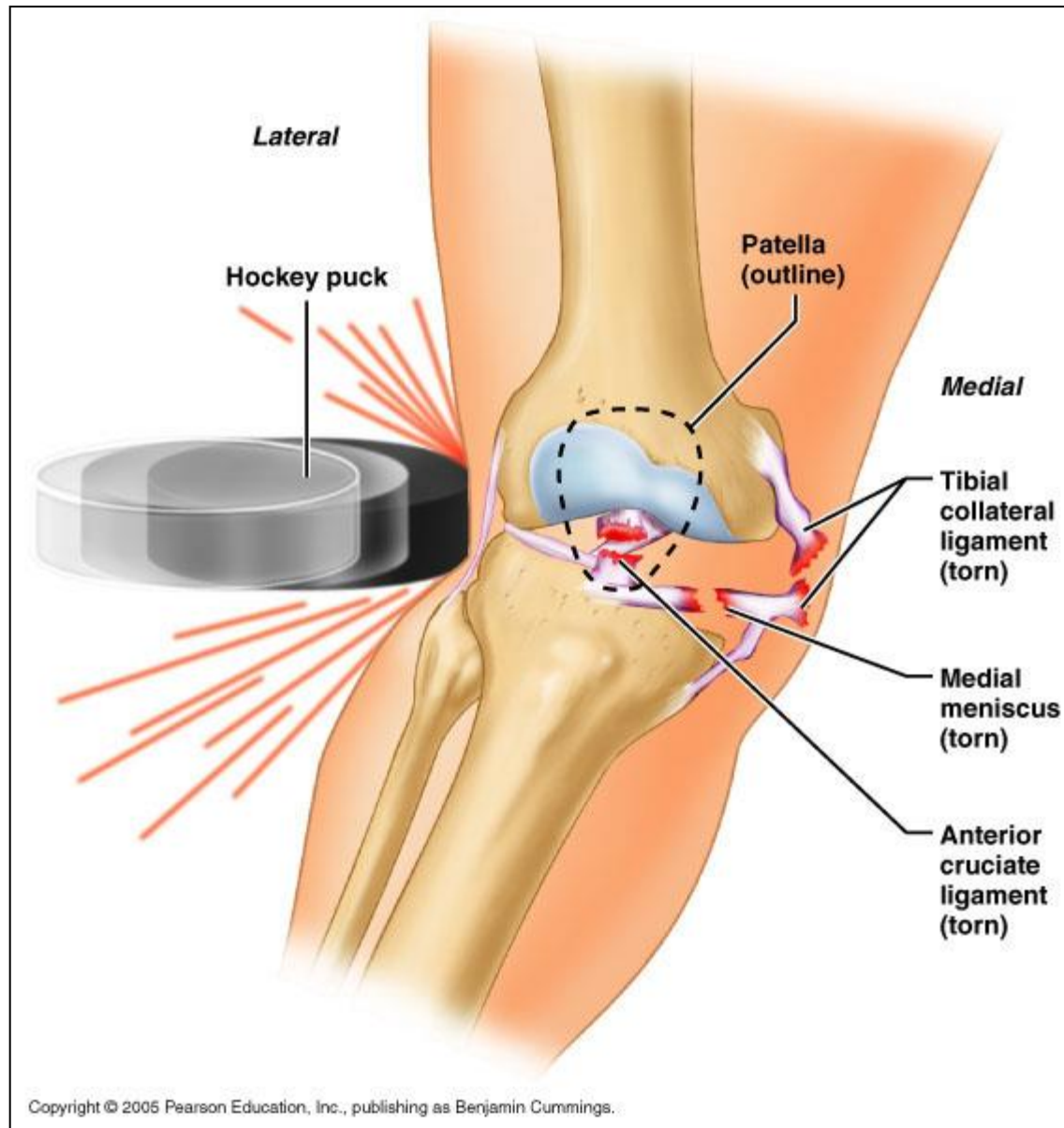


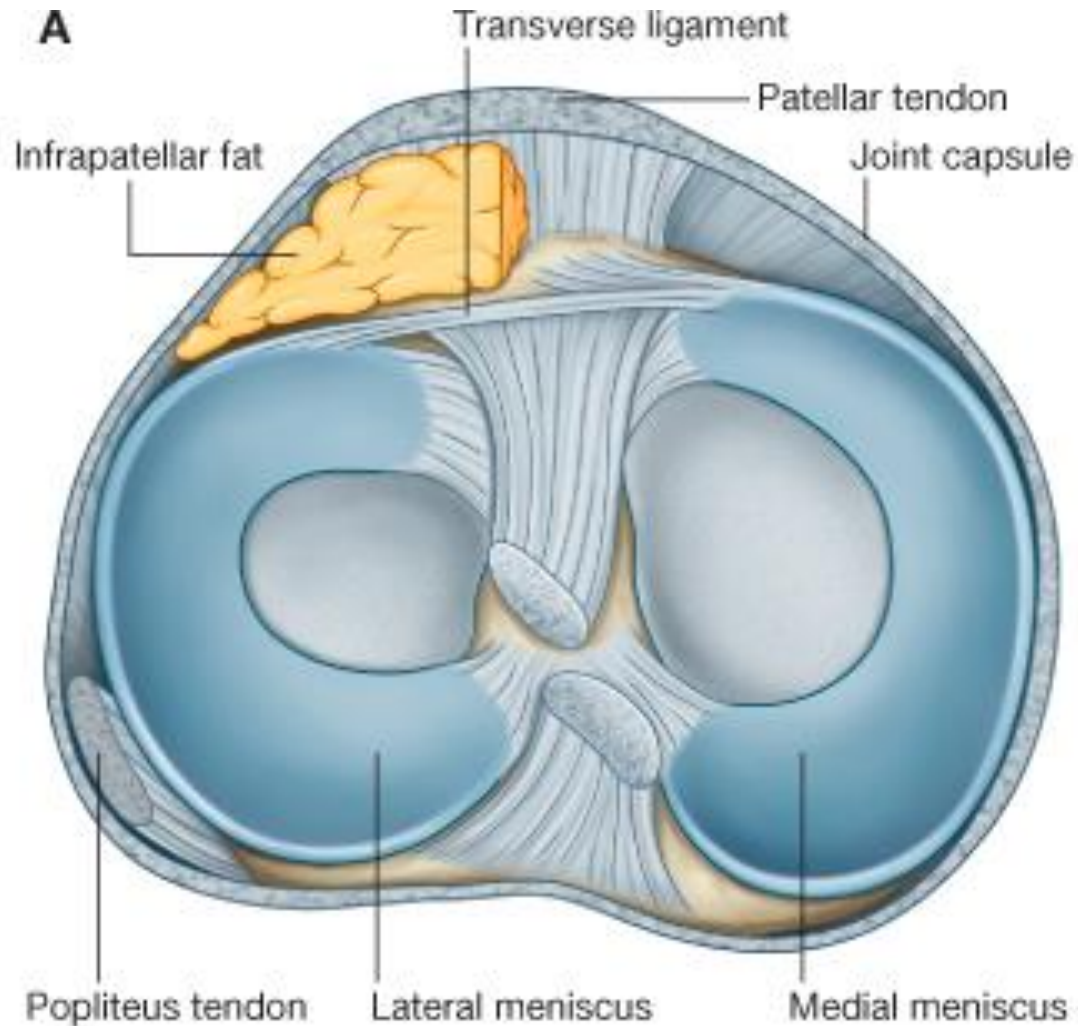
Figure 6.13 Diagram of mean meniscal excursion (mme) along the tibial plateau. Ant, anterior; Post, posterior; mme, mean meniscal excursion (millimeters); P/A, ratio of posterior to anterior meniscal translation during flexion. * $p < .05$ by Student's t test analysis.

Adapted from Thompson et al. 1991.

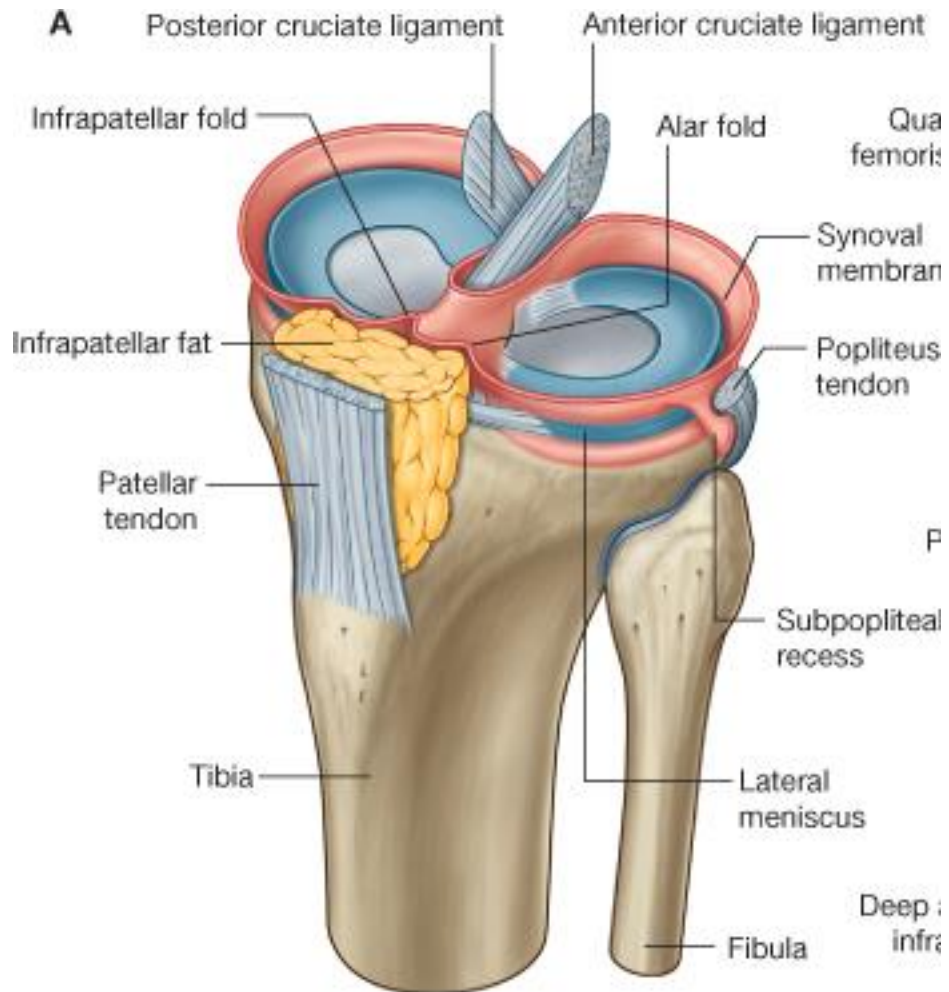
Knee Injury



Menisci



Menisci



Patella as a pulley

- a pulley changes the direction of an applied force
- the patella helps to support the work of the quadricep muscles during the contraction of the quadricep that allows for extension of the knee

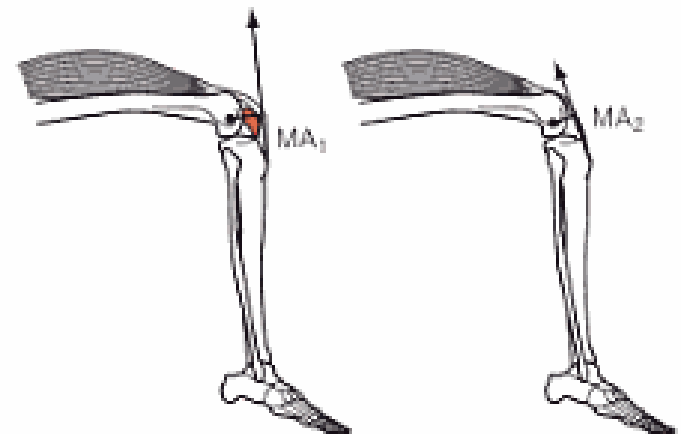
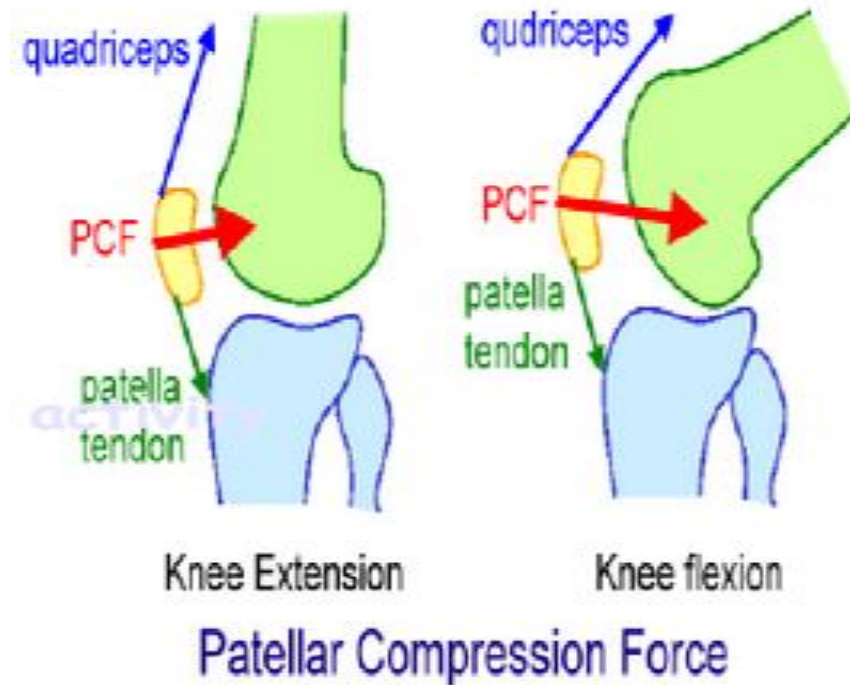
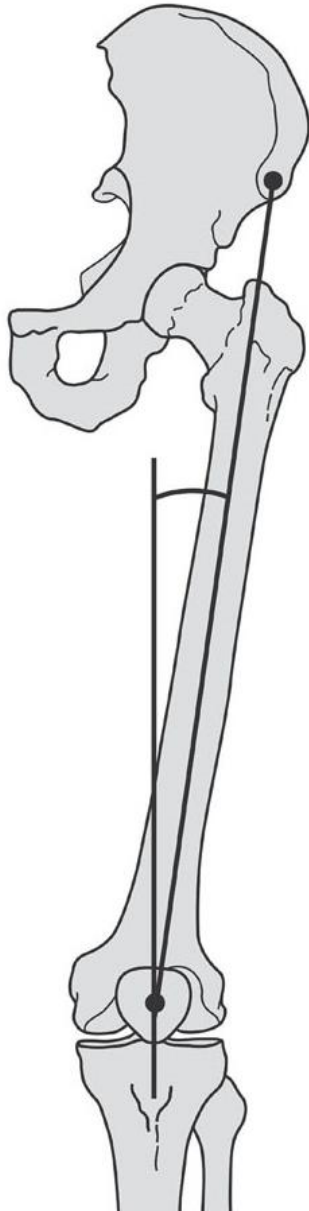
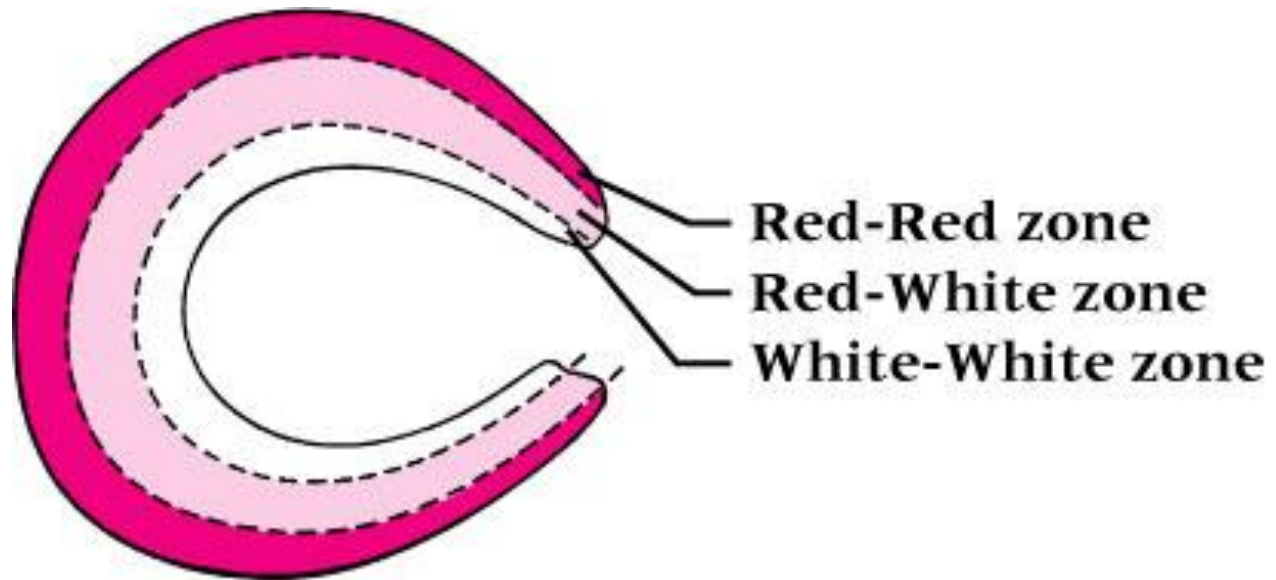


FIGURE 8.24 The angle of pull changes when the patella is removed.

Q Angle





Problem!

- Givens: Quadriceps tendon is inserted on the tibia 5 cm from the knee joint, and is at a 30deg angle. Weight of the lower leg is 48 N. Center of gravity of the lower leg is 0.20 m from the knee joint.
1. Determine F_{quad} required to hold the lower leg in static equilibrium
 2. Determine the joint reaction force of the femur

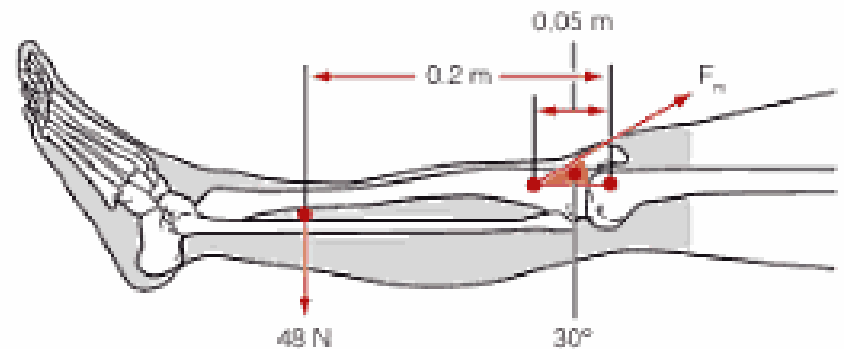


FIGURE 7.1 Solving for quadriceps force. Why is the necessary muscle force so large?

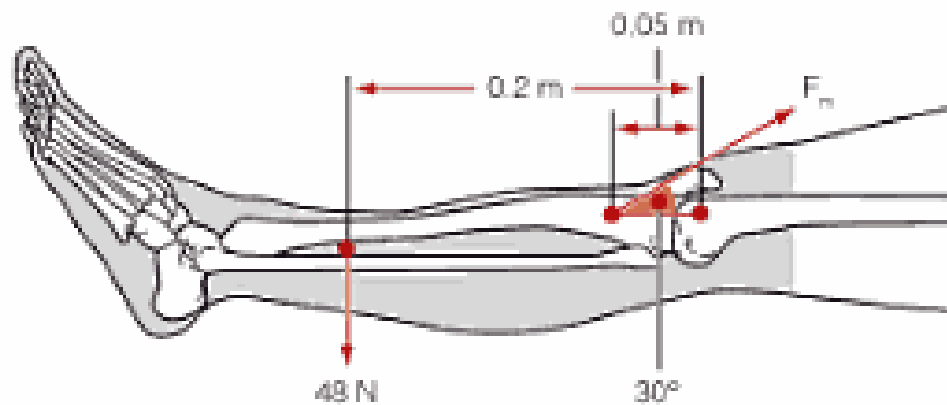
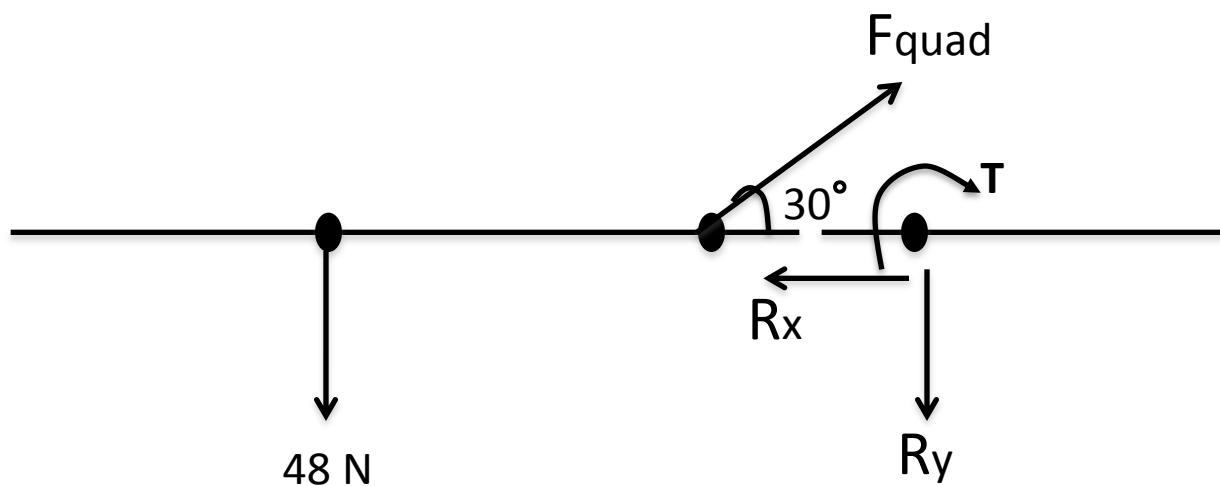
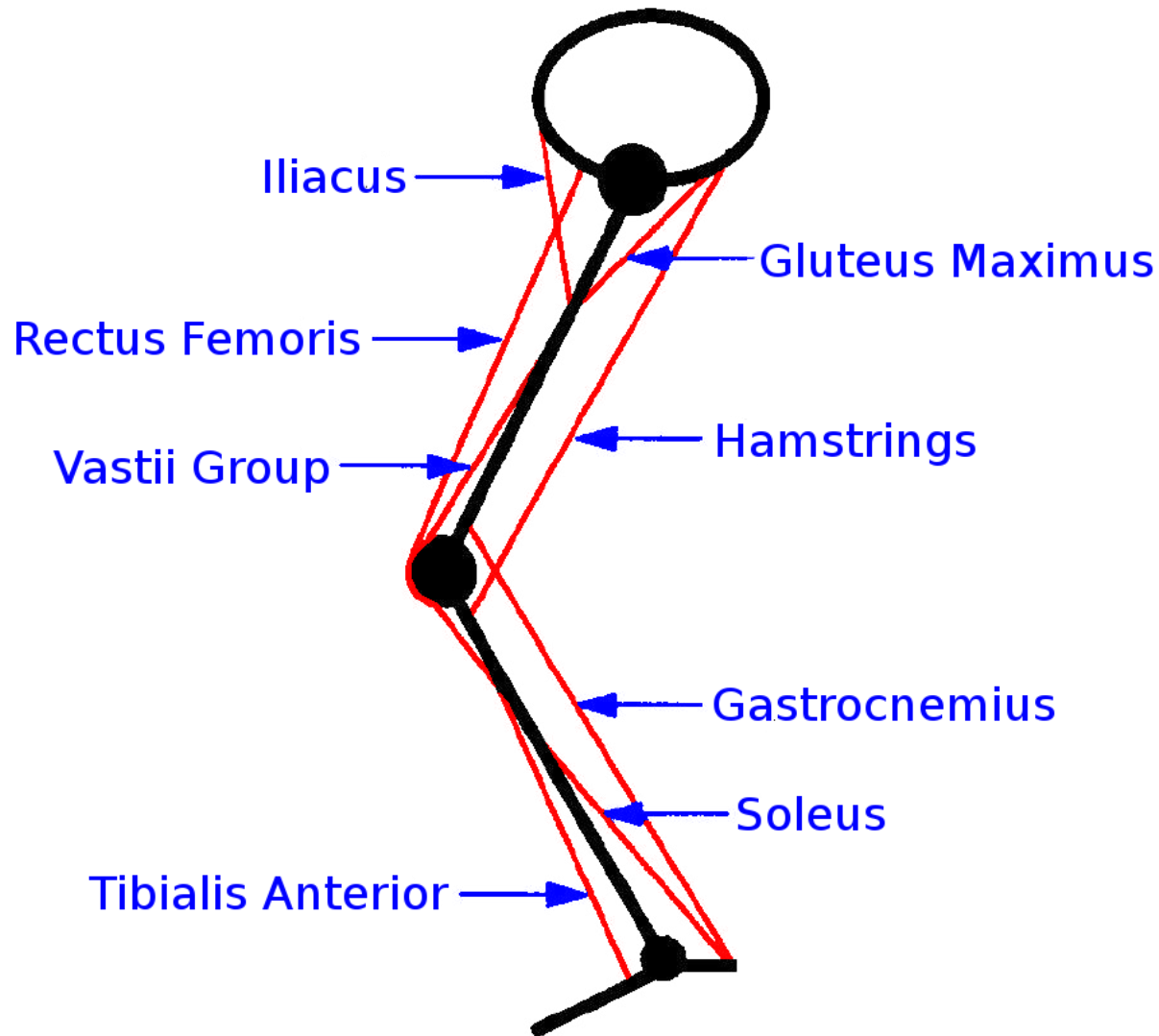


FIGURE 7.1 Solving for quadriceps force. Why is the necessary muscle force so large?

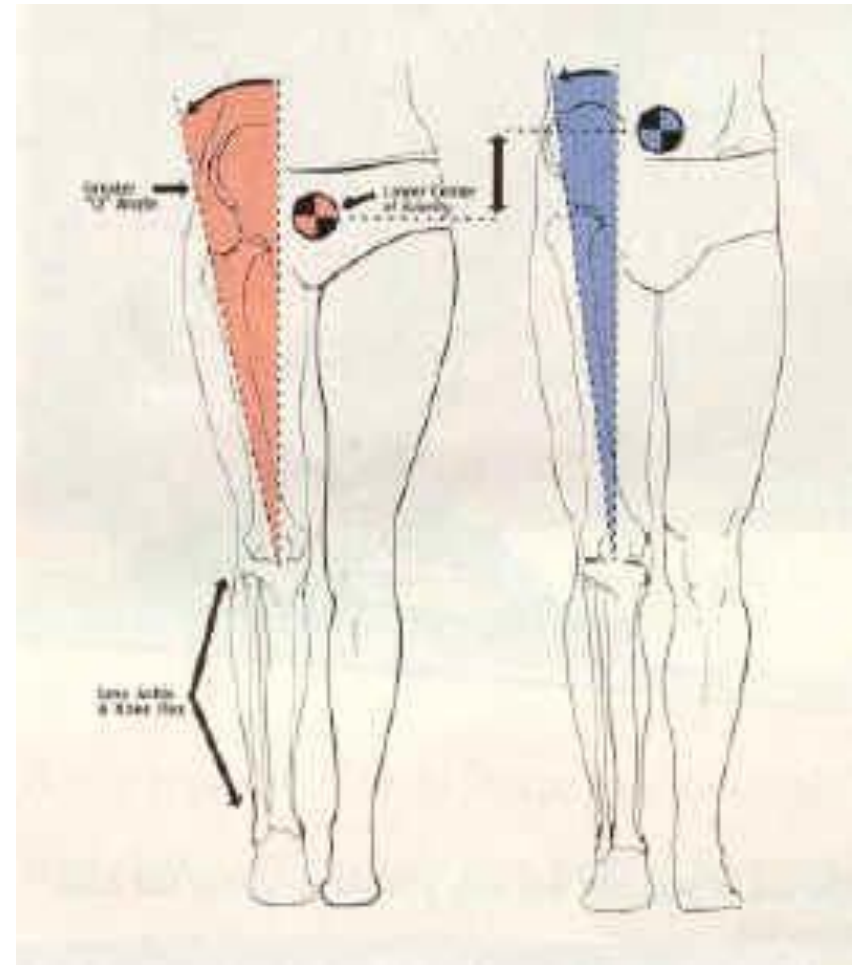


Muscles of the Leg



“Q” Angle

- Normal
 - about 15°
- Males vs. Females
 - wider pelvis



Q Angle

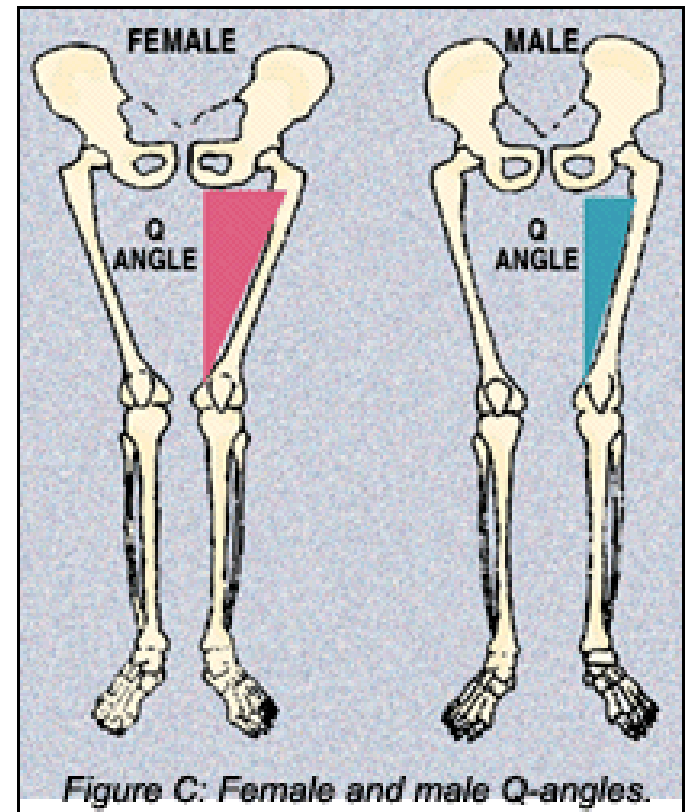
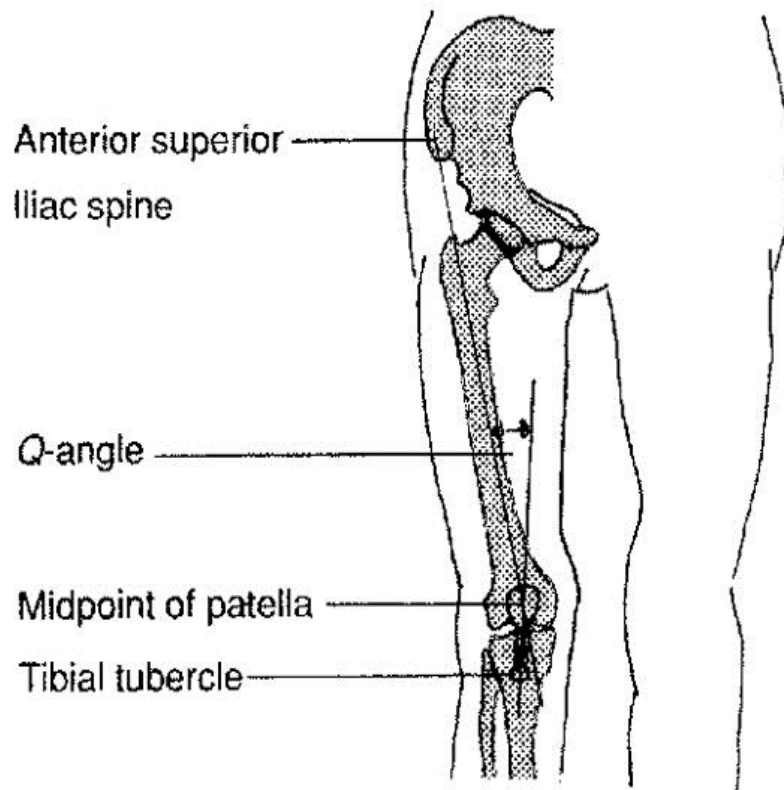


Figure 1: Staff illustration

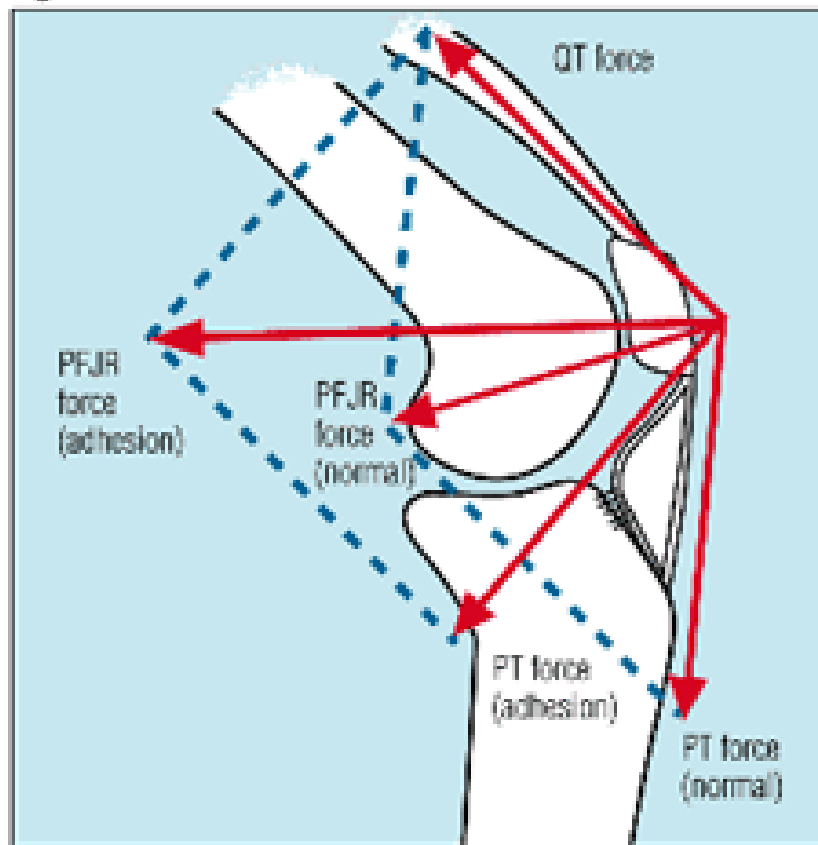
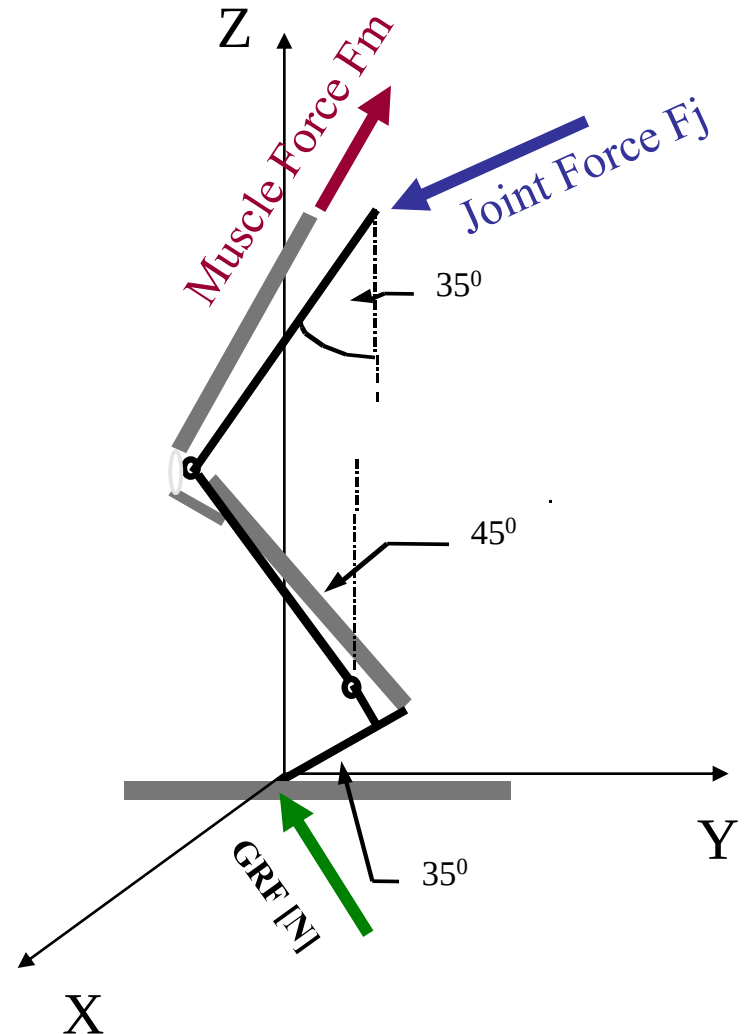


FIGURE 1. Adhesions alter the vector forces and increase loading of the knee. Patellofemoral joint reaction (PFJR) force results from the sum of the vector representing the quadriceps tendon (QT) and the patellar tendon (PT) forces. In the presence of adhesions of the patellar tendon to the tibia, the patellar tendon force is altered, resulting in an increased PFJR and increased loading of the patellofemoral joint.

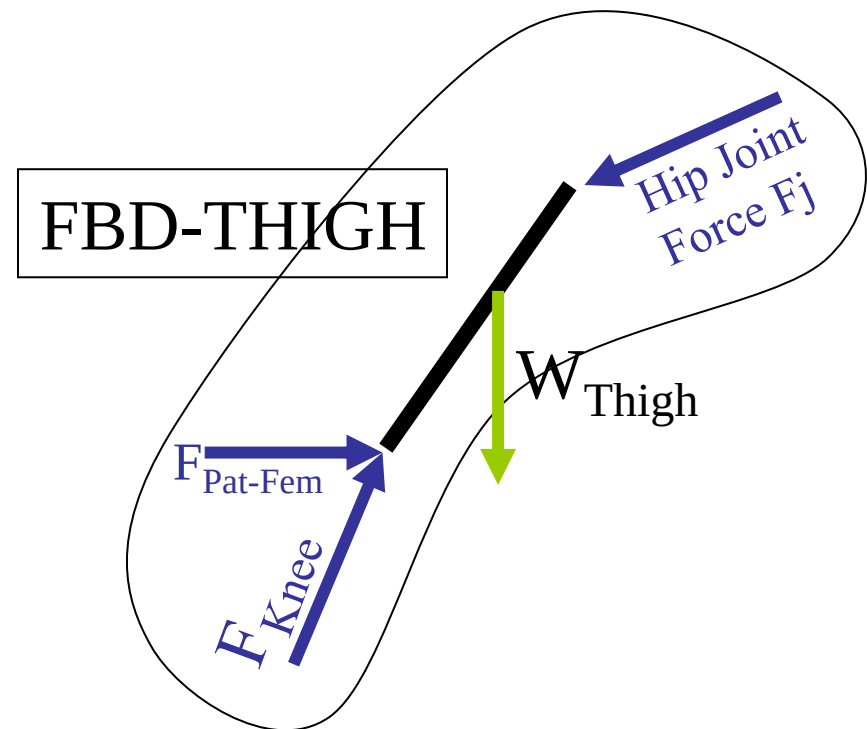
Adapted with permission from Ahmad et al: Effects of patellar tendon adhesion to the anterior tibia on knee mechanics. Am J Sports Med 1998;26(5):715-724.

Free Body Diagram

- To obtain muscle and joint forces: $\Sigma \vec{F} = 0$
 $\Sigma \vec{M} = 0$
- That kind of solution will only provide information on the hip joint and thigh muscle forces

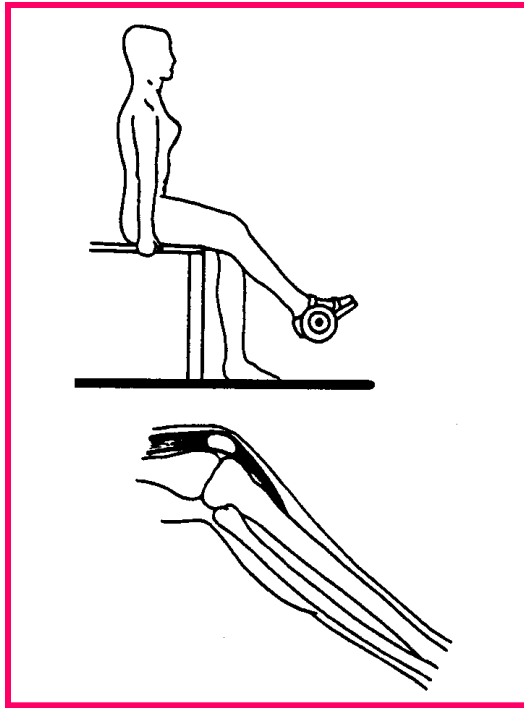


The Thigh



Knee - Example

A person is wearing a weight boot and doing lower leg flexion/extension exercises to strengthen the quadriceps. Determine F_M and F_J for the sitting position shown.



$$W_1 = 0.06W$$

$$W_0 = 0.06W$$

$$c = 0.28H$$

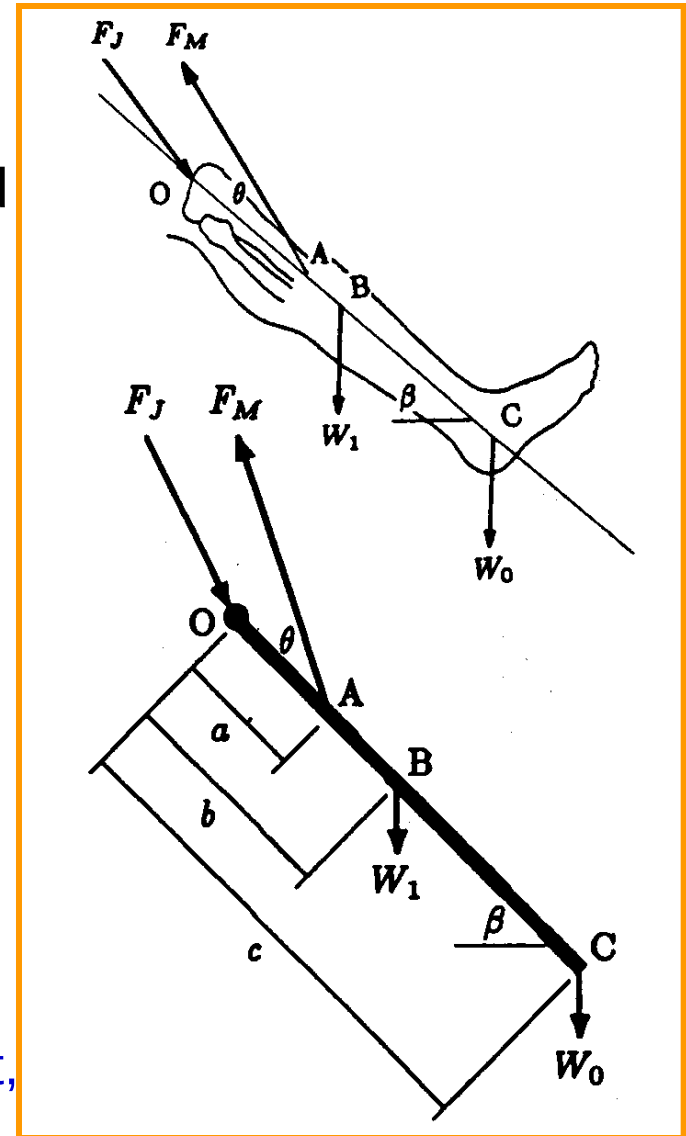
$$b = 0.14H$$

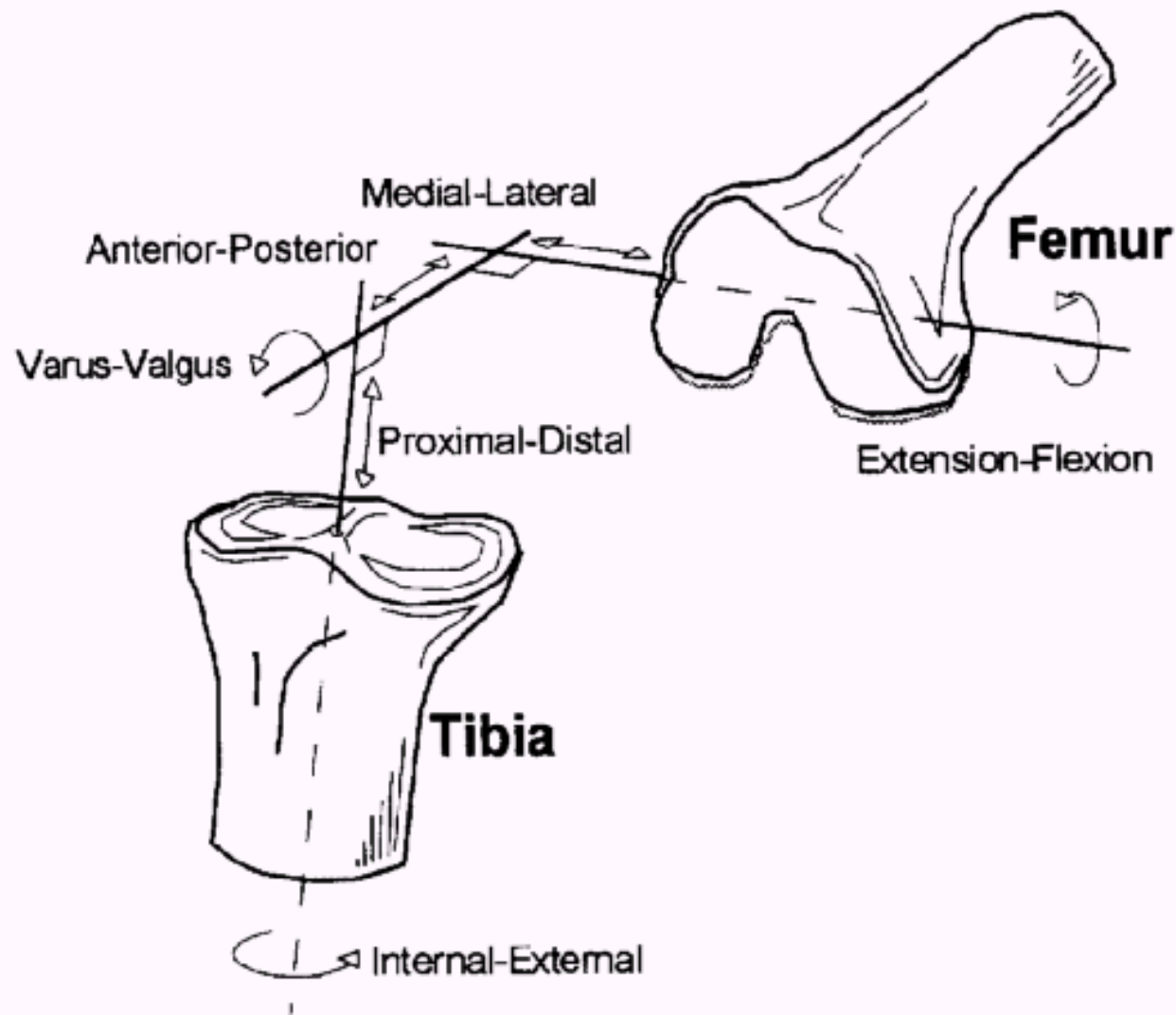
$$a = 0.08H$$

$$\beta = 45^\circ$$

$$\theta = 15^\circ$$

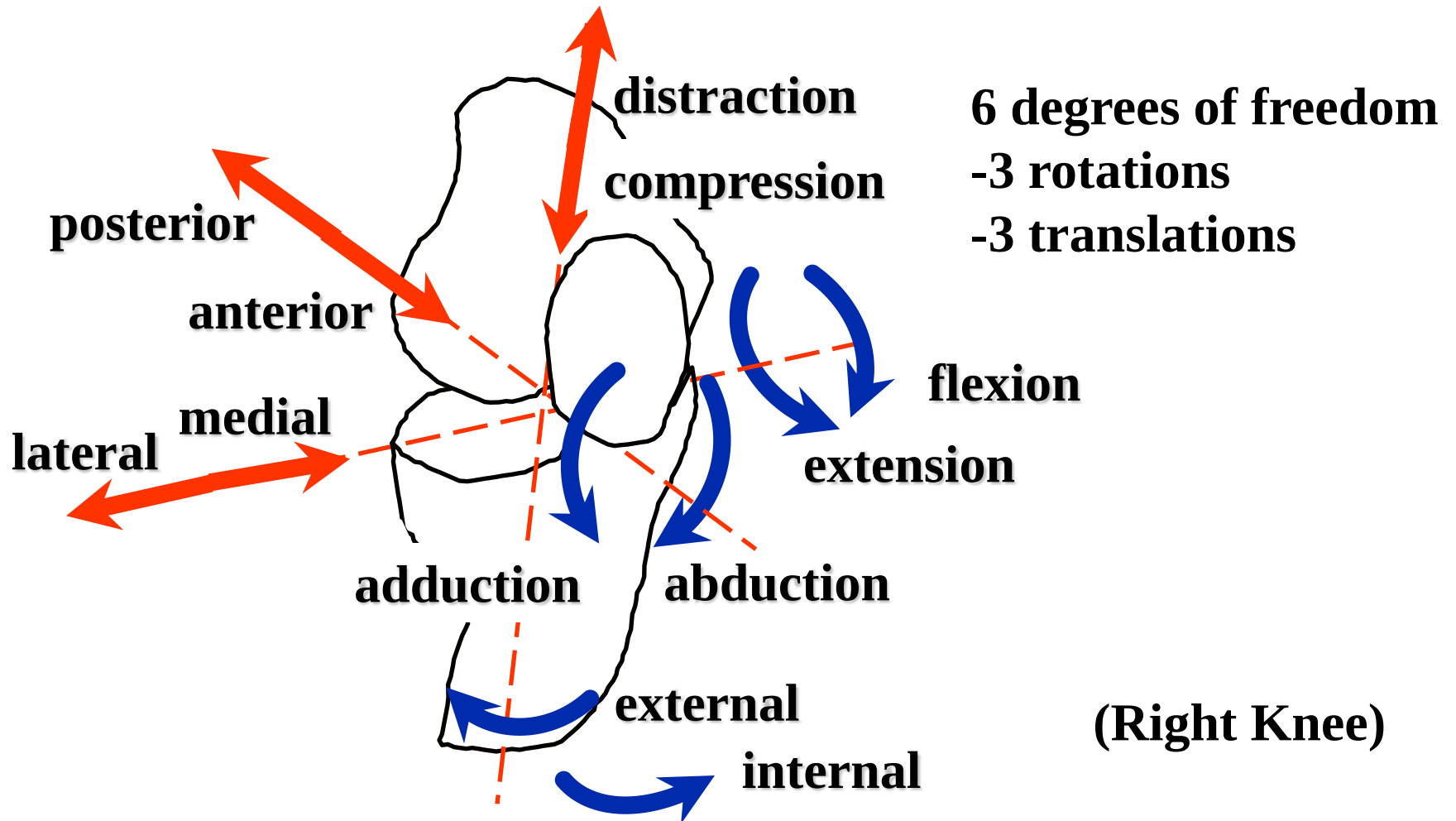
W – Body Weight,
 H – Body Height





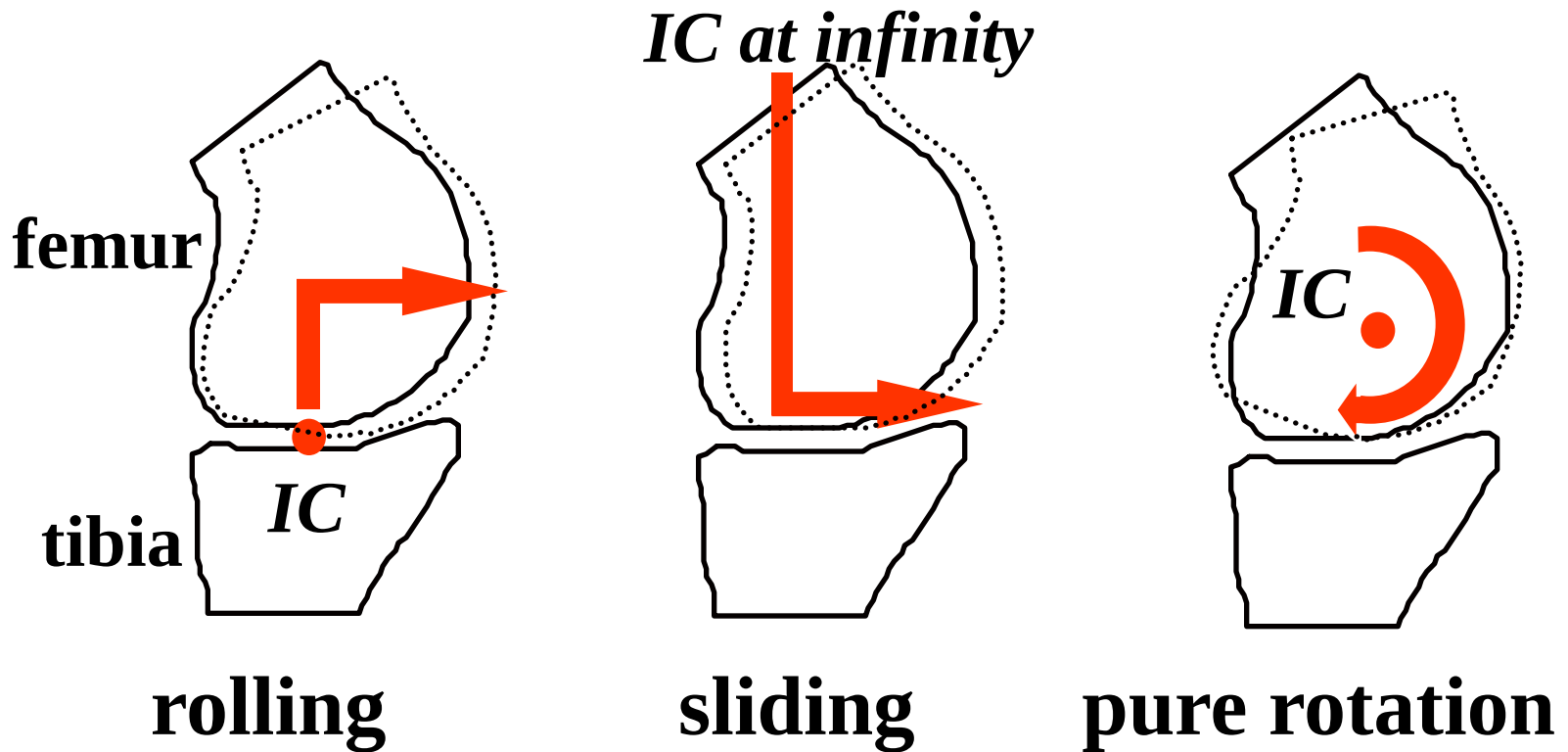
Adapted from Fu, Harner, & Vince (eds). *Knee Surgery*. Baltimore, MD: Williams & Wilkins, 1994.

Knee Joint Motion

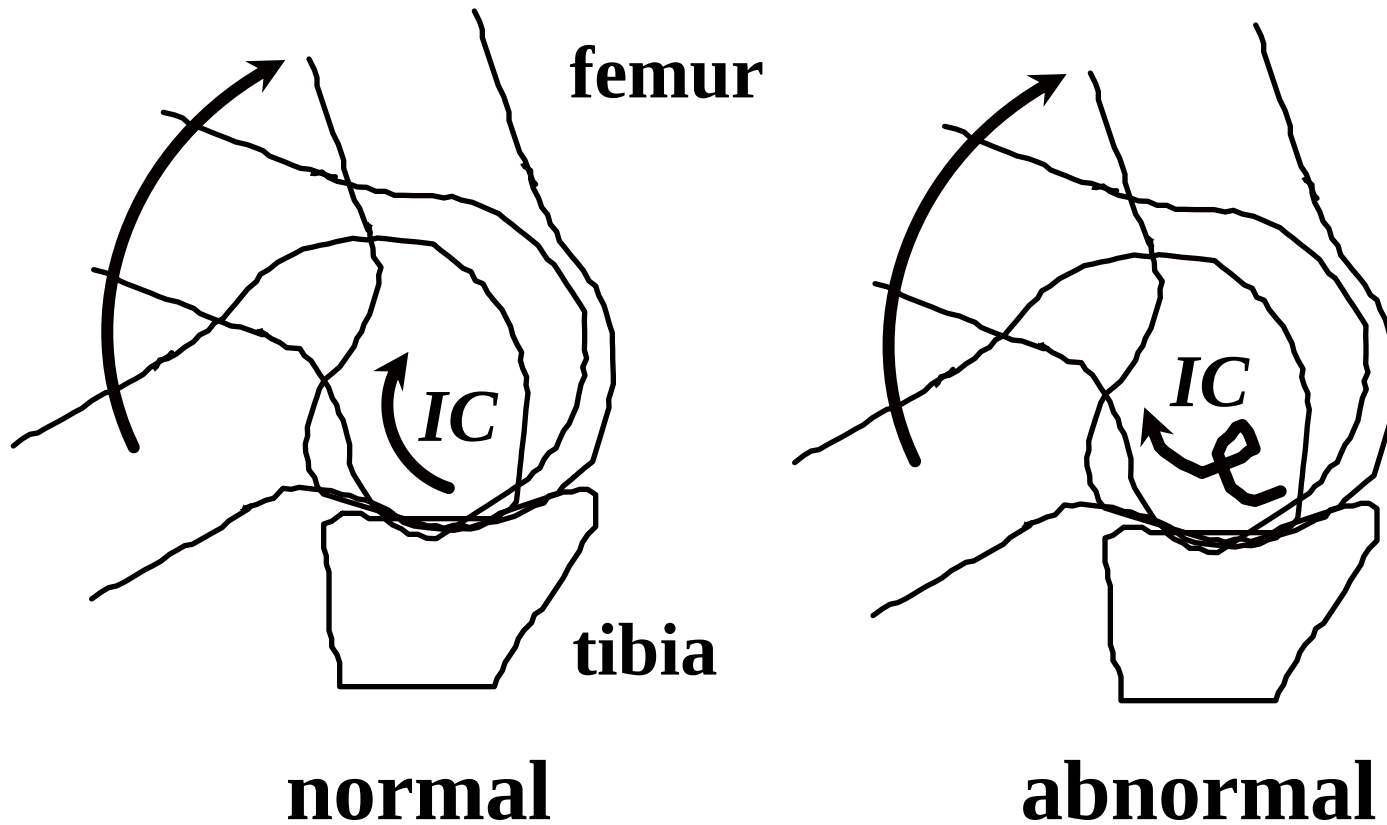


Instant Centres (Centre of Rotation) for Femur Motion during F/E

(sagittal view)

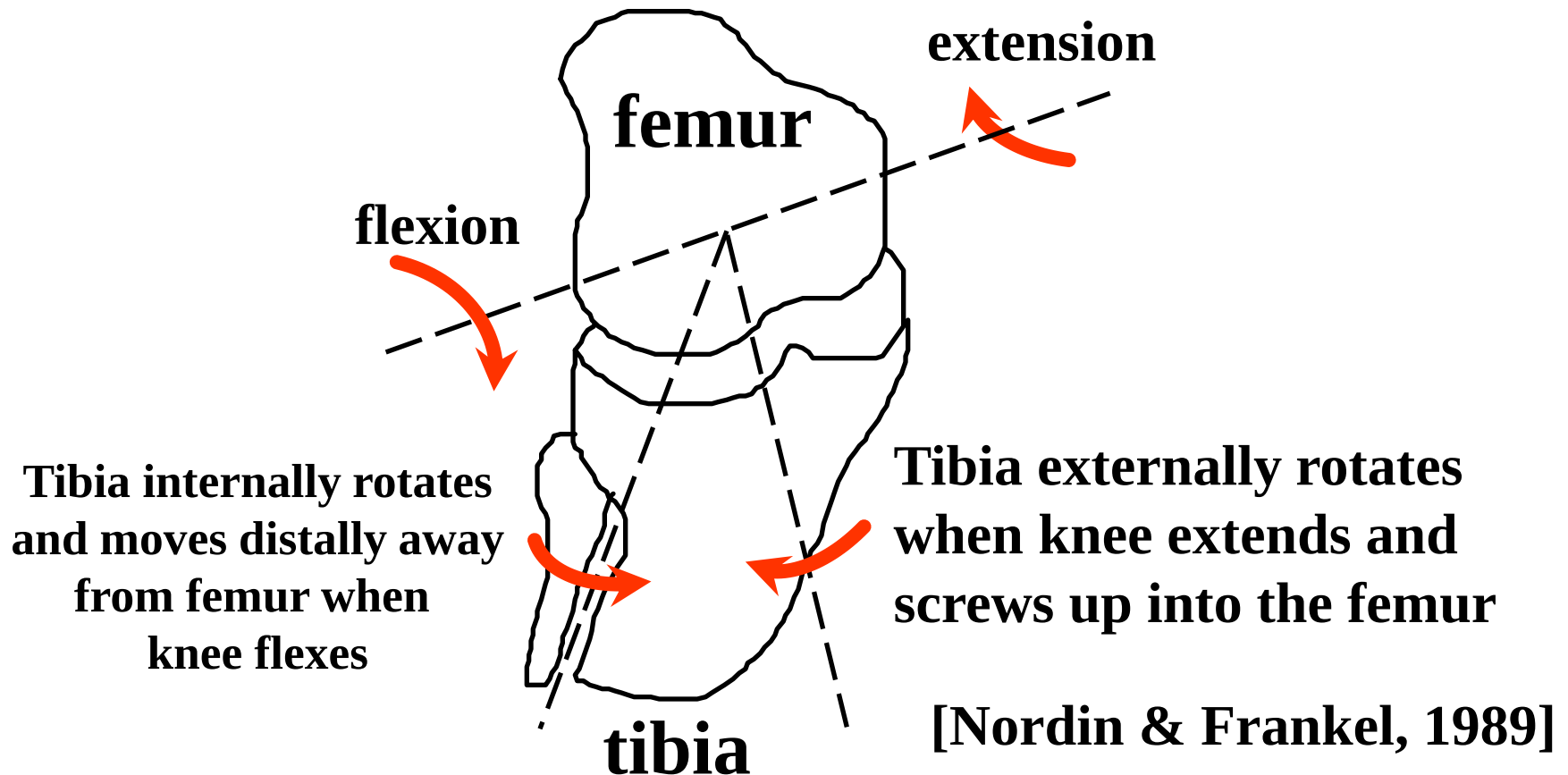


Instant Centre Movement during F/E (sagittal view)

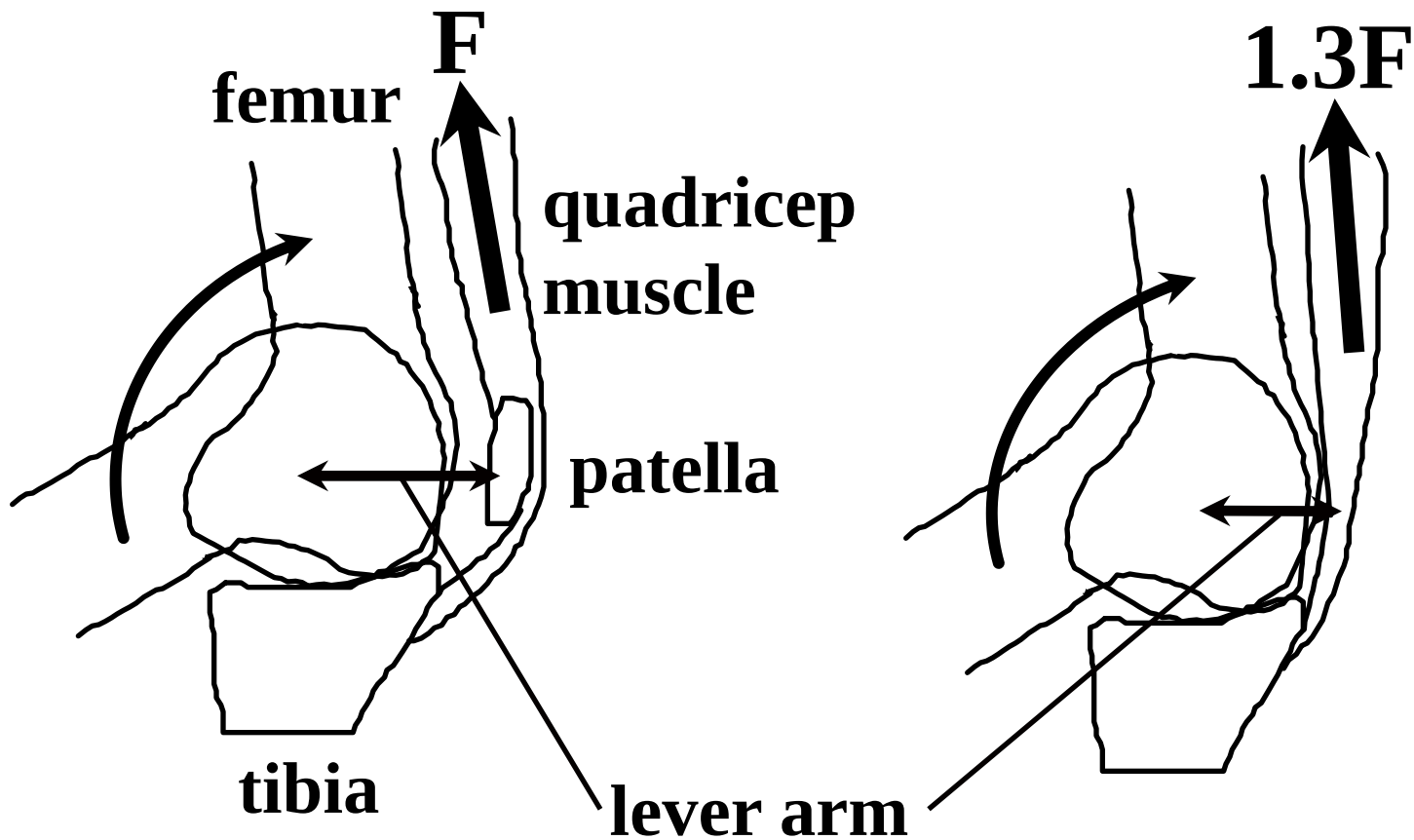


[Nordin & Frankel, 1989]

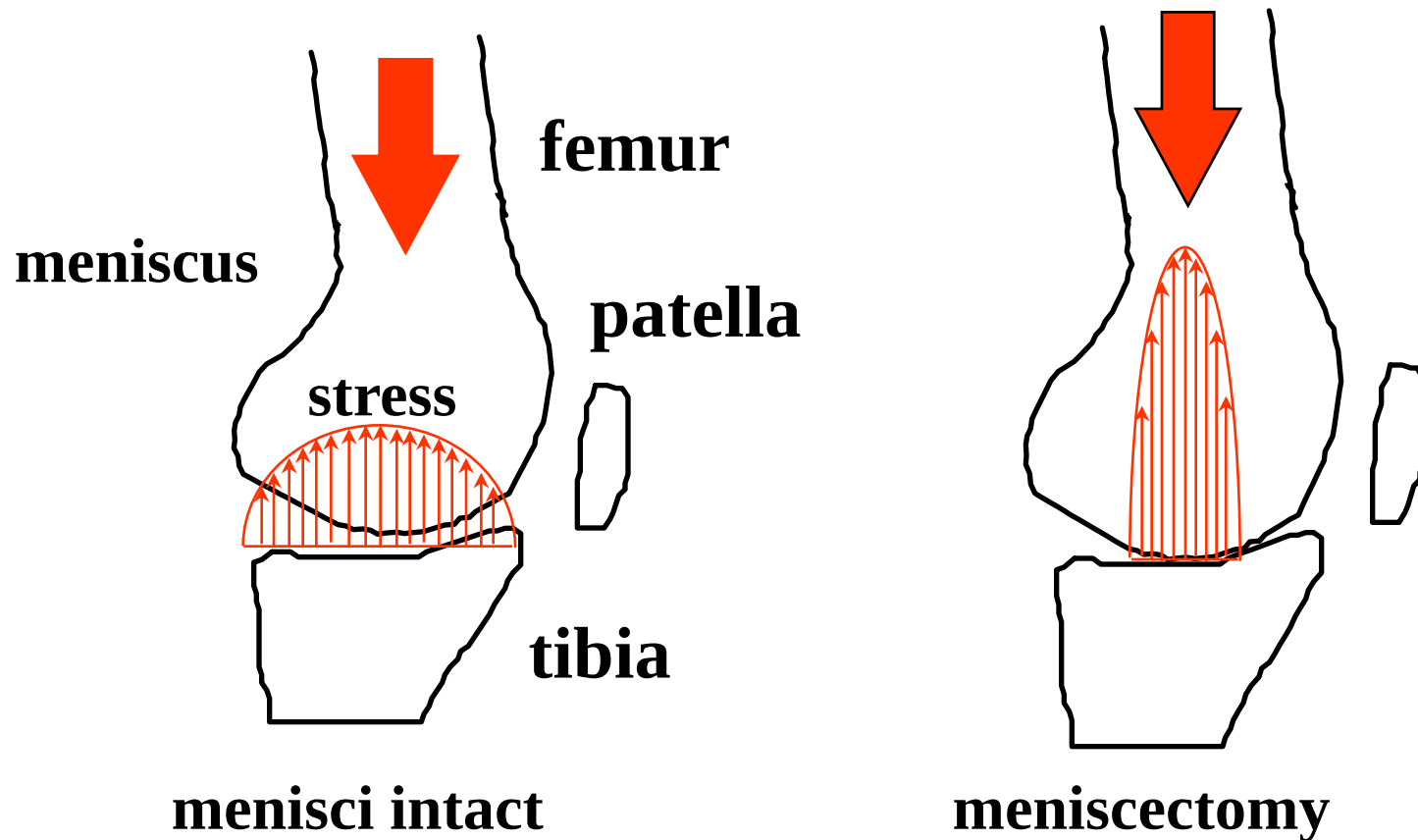
Screw-Home Mechanism



Patella (sagittal view)



Menisci & Stress Distribution



Pathology at Specific Joints

- Knee
 - Limited flexion
 - Hyper or hypo extension
 - Varus/valgus
 - Wobbling
 - Extension thrust



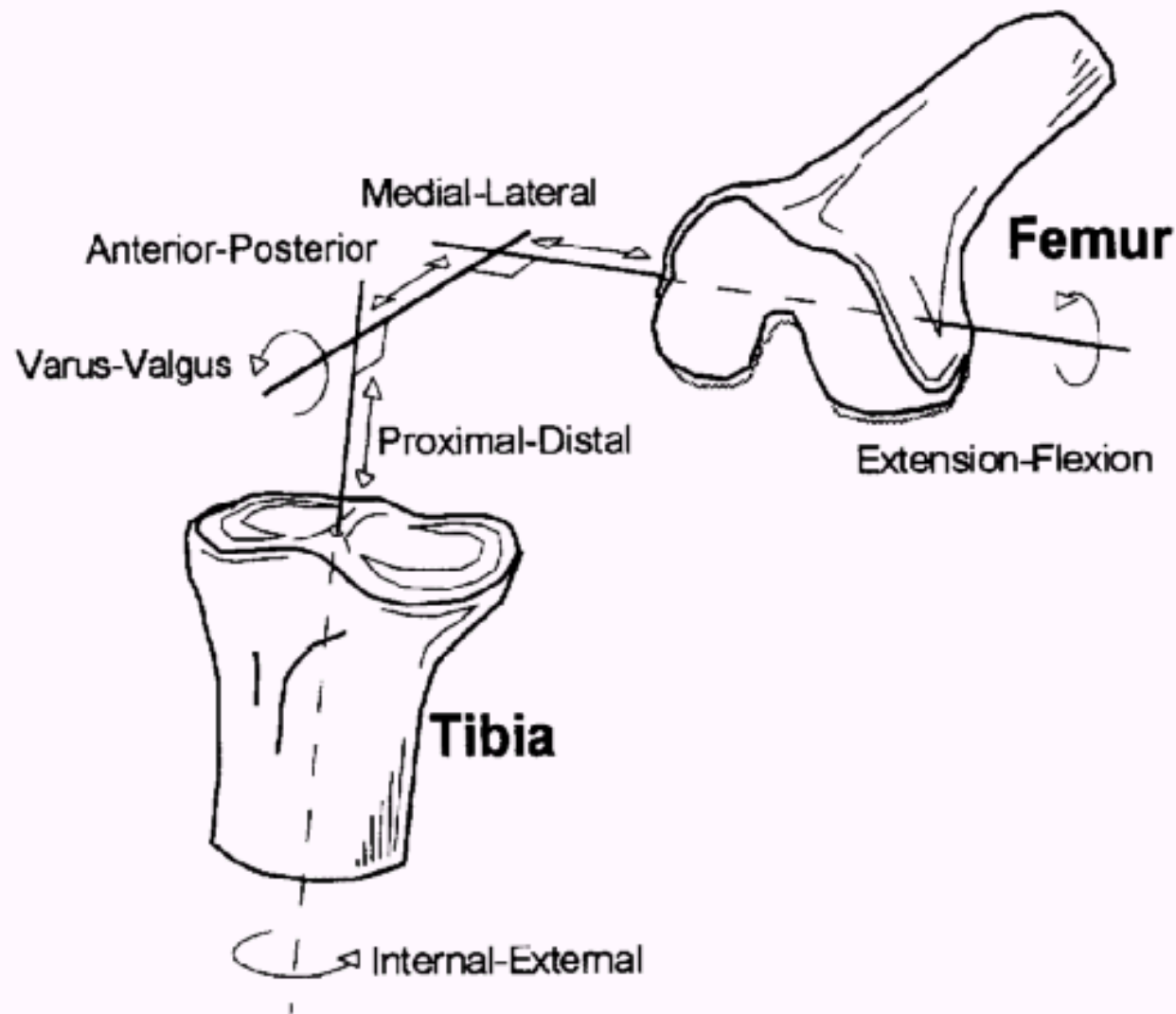
Normal



Bowleggedness (Vargus)



Knock Knees (Valgus)



Adapted from Fu, Harner, & Vince (eds). *Knee Surgery*. Baltimore, MD: Williams & Wilkins, 1994.

Quadriceps Tendon and Patella Force Lines

Compressive force at PFJ is $\frac{1}{2}$ body wt during normal walking, and over 3 times bw during stair climbing
Comp force increases as knee flexion Angle increases

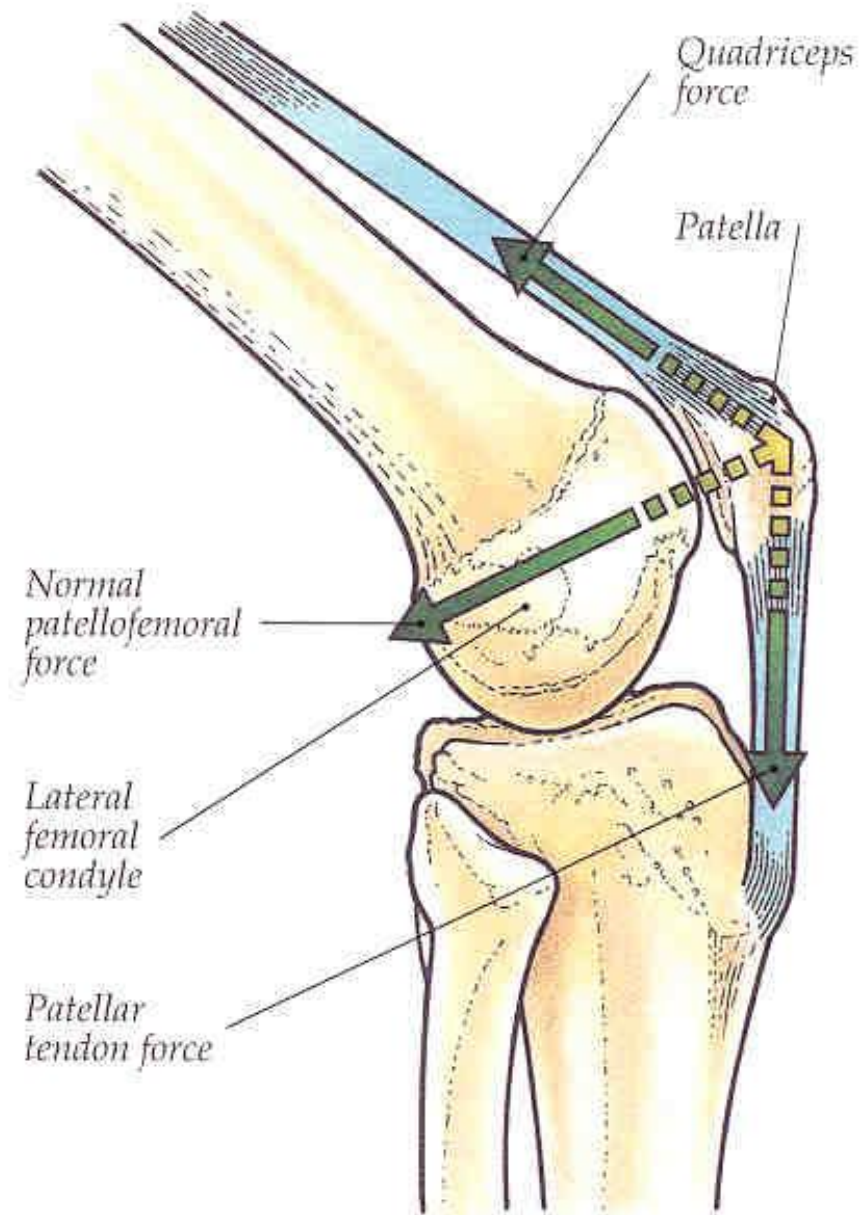
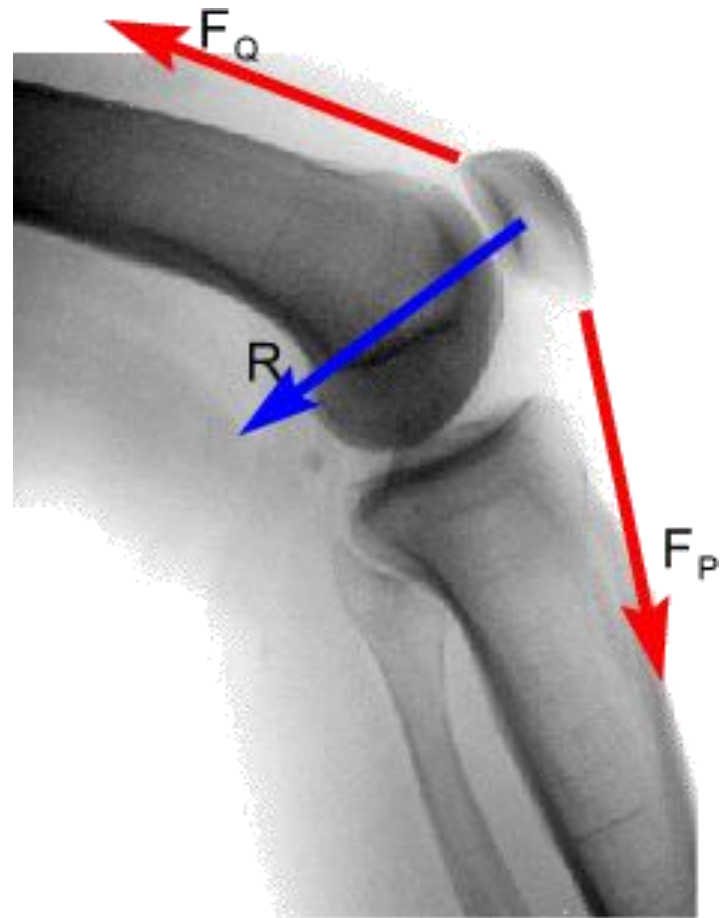
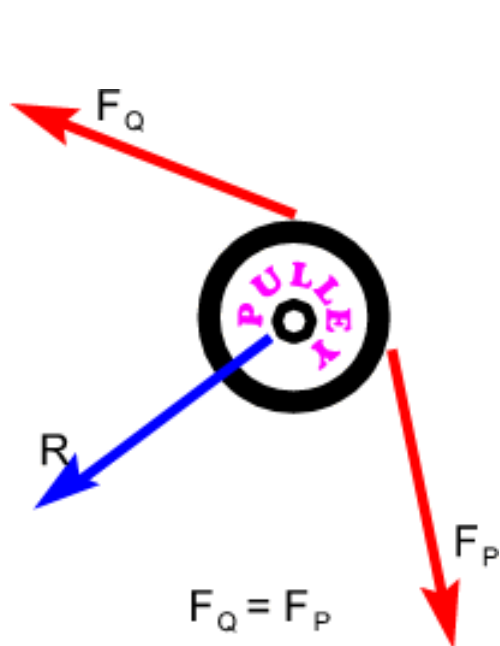


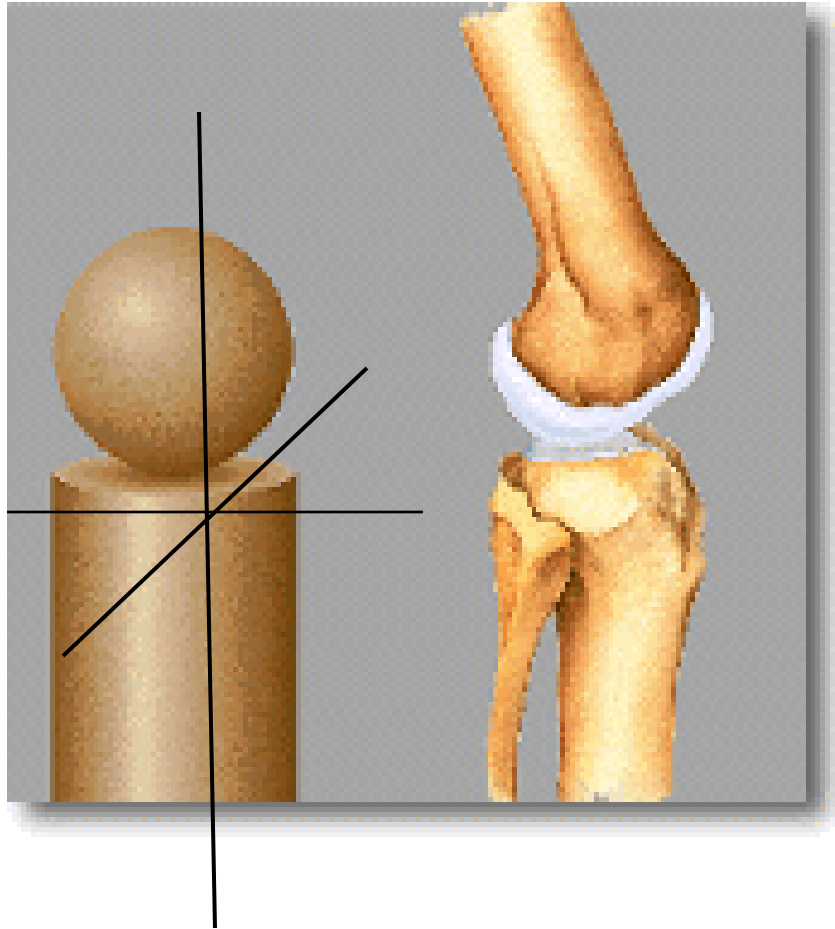
FIGURE 1

The knee joint and the forces acting on the patella

Determine forces acting in the joints and tissue elements during physical activity



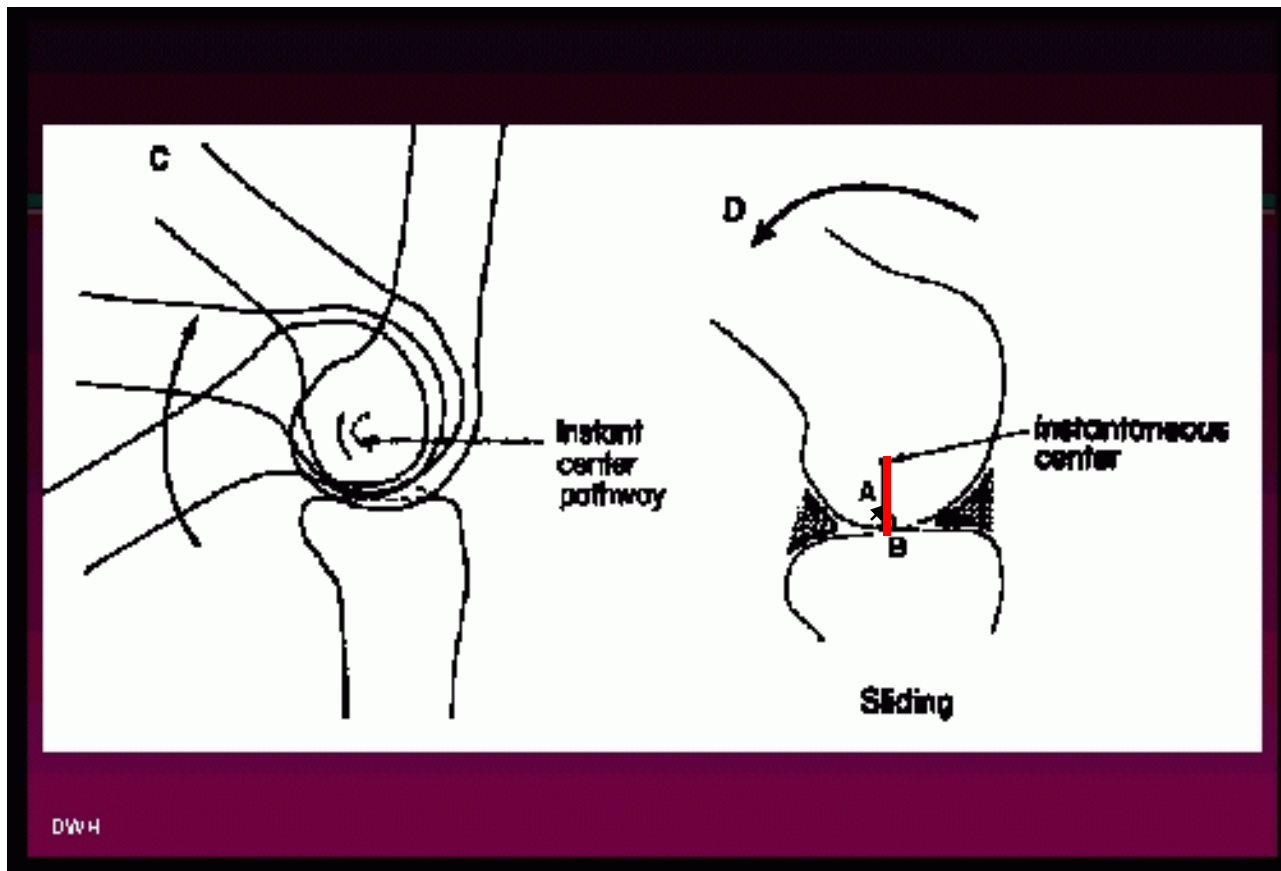
A Hinge Joint ?



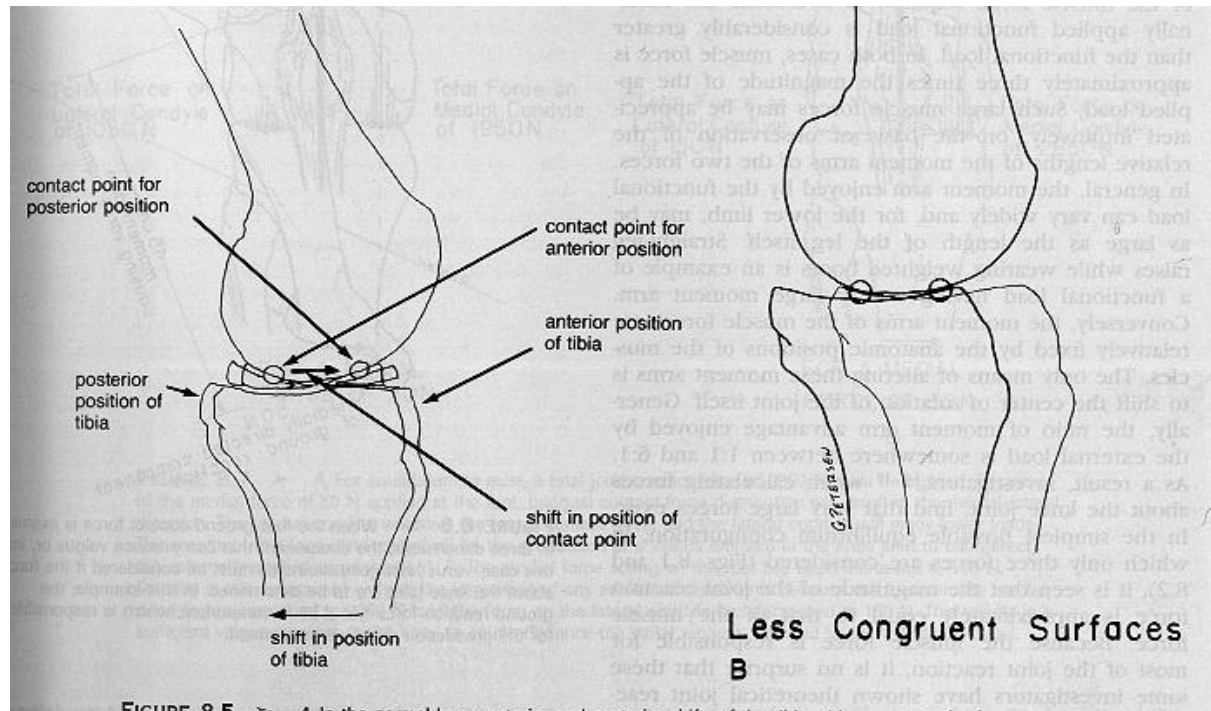
- Complex Hinge joint
- Ball and Plate model

Instant centre pathway

» Perpendicular to
Point of contact
of joint surface



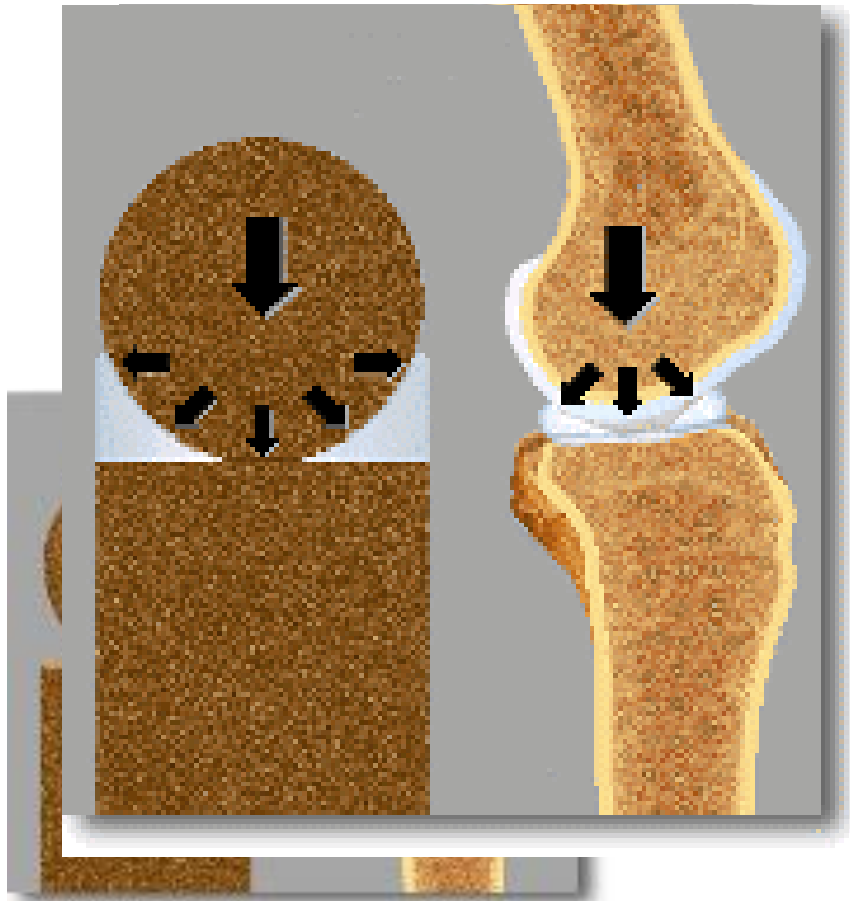
Shift of contact point



congruency

- Cruciates and menisci play important role

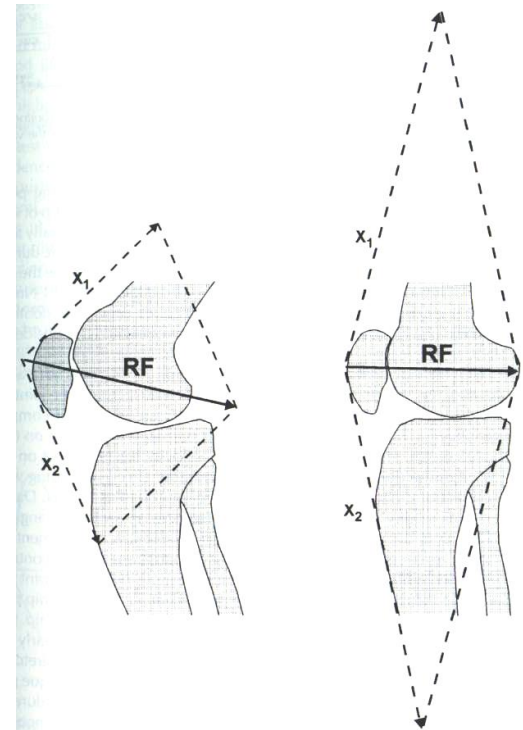
Menisci – “The wedge effect”



- Adds to joint conformity
- Block under a tyre
- Converts tibia onto a shallow socket
- Weight distribution

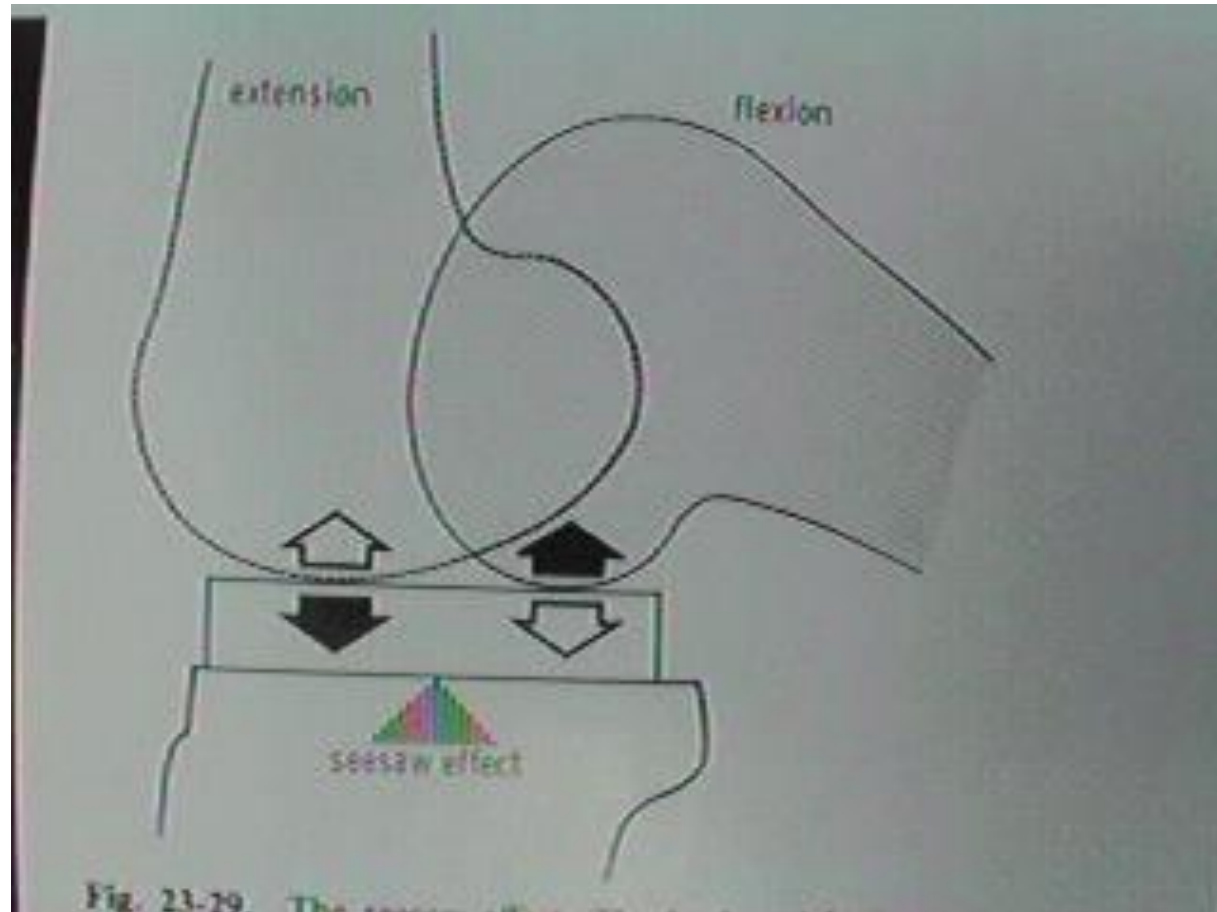
Function of patella

- increase moment arm
- protect femur joint surface
- distribute pressure
- adjust joint force



PCL Retention: Disadvantages

See-saw effect

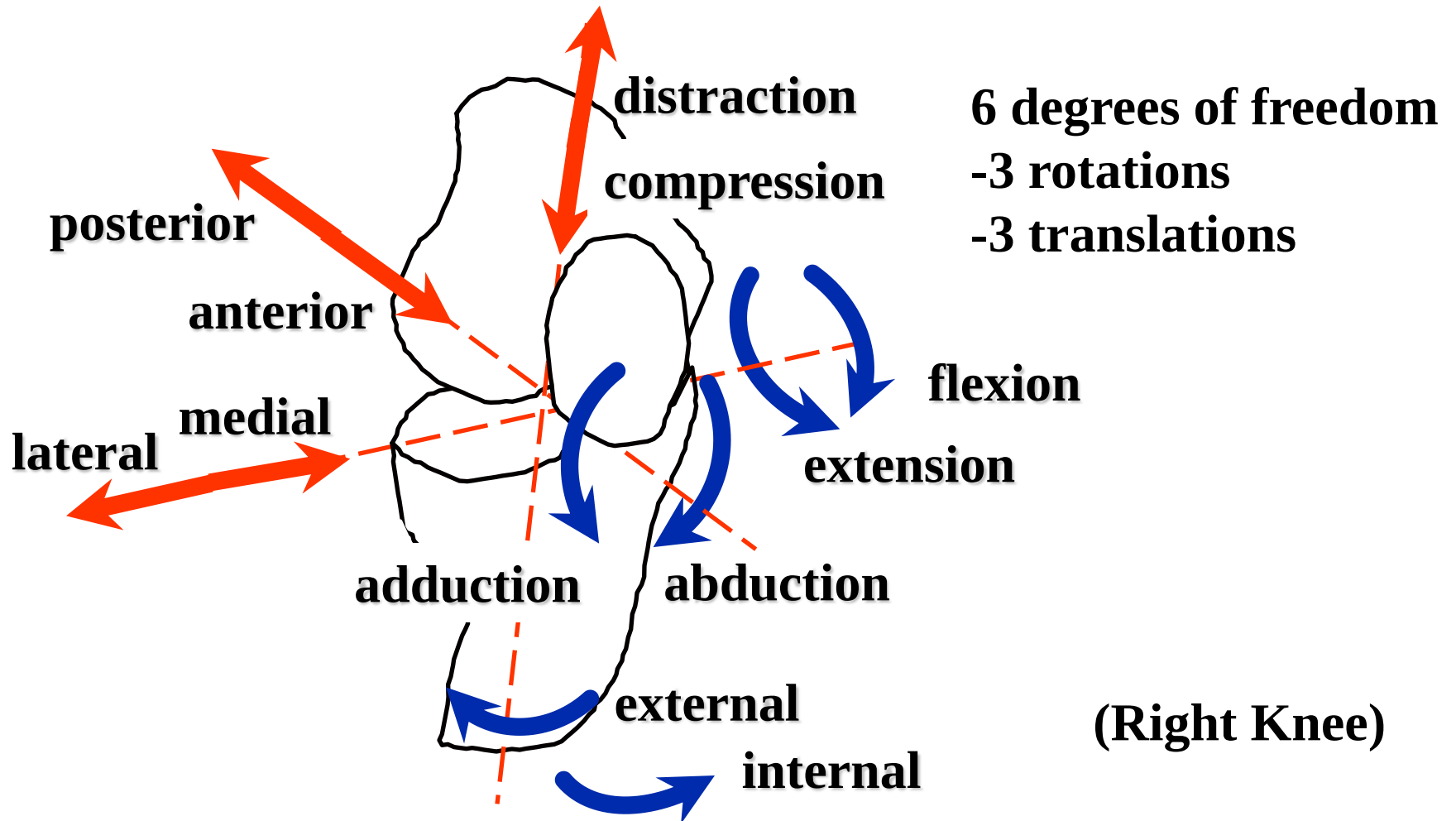


Knee Evaluation (Observation)

- Observation
 - Walking, half squatting, going up and down stairs
 - Swelling, ecchymosis,
 - Leg alignment
 - Genu valgum and genu varum
 - Hyperextension and hyperflexion
 - Patella alta and baja
 - Patella rotated inward or outward
 - May cause a combination of problems
 - Tibial torsion, femoral anteversion and retroversion

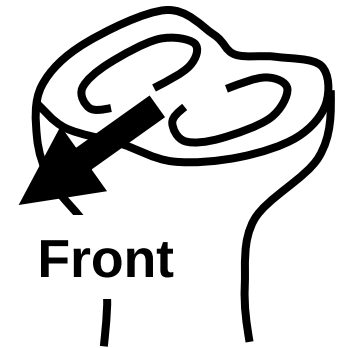


Knee Joint Motion



Tibiofemoral Joint Activities: Flexion & Anterior Shear

• cycling		105 deg	0.05BW
• walking	5	0.4	
• stairs-down	5		0.6
• squat-rise		140	3.0
• squat-down		140	3.6

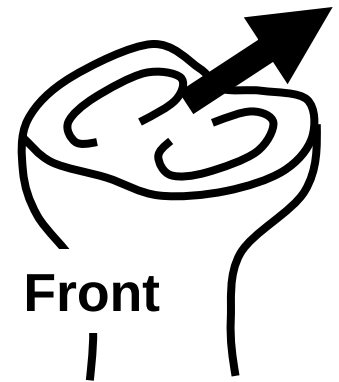


[Mow & Hayes, 1991]

BW = body weight

Tibiofemoral Joint Activities: Flexion & Posterior Shear

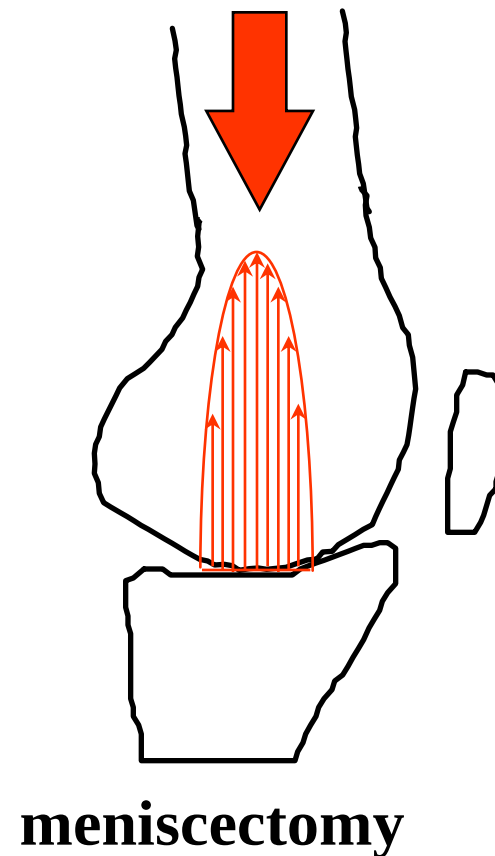
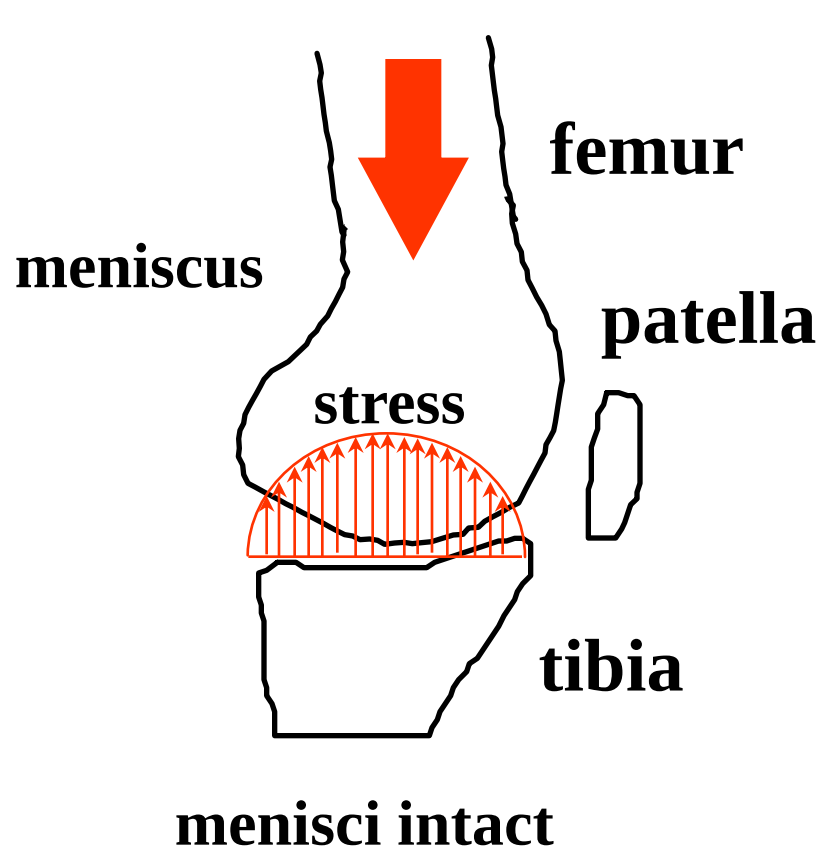
• cycling		65 deg	0.05BW
• stairs-up		30	0.05
• stairs-down		15	0.1
• walking	15	0.2	



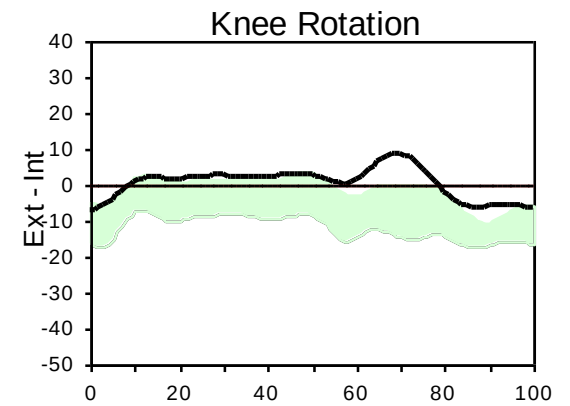
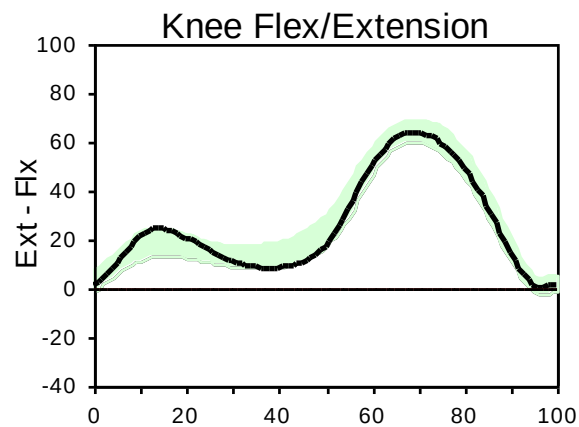
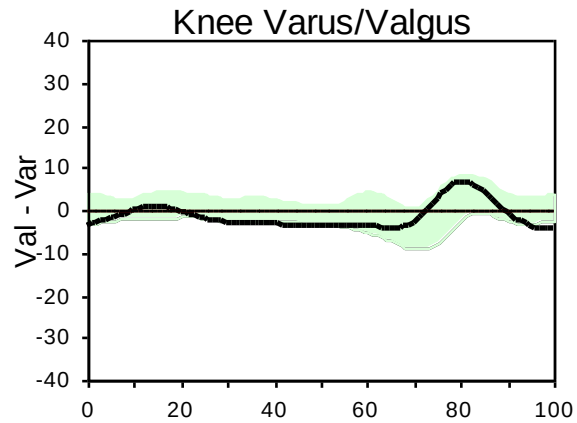
[Mow & Hayes, 1991]

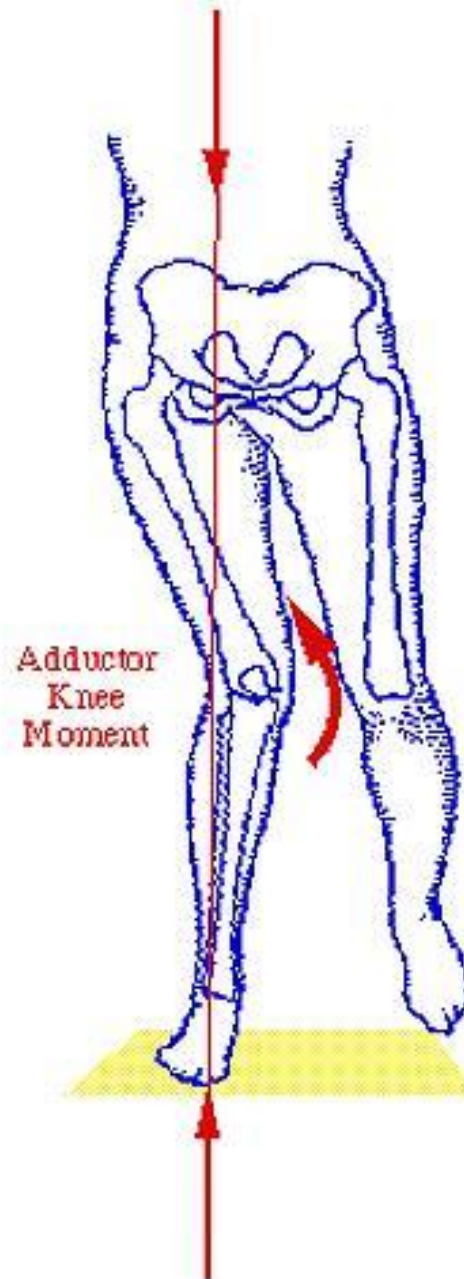
BW = body weight

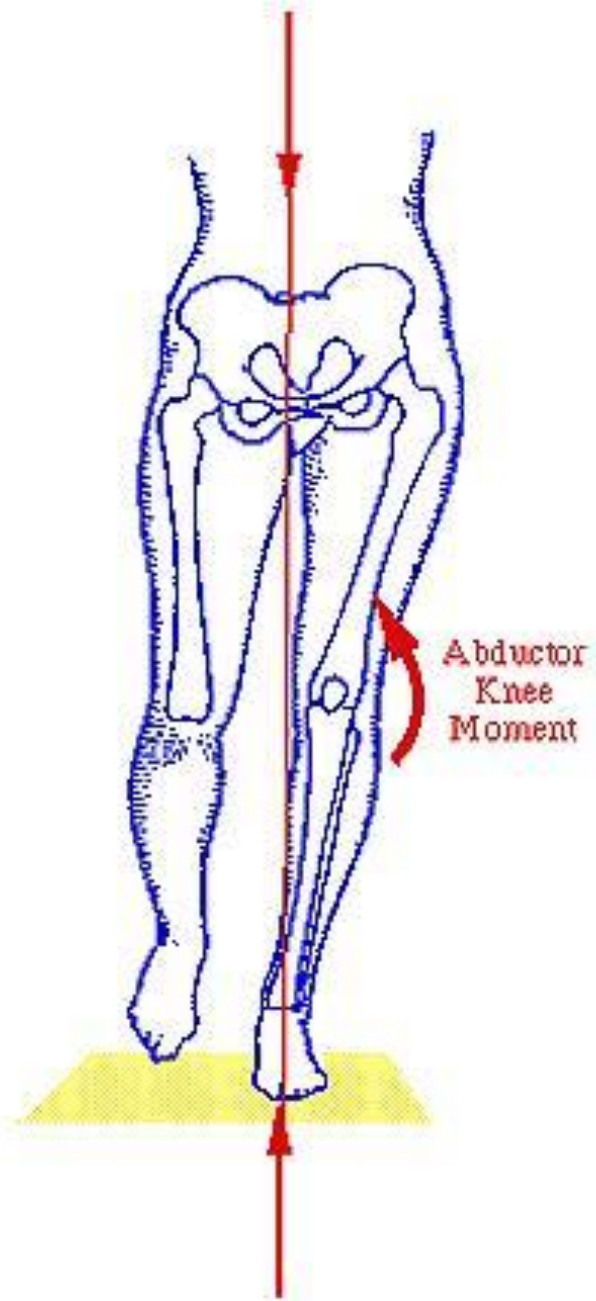
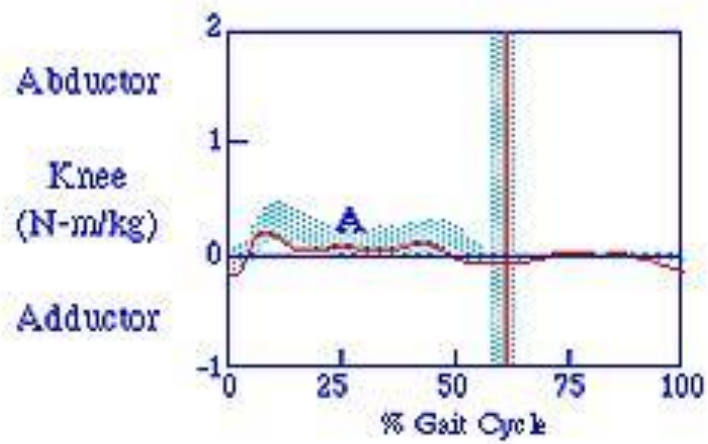
Menisci & Stress Distribution



Knee Motion



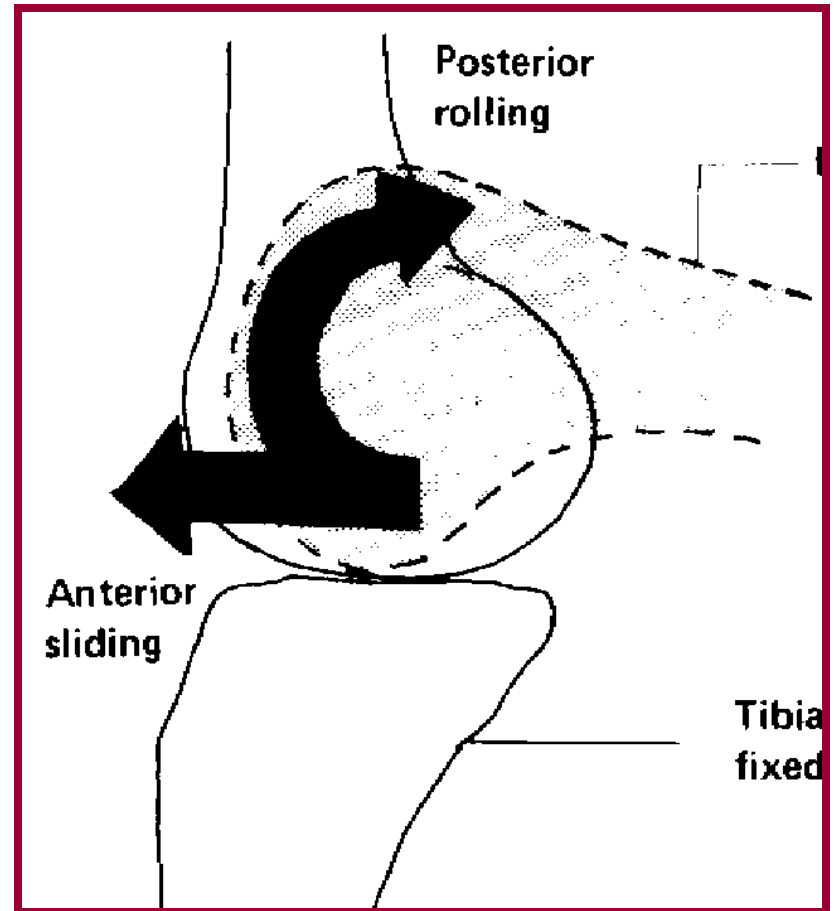
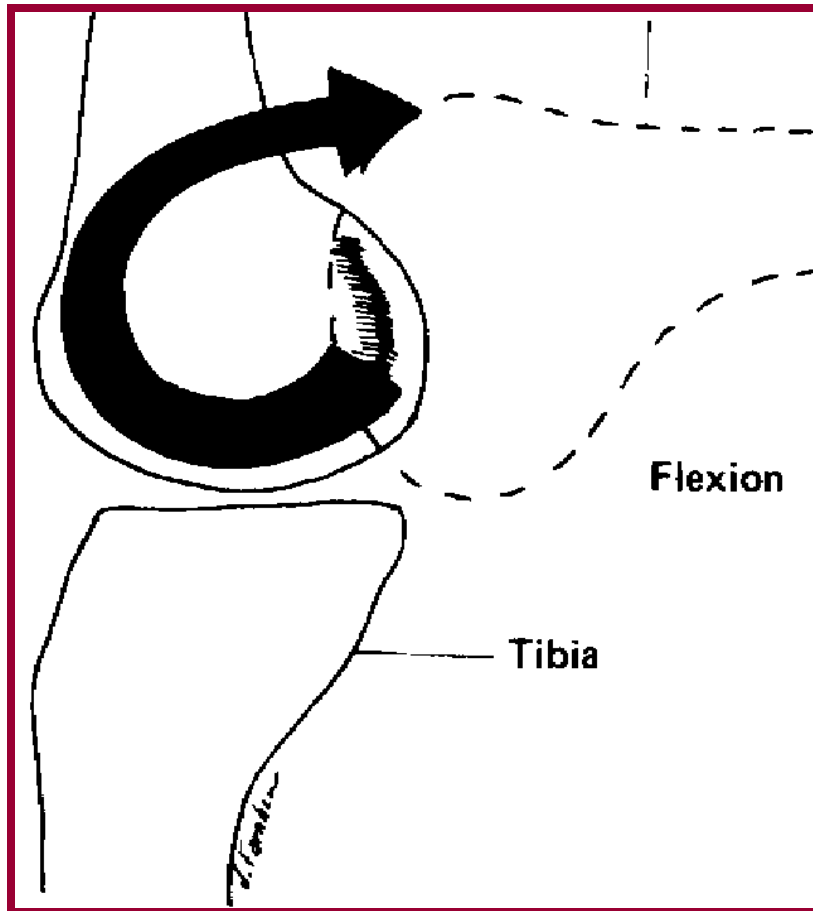




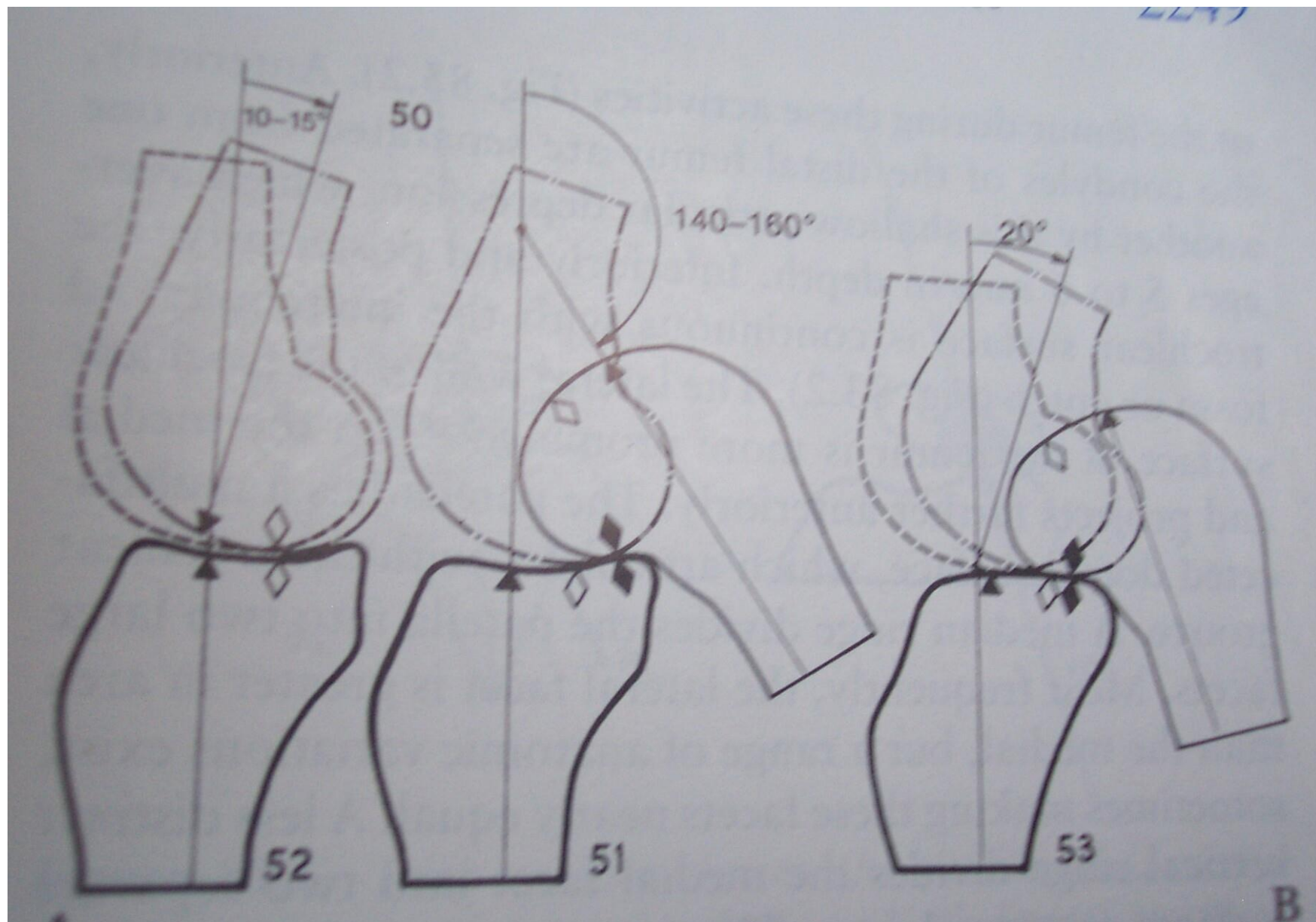
Functional ROM at the Knee

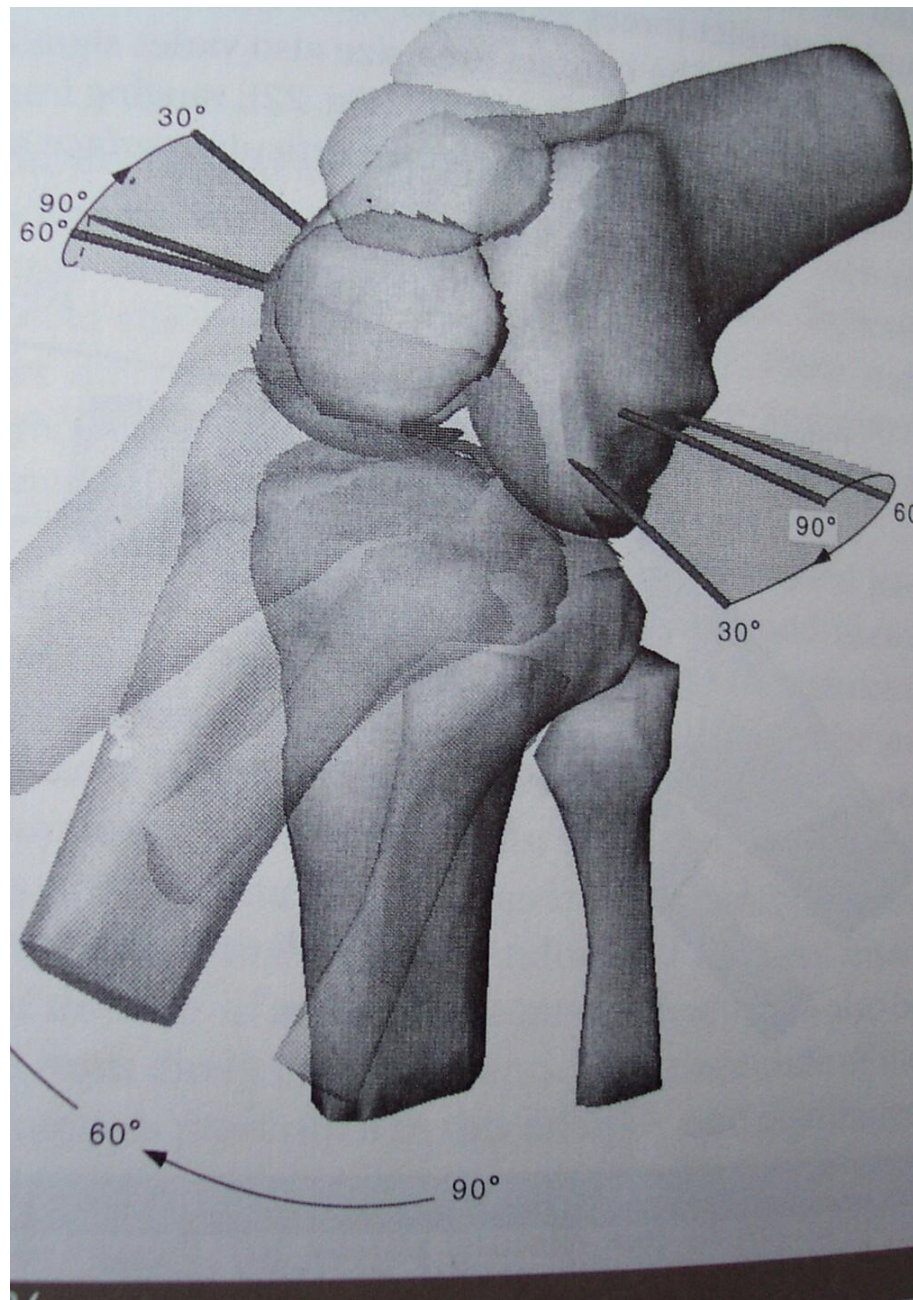
<u>Activities</u>	<u>Knee</u>
<u>Flexion</u>	
• normal gait/level surfaces	60°
• stair climbing	80°
• sitting/rising from most chairs	90°
• sitting/rising from toilet seat	115°

Arthrokinematics: Femoral Condyles in Flexion



1st 25° - mainly roll >25° roll and ant glide



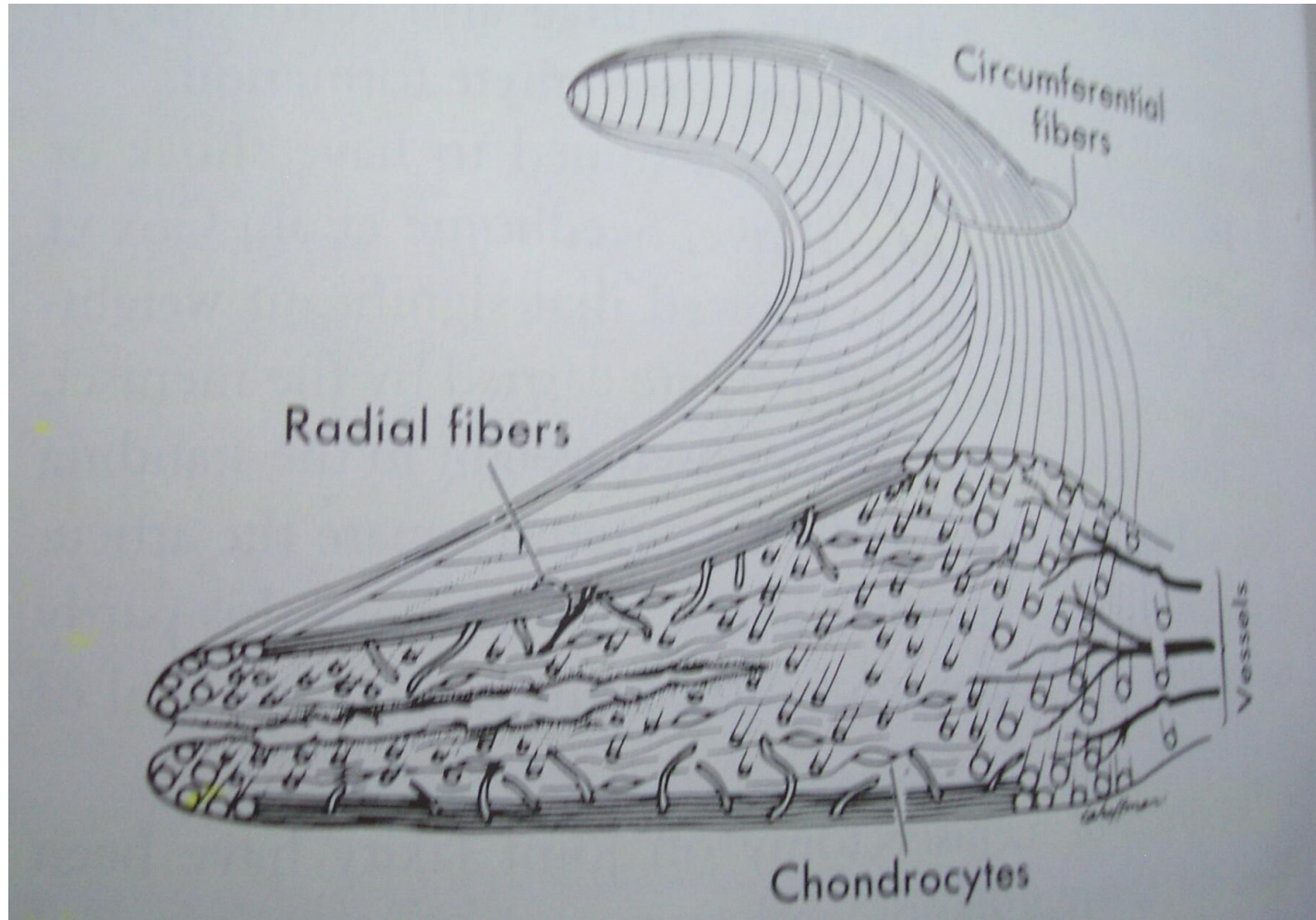


Menisci

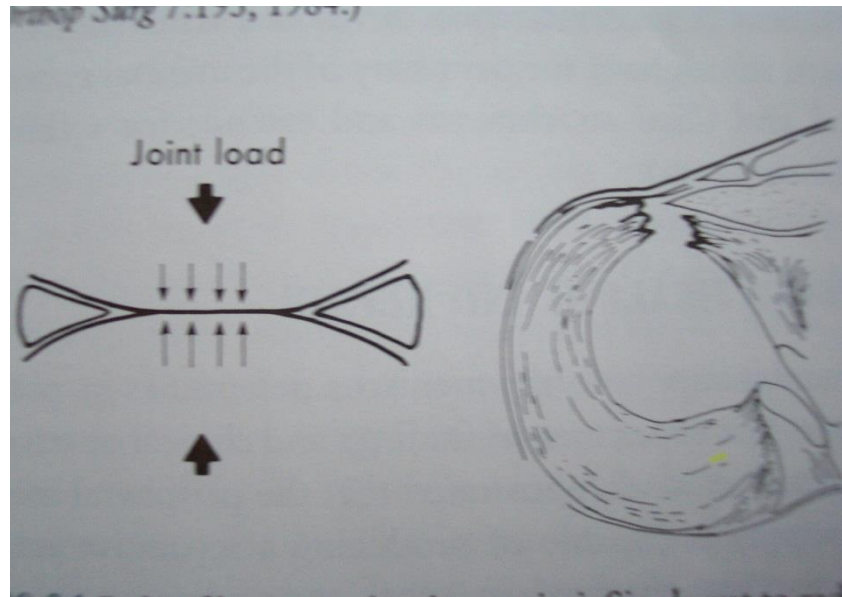
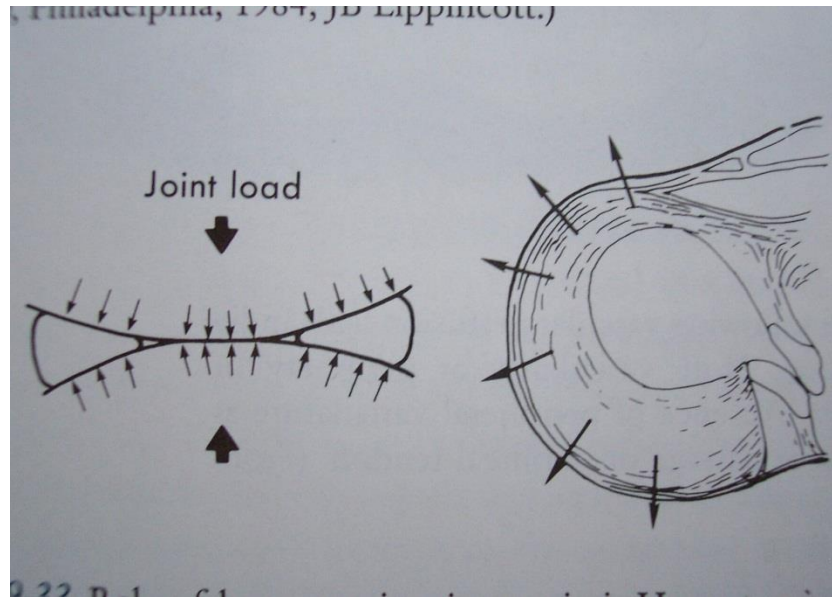
- Fairbank - 1948
- late 60's - poor results of miniscectomy
- mid 70's - load transmission confirmed
 - 40-60% load is on meniscus
 - lateral > medial

- 2 functions
 - load bearing
 - stability
 - also, joint lubrication
 - prevent capsule, synovial impingement
 - shock absorbers

Load bearing
-composition;



Hoop stress



Forces at the tibiofemoral joint

3 main coplanar forces on the knee joint

Ground
reaction
force (equal
to body
weight)(W)

Patellar
tendon force
(P)

Joint
reaction
force (J)

In single leg stance, the leg has a

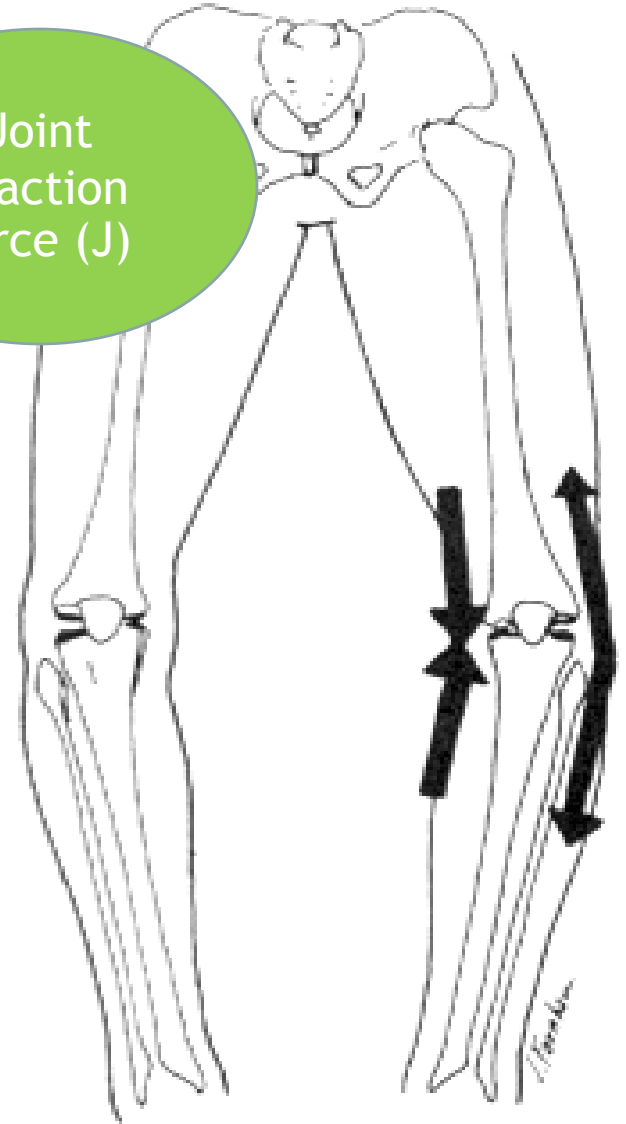


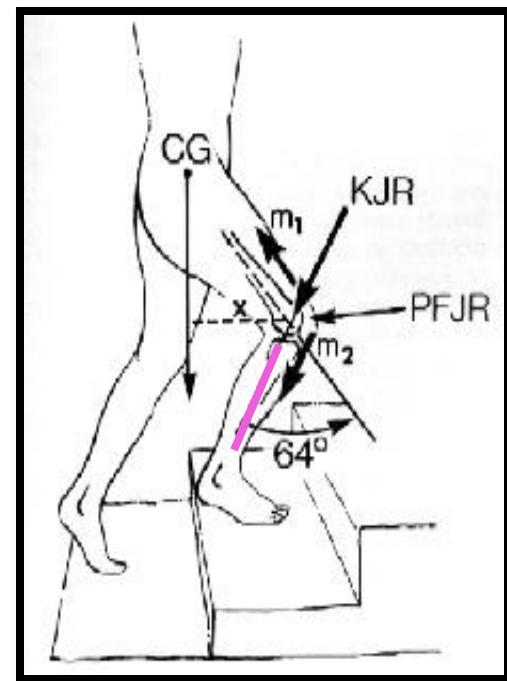
Table 83.2. Patellofemoral Joint Reaction Force

<i>Source</i>	<i>Activity</i>	<i>× Body weight (BW)</i>
Reilly and Martens ^a	Level walking	0.5 × BW
Smidt ^b	Straight leg raise	2.6 × BW
Reilly and Martens ^a	Stair climbing	3.3 × BW
Dahlqvist, et al. ^c	Squatting	7.0 × BW
Nisell ^d	Jogging	7.0 × BW

^a Reilly DT, Martens M. Experimental Analysis of the Quadriceps Muscle Force and Patello-femoral Joint Reaction

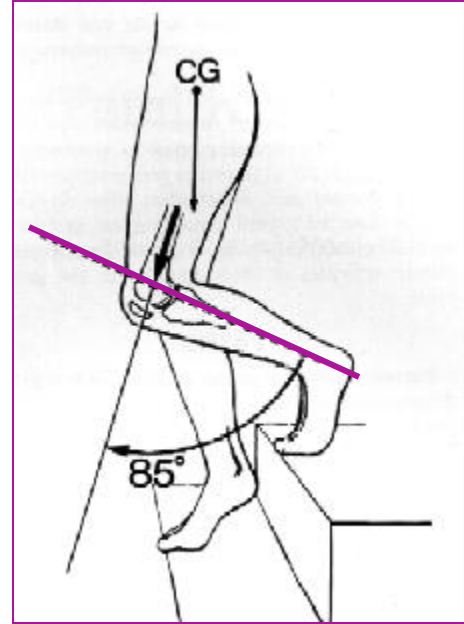
Ascending stairs

- The actual degree of knee flexion required to ascend stairs is determined not only by the height of the step, but also by the height of the patient.
- For the standard 7" step approximately 65° of flexion will be required.
- In climbing stair , lever arm can be reduced by leaning forward. Also, in stair climbing the tibia is maintained relatively vertical, which diminishes the anterior subluxation potential of the femur on the tibia.



Descending stairs

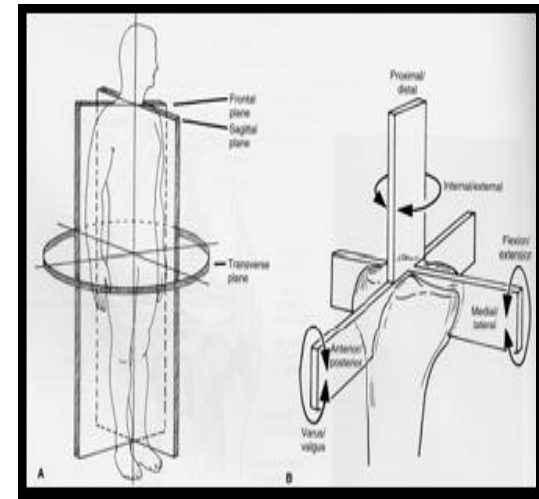
- In standard step 85° of flexion is required.
- The tibia is steeply inclined toward the horizontal, bringing the tibial plateaus into an oblique orientation.
- The force of body weight will now tend to sublux the femur anteriorly.



This anterior subluxation potential will be resisted by the patellofemoral joint reaction force, and the tension which develops in the posterior cruciate ligament.

Kinematics of tibiofemoral joint

Motion(sagittal, transverse and frontal planes).

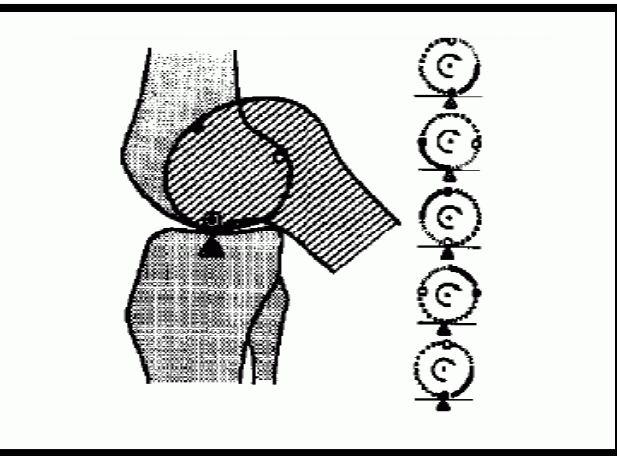


It is greatest in the sagittal plane (0-140 degree), minimal in the transverse and frontal planes.

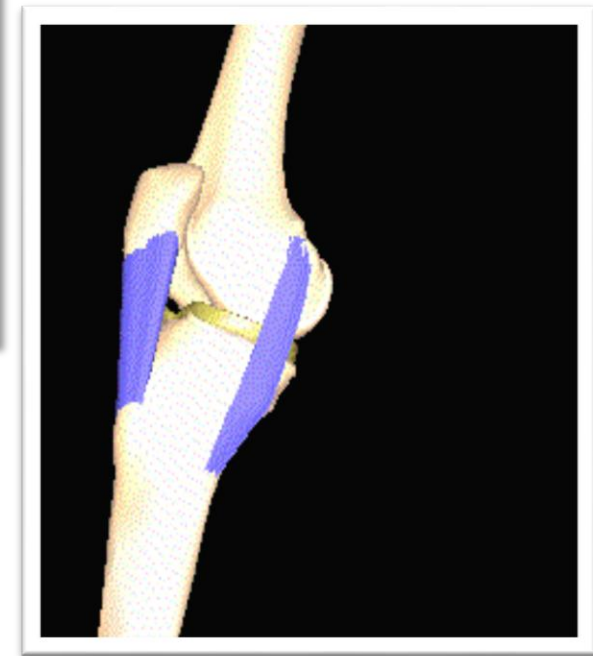
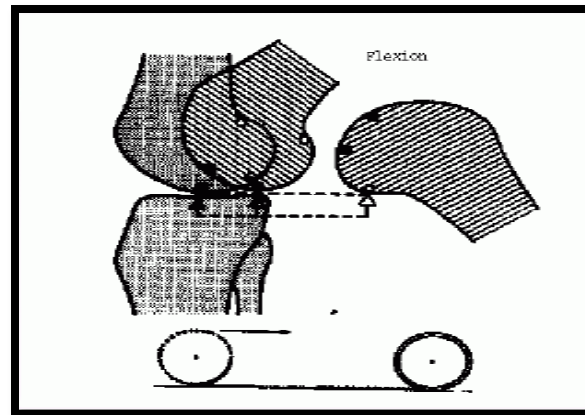
in sagittal plane

(Sagittal plane) Knee flexion/extension involves a combination of rolling and Gliding motion

Rolling Motion: Initiates flexion

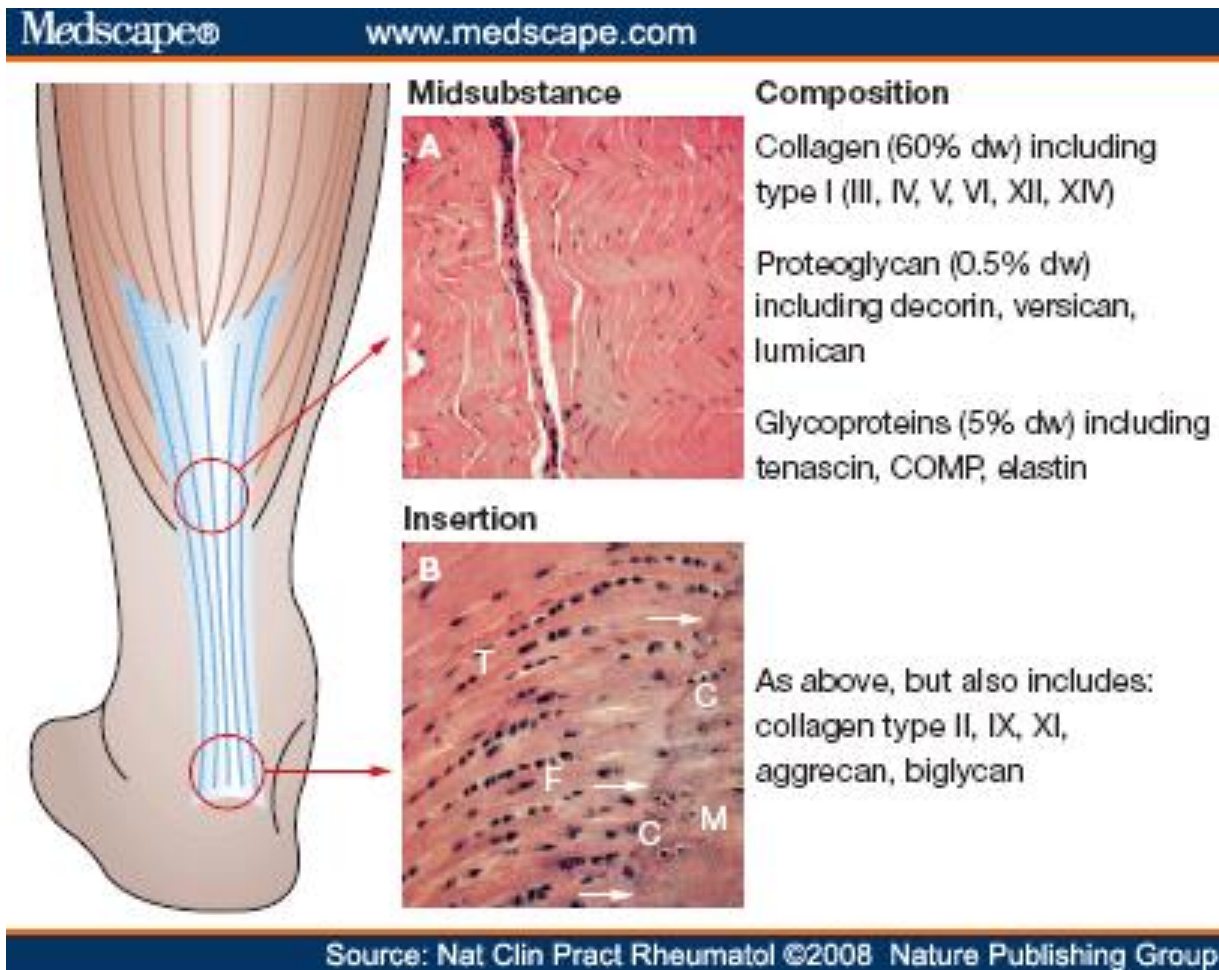


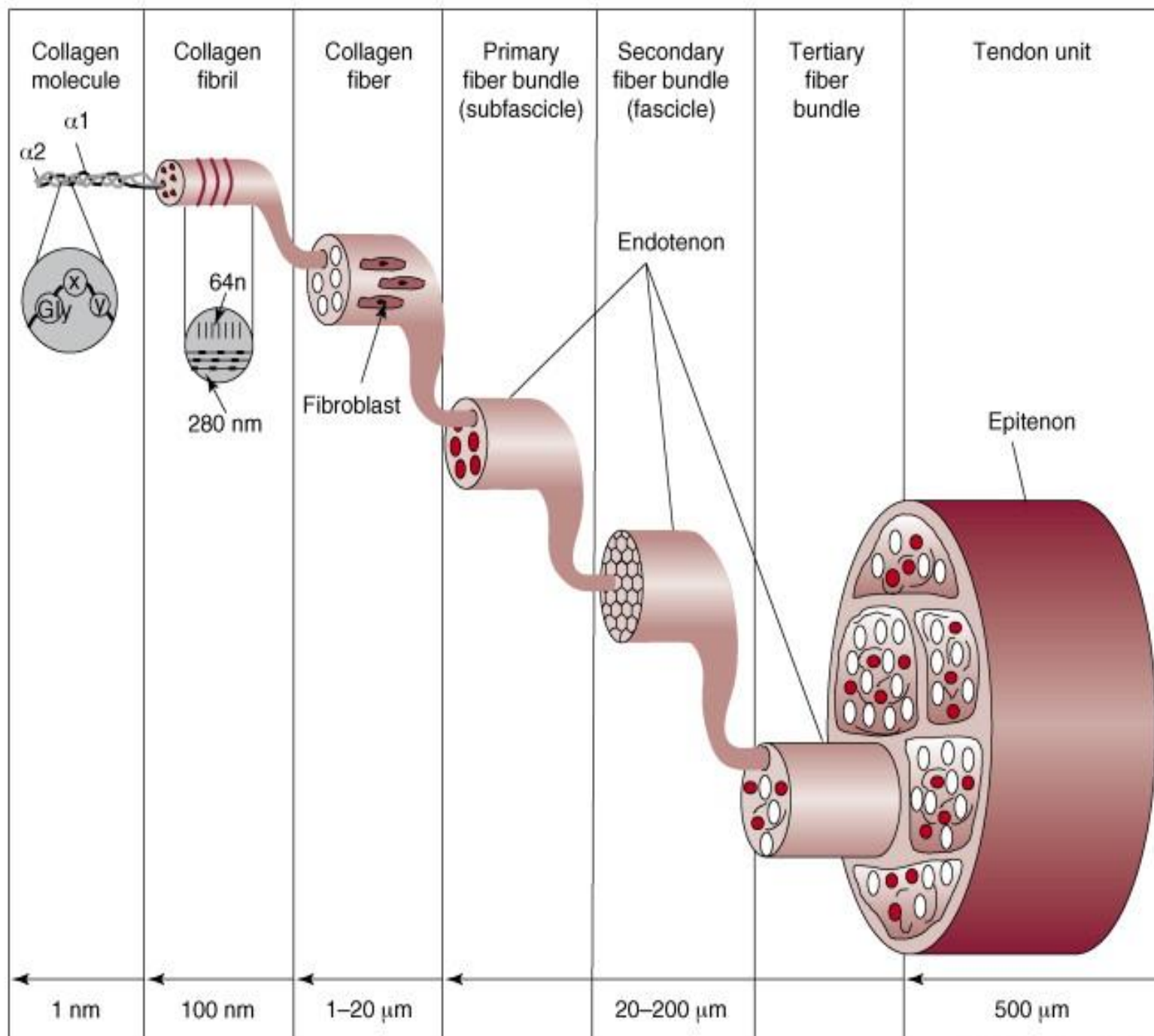
Gliding Motion:
Occurs at end of flexion



What are Tendons?

Tendons are bundles or bands of strong fibers that attach muscles to bones



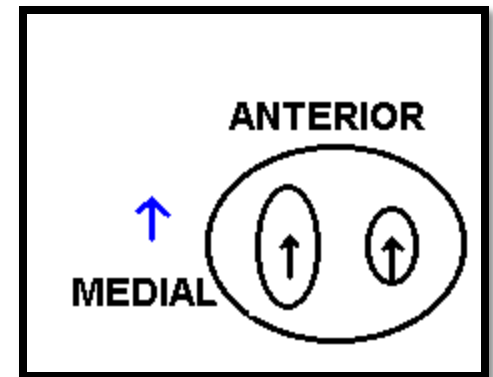


“Screw-Home” mechanism:

Rotation between the tibia and femur.

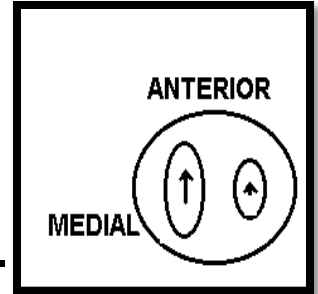
During Knee extension:

- It is considered a key element to knee stability for standing upright.
 - Tibia rolls anteriorly, on the femur, PCL Elongates.
 - PCL's pull on tibia causes it to glide anteriorly.

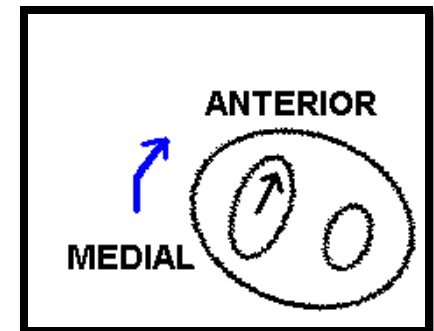


During the last 20 degrees of knee extension

- anterior tibial glide persists on the tibia's medial condyle because its articular surface is longer in that dimension than the lateral condyle's.



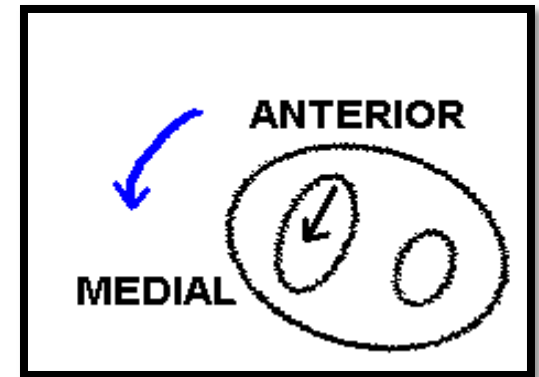
- Prolonged anterior glide on the medial side produces external tibial rotation, the "screw-home" mechanism.



THE SCREW-HOME MECHANISM REVERSES DURING KNEE FLEXION.

When the knee begins to flex from a position of full extension .

- Tibia rolls posterior, elongating ACL.
- ACL's pull on tibia causes it to glide Posterior.

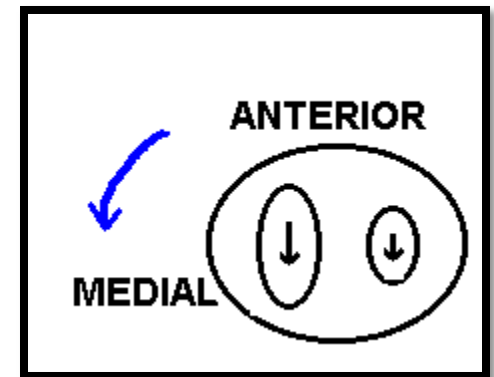


- Glide begins first on the longer medial condyle.

Between 0° extension and 20° flexion

Posterior glide on the medial side produces

- Relative tibial internal rotation.
- A reversal of the screw - home mechanism.



In transverse plane:

- In full extension almost no motion, because of interlocking of the femoral and tibial condyles.
- At 90 degrees of flexion:
 - external rotation of the knee ranges (0 -45)degrees
 - internal rotation ranges (0 to 30) degrees.
- > 90 degrees of knee flexion:
the range of motion ,because of the restriction function of the soft tissues.

:In frontal plane

- In fully extended knee almost no abduction or adduction is possible.
- knee is flexed up to 30 degree:
only a few degrees in either passive abduction or passive adduction.
- > 30 degrees of flexion:
- Motion ↓ ,because of the restriction function of the soft tissues.

■ Maximal knee flexion occurred during lifting, A significant relationship between the length of lower leg and the range of knee motion. The longer leg was, the greater the range of motion.

In the double stance phase of gait

- When the body weight is borne equally on both feet the force which passes through the knee is only a fraction of body weight.
- There is no bending moment around either knee.

in single leg stance

➤ Body weight passes onto the single leg, the center of gravity moves away from the supporting leg and up, this shift occurs because the weight of the supporting leg is not included in the body mass to be supported by the knee while the suspended leg is included.



➤ To minimize movement of the body mass from side to the midline at heel strike as the center of gravity is displaced slightly towards the support side.

- In man, with upright single leg stance, this orientation is accomplished by the overall valgus orientation of the lower extremity which naturally brings the foot toward the midline.
- In single leg stance, therefore, the leg has a valgus orientation. This situation exerts a bending moment on the knee which would tend to open the knee into varus, the ligaments and capsule are tight, in part because of the "screw-home" mechanism. These structures resist this bending moment.

During gait

- Multiple muscles which cross the joint in the center or to the lateral side of center combine to provide a lateral resistance to opening of the lateral side of the joint. These include the quadriceps-patellar tendon forces, the lateral gastrocnemius, popliteus, biceps and iliotibial tract tension.
- With increasing knee varus the medial lever arm increases requiring an increased lateral reaction to prevent the joint from opening.

In total joint replacement a single cane in the opposite hand does much to unload the knee and particularly to reduce the magnitude of the varus bending moment, a cane in the opposite hand will reduce knee loading by 46%.

narrow base gait is the norm, and the most energy efficient.

The side to side deviation of the center of gravity is reduced to approximately 2 cm in each direction toward the support side or a total of 4 cm through the gait cycle involving both legs.

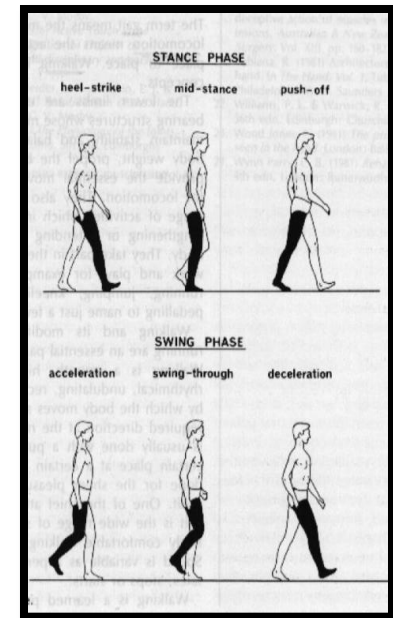
waddling or broad based gait

lateral displacement will be accentuated requiring greater energy for walking.

the orientation of the lower extremity to the vertical and to the center of gravity will be the same during the single leg support phase of gait.

Normal gait is divided into two phases: stance phase and swing phase.

- Quadriceps contraction begins just before heel contact, to stabilize the knee for heel contact.
- Between HS and FF, the knee flexes 20° → hamstring muscles contract to stabilize the knee during the 20° of flexion, and lengthening of quadriceps.
- During mid stance, the quadriceps is again contract.
- The hamstrings contract Just at and after toe off to add additional flexion for clearance of the foot during the swing phase.



- In the absence of a posterior cruciate ligament, only the collateral ligaments are available to assist the patellofemoral joint reaction force in providing anterior-posterior stability.
- ❖ Many patients with arthritis will report difficulty descending stairs normally, this will also be true after total knee replacement. A simple remedy is to have them descend either sideways or backward, which is biomechanically the equivalent of ascending the stairs with its decreased mechanical and range of motion demands.

Patellofemoral joint:

Patellofemoral joint consist of the articulation of the triangularly shaped patella, encased in the patellar tendon. The posterior patella is coverd with articular cartilage, which reduces friction between the patella and the femur.



Function of patella

- Increase the angle of pull of the quadriceps tendon
- Increase the area of contact between the patellar

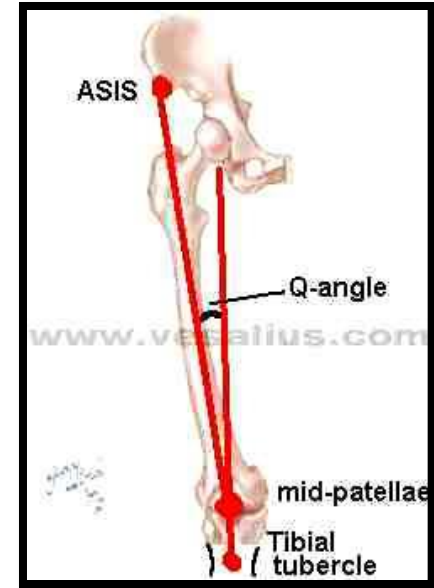


The Q-angle (or "quadriceps angle") is formed in the frontal plane by two line segments:

Angle formed at the knee joint

➤ By connecting a line from the anterior iliac crest to the center of the patella.

➤ And a second line from the center of the patella to the center of the patellar tendon insertion into the tibial tubercle.

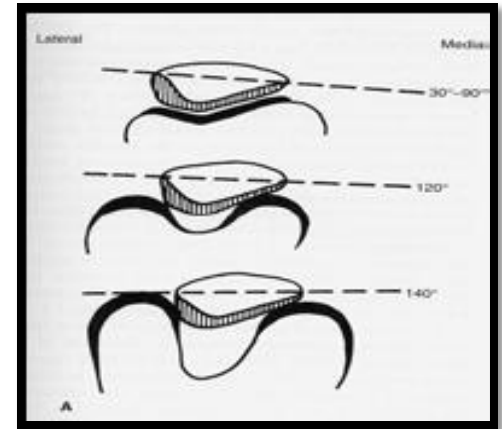


❖ the Q-angle is normally less than 15 degrees in men and less than 20 degrees in women. An abnormally large Q-angle usually results in a disorder called abnormal quadriceps pull

Kinematics of patellofemoral joint

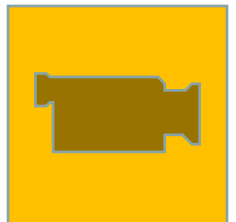
➤ Motion occurs in two planes: Frontal and transverse.

➤ At full extension both medial and lateral femoral facet articulate with the patella.



➤ > 90degrees of flexion the patella rotate externally, and only the medial femoral facet articulate with the patella.

➤ At full flexion patella sinks into intercondylar groove.



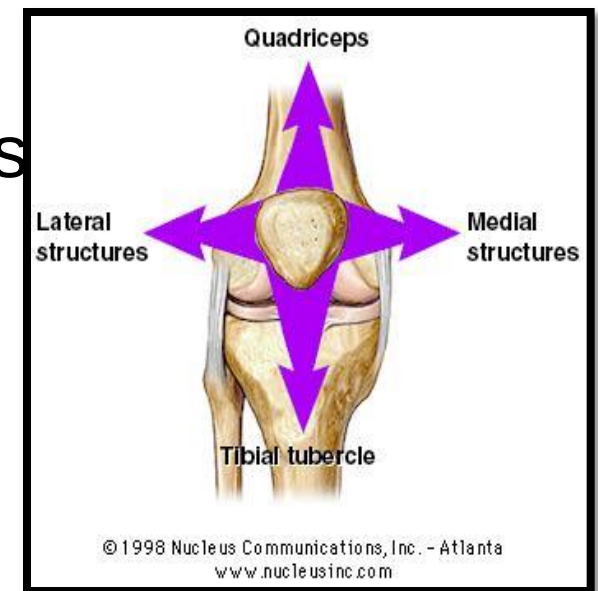
Forces acting on the Patella:

Laterally- lateral retinaculum, vastus lateralis m, iliotibial tract.

Medially- medial retinaculum and vastus medialis m.

Superior- Quadriceps via quadriceps tendon.

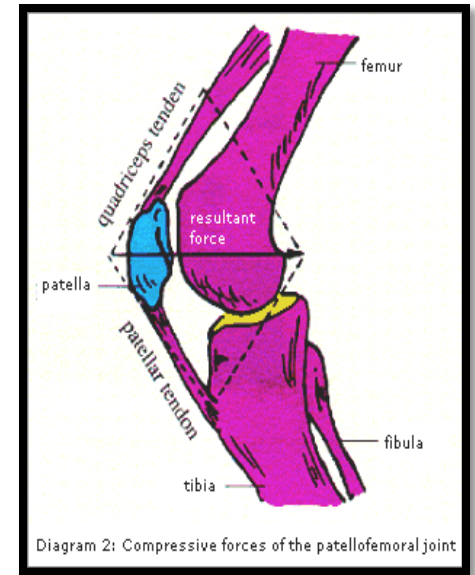
Inferior- Patellar tendon.



❖ Compressive force is additional force at patellofemoral joint.

PF Compressive Force Function

- Stabilizes patella in trochlea groove.
- Patella assures “some” compression in full extension.



■ Patellofemoral compression ↑ with knee flexion during weight bearing, because as flexion increases, a large amount of quadriceps tension is required to prevent the knee from buckling against gravity.

- Squat exercise stressful to the knee complex, produces a patellofemoral joint reaction force 7.6 times body weight.



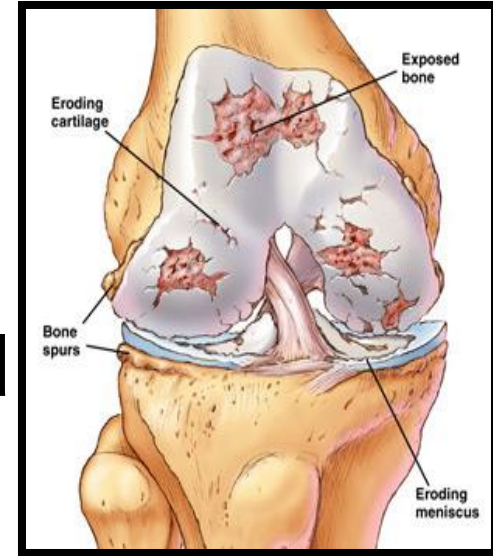
- It one-half of body weight during normal walking, increasing up to over three times body weight during stair climbing.

Common Knee Injuries and Problems

Osteoarthritis

the cartilage gradually wears away and changes occur in the adjacent bone. Osteoarthritis may be caused by joint injury or being overweight.

It is associated with aging and most typically begins in people age 50 or older.

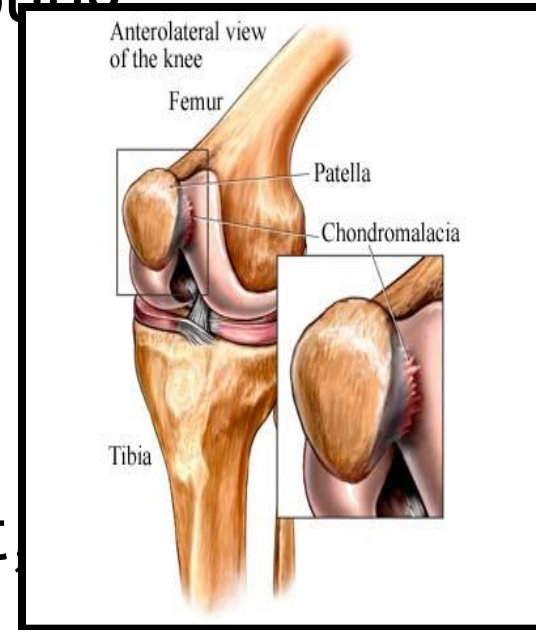


Chondromalacia

Also called chondromalacia patellae, refers to softening of the articular cartilage of the kneecap.

This disorder occurs most often in young

adults and can be caused by injury, overuse, misalignment of the patella, or muscle weakness. Instead of gliding smoothly across the lower end of the thigh bone, the kneecap rubs against it, thereby roughening the cartilage underneath the kneecap.



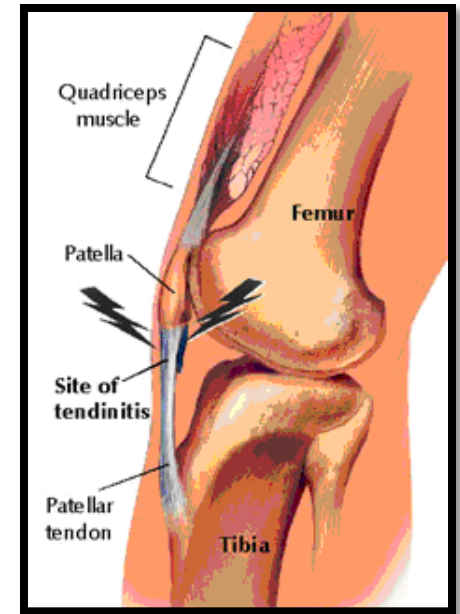
Meniscal Injuries

The menisci can be easily injured by the force of rotating the knee while bearing weight. A partial or total tear may occur when a person quickly twists or rotates the upper leg while the foot stays still. If the tear is tiny, the meniscus stays connected to the front and back of the knee; if the tear is large, the meniscus may be left hanging by a thread of cartilage. The seriousness of a tear depends on its location and extent.



Tendon Injuries

Knee tendon injuries range from tendinitis (inflammation of a tendon) to a ruptured (torn) tendon. If a person overuses a tendon during certain activities such as dancing, cycling, or running.

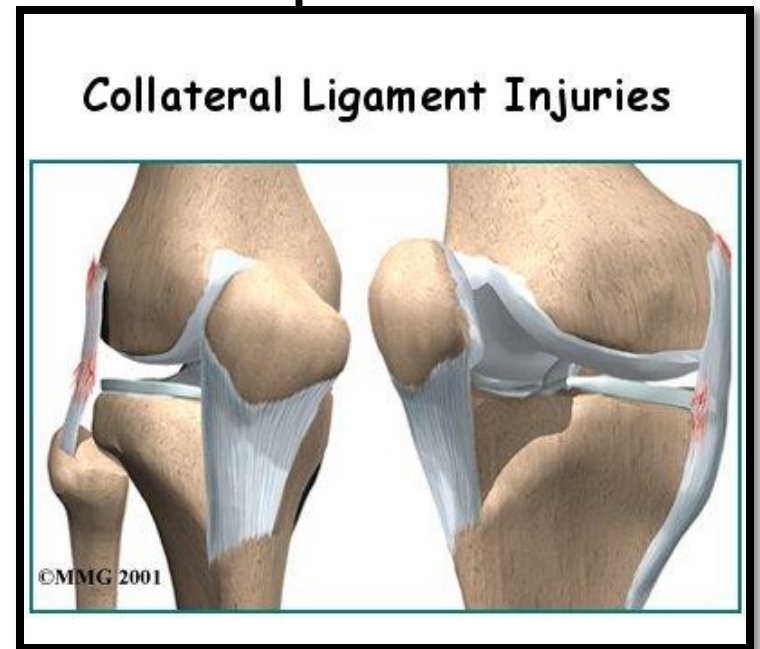


, the tendon stretches and becomes inflamed.

Tendinitis of the patellar tendon is sometimes called “jumper’s knee” because in sports that require jumping, such as basketball, the muscle contraction and force of hitting the ground after a jump strain the tendon. After repeated stress, the tendon may become inflamed or tear.

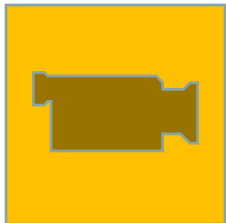
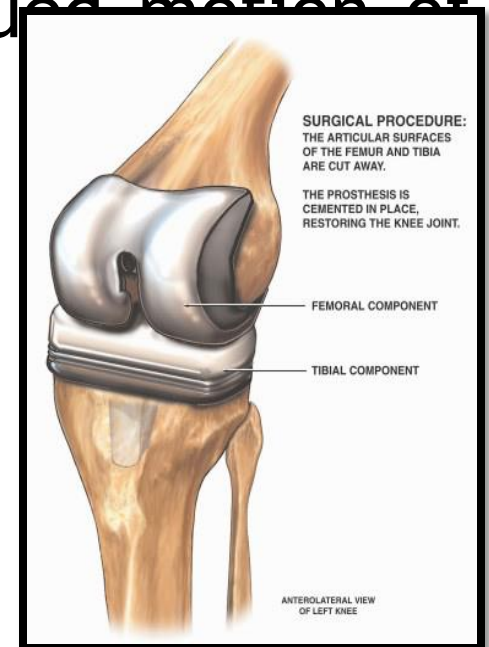
Medial and Lateral Collateral Ligament Injuries

The medial collateral ligament is more easily injured than the lateral collateral ligament. The cause of collateral ligament injuries is most often a blow to the outer side of the knee that stretches and tears the ligament on the inner side of the knee. Such blows frequently occur in contact sports such as football or hockey.



Knee replacement, or knee arthroplasty

is a common surgical procedure most often performed to relieve the pain and disability, Knee replacement surgery can be performed as a partial or a total knee replacement, the surgery consists of replacing the diseased or damaged joint surfaces of the knee with metal and plastic components shaped to allow continued motion of the knee.



SUMMARY

- The knee is a two-joint structure composed of the tibiofemoral joint and the patellofemoral joint.
- In the tibiofemoral joint, surface motion occurs in three planes, greatest in sagittal plane. In the patellofemoral joint, surface motion occurs in two planes frontal and transverse.
- The screw home mechanism of tibiofemoral joint adds stability to the joint in full extension.
- Both the tibiofemoral joints and patellofemoral joints are subjected to high forces. The magnitude of the joint reaction force on both joints can reach several times body weight.
- Although the tibial plateaus are the main load bearing structures in the knee, the cartilage, menisci, and ligaments also bear load.
- The patella aids knee extension by lengthening the lever arm of the quadriceps muscle, and allows a better distribution of compressive stress on the femur.

THANK YOU

Introduction

- What kind of joint is it?
- Limits of motion
- Normal kinematics of a step
- Plateau & condyles
- Patello Femoral articulation
- Menisci
- Medial, lateral and anterior stability
- ACL & PCL

Knee joint

- Ginglymus (hinge) ?
- Arthodial (gliding) ?
- 6 degrees of freedom
 - 3 rotations
 - 3 translations

- Rotations
 - flex/ext - -15 to 140 deg
 - varus valgus - 6-8 deg in extension
 - int/ext rotation - 25 - 30 deg in flexion

- Translations
 - AP 5 - 10mm
 - comp/dist 2 - 5mm
 - medio-lateral 1-2mm

Taking a step

- Just prior to heel strike - max extension & max external rotation
- heel strike - max valgus
- flat foot - flexion & intrenal rotation progress
- swing phase - internal rotation continues, max flexion, max anterior translation.

Condyles and plateau

Medial intercondylar tubercle

Articular surface of
medial condyle

Lateral intercondylar tubercle

Articular surface of lateral condyle

Posterior intercondylar area

A

Anterior cruciate ligament

Medial meniscus

Bursa in tibial collateral ligament

Coronary ligament

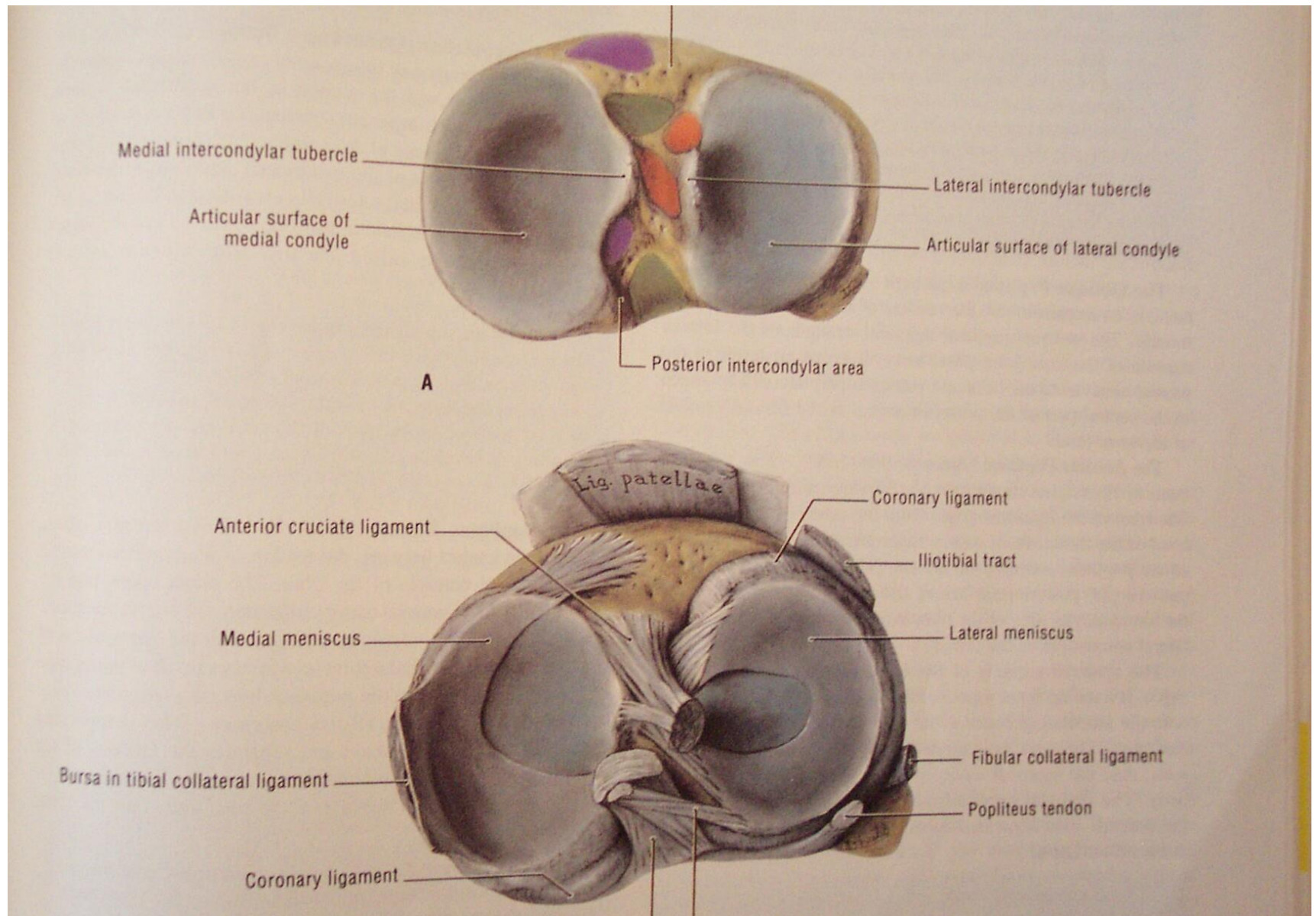
Coronary ligament

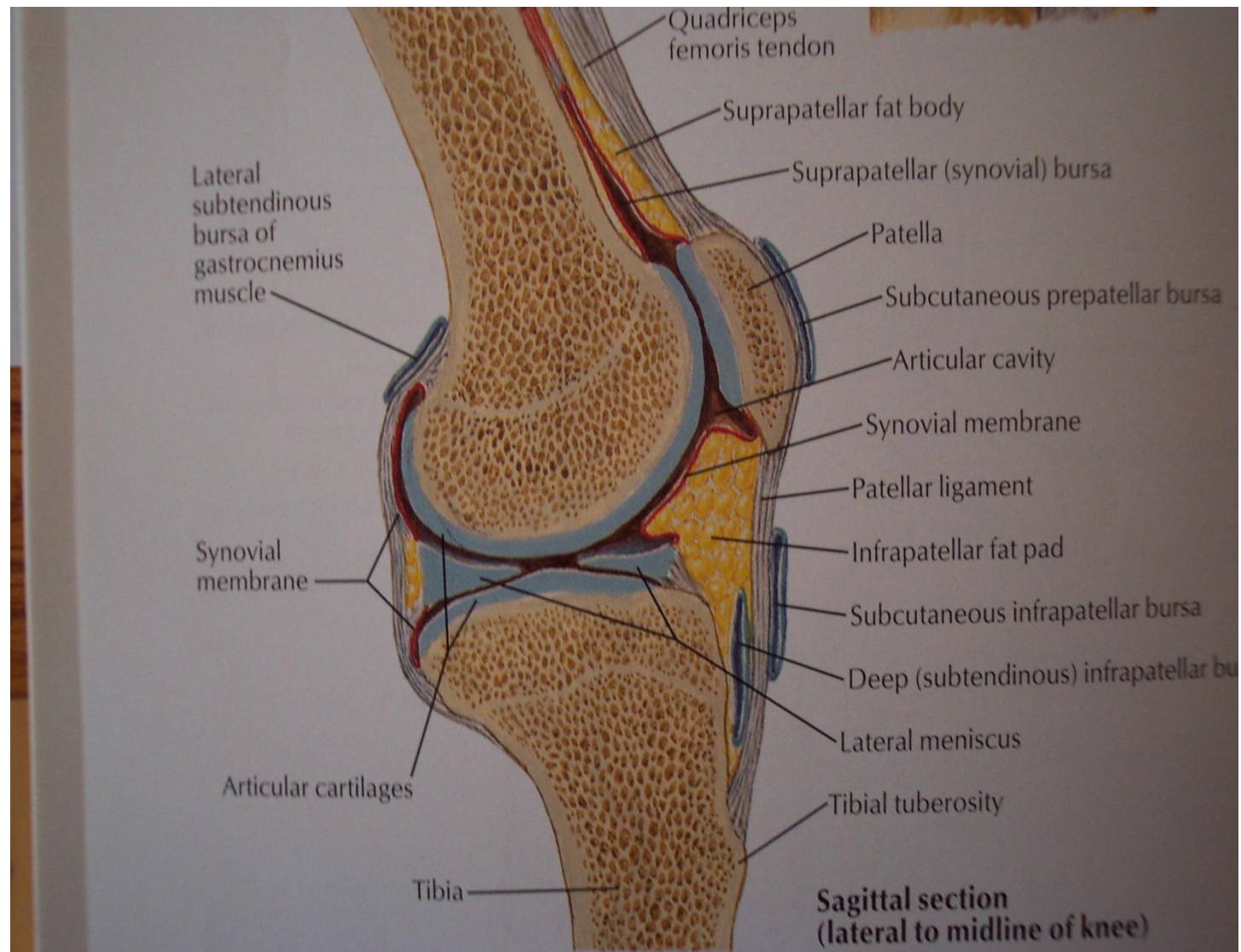
Iliotibial tract

Lateral meniscus

Fibular collateral ligament

Popliteus tendon

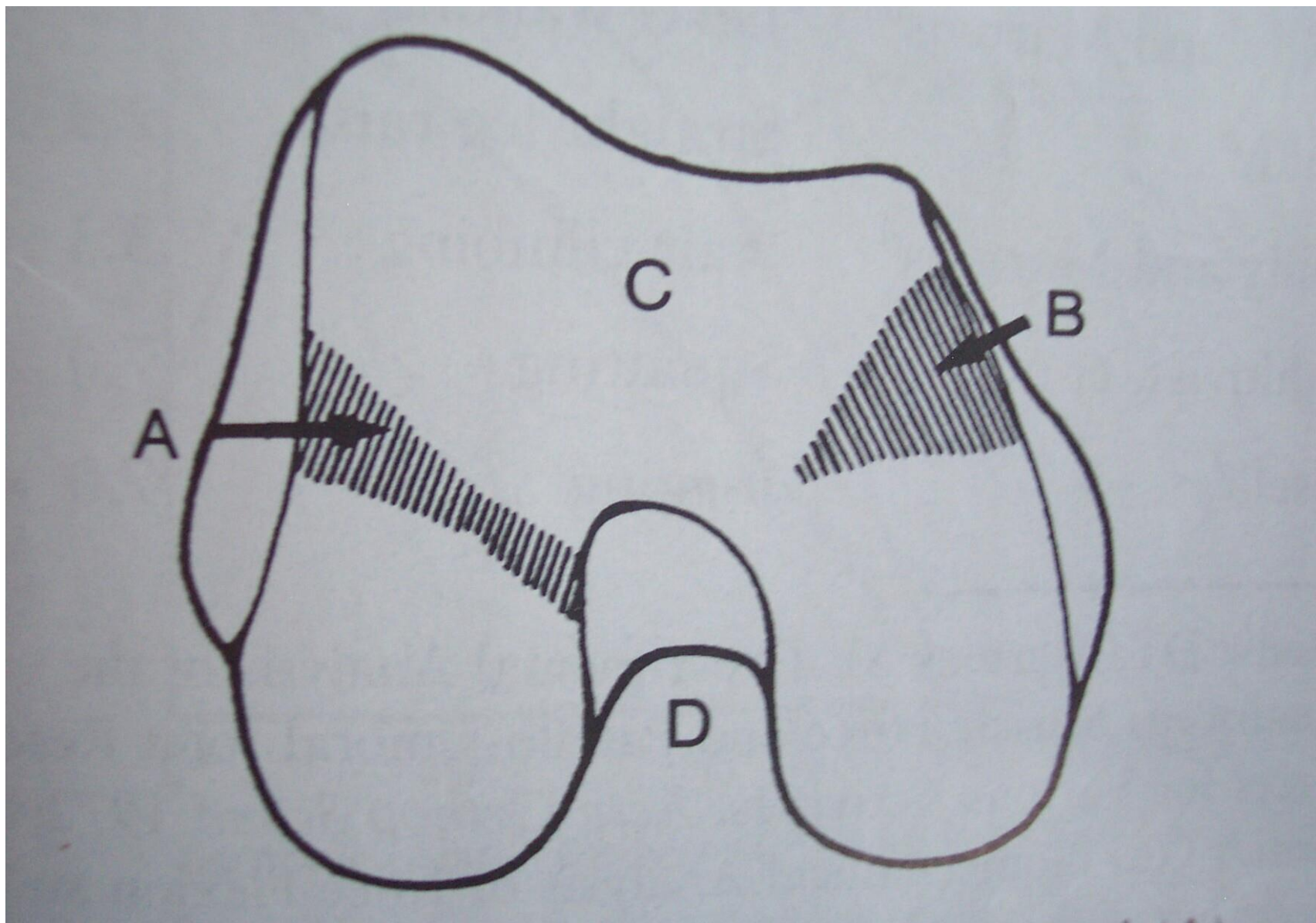




Patellofemoral articulation

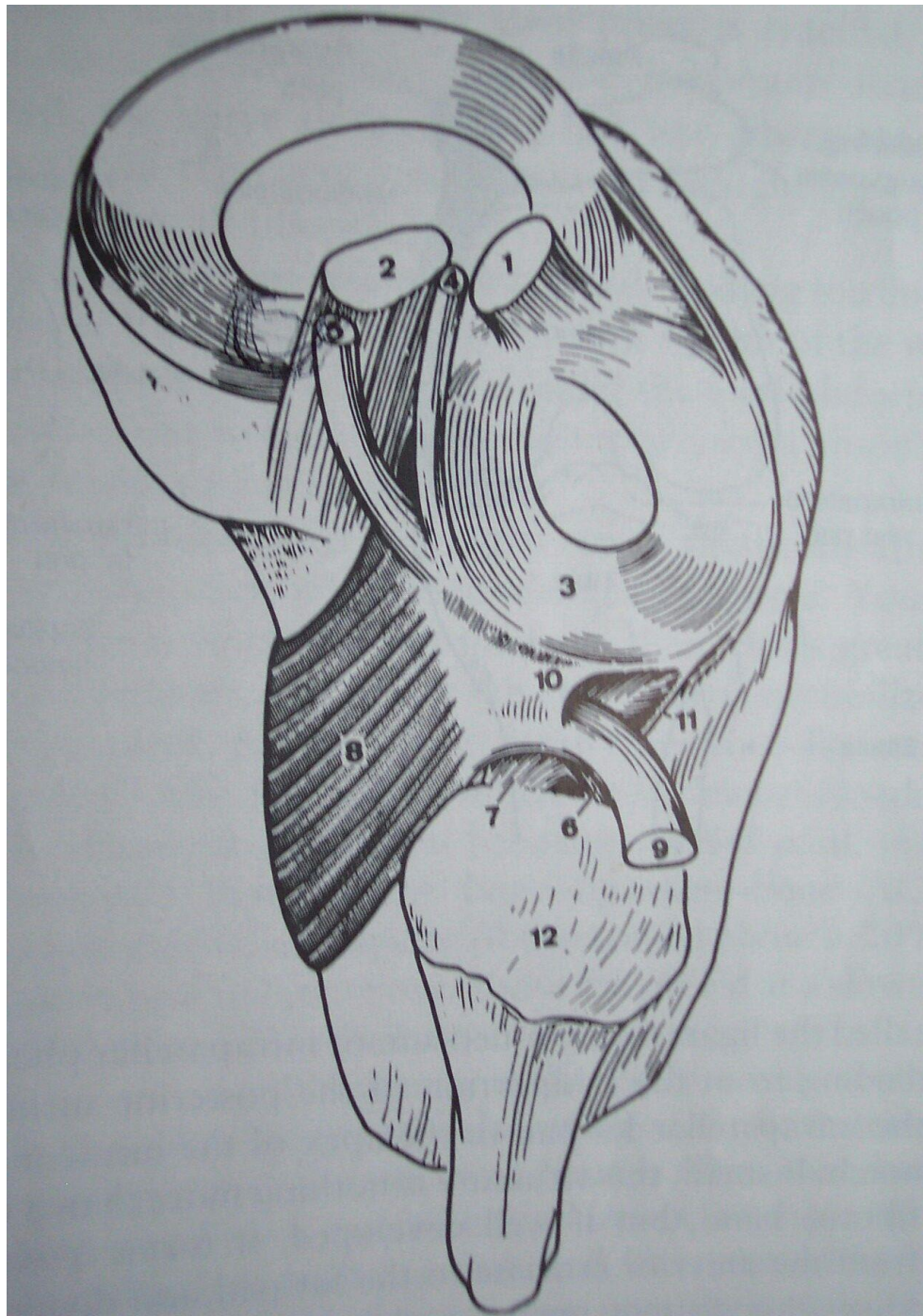
- Shape/anatomy of patella
- Anatomy of intercondylar groove
- direction of force
- PFJR vs flexion angle and quads force
- Contact area vs PFJR and stress
- chondromalacia of patella

- Patella functions
 - Increases moment arm (increases rotational torque) 0 - 45 deg
 - lever at > 45 deg
- Patellectomy?



- Knee joint stability
 - mainly rotational
 - miniscectomy +/- ACL and translation
 - why differences in lat vs med ?
 - Structure
 - attachments

Attachment sites

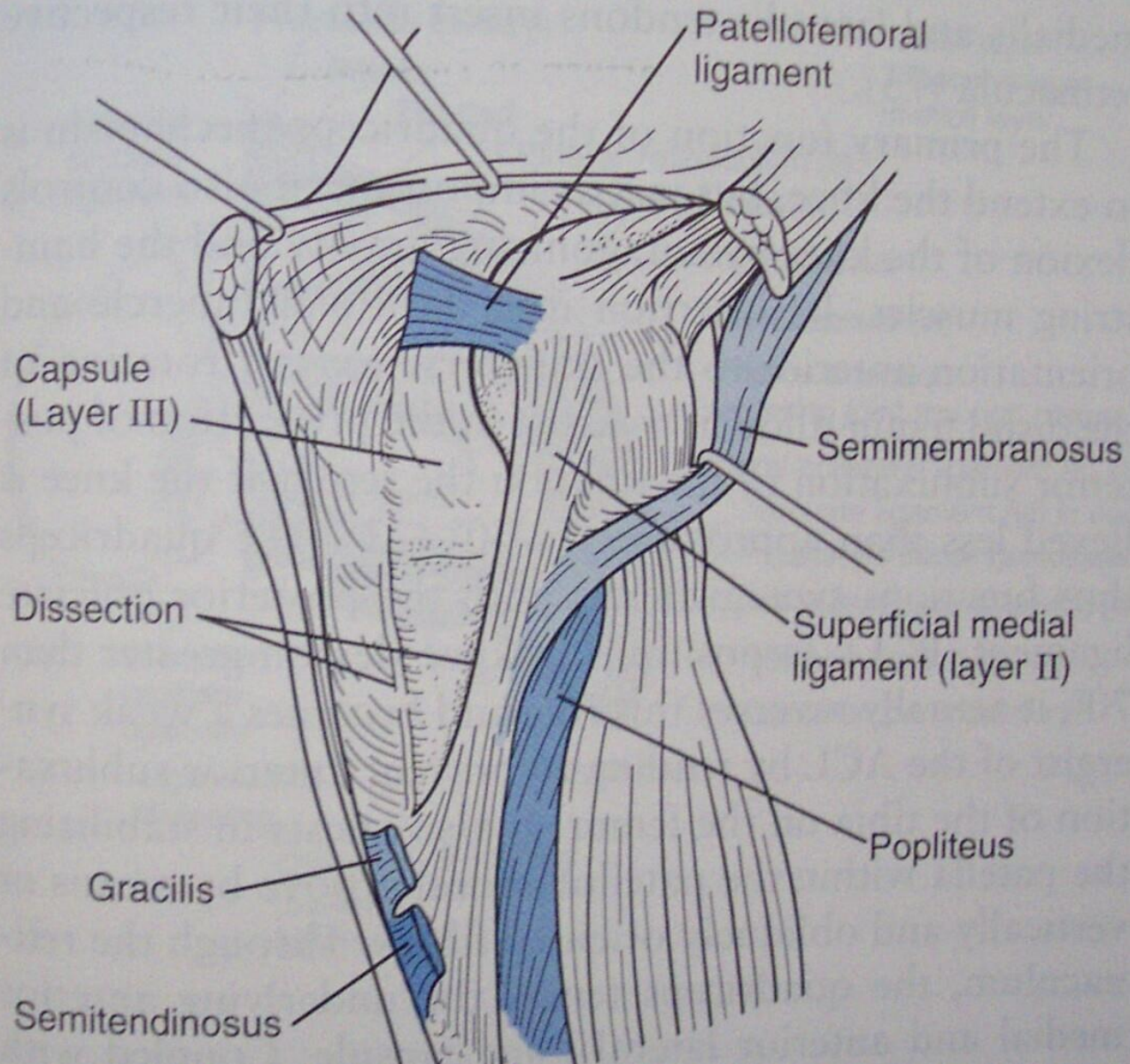


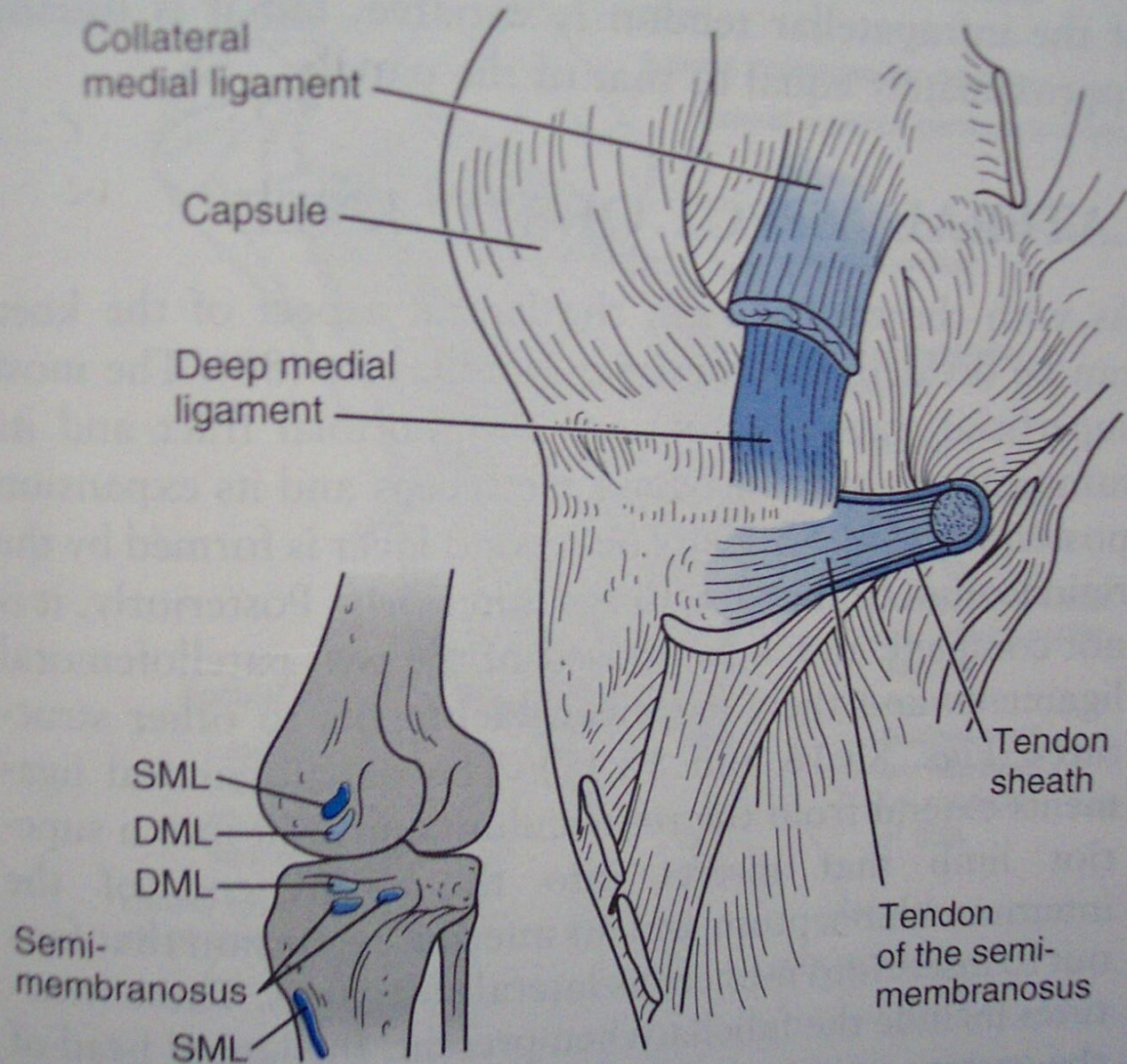
Medial & lateral stabilizers (mostly ligaments)

- Ligaments
 - most important static stabilizers
 - tensile strength - related to composition

Medial side

- Superficial MCL
 - Primary valgus restraint -57-78% restraining moment of knee
 - femoral attachment fans out around axis of rot.
 - Lax in flexion
- Semimembranosus (expansion)
 - internally rot's tib on femur
 - tenses post/med capsular structures that are lax in knee flexion
 - acts with ACL





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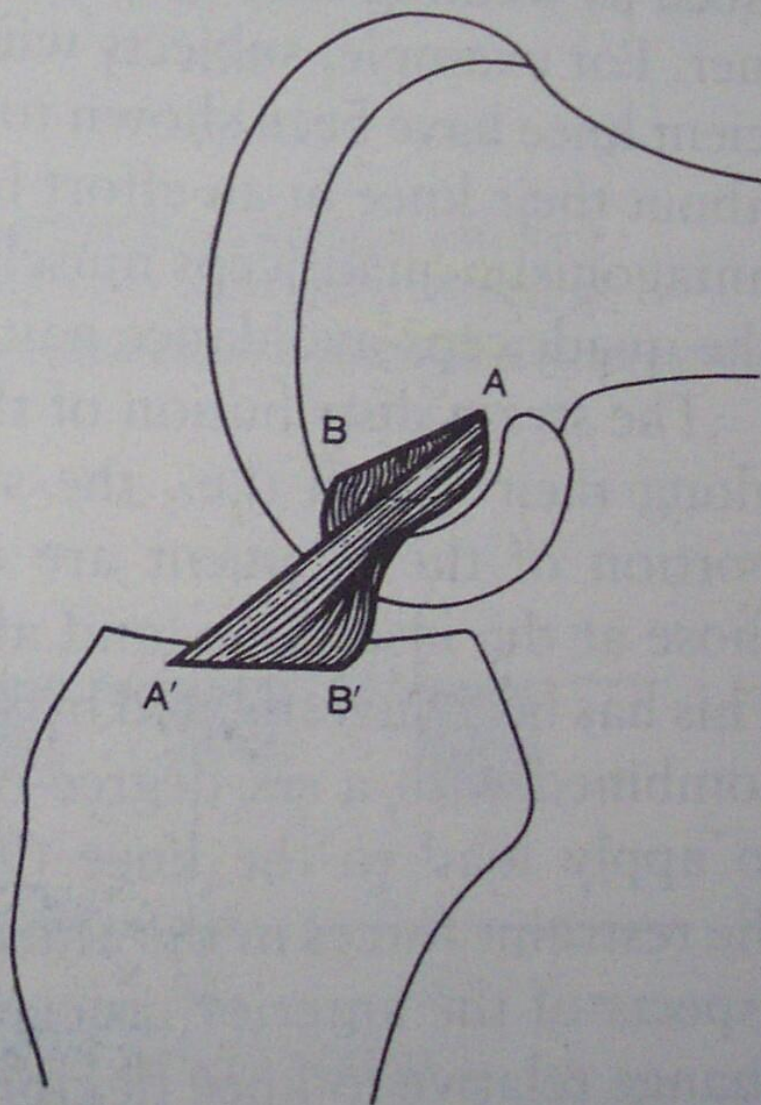
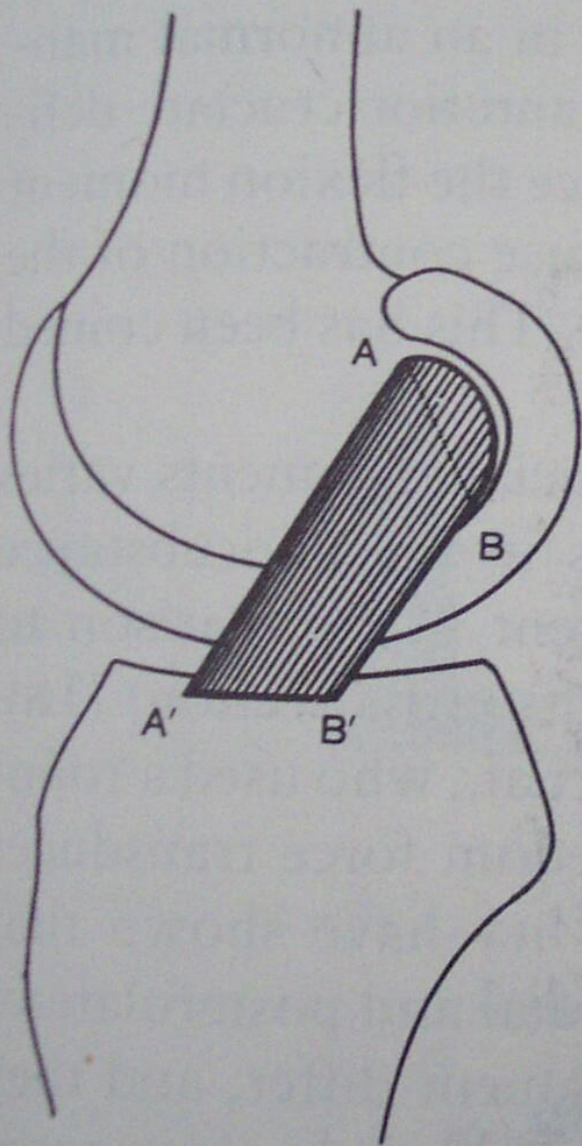
Lateral side

- LCL
 - Primary varus restraint
 - lax in flexion
 - Biceps passes it and blends with insertion
 - maintains tension?

- Biceps
 - flexor(with semimembranosis and pes)
 - externally rotates tibia
 - tenses LCL
 - dynamic assistor of PCL

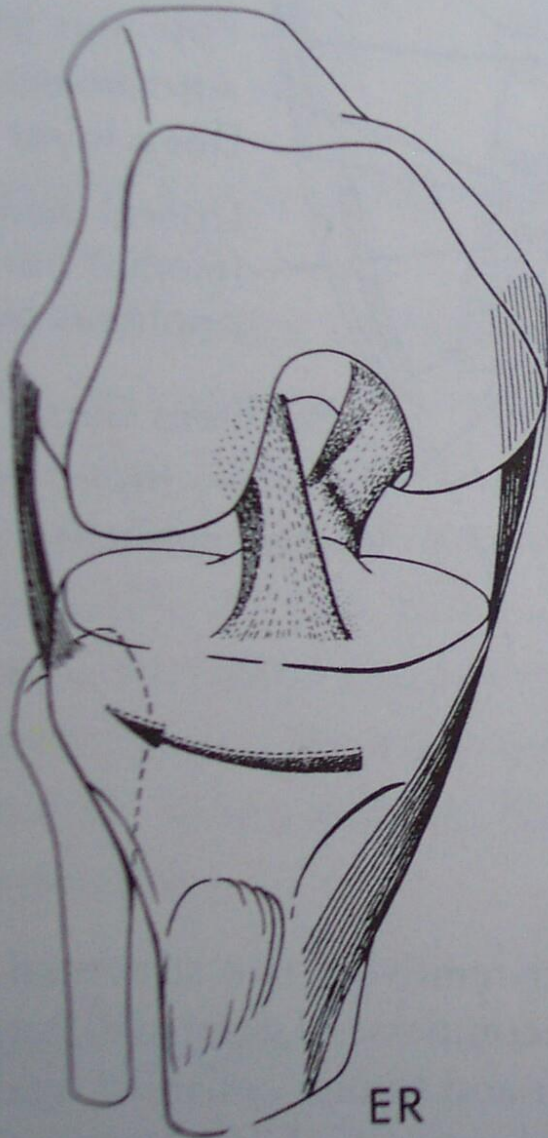
Cruciates

- ACL
 - Primary static restraint to anterior displacement
 - tense in extension, 'lax' in flexion

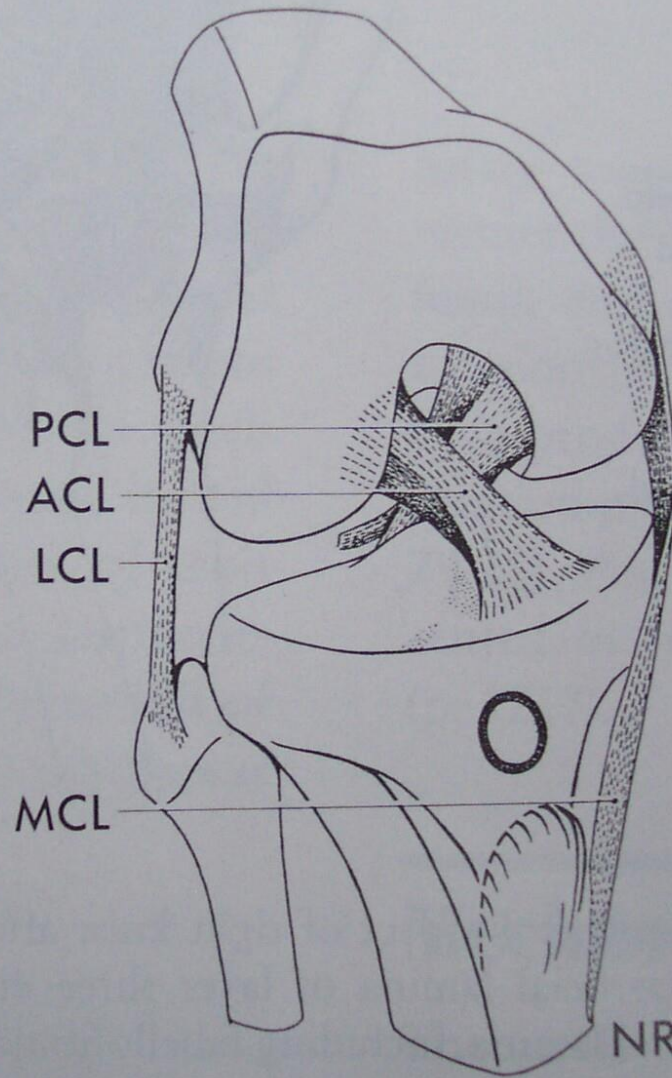


- PCL
 - Primary restraint to post. Displacement - 90%
 - relaxed in extension, tense in flexion
 - reinforced by Humphreys or Wrisberg
 - restraint to varus/valgus force
 - resists rotation, esp.int rot of tibia on femur

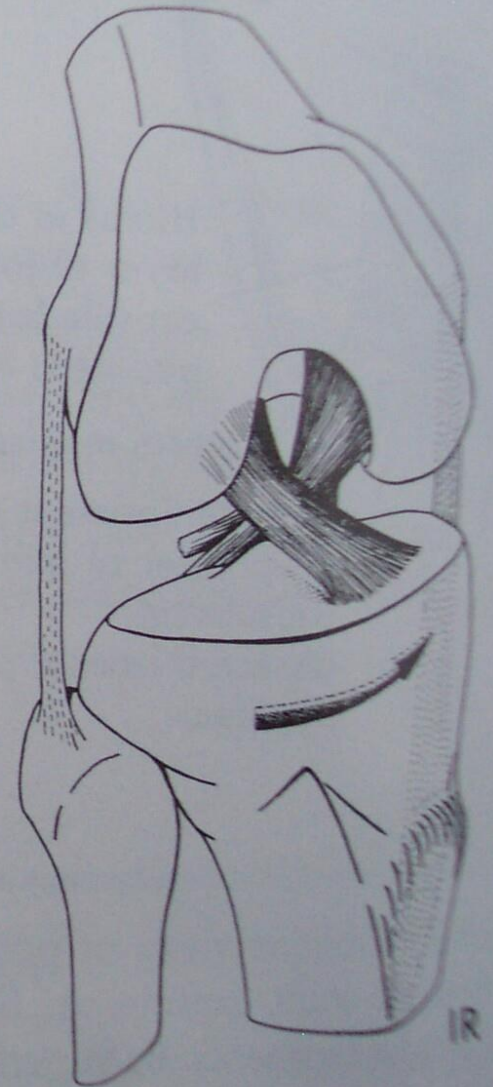
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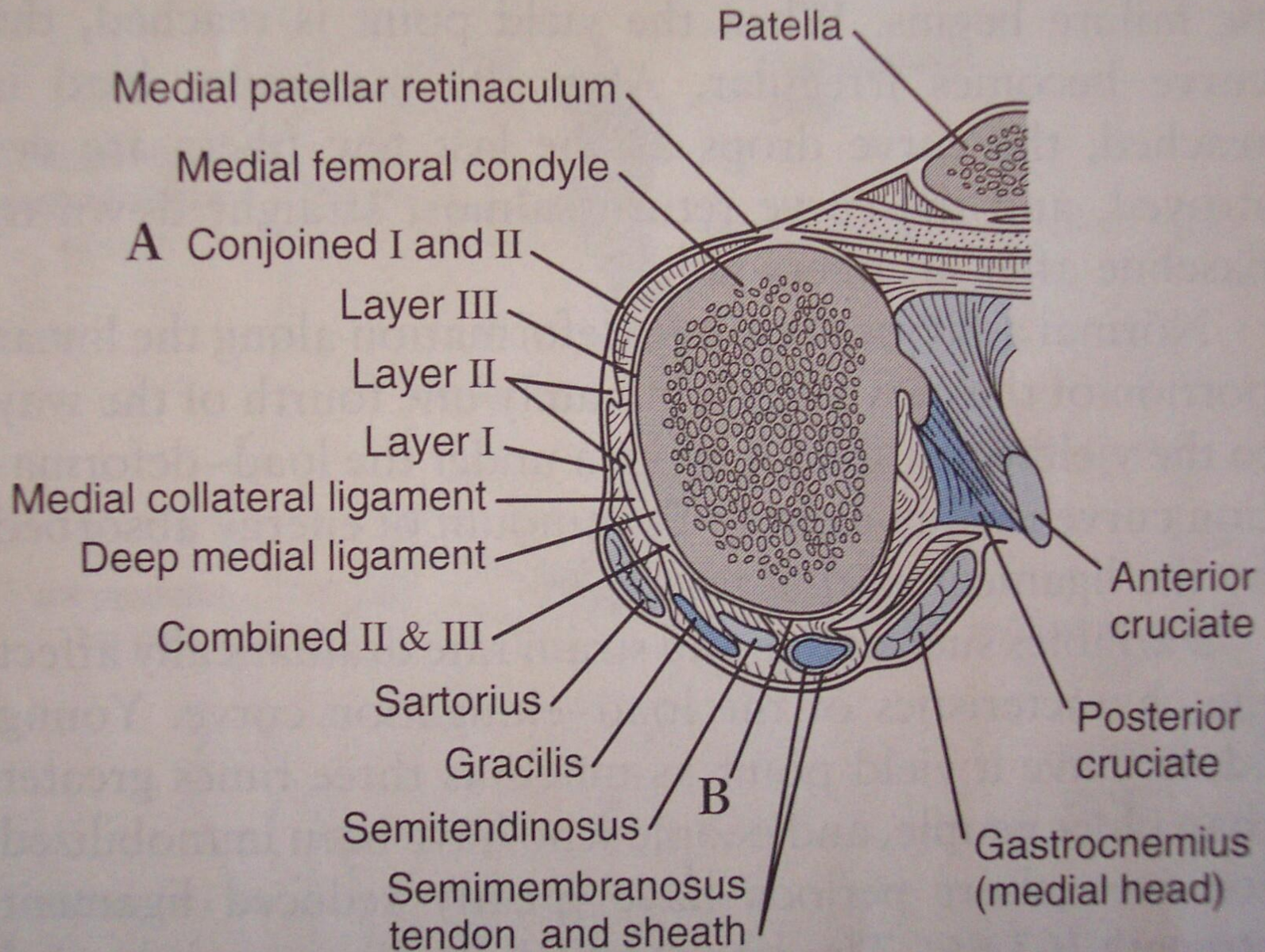


B



C





Overview

- Knee provides mobility and support during dynamic and static activities
- Support during weight bearing
- Mobility during non-weight bearing
- Involved with almost any functional activity of the lower extremity



Knee Joint Motion: Flexion and Extension

- Knee flexion - normal ROM is 130-140°
 - Routine ADL's require 115°
 - Can be as high as 160° in squatting
- Extension - 5-10° hyperextension can be normal

Knee Joint Function: Muscle Action

Muscle Action: Flexors

- Semimembranosus, Semitendinosus, Biceps femoris, Sartorius, Gracilis, Popliteus, Gastrocs
- All are 2 jt. muscles except popliteus & short head of biceps femoris



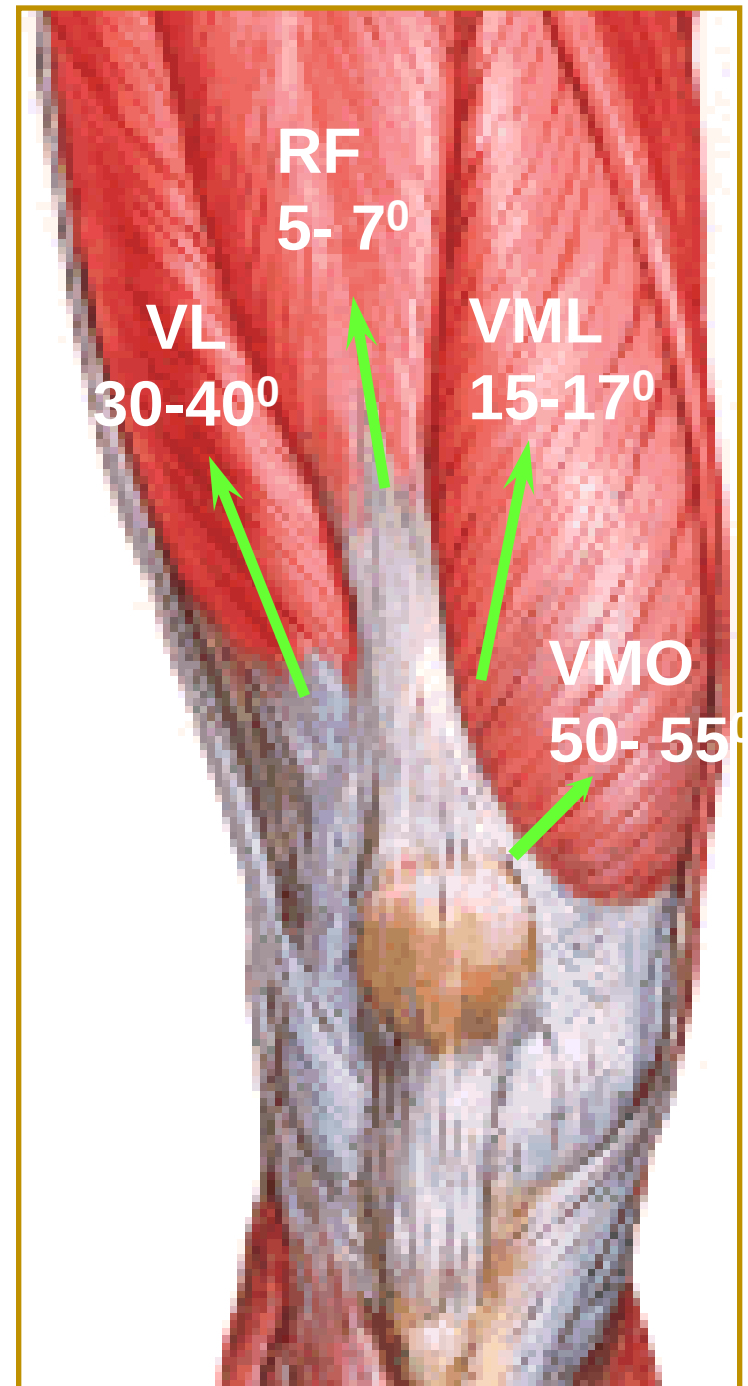
Muscle Action: Flexors

- Gastrocnemius - 2 jt. muscle
 - Small contribution to flexion
 - Very susceptible to active insufficiency
 - During PF with knee flexed, most work done by soleus
 - At the knee, appears to be more of a **dynamic stabilizer** than mobility muscle



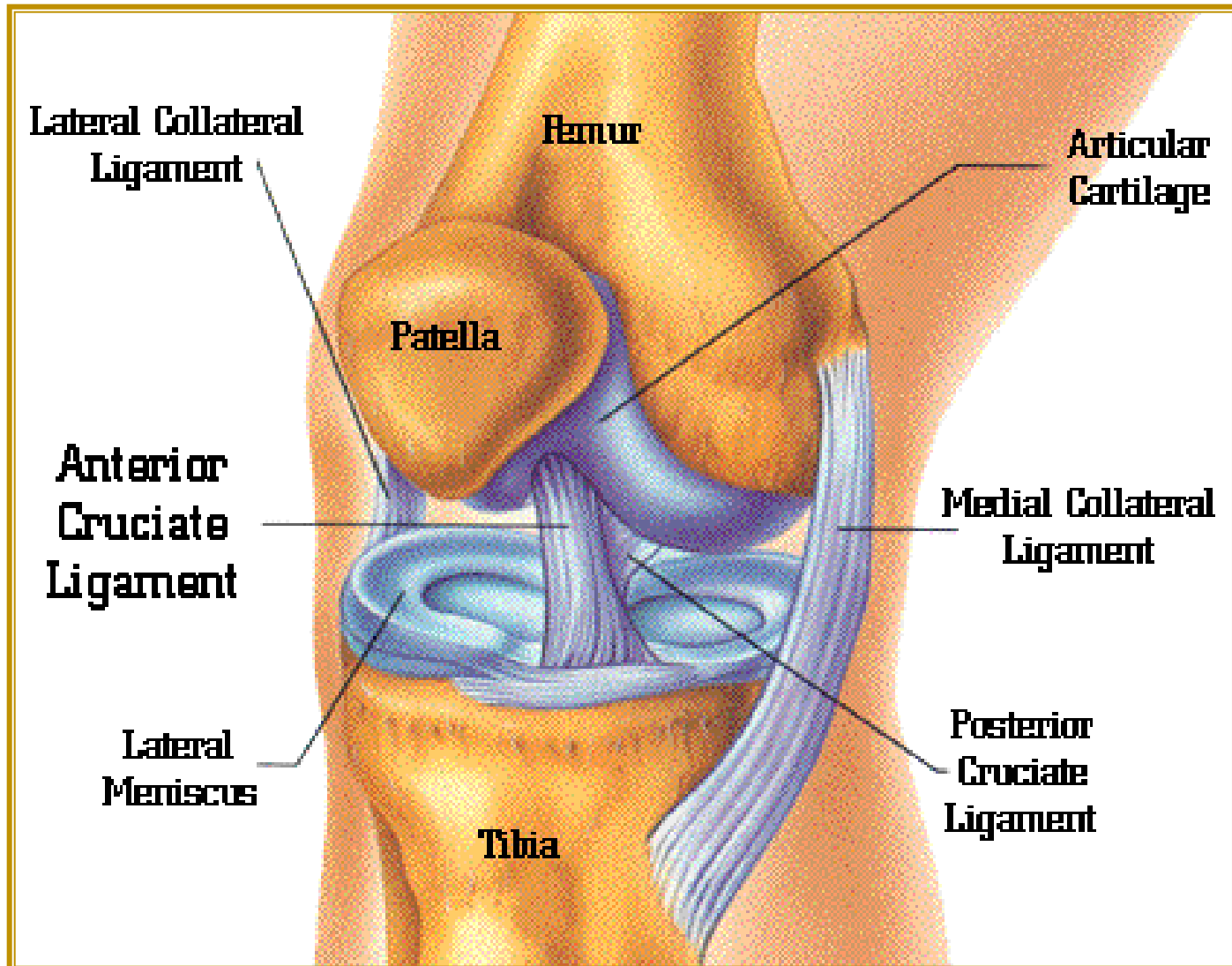
Muscle Action: Extensors

- Quadriceps
 - Rectus femoris - 2 jt.
 - Vastus intermedius, lateralis, medialis
- Resultant pull:
 - Lateralis
 - Medialis
 - Rectus femoris



Patellofemoral joint

Patellofemoral Joint



Functions of the patella/PFJ

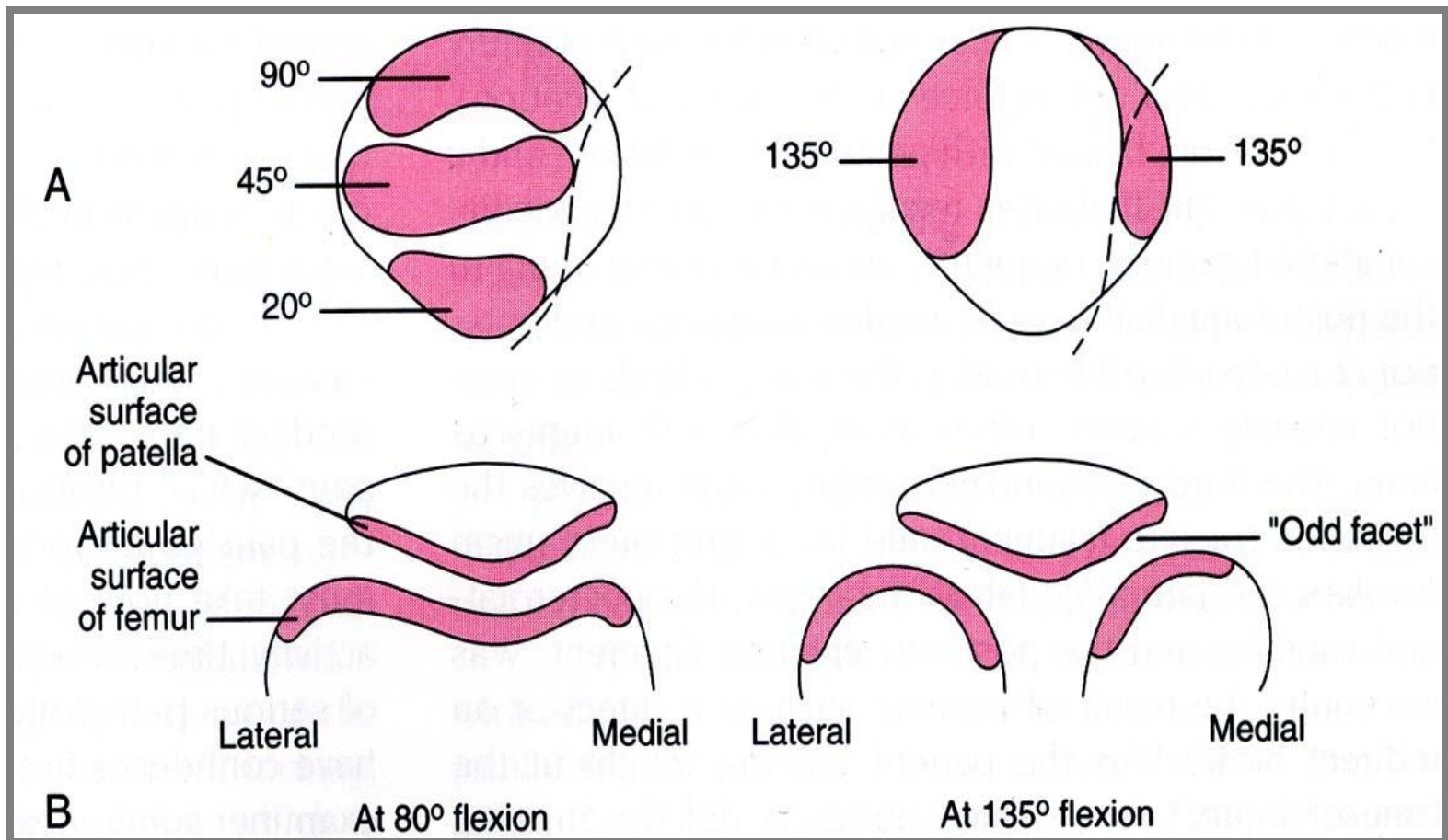
- Increase the mechanical advantage of the quadriceps muscle group
- Decreases friction - quad tendon & femoral condyles
- Helps to distribute the compressive forces that are placed on the femur

Patellofemoral Joint:

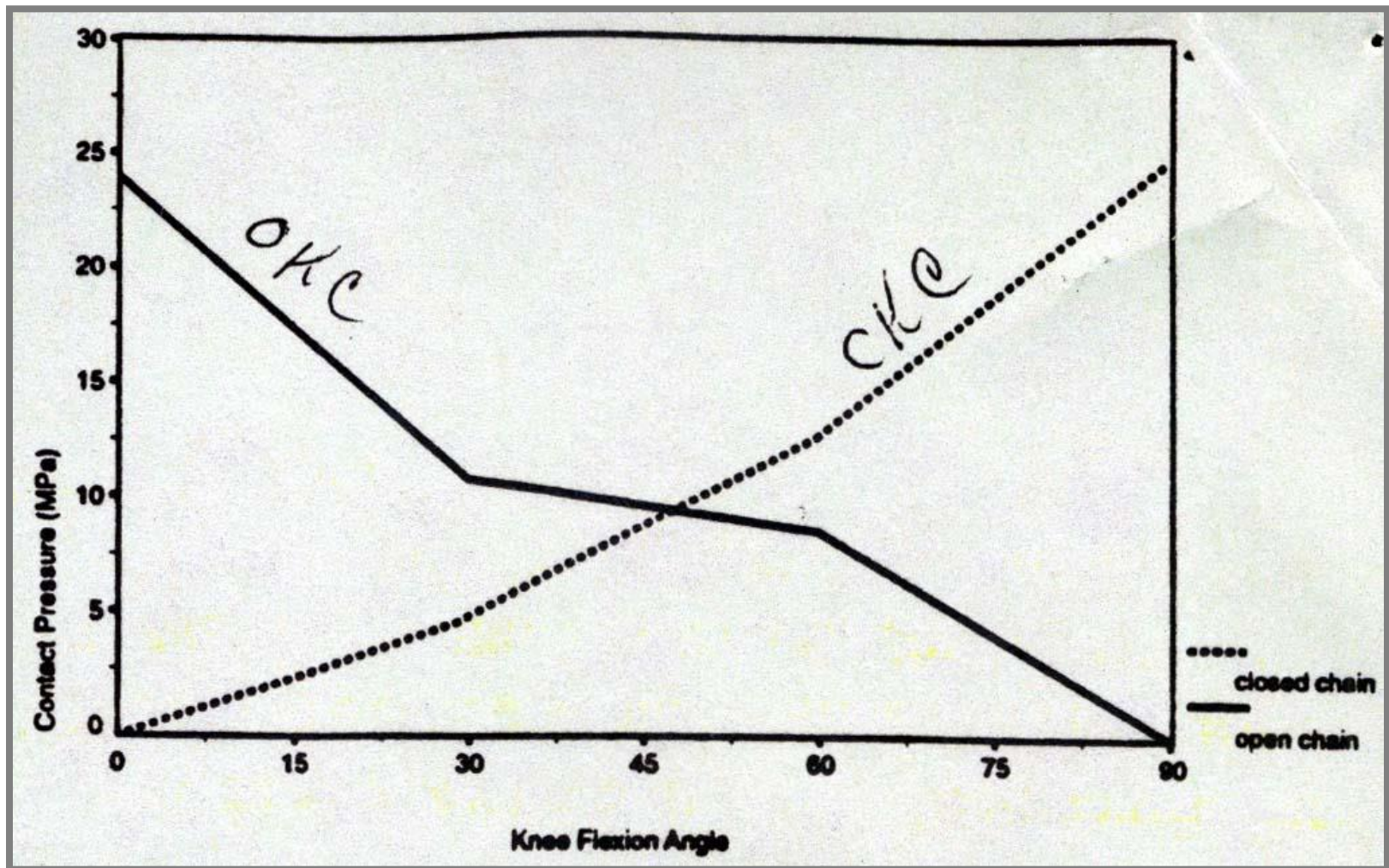
Patellofemoral Joint Congruence

- 1st consistent PF contact is between 10-20° flexion with increased contact as flexion increases
- By 90° all aspects of facets have made contact except odd which contacts at >90°
- At 135° - contact is on odd & lat facets

Patellofemoral contact points



Patellofemoral joint stress in weight-bearing and non-weightbearing



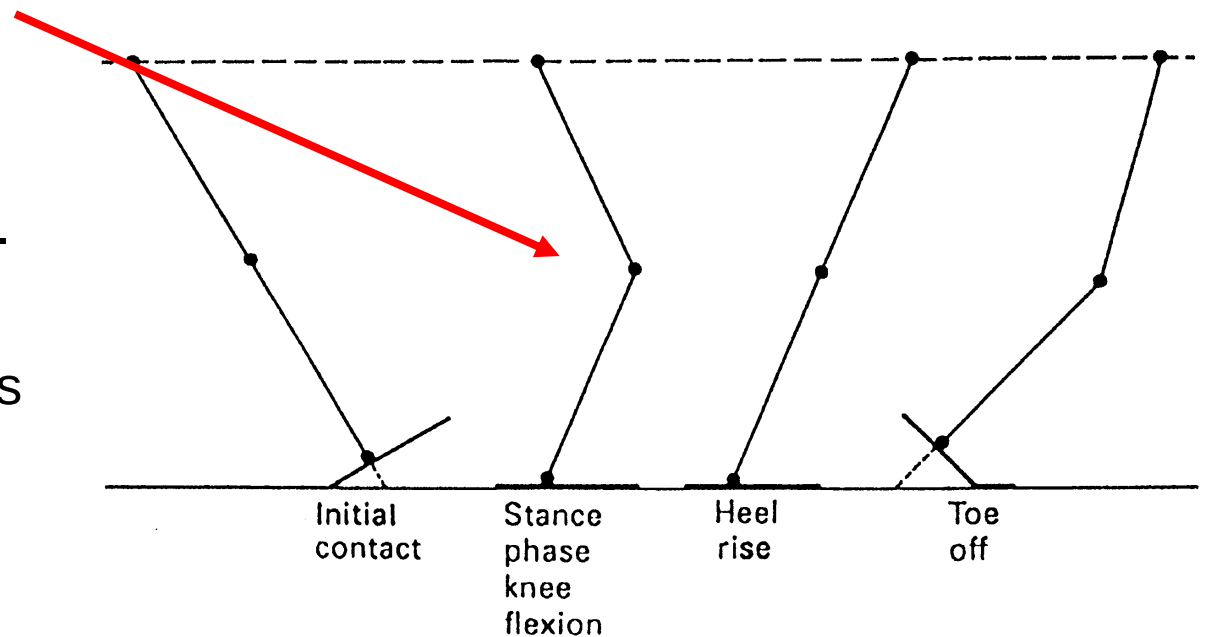
Biomechanics: clinical implications

(Grelsamer & Klein, 1998)

- Quad strengthening can be safely performed in the 0-90° range by varying the mode of exercise if ROM restrictions are in place.
- Specifically, open chain (NWB) exercises are most safely carried out from 25-90°, and SLRs with the knee at 0° of extension are equally safe.
- Closed chain (NWB) exercises are safest in the 0-45° range.

DG3: Knee flexion in stance phase

- As the hip joint passes over the foot during the support phase, there is some flexion of the knee.
- This reduces vertical movements at the hip, and therefore of the trunk and head.



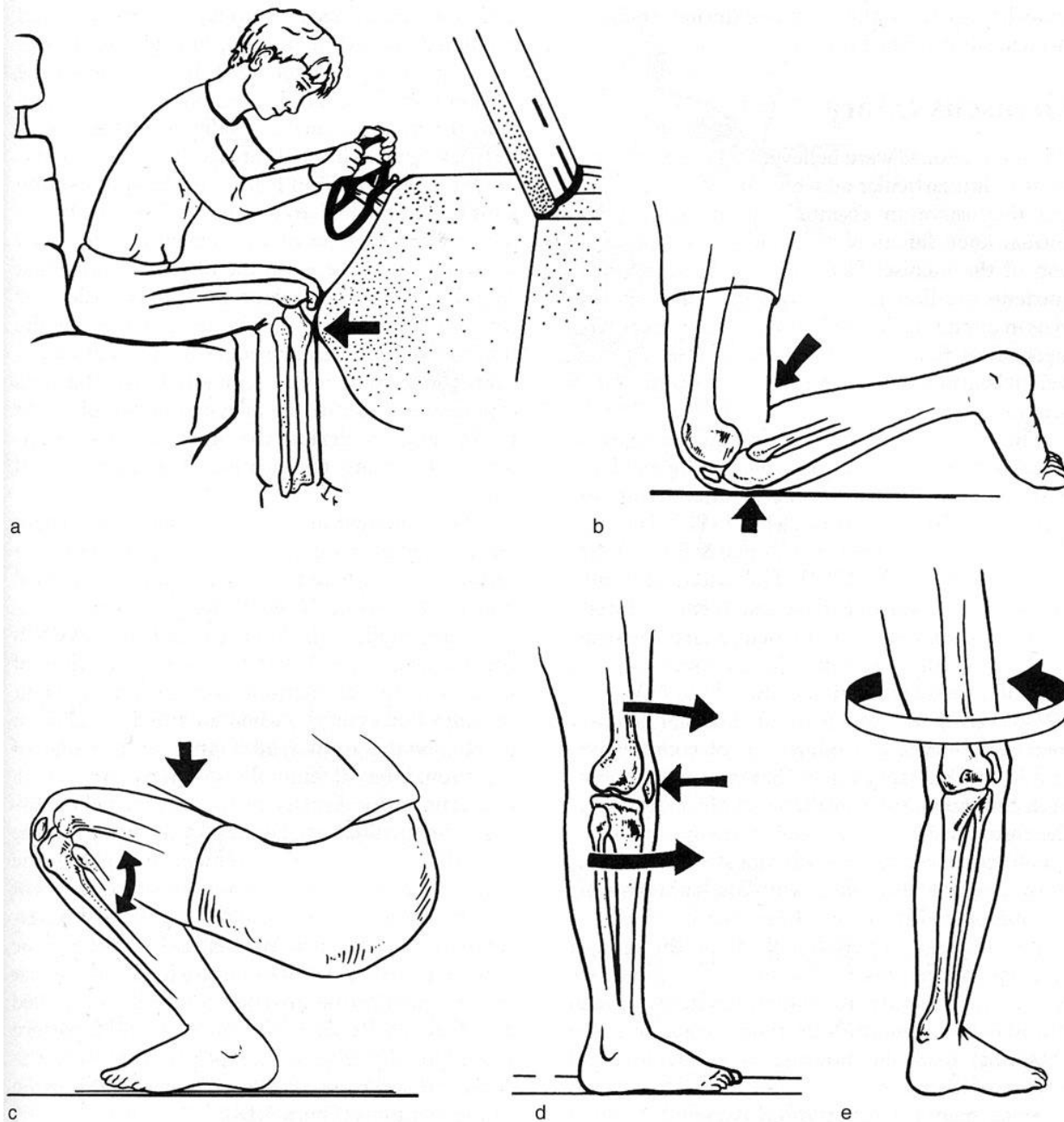
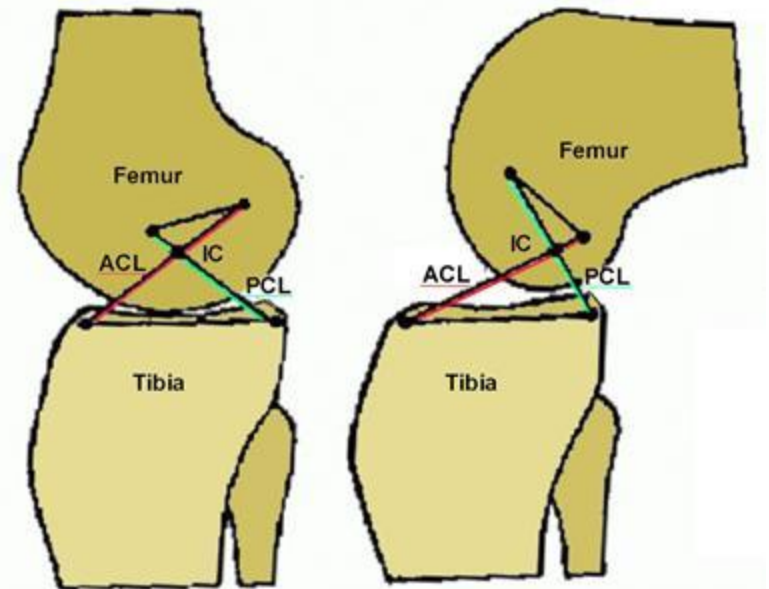


Figure 6.17 Mechanisms of posterior cruciate ligament injury. (a) Motor vehicle collision in which impact with the dashboard forces the tibia posteriorly relative to the femur. (b) Fall on a flexed knee pushing tibia posteriorly. (c) Forced knee flexion. (d) Forced knee hyperextension. (e) Cutting (changing direction) on a minimally flexed knee.

Stability in motion during stair climbing

- In **anterior** direction of knee joint
 - By Anterior Cruciate ligament (**ACL**)
- In **posterior** direction of knee joint
 - By Posterior Cruciate ligament (**PCL**).



Stability in motion during stair climbing (cont'd)

- **Rotational stability**

is due to

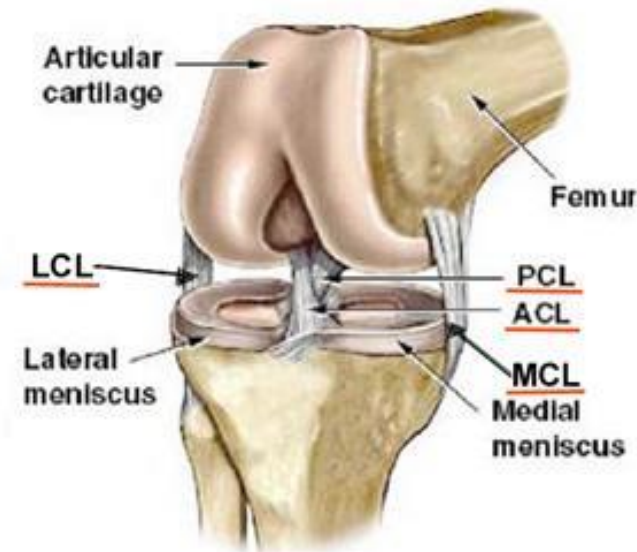
- Medial Collateral Ligament (**MCL**)
- Lateral Collateral Ligament (**LCL**)

- Stability in **valgus**

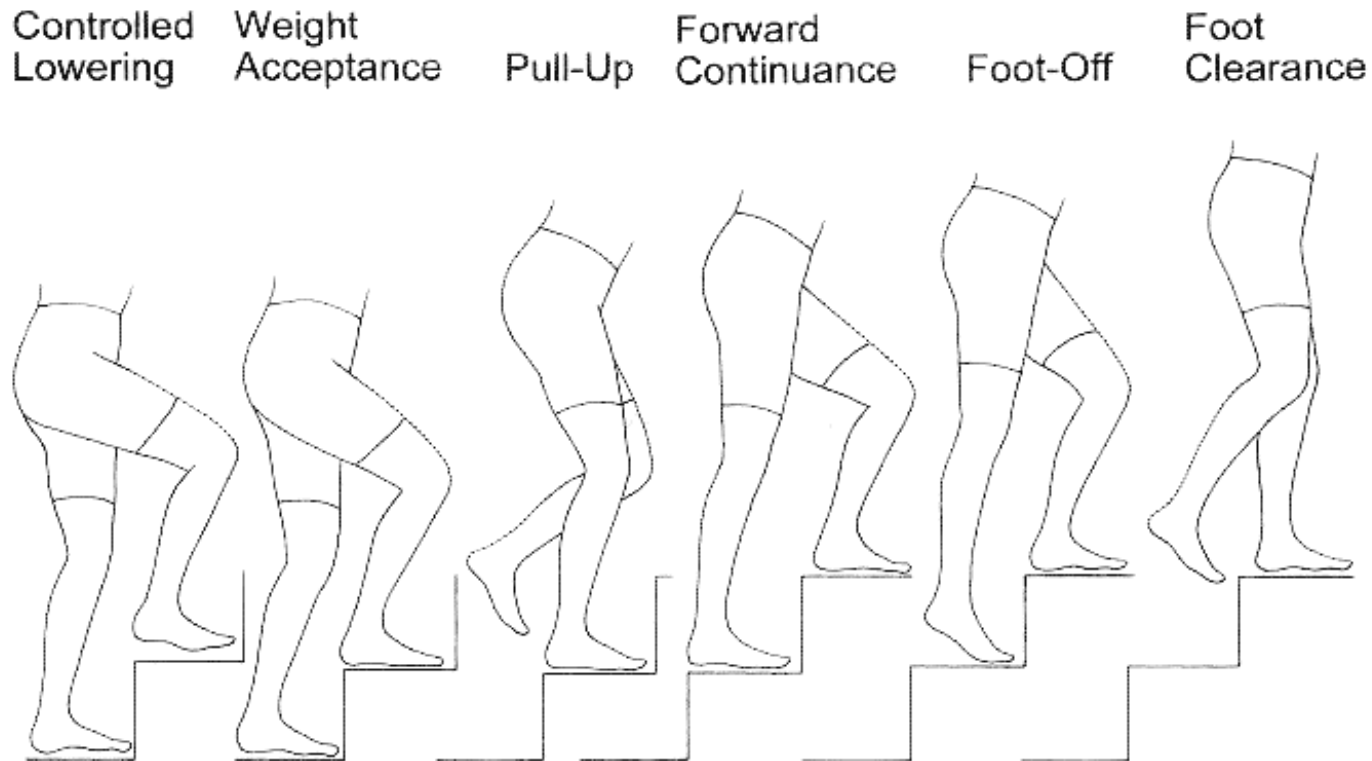
- MCL

- Stability in **varus**

- LCL



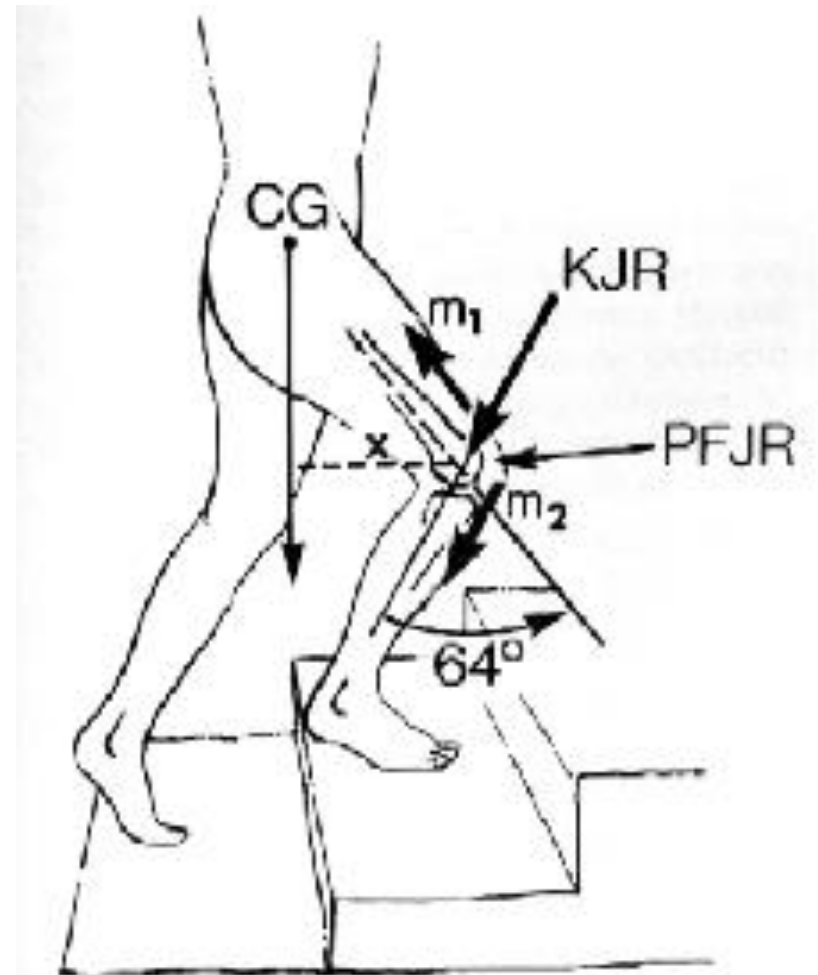
Phases in Stair Climbing



- **Stance Phase:** Weight Acceptance to Foot-off
- **Swing Phase:** Foot off to foot strike

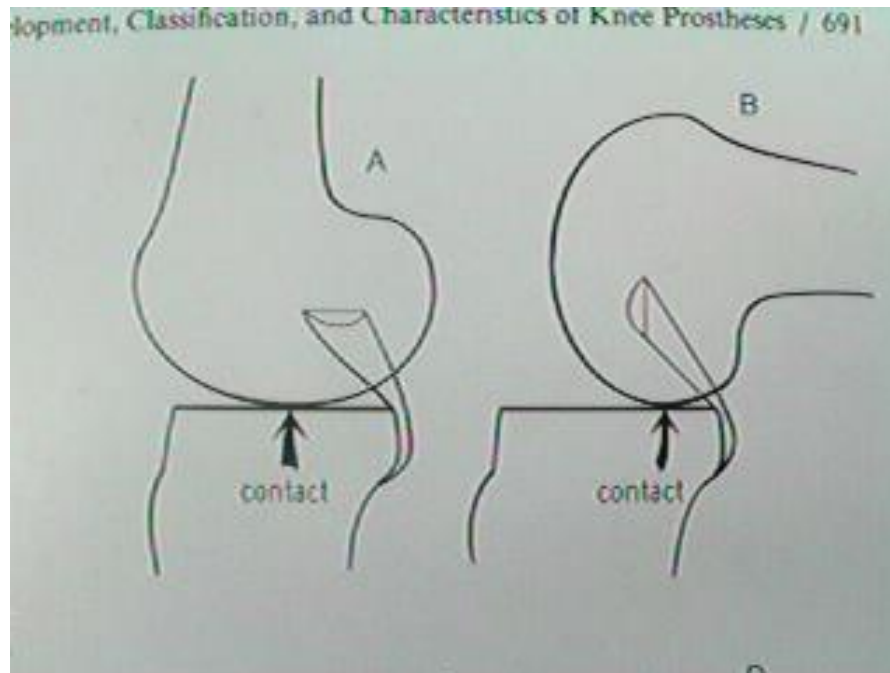
Joint Moments

- Product of instantaneous equivalent muscle force multiplied by equivalent lever arm at a joint
- Instantaneous estimates of required strength for doing motion



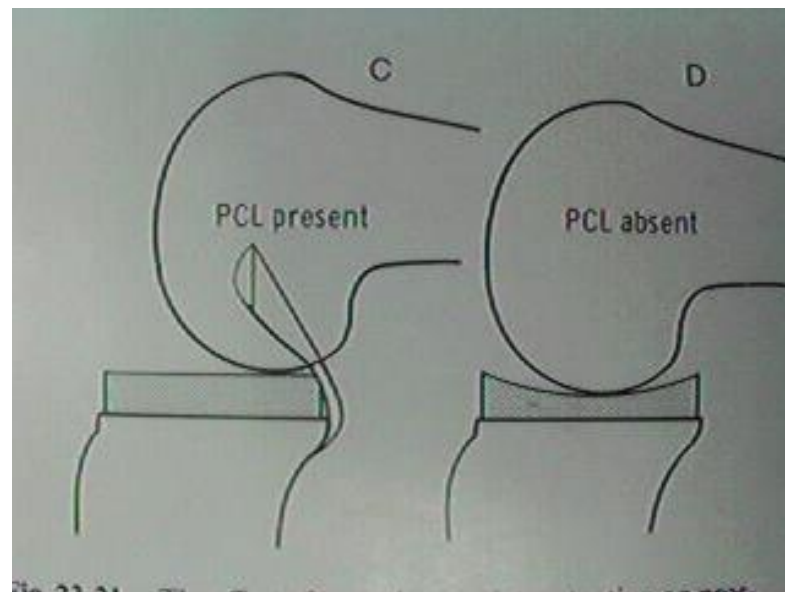
Posterior Cruciate Ligament

- Allows femoral condyles to glide and rotate posteriorly.



Posterior Cruciate Ligament

- If the PCL is retained the tibial component needs to be flat in AP plane. Why ???



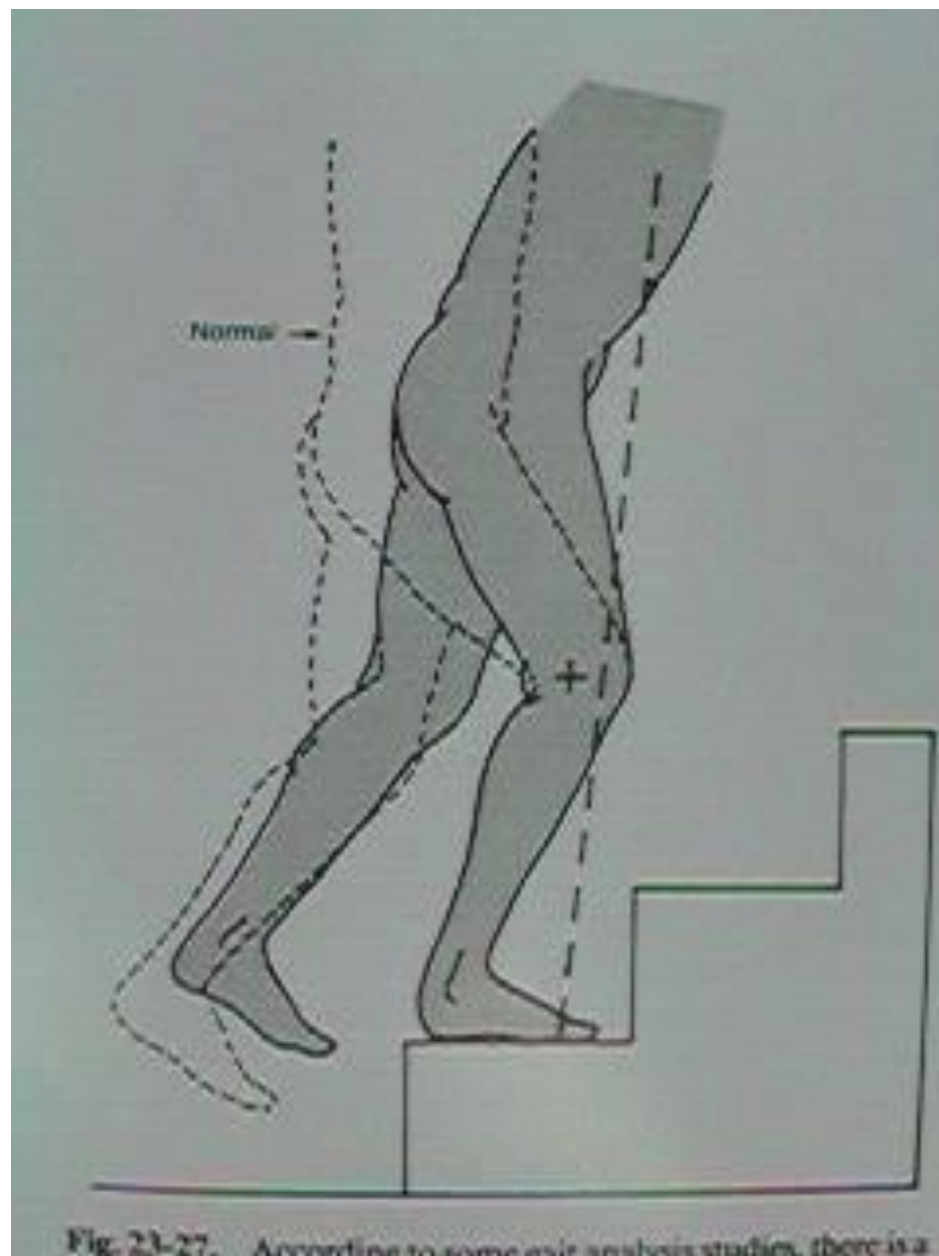
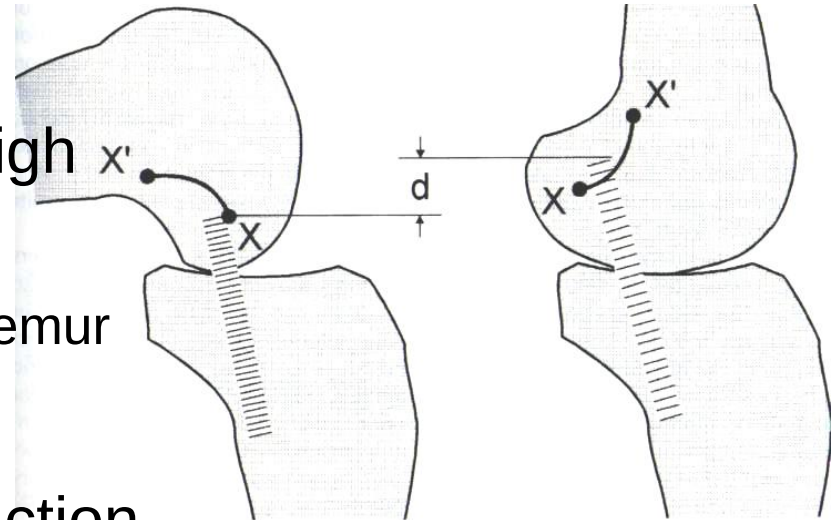


Fig. 23-27. According to some axis analysis studies, there is a

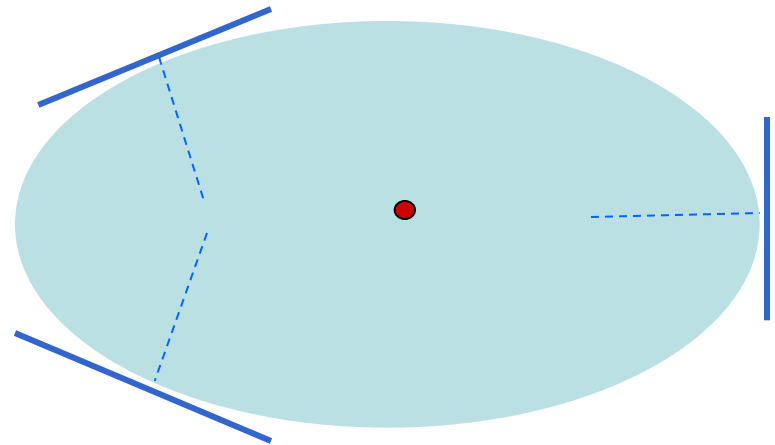
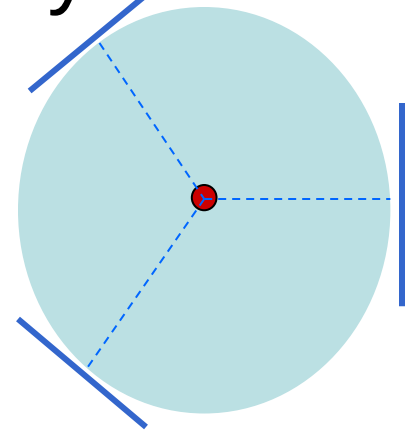
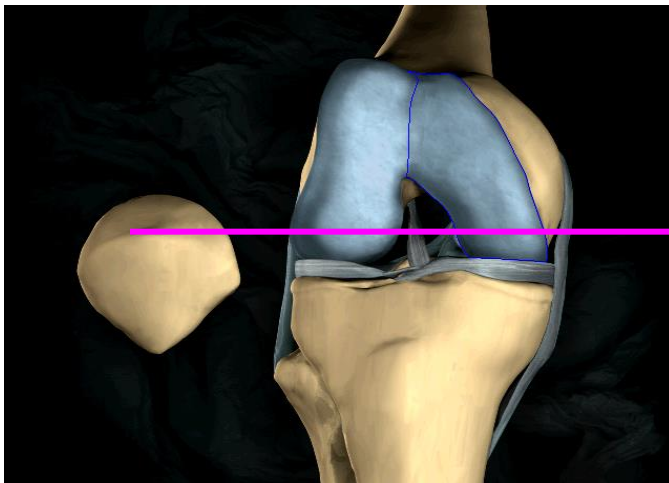
Collateral ligaments

- flexion and extension
 - tension
 - knee flex, low
 - knee extension, high
 - reason?
 - Contact point of femur
- abd/adduction
 - medial, prevent abduction
 - lateral, prevent adduction
- both
 - prevent over rotation



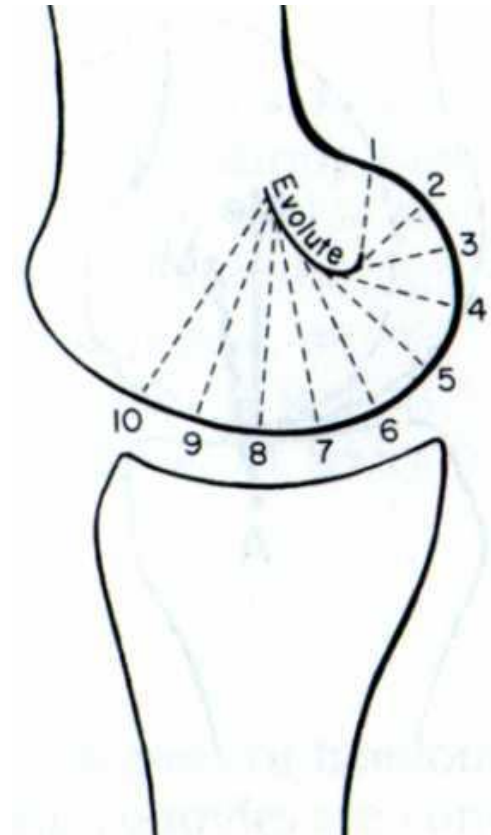
Axes for flexion/extension

- a few centimeter above joint line passing through femoral condyles
- mechanics center of rotation
 - perpendicular to curvature
 - relative zero velocity



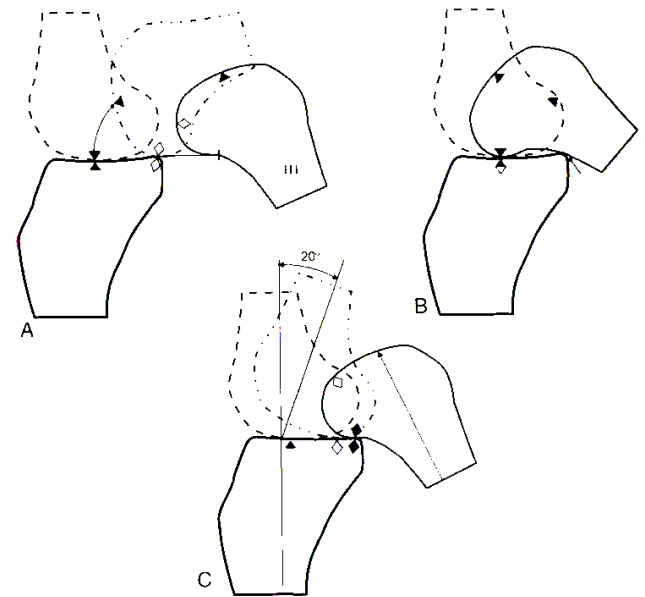
Methods of defining knee joint axis

- x-rays (perpendicular to curvature)
- instant center of rotation (zero velocity)
- results
 - extension 10
 - flexion 1
- problems
 - devices
 - goniometer
 - isokinetic dynamometer
 - orthotic knee joint



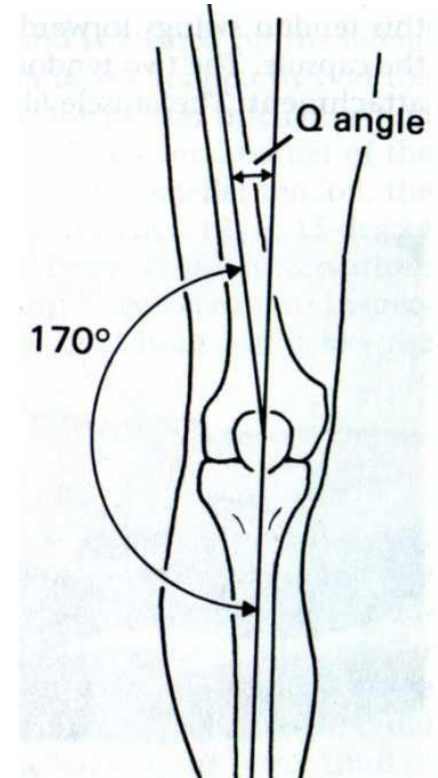
Anatomic basis for joint motion

- Arthokinematics
 - combining rolling and sliding
 - flexion to extension
 - rolling predominant first
 - sliding predominant later
 - convex surface rotate on concave surface
 - different direction
 - conversely

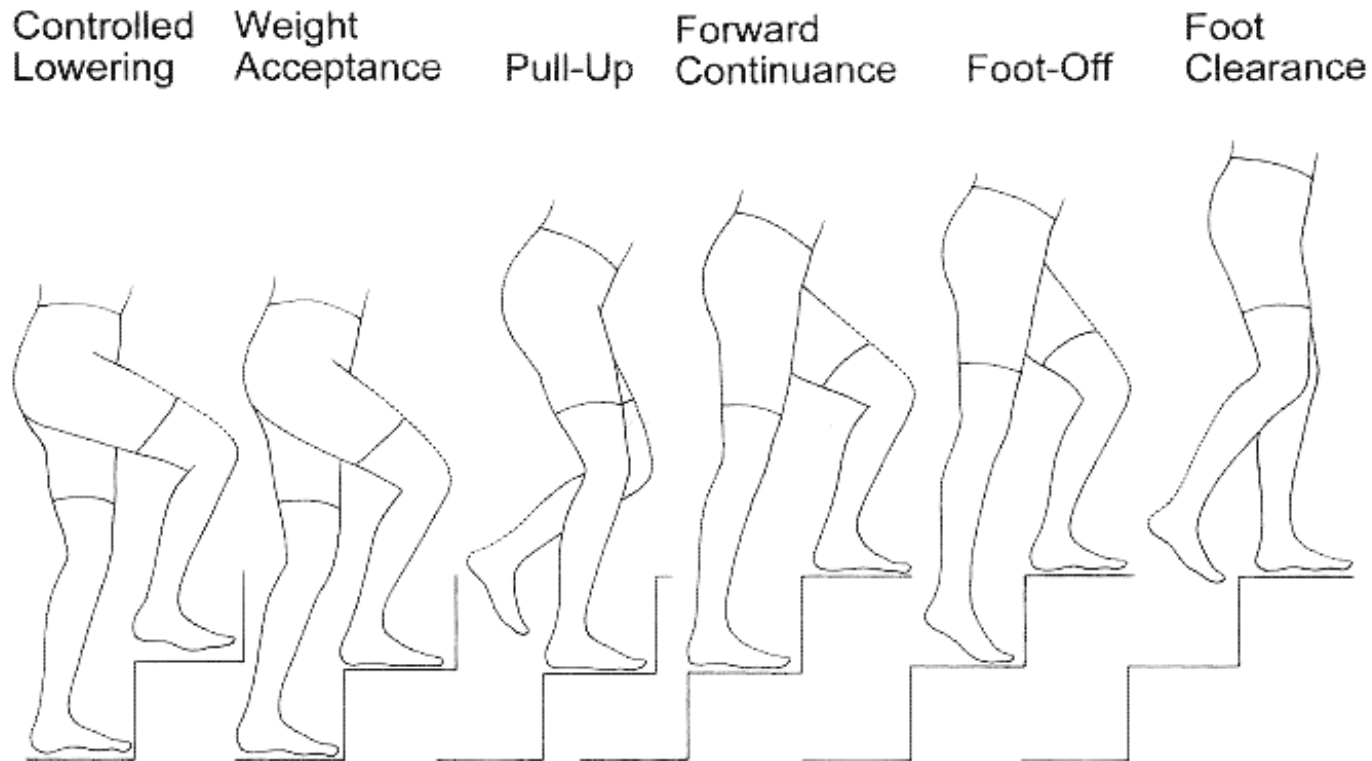


knee alignment and deformities

- femur-tibial angle
 - normal, 170 degrees
 - small, genu valgum(knock knee)
 - large, genu varum(bowleg)
- Q-angle
 - between femur line and extended tibial line
 - 180 degree - femur-tibial angle



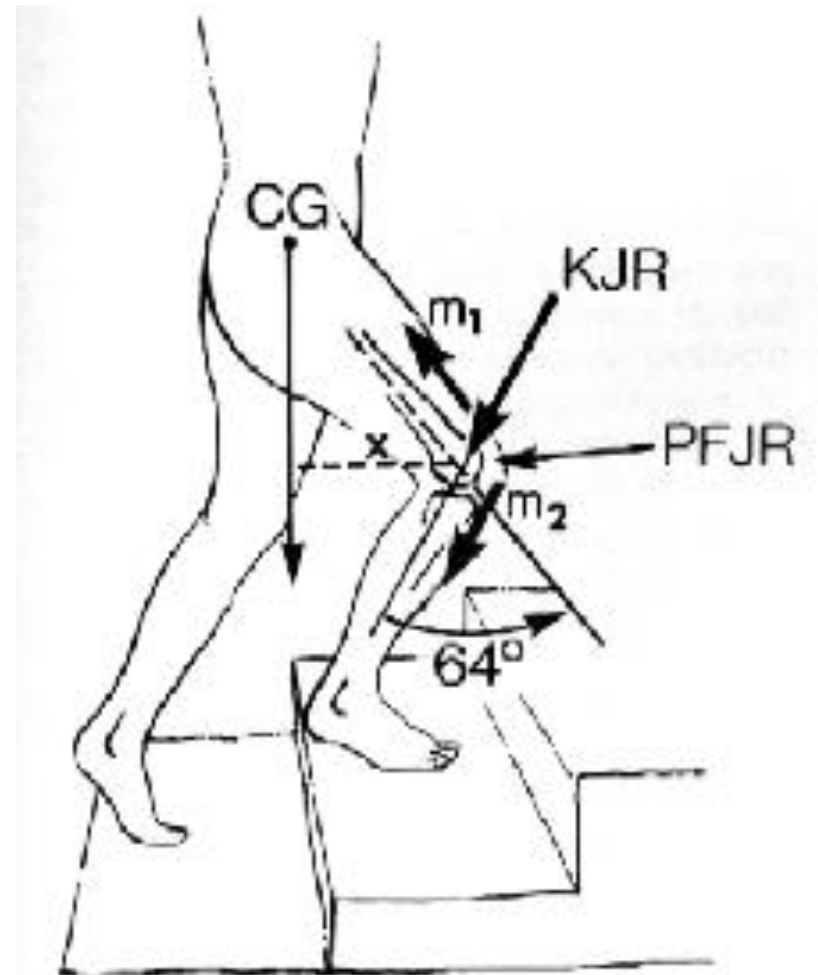
Phases in Stair Climbing



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- **Swing Phase:** Foot off to foot strike

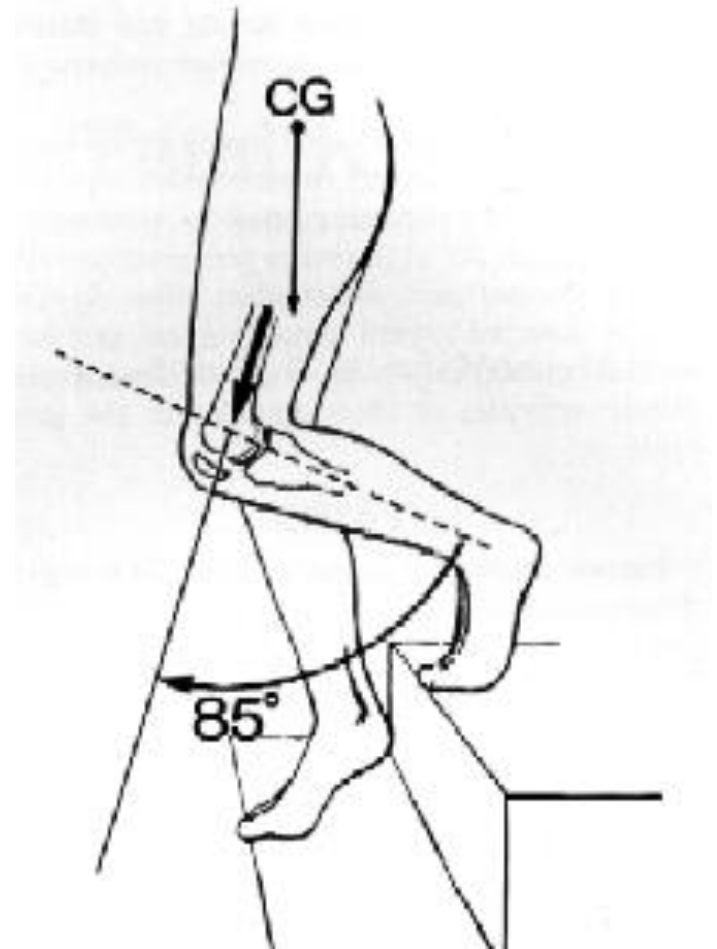
Joint Moments

- Product of instantaneous equivalent muscle force multiplied by equivalent lever arm at a joint
- Instantaneous estimates of required strength for doing motion



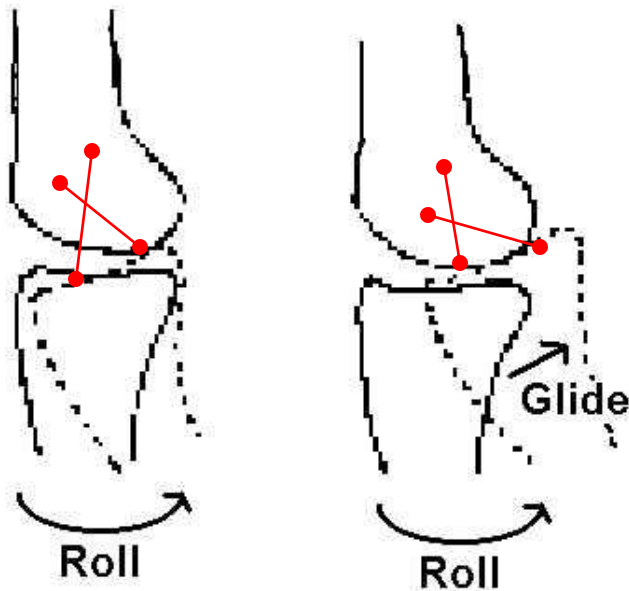
Joint Moments (cont'd)

- Maximum knee flexion angle
 - During **swing** phase
- Maximum knee flexion moment
 - During **stance** phase
 - During descending stairs ($49.2^\circ \pm 9.5^\circ$) was nearly **6 times** (external moment) that when ascending stairs ($59.5^\circ \pm 16.8^\circ$)

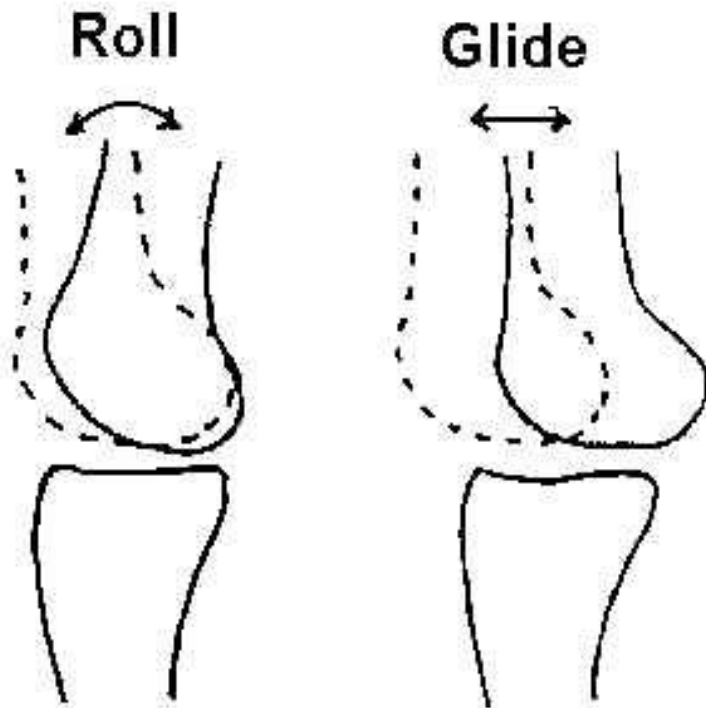


Crossed four-bar linkage

- Mechanical model
- ACL and PCL as two crossed bars
- Accounts for changing roll/glide ratio with knee flexion

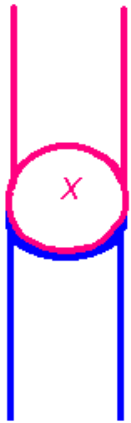


Arthrokinematics of the knee

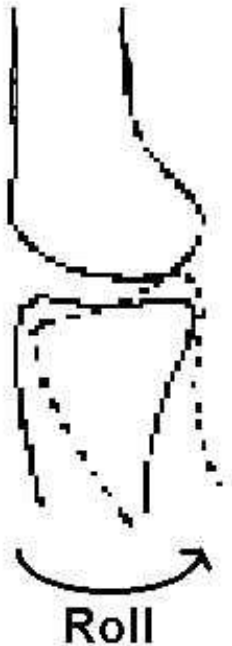


- Roll-tyre rolling on road
- Glide-tyre slipping on road
- Spin

Rule of concavity and convexity

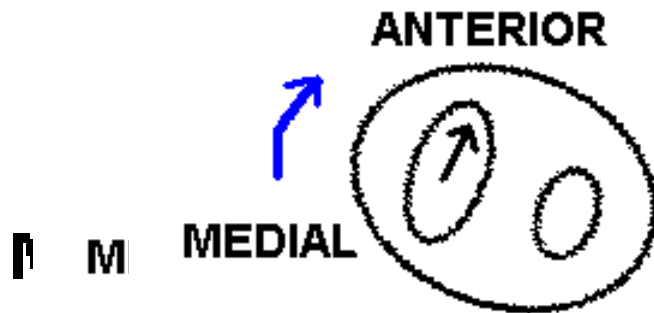


- Roll and glide simultaneously, in opposite and same direction
- Helps prevent subluxation and impingement
- Rolling initiates flexion
- Gliding occurs with final flexion



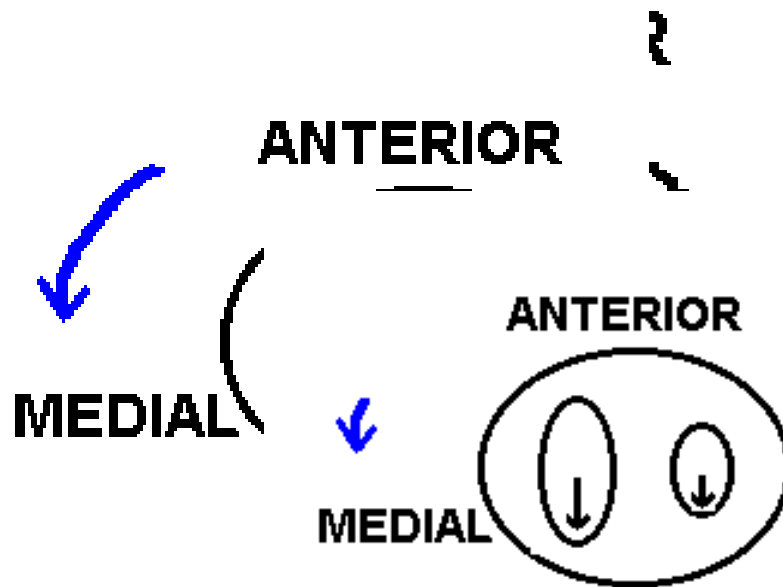
Screw-home mechanism

- From 20 degrees of flexion to full extension
- Anterior tibial glide
- Tibia rotates externally

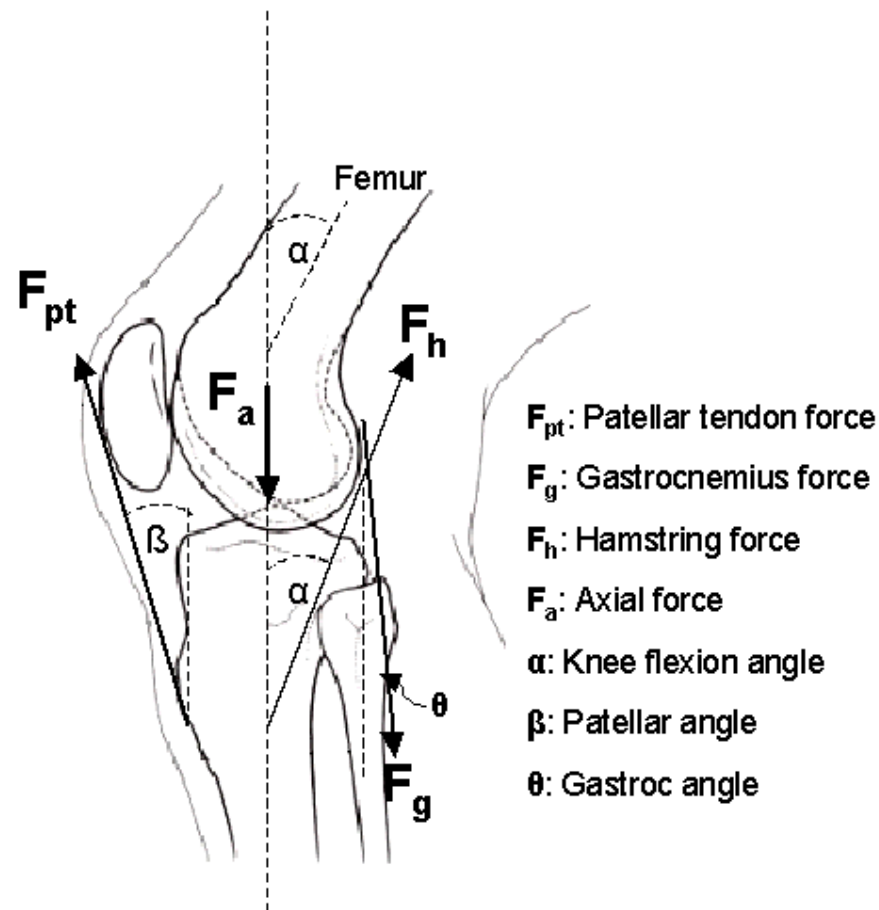


Screw-home mechanism

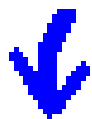
- Reverses in knee flexion
- From full extension to 20 degrees of flexion
- Posterior tibial glide
- Internal rotation of tibia



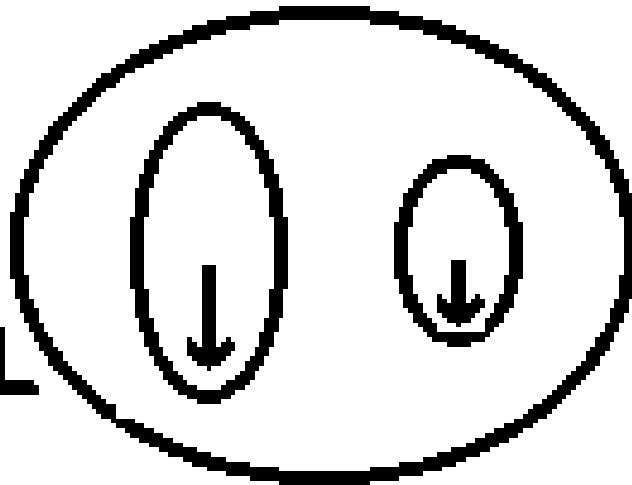
Forces added to the model



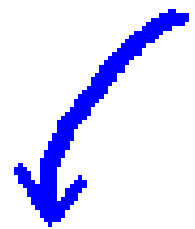
ANTERIOR



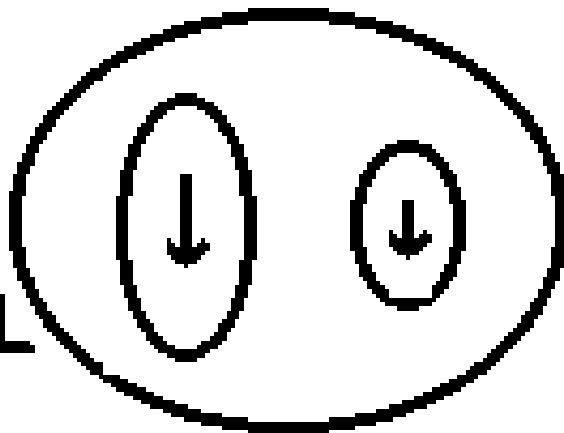
MEDIAL

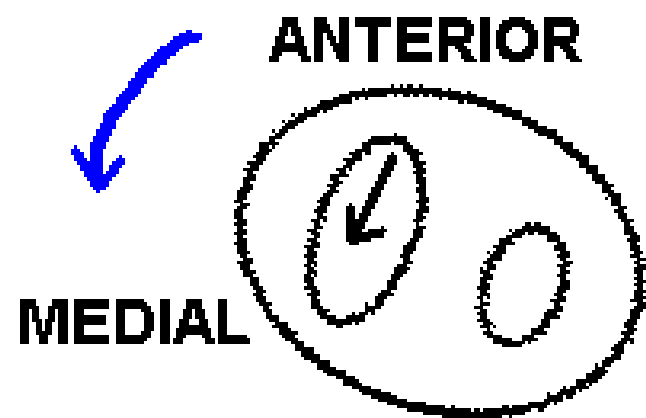


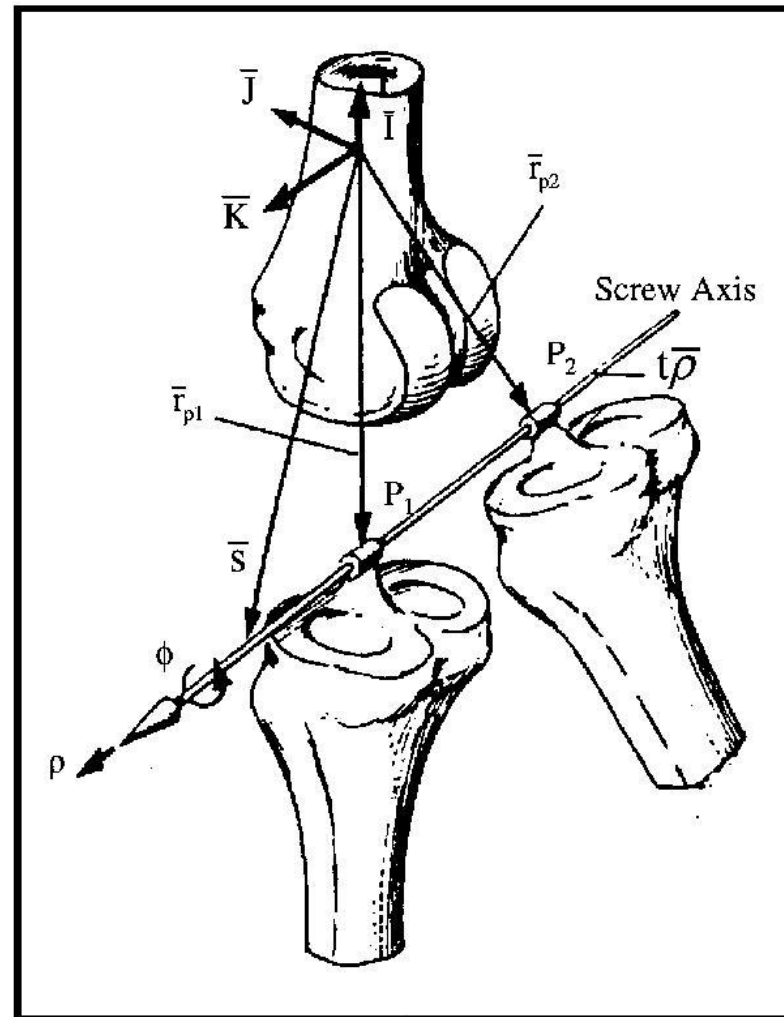
ANTERIOR

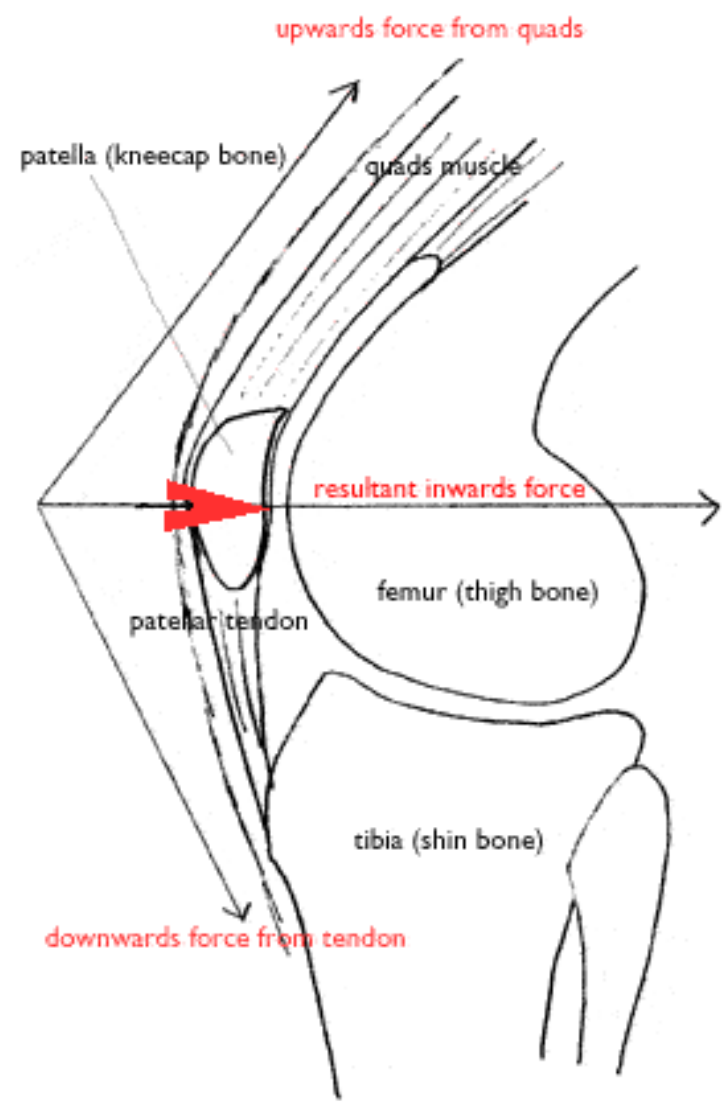


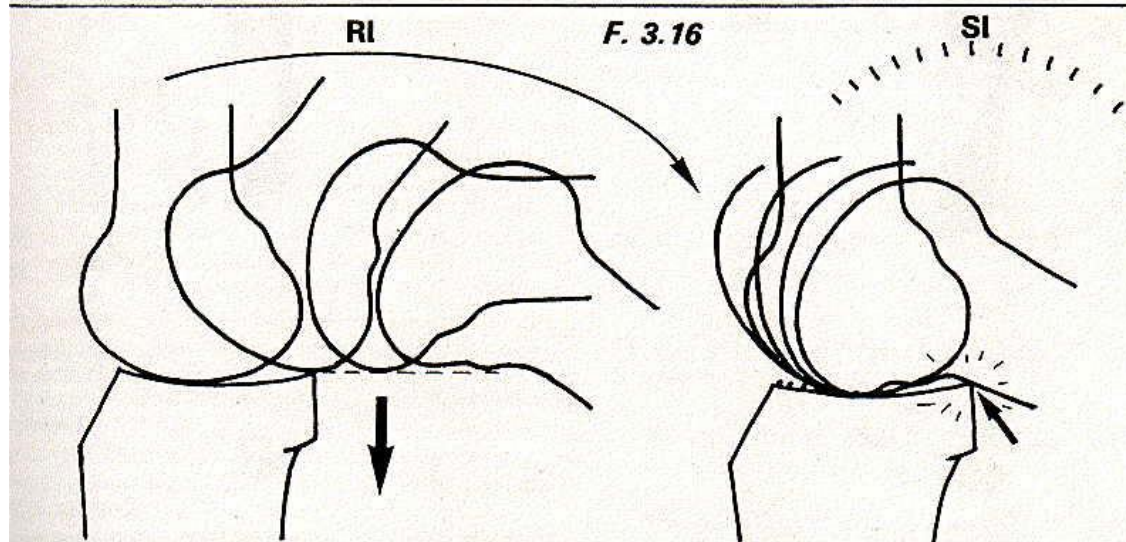
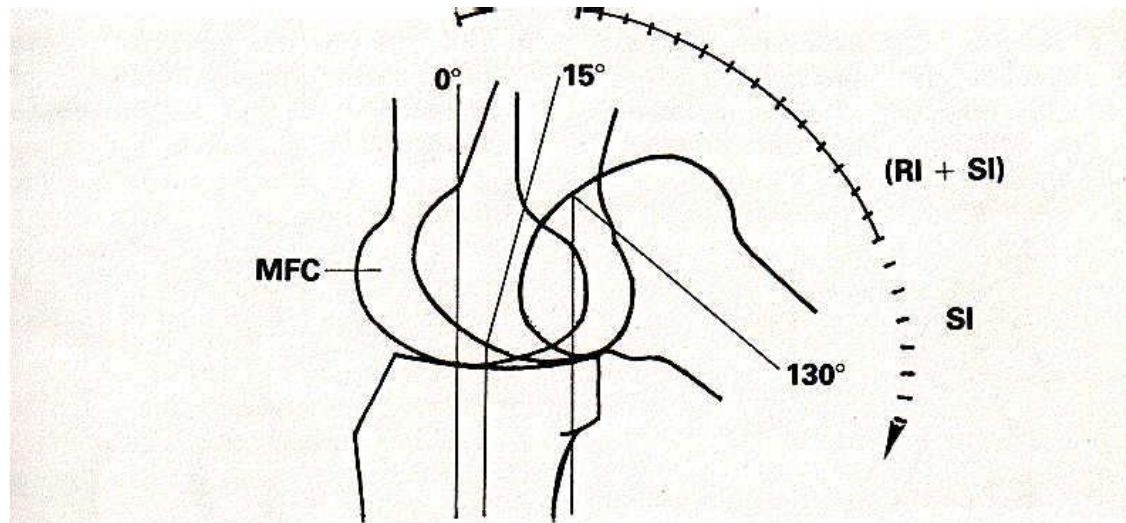
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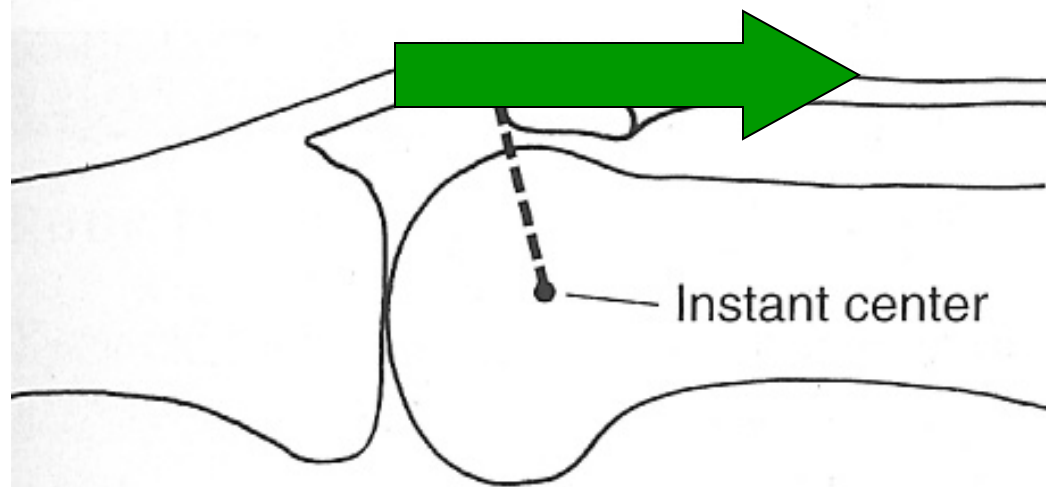




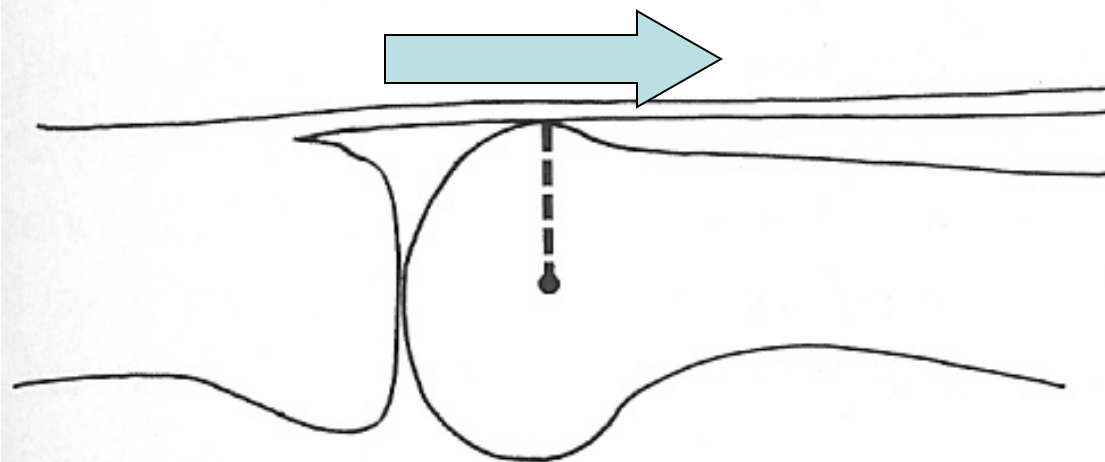








Normal



After Patellectomy

